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SAB Review Draft

Multimedia, Multipathway, and Multireceptor Risk Assessment (3MRA) Modeling System

Volume I: Modeling System and Science

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Multimedia, Multipathway, and Multireceptor Risk Assessment (3MRA) Modeling System

Volume I: Modeling System and Science

prepared by

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This document is the first volume of a five-volume set. This volume describes the conceptual design, scientific rationale, and supporting data that are the foundation for the 3MRA modeling system. Volume II describes the data developed and used to run the 3MRA modeling system. Volume III describes the approach to quality assurance, including verification and validation activities ranging from extensive peer reviews to multimedia model comparisons. Volume IV describes the methodology used to evaluate sensitivity of model parameters and characterize different types of uncertainty in the 3MRA modeling system. Volume V describes the technology design and includes a user's guide.

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1.0 Introduction

1.1 Background

The U.S. Environmental Protection Agency (EPA) Office of Solid Waste (OSW) is responsible for managing solid and hazardous waste as specified by the Resource Conservation and Recovery Act (RCRA) of 1976 and subsequent legislation, such as the Hazardous and Solid Waste Act (HSWA) of 1984. These acts and the programs developed to implement them were designed to protect human health and the environment. Thus, many of the regulatory decisions within the RCRA programs are based, at least in part, on the human health risk and environmental impacts of the regulatory options under consideration.

In recent years, as the RCRA program has evolved, and as new risk assessment methods have been developed, EPA's need for improved risk assessment models has greatly increased. The RCRA programs initially addressed only releases to ground water from land disposal operations and releases to air from waste incinerators and other types of boilers and industrial furnaces. However, the RCRA programs have expanded in scope over the years to encompass hundreds of constituents, thousands of waste streams, and many types of waste management practices, ranging from recycling and reuse to disposal and destruction techniques. Thus, new risk assessment models were needed to assess the types and magnitude of risks that fall under the broad purview of the RCRA programs.

In addition, in the mid-1990s, several groups within and outside of EPA came forward with recommendations or guidance for improving risk assessment methods. In 1996, EPA issued new guidelines for conducting exposure assessments and risk assessments (U.S. EPA, 1996f,g). These guidelines focused on improving the science underpinning the risk or exposure assessments that were being conducted, as well as improving the methods for characterizing the uncertainty in the risk estimates that are generated. In 1997, the Presidential/Congressional Commission on Risk Assessment and Risk Management (CRARM, 1997) issued a report on improving risk assessment methods used by the federal government. Also, EPA's Science Advisory Board (SAB) reviewed and commented on a number of EPA risk assessments and models, including the dioxin and mercury risk assessments (U.S. EPA, 1996e, 2000).

One of the assessments that the SAB reviewed was a national-level risk assessment effort conducted by OSW to support the proposed exit levels in the 1995 Hazardous Waste Identification Rule (HWIR). The proposed HWIR (60 FR 66344, December 21, 1995) was, in part, designed to establish contaminant-specific exit levels for low-risk solid wastes. Under this proposal, generators of listed hazardous wastes that could meet the new concentration-based criteria defined by the HWIR methodology would no longer be subject to the hazardous waste management system specified under Subtitle C of RCRA. This established a risk-based "floor" for low-risk hazardous wastes that would encourage pollution prevention, waste minimization, and the development of innovative waste treatment technologies. The rule also sought to reduce possible overregulation arising from the "mixture" and "derived-from" rules promulgated earlier.

The SAB concluded that the methodology proposed for the HWIR assessment "lacks the scientific defensibility for its intended regulatory use," and the subcommittee made the following recommendations for establishing a scientifically defensible risk-based methodology applicable at the national level for the waste program (U.S. EPA, 1996a):

- Develop a true multipathway risk assessment in which a receptor receives a contaminant from a source via all pathways concurrently, a receptor is exposed to the contaminant via different routes, and the dose corresponding to each route is accounted for in an integrated way;
- Use a methodology that maintains mass balance;
- Before implementing the model, focus a validation effort on a few key exposure pathways of concern and those parameters that have a major impact on risks to human health and the environment;
- Examine model parameters systematically to ensure a consistent and uniform application of the proposed approach, and further, to ensure that the full suite of uncertainties is addressed in the final methodology;
- Discard the proposed screening procedure for selecting the initial subset of contaminants for ecological analysis and instead require that a minimum data set be satisfied before ecologically based exit criteria are calculated;
- Seek substantive participation, input, and peer review by EPA scientists and outside peer-review groups as necessary to evaluate the individual components of the methodology in much greater detail; and
- Reorganize and rewrite the documentation for both clarity and ease of use.

The SAB review findings led to a joint decision between OSW and EPA's Office of Research and Development (ORD) to develop an integrated and improved source, fate and transport, exposure, and risk modeling tool that could be used to support national assessments and regulatory actions. In 1997, OSW and ORD began working together on the development of such a tool: the Multimedia, Multipathway, and Multireceptor Risk Assessment (3MRA) modeling system. The design of the 3MRA modeling system

- Follows the risk paradigm and recent EPA guidance and scientific recommendations;
- Addresses major review comments of the HWIR analysis in 1995, specifically those concerning the need for a true multipathway risk assessment methodology, the conservation of mass, the validation of the methodology and its components,

and the substantive participation of EPA scientists and outside peer-reviewer groups in evaluating the methodology and its components;

- Is based on current science and a state-of-the-art modeling framework that facilitates consistent use of sound science models, controls model sequencing, facilitates data exchange, and provides data analysis and a results visualization tool;

- Has multimedia and multipathway exposure and risk assessment capabilities;

- Has human health and ecological exposure and risk characterization capabilities;

- Is based on a mass balance approach within the waste management units (WMUs);

- Represents variability in environmental fate and transport, exposure, and risk;

- Can quantitatively define the degree of protectiveness for a specific risk value (e.g., protective of 95 percent of the exposed population at a risk level of 10^{-6});

- Uses site-based data (i.e., actual geographic locations and associated environmental and population characteristics) to the extent available;

- Is applicable to multiple scales of analysis, including regional and national;

- Is capable of assimilating new science and component modules in the software system;

- Reflects quality assurance and control protocols and is reproducible; and

- Has a verified approach and components that have been compared with other analytical solutions, numerical models, and/or field data.

In addition to the above discussion on the background of the 3MRA modeling system, this section is intended to provide the necessary context to review the science modules (Sections 2 through 16 of this Volume), the databases developed to support the modeling system (Volume II), the quality assurance approach for the technology, science modules, and data (Volume III), and the sensitivity and uncertainty analyses conducted to date (Volume IV). The 3MRA technology is discussed in detail in Volume V of this report, *Technology Design and User's Guide*. The remainder of Section 1 provides an overview of the overall design goals, system technology and architecture of the science modules, and output generated by the modeling system to support national-scale, risk-based decision making. In keeping with EPA's goals for easy accessibility and clear and transparent documentation, EPA has released Version 1 of the 3MRA modeling system available on CD ROM from OSW, and on ORD's Center for Exposure Assessment Modeling (CEAM) Web site (http://www.epa.gov/ceampubl/mmedia/index.htm), along with all technical documentation and data files (http://www.epa.gov/epaoswer/hazwaste/id/hwirwste/risk.htm).

1.2 Overall Design Goals of the 3MRA Modeling System

The overall design goals for the 3MRA modeling system reflect one of the most important regulatory goals for OSW: determining when the chemical constituent concentrations in a waste stream, as managed, may pose unacceptable risks to human health and the environment. For example, certain generic waste streams are currently being managed as hazardous waste because of the way they were captured by the hazardous waste regulations (under Subtitle C of RCRA). However, some subset of these waste streams may not pose significant health or ecological risks if they are disposed of as non-hazardous industrial wastes (under with Subtitle D regulations). To quantify specific concentration-based criteria for determining which waste streams may “safely” exit the hazardous waste “cradle-to-grave” program, EPA could assess the potential health and ecological risks related to the management in Industrial Subtitle D units of certain wastes with low concentrations of hazardous contaminants. Waste streams containing concentrations below EPA-specified thresholds could exit the hazardous waste system; those containing concentrations above the thresholds would remain in the hazardous waste program.

Using the 3MRA modeling system, constituent-specific distributions could be generated of cancer risks or hazards to humans and hazards to ecological receptors living in the vicinity of Industrial D waste sites that might manage exempted wastes. The 3MRA modeling system can produce national-level statistics that characterize risks and provide exit-level waste concentrations for a contaminant that meet specific criteria of policy makers. These policy criteria might include the following:

- The level of acceptable risk (e.g., 1 in 1 million),
- The probability of protection (e.g., 95 percent of the exposed population at a site),
- The probability that a particular site is protective at the risk level and population protection level specified,
- The risk to the exposed population at various distances from the site (e.g., 1,000 m),
- The risk to various receptor types (e.g., children, farmers, rare and endangered species), and
- The risk to each receptor type by each exposure pathway (e.g., inhalation of ambient air, ingestion of ground water used as drinking water).

To ensure that the modeling system could provide this type of information, OSW and ORD engaged in a series of discussions to identify the key functional requirements for the model, as well as the requirements for scientific defensibility that would shape the 3MRA modeling framework. The requirements are summarized below represent the core design decisions for the 3MRA modeling system.

1.2.1 Key Functional Requirements

The 3MRA modeling system is intended to be one of EPA's next generation of multimedia exposure and risk models to support regulatory decisions. EPA designed the modeling system specifically to meet the needs of OSW programs, but the model has the flexibility to be used for many EPA applications. In particular, it was designed to provide risk managers with information, at a national level, on exposure and risk to human and ecological receptors from the release of hazardous contaminants from the management of industrial wastes. Specific functional requirements for the design of the 3MRA modeling system include the following:

- ***Multiple Contaminants***. Many different constituents may be present in wastes regulated under RCRA. Therefore, the 3MRA modeling system was designed to use current science to model more than 400 constituents with very diverse physical-chemical properties and effects on humans and the environment. The science on which the fate and transport of contaminants in the environment is based differs for various types of constituents. In addition, the nature of national assessment methodologies requires the ability to adjust some physical-chemical properties that are dependent on environmental variables such as temperature. Within specified constraints, constituents were grouped in the following five categories: metals, organic chemicals, dioxin-like chemicals, mercury, and other special case chemicals, such as those that are miscible in water or are metabolized quickly. Even within these categories, distinctions must be made regarding bioaccumulation, metabolism, and transformation for organics and regarding congener-specific properties for dioxin-like chemicals.

- ***Multimedia.*** Traditionally, land disposal (i.e., landfilling) and destruction by combustion sources have been the predominant waste management scenarios used in the RCRA program. Release of a constituent to ground water is typically the primary pathway modeled for landfilling, and releases to air are the primary pathway modeled for combustion sources. However, a large quantity of hazardous and industrial wastes are managed as wastewaters in surface impoundments and tanks. Releases to air can be significant for some constituents managed in these types of WMUs. Similarly, the ash from combustion and the sludge from tanks and impoundments are sometimes applied to land, as are some organic wastes. For the 3MRA modeling system, new source models for surface impoundments, tanks, landfills, land application units (LAUs), and waste piles were developed to provide the ability to model releases to all environmental media (air, soil, and ground water).

- ***Multipathway.*** Once a contaminant is released to the environment, it is important to follow the transport and fate of the contaminant through all environmental media in order to capture all relevant pathways of exposure. In a national application of such a model, what may be a driving pathway in one type of environmental setting may be a small contributor to exposure in another. Also, given the broad range of chemical properties being considered, the ground water ingestion pathway may be important for one contaminant, the inhalation of ambient air for another contaminant, the ingestion of contaminated food crops or prey for another, and the ingestion of fish from contaminated streams for another. Therefore, the modeling system needed to include as many pathways as was feasible and scientifically defensible.

- ***Multireceptor***. The 3MRA modeling system was designed specifically to quantify the risk to ecological and human receptors. The 3MRA modeling system uses a site-based approach, in which various types of human and ecological receptors are spatially delineated around a WMU. This spatial component of the 3MRA modeling system is critical in analyzing the differential effects on human and ecological receptors in all media across the study area.

- ***Variability and Uncertainty***. Quantifying variability and uncertainty in exposure and risk estimates is an important capability of any modeling system. The 3MRA modeling system was designed with a two-stage Monte Carlo analysis capability, which enables users to distinguish between variability and uncertainty in input variables. In addition, EPA has conducted extensive sensitivity analyses and benchmarking against other similar models or model in order to understand the limitations and uncertainty of this modeling system.

- ***Programmatic Needs***. The 3MRA modeling system was designed with several specific science and technology requirements agreed upon by OSW and ORD. Key requirements included scientific defensibility (see Section 1.2.2) and a technology design that was adaptable to a wide range of future applications and the capability to incorporate into the system legacy codes that have been extensively peer reviewed and used in support of regulatory activities at EPA (see Volume V of this report). The following regulatory models or components of regulatory models have been incorporated into Version 1 of the 3MRA modeling system: Industrial Source Complex Short-Term Model, Version 3 (ISCST3) (U.S. EPA, 1995) for air dispersion and deposition; EPA's Composite Model for Leachate Migration with Transformation Products (EPACMTP) (U.S. EPA, 1996b,c,d, 1997) for subsurface transport; and EPA's Exposure Analysis Modeling System II (EXAMS II) (Burns et al., 1982; Burns, 1997) for surface water transport.

1.2.2 Scientific Defensibility Requirements

To address scientific defensibility, EPA implemented a systematic quality assurance program throughout the conceptual design, development, and application of the 3MRA modeling system. The program was designed to build confidence in the underlying science and technology, ensure that the system could produce reliable risk results, and to characterize the uncertainty and variability inherent in multimedia modeling. The major components of that program are described briefly below.

- ***Technical Review.*** The development of the 3MRA modeling system evolved from the ORD/OSW Integrated Research and Development Plan for the Hazardous Waste Identification Rule (U.S. EPA, 1998). This report represented the collaboration between ORD and OSW to define (1) the assessment strategy for national-scale risk analyses and (2) the design specifications of the 3MRA modeling framework. A critical component of this blueprint for the 3MRA modeling system was the technical review cycles required for the databases, science modules, and system technology. This iterative process ensured that the conceptual approach and implementation of the 3MRA modeling system was consistent and defensible across all data sources and science modules, as well as the software technology.

- ***Peer Review.*** The overall technical approach and each science-based module included in the 3MRA modeling system were peer reviewed. Teams of peer reviewers (at least three per module) provided critical feedback about the science-based modules. More than 45 independent experts reviewed the science modules to ensure that the theoretical concepts describing the processes within the release, fate, transport, uptake, exposure, and risk components were adequate representations of the processes to be evaluated. The peer review cycle has been an ongoing process since 1999, and is described in Volume III of this report. The process has recently included the acceptance of *The 3MRA Risk Assessment Framework - A Flexible Approach for Performing Multimedia, Multipathway, and Multireceptor Risk Assessments Under Uncertainty* (in press) by the International Journal of Human and Ecological Risk Assessment (Marin, et al., n.d.).

- ***Verification.*** All software components and databases underwent a series of tests to verify that the software and data were performing properly. At the heart of this protocol was the requirement that each component of the modeling system include a designed and peer-reviewed test plan that was executed by both the module developer and a completely independent modeler (i.e., someone who did not participate in the original module development). Prior to testing, each of the test plans was reviewed and revised, as appropriate, to ensure that the tests were comprehensive and addressed all of the major functions of the software and supporting databases. These procedures, test plans, test packages, and test results are fully documented and available to the public. Verification efforts for the 3MRA modeling system are described in Volume III of this report.

- ***Validation.*** True validation of the 3MRA modeling system for a national application would require validation over the full range of environmental settings that are relevant to the application. However, determining whether the modeling system is valid for the full range of settings was not possible because EPA was unable to find such a data set. Instead, individual modules and data sets were validated when appropriate data could be identified. A number of science modules were based on existing models/methods that had already been validated using field data and, therefore, prior validation studies and data were evaluated to determine their relevance to the 3MRA application. The use of legacy models developed by EPA was considered to be an important aspect of the overall validation efforts. A description of the validation efforts of the 3MRA modeling system, including the system, science modules, and supporting data, is documented in Volume III of this report.

- ***Model Comparison.*** The 3MRA modeling system is undergoing a comparative analysis with EPA's Total Risk Integrated Methodology (TRIM) (U.S. EPA, 2002), which is currently under development. The objective of the model comparison effort is to increase confidence that the 3MRA modeling system produces estimates consistent with other multimedia models. A description of the model comparison efforts is documented in Volume III of this report.

- ***Environmental Data Comparison.*** Modeled results from the 3MRA modeling system are being compared with a multimedia data set for an actual industrial site for which media monitoring data and field data (e.g., fish concentrations) were available for mercury. These data represent a snapshot of mercury in the surrounding environment rather than a continuous measure of its presence during the operation of an industrial facility. Although the data set does not vary as a function of time or include estimates of exposure and risk, the comparison will yield important insights into the performance of multimedia models. A description of this comparison effort is documented in Volume III of this report.

- ***Representative National Data Set.*** A comprehensive data collection approach was developed to parameterize the modeling system for 201 sites in accordance with the site-based approach described in this document. The site-based data are intended to provide a representative data set for a national-level assessment and, to a large degree, serve as the test data set for the 3MRA modeling system. This data collection plan described the general collection methodology for the major types of data (for example, facility location, land use, soil characteristics, and receptor locations), including quality assurance and quality control procedures and references for data sources. The data collection effort for the representative national data set is documented in Volume II of this report.

- ***Sensitivity and Uncertainty Analyses.*** As the level of assessment complexity grows, it becomes both more difficult and more important to establish comprehensive and quantitative expressions of uncertainty. Uncertainty analysis can be a difficult task, especially for complex, integrated, multimedia models such as the 3MRA modeling system. Thus, a formal program focusing on sensitivity and uncertainty analysis for complex modeling systems was initiated at ORD. The early focus of this program is the investigation of parameter sensitivities and system uncertainties within the 3MRA modeling system. To facilitate this evaluation, ORD has recently developed a Windows-based parallel computer cluster. This supercomputer for Modeling Uncertainty and Sensitivity Evaluation (SuperMUSE), comprising 176 client PCs and supporting software infrastructure that allows exhaustive experimentation of the 3MRA modeling system. A complete description of the computational and software framework for conducting evaluation strategies for the 3MRA modeling system, the SuperMUSE system, and initial results of the evaluation efforts are presented in Volume IV of this report.

1.3 Overview of the 3MRA Modeling System Technology

The 3MRA modeling system was developed as a predictive tool to provide risk assessment support for the types of risk management decisions that are made within OSW. OSW applies risk assessment modeling tools in a variety of situations; one application is the conduct of site-based national-level risk assessments to support rulemaking for the identification of hazardous waste. Consequently, the 3MRA modeling system needed to be able to model waste management environmental settings that are representative of the range of environmental settings found in the United States, and within this broad range of settings, to simulate the release, fate, and transport of many contaminants in waste undergoing a range of physical and biochemical processes. More than 400 constituents are regulated under the RCRA programs. EPA needs to consider the impacts of these released contaminants on humans and the environment within the broad range of environmental settings. This requires a modeling tool that encompasses releases to all media, transport within those media, uptake in terrestrial and aquatic food webs, and exposure of specific receptors to contaminated media and food items.

The 3MRA modeling system predicts human and ecological risks at a statistically selected number of waste management sites across the United States, and accumulates statistics on the estimated risks at each of these sites in order to generate national distributions of risks. This site-based modeling is applied for each combination of site, WMU, chemical, and concentration level in a given waste stream. The assessment methodology is national in scale and site-based; that is, risks are assessed at individual sites across the United States and rolled up to represent a national distribution of risks. The resulting national distribution of risks forms the basis for determining waste stream constituent concentrations that satisfy criteria reflecting the percentage of nationwide receptors and sites that are "protective." When a simulation is complete, the risk estimates are organized into expressions of the probability of protection for different constituent concentration levels in waste streams. These risk outputs can be expressed as a function of several dimensions, for example, waste unit type, human receptor type, or the distance from the WMU. This ability to express risks and hazard as a function of different attributes provides the decision makers significant flexibility to ask "what if" questions leading up to final decisions. The primary technology requirements related to the application of the 3MRA modeling system to national level human and ecological risk assessments are as follows:

- Facilitate "plug and play" functionality throughout the modeling system to allow new science, modules, and data to be integrated into the system as the state of multimedia modeling science and software development continues to evolve;

- Conduct simulations using the 3MRA modeling system science modules and databases; each simulation representing a combination of site, chemical, WMU, and waste constituent concentration;

- Generate and store risk matrices, i.e., risk estimates as a function of site, exposure area, exposure pathway, exposure route, contact medium, receptor, receptor type, and Monte Carlo realization; and

- Provide a flexible framework that can accommodate alternate policy formulations including different measures of protection, and both waste and leachate concentration regulatory limits.

Figure 1-1 provides an overview of the 3MRA modeling system design. As suggested in this figure, the system is constructed of components that manage the processing, flow, and storage of information through the system, including input/output files and a variety of supporting databases. At the top of Figure 1-1, the looping structure used to conduct national-scale analyses is summarized, including the site location loop, the WMU loop, the number of iterations, and the number of chemicals (which are simulated individually). The function of each component that serves to manage, process, and store information is summarized below. For additional information and details on the 3MRA technology, see Volume V of this report, *Technology Design and User's Guide*.

- **System User Interface (SUI).** This processor represents the user access point to the technology. Via the SUI, the user selects which combinations of sites, WMUs, constituents, and constituent concentrations in waste streams to be simulated, and the number of Monte Carlo simulations to be executed per site. The SUI also provides the user with the ability to configure the computer directory structure where individual components of the system are stored. Finally, the SUI manages the overall execution of the user-defined national assessment.

- **Site Definition Processor (SDP).** This processor performs all data retrieval from the site, regional, national, and chemical databases and organizes the data into a series of "site simulation files" that contain the input data for each of the seventeen science models. The site definition in the figure includes both the selection of site characteristics from data sources at multiple spatial scales (i.e., local, regional, and national), as well as the estimation and selection of chemical properties.

- **Multimedia Simulation Processor (MMSP).** This processor manages the invocation, execution, and error handling associated with the seventeen individual science models that simulate source release, multimedia fate and transport, food web dynamics, and human/ecological exposure and risk. The multimedia, multipathway simulation includes all of the science modules linked together to predict behavior of constituents from source release through exposure and risk.

- **Chemical Properties Processor (CPP).** This processor accesses the chemical properties database and either transfers or calculates all requested data. The CPP provides a single location within the modeling system where chemical data is available.

- **Exit Level Processor I (ELP I).** This processor assimilates the individual site risk results and builds a risk summary database containing data used to assess national protection criteria.

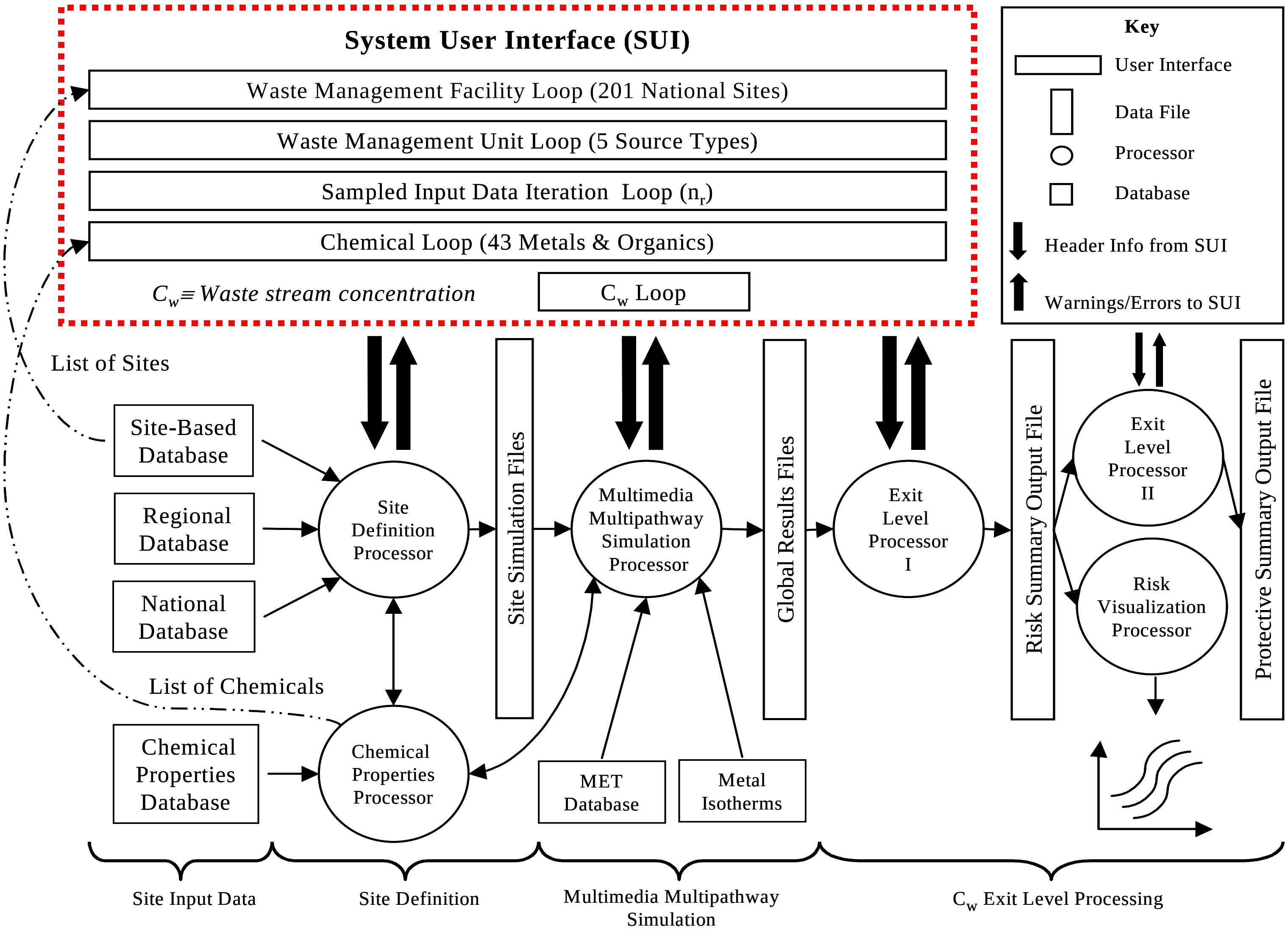


Figure 1-1. 3MRA modeling system design.

- **Exit Level Processor II (ELP II).** This processor reads the risk summary database created by the ELP I and generates, based on regulatory criteria, specific national exemption levels.

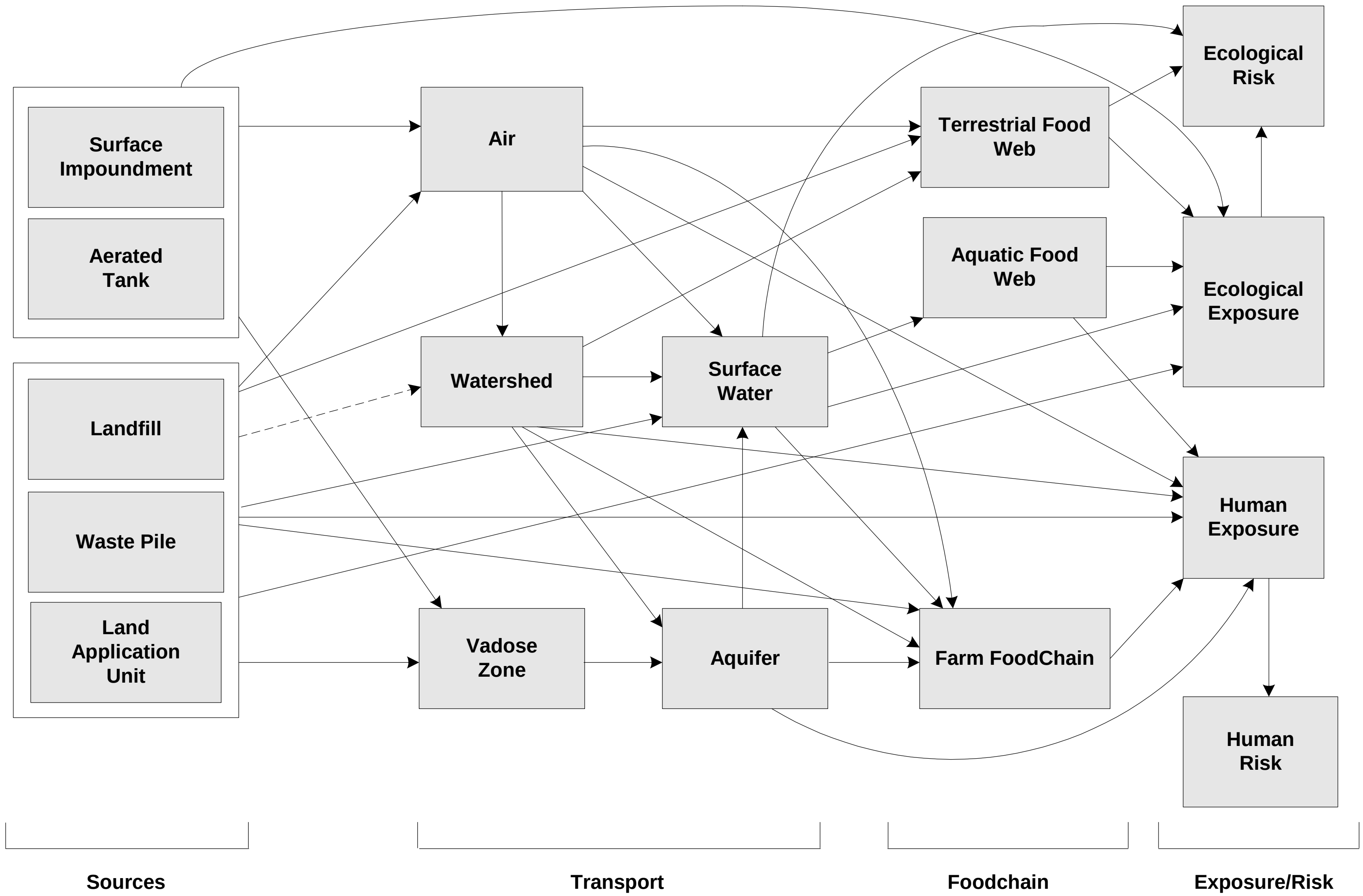
- **Risk Visualization Processor (RVP).** This processor is identical to the ELP II except that it presents results in graphical form.

1.4 Overview of the 3MRA Science Module Architecture

The MMSP can be viewed as an integration of science-based modules within an assessment strategy, i.e., procedures by which the modules are combined and applied to perform the risk assessment. The 3MRA modeling system simulates contaminant releases from a WMU to the various media (air, water, soil) based on the physical-chemical properties of the constituent, the characteristics of the modeled WMU, and the environmental setting (e.g., meteorological region) in which the facility is located. Once released from the WMU, the constituent is transported through environmental media and into biological compartments such as produce, beef, and fish. Human and ecological receptors included in the simulation may be exposed concurrently to contaminated media and food through multiple pathways and routes of exposure. For each receptor that is included in the simulation, the 3MRA modeling system performs risk/hazard calculations based on aggregate exposures modeled through space and time. The linkages among the science modules are depicted in Figure 1-2.

Figure 1-3 illustrates the conceptual layout of a typical 3MRA modeling system site. As suggested by this figure, the 3MRA site model represents a comprehensive multimedia approach to assessing the potential impacts of chemical releases from land-based WMUs. Shown in Figure 1-3 are the primary site layout features including the WMU at the center of the Area of Interest (AOI), which extends 2 kilometers from the unit boundary by default. The concentric distance "rings" are used to characterize risk as a function of distance. Other physical features of the site layout include watersheds, surface water networks, aquifers, ecological habitats, home ranges of resident ecological species, and human population distributions. These features are explicitly delineated and the relative connectivity determined for each site included in the assessment.

To the extent possible, the site layout data are based on site-specific information. However, site data were not available for all sites, and resource limitations associated with collecting the data are prohibitive. Consequently, the supporting data for the representative national data set are based on a tiered approach to data collection that includes site-specific-, regional-, and national-level data. During any execution of the modeling system, the most preferred data source is site-specific followed by regional. Finally, lacking either a site-specific or regional source of data, a national-scale statistical distribution of the variable is sampled and assigned to the site. Each of these databases is available to the system and, as new data become available, the system automatically acts on this hierarchy. In all, several hundred variables are required to model any given site.



The dashed line indicates that soil concentrations for the local (land-based source) and regional watersheds may be added together to estimate total soil concentrations for areas (e.g., habitats) that include both regional and local watershed components.

Figure 1-2. Linkages among the source, fate, transport, exposure, and risk modules fo the 3MRA modeling system.

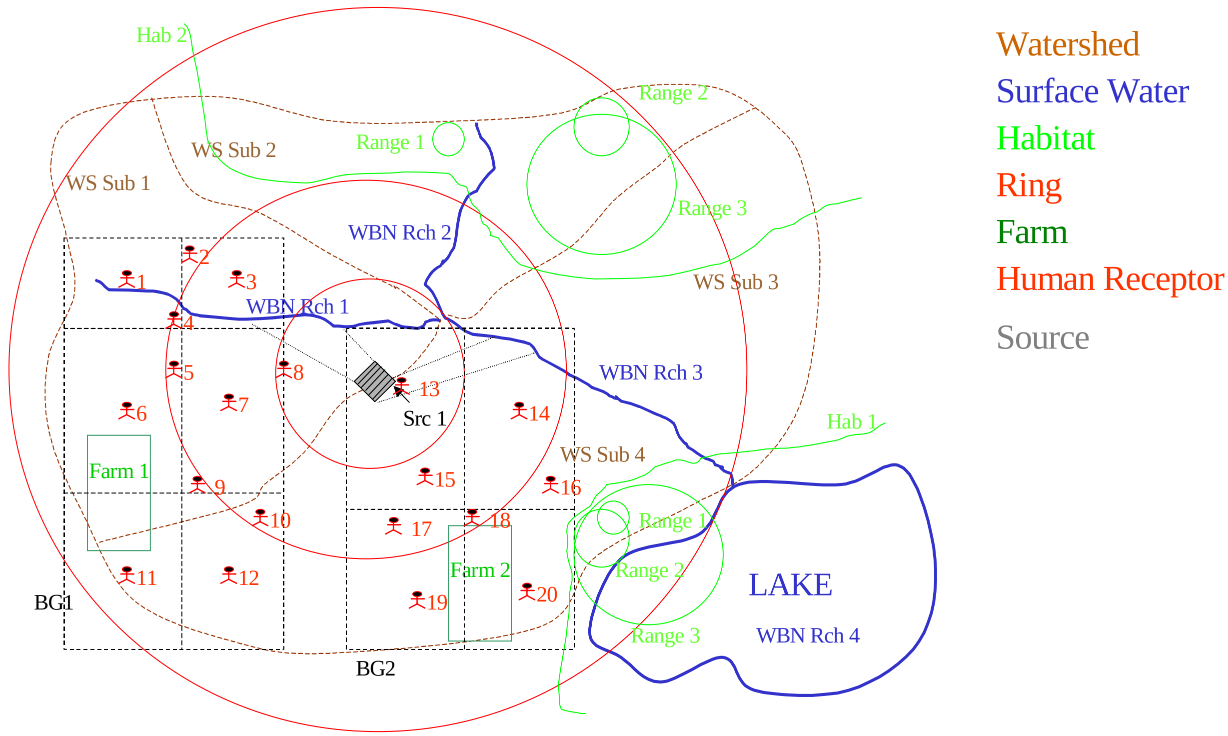


Figure 1-3. Conceptual layout of typical 3MRA modeling system site.

The loading of chemical constituents into the WMU, and the release of these constituents is simulated given the specific environmental characteristics delineated for site layout. The general steps in the site-based modeling are as follows:

- Simulate the loading of waste streams to land-based WMUs (including surface impoundments, landfills, land application units, waste piles, and aerated tanks) over the lifetime of the WMU;

- Simulate the release of constituents from the WMU to air (volatilization, particle re-entrainment), vadose zone (leaching), and watersheds and surface waters (overland runoff/erosion);

- Simulate the fate and transport of contaminants in and between major environmental media (air, watershed soils, vadose zone, ground water, surface water, and sediments);

- Simulate movement of contaminants through the farm food chain and aquatic and terrestrial food webs;

- Simulate human and ecological exposure via selected pathways (for human receptors, the pathways include air inhalation, shower air inhalation, ground water ingestion, soil ingestion, produce ingestion, beef ingestion, milk ingestion, fish ingestion, and breast milk ingestion for infants);

- Estimate human and ecological risk per receptor per pathway, and aggregate pathways and receptors as appropriate; and

- Repeat this sequence for each of a series of waste concentrations (C_w) to establish a quantitative relationship between C_w and risk or hazard.

1.5 Overview of Results Generated by the 3MRA Modeling System

The 3MRA modeling system produces two measures of protection that can be used in regulatory decision making. The first measure of protection is the nationwide distribution of risks for receptors of concern. Specifically, a concentration limit can be defined by the percent of nationwide receptors of concern that exceed a given risk level falling below a specific percent (e.g., 95 percent of all receptors across all sites, all pathways, and all WMU types within 2 km of the WMU incur a risk of 10^{-6} or less). Because the risk/HQ data at the site level is stored by indices including receptor type, exposure pathway, exposure ring distance, and WMU, it is possible to construct "views" of the national-scale protectiveness that reflect varying combinations of the indices. For example, protection measures can be applied to individual receptor types, combinations of receptor types, individual WMUs, etc. The second measure of protection is the nationwide distribution of sites that are protected. The limit can be defined under this protection measure by the percentage of protected sites nationwide that is greater than a given target level. A site is protective if the percentage of site-based receptors incur a risk/HQ less than a specified target value.

These measures of protection are combined in the 3MRA modeling system to allow a user to specify both the percentage of receptors nationwide as well as the percentage of sites that are to be protected (e.g., 95 percent of the sites are protective of 99 percent of the site-based receptors). To transform the site-based risk/HQ data into these national measures of protectiveness requires that the population counts be converted to population percentages and accumulated across all sites. With the risk/HQ indices preserved from the site results (i.e., receptor, pathway, ring distance, WMU, etc.), one can query the database containing results and generate the desired estimates of national protectiveness. Figure 1-4 presents an example corresponding to a query for a target risk level of 10^{-6} from the iterations corresponding to a waste concentration of 10^{-3} mg/kg. The figure indicates that there is a 5 percent chance that the level of protection (percent of receptors that would be protected at the target risk level for the given waste concentration) would be less than or equal to 85 percent. Similarly, there is a 25 percent chance that less than or equal to 93 percent of the receptors would be protected at the target risk level for the given waste concentration. Querying the output data base for different waste concentrations can produce the set of graphs such as those shown in Figures 1-5(a), (b), and (c). The figures show how the percent protection varies as a function of the target risk, the waste concentration, and the confidence limit, and can be used to select the waste concentration that meets a specified protection measure. These types of figures could also be produced for subsets of receptors to investigate the effects of selecting a waste concentration on secondary protection measures.

These risk results are produced by three processors (ELP I, ELP II, and RVP) that collectively accumulate, transform, and present information generated by the individual site-based risk assessments in the form of measures of site and population protectiveness at the national level. The ELP I reads individual site-based human-health risk and hazard and

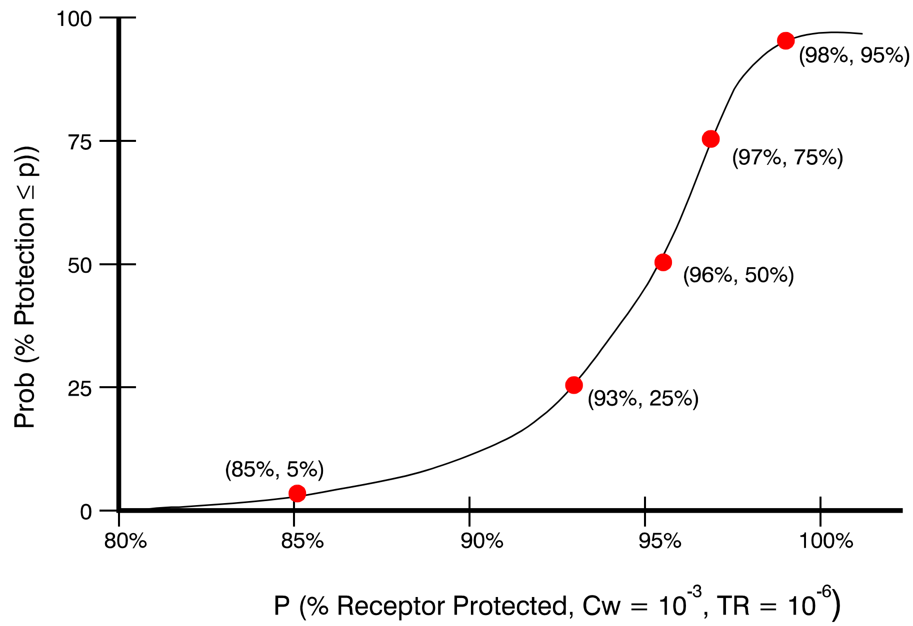


Figure 1-4. Probability that percent protection is less than P for a given waste concentration and target risk level.

ecological hazard results from the Human and Ecological Risk Modules, transforms the population counts into percentages of receptor populations protected, and stores the resulting risk and hazard information in a series of national Risk Summary Output Files (RSOFs). The RVP and ELP II processors query the RSOF in response to specific criteria for protective measures and produce graphical and tabular views of the trade off between levels of constituent concentration in waste streams and levels of protectiveness. With the combination of the RSOF database and the RVP/ELP II, the following type of question can be asked: "What is the maximum allowable constituent concentration for waste streams entering landfills such that at least 90 percent of all receptors at 95 percent of the sites nationwide incur a risk less than 10^{-6}?" In addition, the RVP allows the user to query and summarize the information stored in the RSOFs and graphically view the results of such queries. The user is prompted for a set of scenario attributes (that is, WMU type, constituent, distance, receptor type, cohort, etc.) and a level of protection (for example, risk of 1.0×10^{-6} or HQ of 1.0, etc.). The RVP uses this information to construct plots showing the probability of protecting human or ecological receptors (in the group of receptors defined by the scenario attributes) as a function of C_w. Using information obtained from the RSOFs, the probability of protecting a given receptor is determined by taking the group of receptors defined by the scenario attributes across all sites and plotting the percentage of those receptors that are protected (that is, have a risk or hazard value equal to or below the level of protection). Because each site is simulated with multiple realizations, several probability curves are plotted, each corresponding to the chosen confidence levels (for example, 5 percent, 50 percent, 95 percent). These plots are called Protective Summary Output curves. The RVP creates a Protective Summary Output curve for human risk, human hazard quotient (HQ), and ecological HQ, depending on the appropriateness and existence of the data for the constituent chosen. Figures 1-6 and 1-7 illustrate the human risk protective summary and ecological hazard protective summary plots produced by the RVP, respectively.

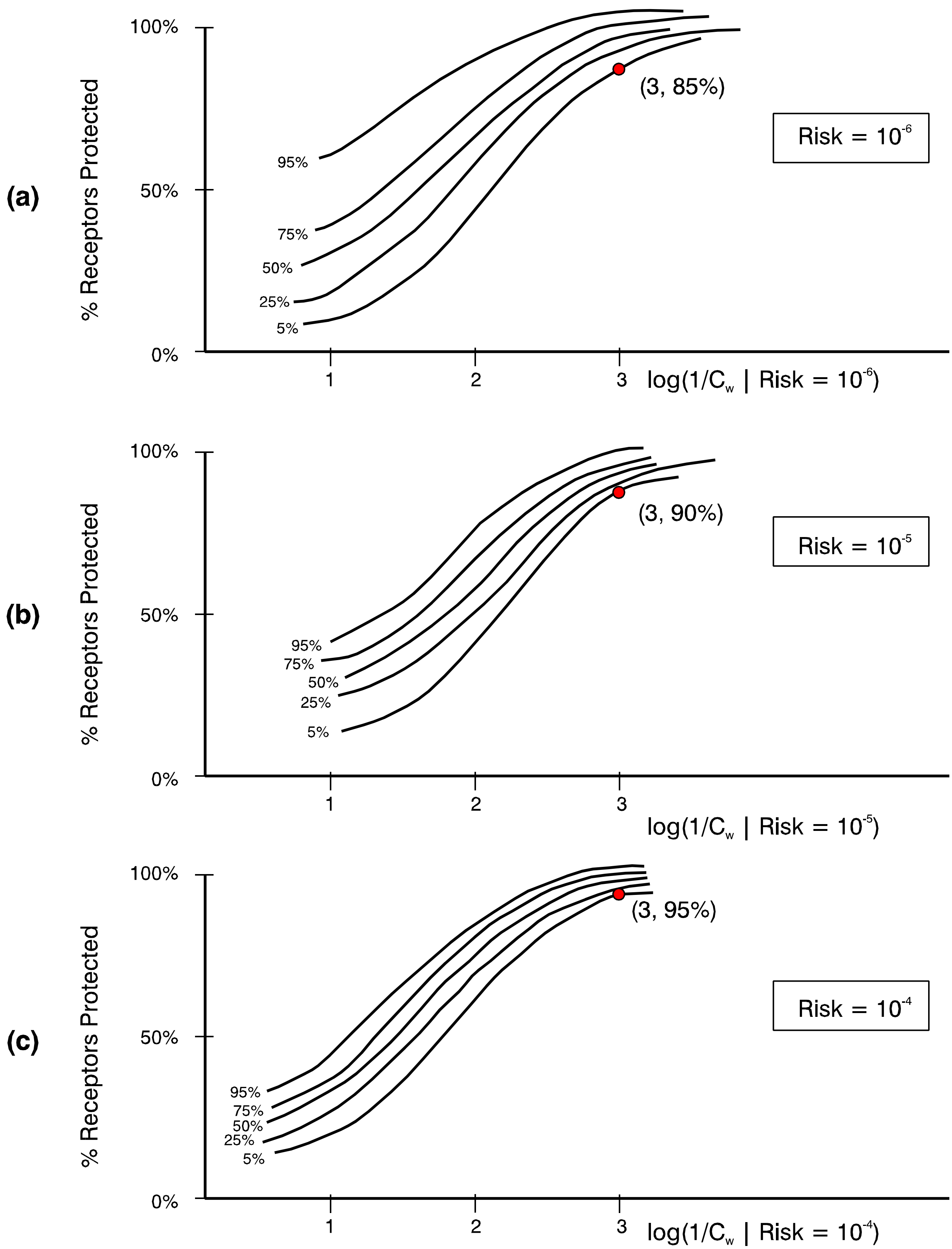


Figure 1-5. Percent of receptors protected for different waste concentrations and risk levels.

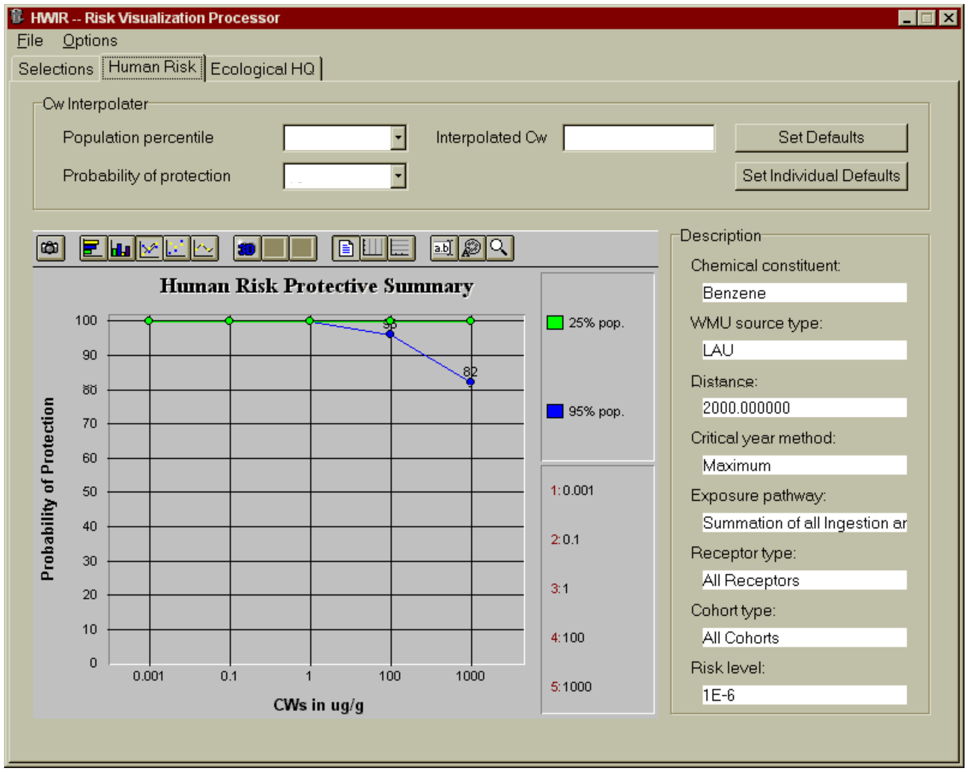


Figure 1-6. Protective Summary Output figure for human risk.

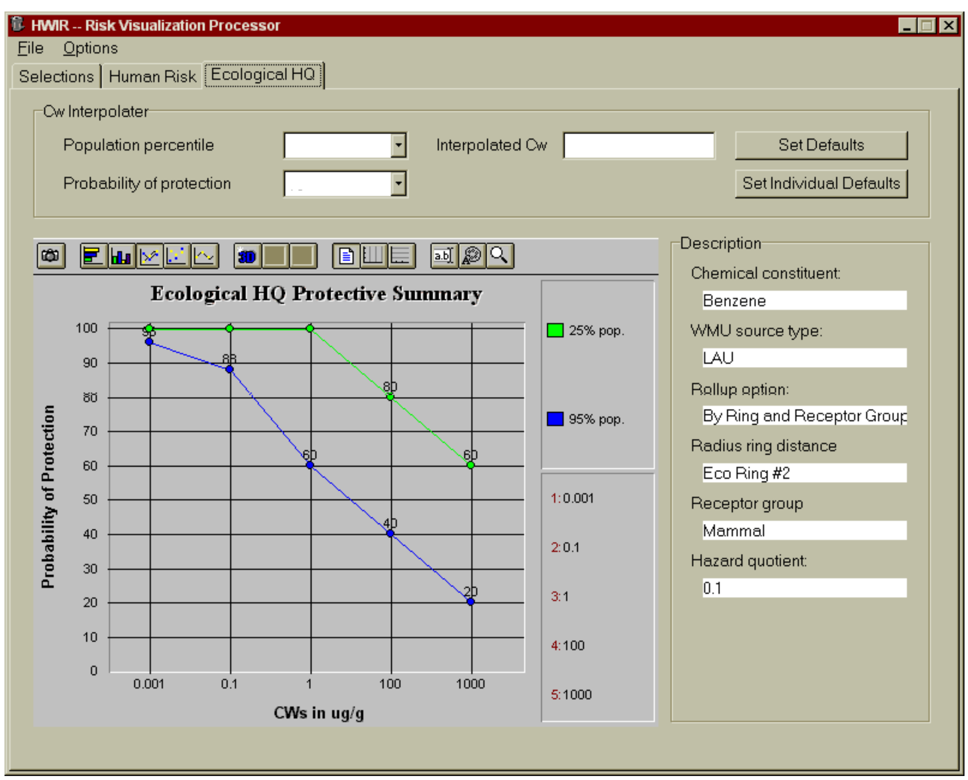


Figure 1-7. Protective Summary Output figure for ecological HQ.

1.6 Organization of This Document

The first three sections of this document (including this introduction) discuss the modeling system as a whole. Section 2 presents an overview of the modeling approach from a conceptual viewpoint and describes the assessment strategy and the general approach for its implementation. Section 3 provides a discussion of the development and use of spatial data in this model, which are fundamental to developing variability in exposure and risk estimates across a site.

The remaining sections describe the WMU source, fate and transport, exposure, and risk modules. Figure 1-8 shows which section describes each module. The five source modules are covered in two sections, one for land-based sources (LAUs, landfills, and waste piles) and one for wastewater systems (surface impoundments and tanks).

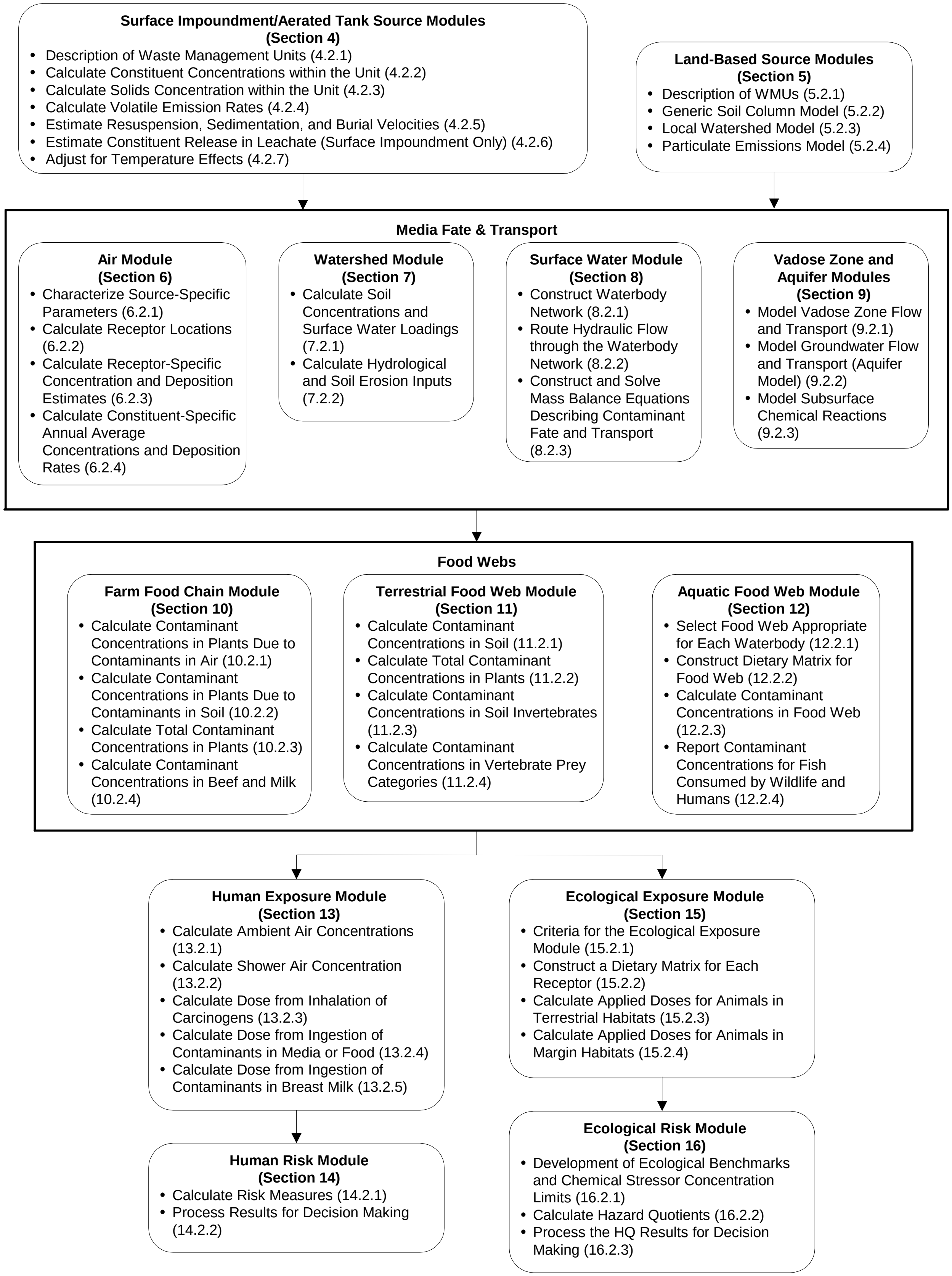


Figure 1-8. Document organization.

1.7 References

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2.0 Modeling Approach

The 3MRA modeling system is intended to be one of EPA's next generation of multimedia exposure and risk models, capable of modeling multimedia, multipathway, and multireceptor exposures. The 3MRA modeling system models multimedia exposures by simulating releases of a constituent to air, soil, and ground water, and then modeling the transport and fate of the constituent in each of these media and in the food webs associated with them. It models multipathway exposures by calculating simultaneous exposures of a receptor through multiple pathways, such as the ambient air, soil, food items, and drinking water, and summing them, when appropriate. It models multireceptor exposures by simulating a set of human receptors and a set of ecological receptors that characterize the receptor populations and behaviors of those receptors within an area of interest (AOI).

This section

- Presents an overview of the 3MRA framework—the conceptual approach and science that is implemented in the 3MRA modeling system;
- Describes the spatial and temporal scales that provide the context for the 3MRA framework; and
- Describes the design of the 3MRA modeling system as it pertains to constituents assessed and sources, exposure pathways, receptors, and human health and ecological exposures and risks modeled.

2.1 Overview and Conceptual Approach

Figures 2-1 and 2-2 illustrate the conceptual framework for human receptors and ecological receptors, respectively, in the 3MRA framework. This conceptual framework is further depicted in the flow diagram in Figure 2-3. These figures show the conceptual models for human and ecological receptors, beginning with the multimedia release of constituents from a waste management unit (WMU). Once released, constituents travel through environmental media that determine the exposure pathways for the analysis: air, ground water, watershed soils, and surface water. In addition, plants and animals take up contaminants either directly from the different media or through bioaccumulation of constituents in terrestrial and aquatic food webs. Eventually, human receptors are exposed to contaminants through inhalation or ingestion of contact media (i.e., contaminated air, soil, ground water, homegrown produce, beef and milk produced on a farm, and fish). The ecological receptors are exposed to contaminants through direct contact or ingestion of contact media (i.e., contaminated air, soil, surface water, and flora and fauna).

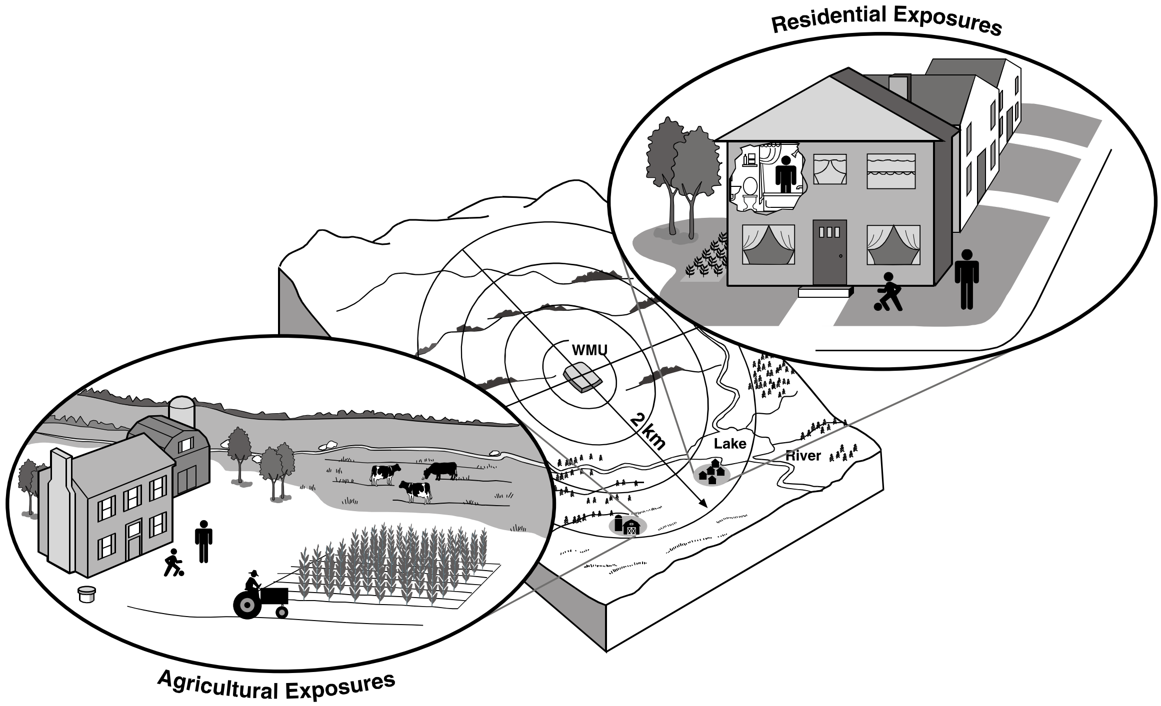

Figure 2-1. Conceptual framework for human receptors.

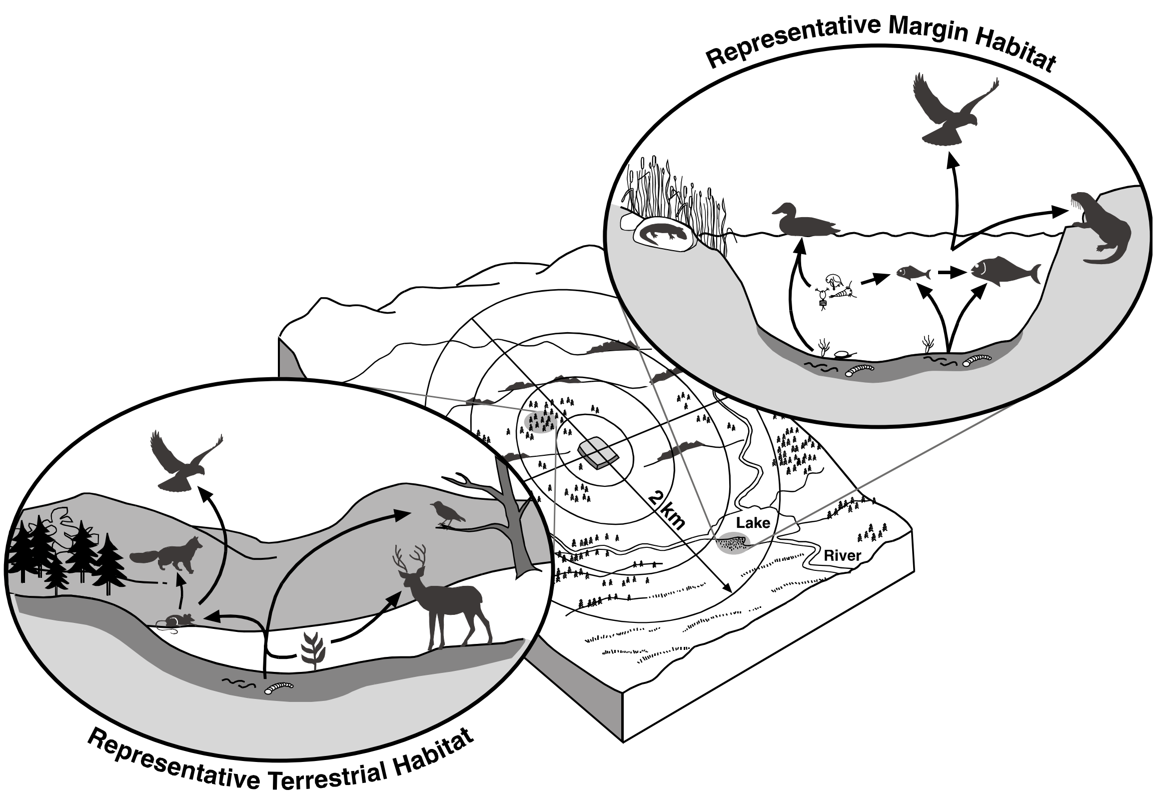

Figure 2-2. Conceptual framework for ecological receptors.

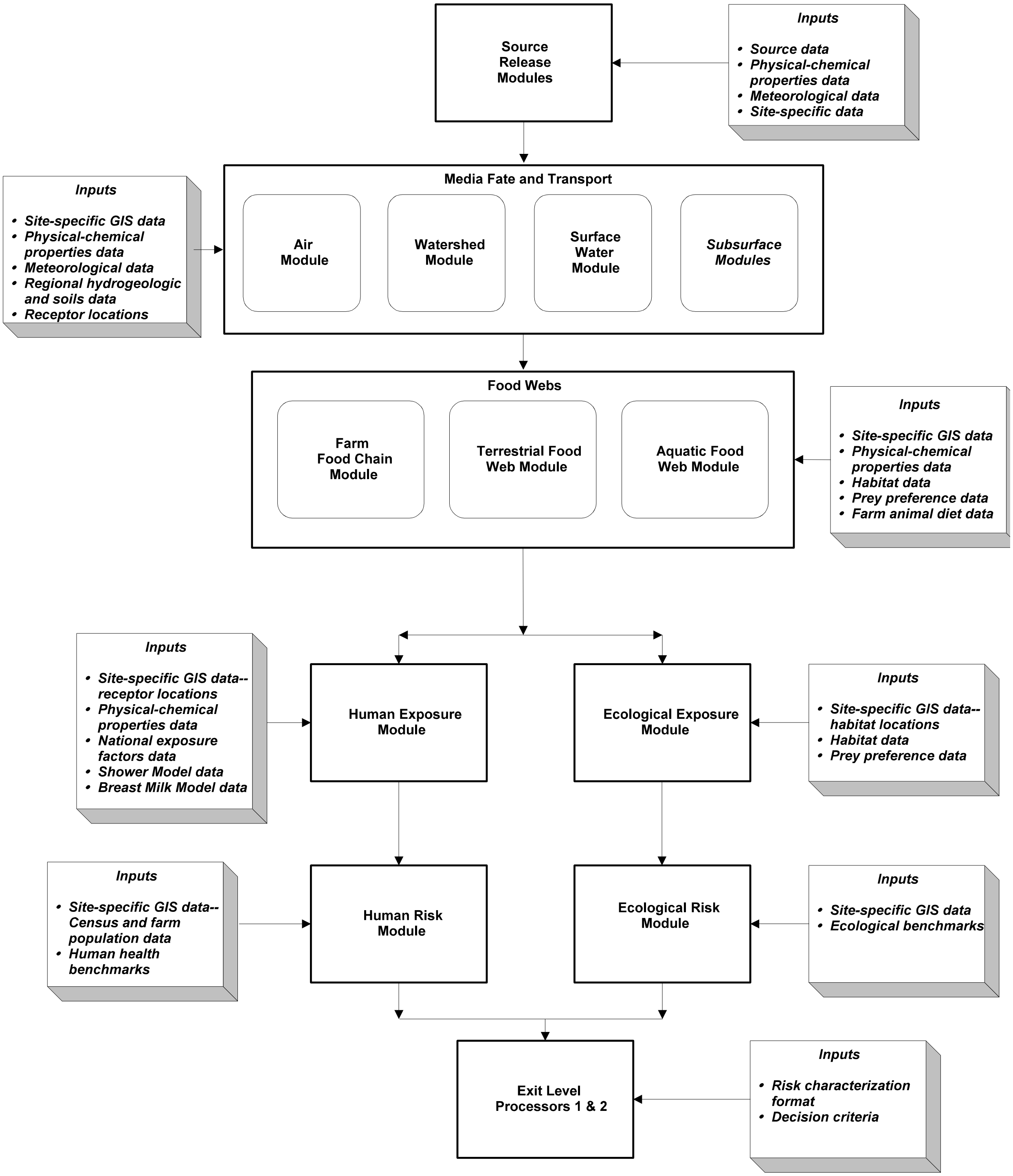

Figure 2-3. Conceptual framework of the 3MRA modeling system.

Several major strengths of the 3MRA modeling system and associated components are that they provide true multipathway functionality, maintain mass balance within the WMUs, model sorption and degradation processes, account for timing of exposures, use a regional site-based approach, and capture variability and uncertainty. These are discussed below.

True Multipathway Functionality. An overriding consideration in the development of the 3MRA modeling system was to provide a system with true multipathway functionality in which a receptor receives constituents from a source via all relevant pathways simultaneously. Constituents are released from a source to various media, depending on the characteristics of the source, the waste, and the surrounding environment. Once the constituents are in the air, soil, and ground water, they are transported through these media to the farm food chain, terrestrial food web, and aquatic food web, or they come into direct contact with a receptor via air, soil, or drinking water. At any given time, the contaminant concentrations to which a receptor is exposed in each medium or food item are evaluated simultaneously.

Mass Balance. The 3MRA modeling system maintains mass balance for a constituent within each type of WMU source. The system starts with a concentration of a constituent in the WMU and partitions the mass to vapor, liquid, and sorbed phases. The 3MRA modeling system continues this partitioning over the life of the WMU as wastes are added and constituents released. Constituent mass released via each phase is no longer available for partitioning to and release through other phases. The data used in partitioning include a range of constituent-specific partition coefficient (K_d) values representing various waste forms that are intended to reflect the full range of waste types (see Sections 4 and 5 for further discussion).

Sorption. The mobility of metals in the environment is dependent on the geochemical properties of the soil, surface water, and ground water to which they are released. To account for the metal-specific interactions with various environments modeled (wastes, soil, surface water, and ground water), the 3MRA modeling system uses nationwide distributions of key geochemical parameters, including metal sorption coefficients. See Volume II of this report for a discussion of the methods and data used to generate the sorption coefficients.

Degradation Processes. Both the source and the fate and transport modules in the 3MRA modeling system were designed to simulate biodegradation and constituent degradation. Full implementation of these processes is limited, primarily because of the limited availability of constituent-specific data on aerobic and anaerobic biodegradation rates. EPA has invested in the development of additional biodegradation rate data specifically to enhance the use of the 3MRA modeling system to support specific RCRA programs. Because of the large number of constituents regulated under RCRA, some constituents were designated as high priority because of their presence in high-volume wastes. Those high-priority constituents were addressed first. Volume II of this report provides a full discussion of the constituent-specific degradation data developed for the 3MRA model (see Sections 4 through 9 for further discussion of the degradation processes included in each of the modules).

Time Series Management. The 3MRA modeling system is designed to estimate the potential risk or hazard quotient (HQ) associated with a constituent that will be managed in a WMU throughout the WMU's operational life and beyond. Average annual exposures are specific to the constituent and environmental setting, and are estimated to occur for up to

10,000 years in some cases. The source modules simulate release of a constituent until the concentration in the WMU decreases to 1 percent of the maximum concentration or until a period of 200 years has been simulated; however, media concentrations are simulated until the constituent concentration in a particular medium (e.g., ground water) decreases to 1 percent of the maximum concentration for that medium or until a period of 10,000 years has been simulated. Therefore, modeling simulations can range from a few hundred years (for contaminants that move quickly in the environment, are not persistent, and do not bioaccumulate) to 10,000 years (for the most persistent and least mobile contaminants, such as some metals).

The risk results are evaluated over the entire modeling period, which could be up to 10,000 years. Evaluating peak doses over this time horizon allows the system to capture the slow movement of certain contaminants through the subsurface. Although the time frame for such travel may be long, such contamination could be a serious problem when the contaminant reaches the receptor wells. Particularly for contaminants that do not degrade, it is important to determine the magnitude of risk that would be experienced once the contamination does reach a drinking water well. The selection of a long time frame assumes that peaks are more likely to be considered in the assessment (see Section 2.2.1 for further discussion).

Regional Site-Based Data. The 3MRA modeling system was designed to be implemented using a site-based approach. In this approach, site-based data are used as inputs to the system when available. When site-based data are not available, data collected on a regional level, followed by data collected on a national level, are used for the evaluation. Using site-based and regional data to create data sets that represent actual site and environmental conditions ensures that implausible exposure pathways and scenarios are not included.

Detailed documentation regarding the methodology for gathering data, what data were collected, where they were obtained, how they were collected and processed, and other issues and uncertainties is provided in Volume II. Examples of the types of data collected as site-based characteristics include facility locations and the physical and environmental characteristics of the sites and surrounding areas (e.g., land use, human receptor locations, and ecological habitats). Regional data collected include meteorological data, some soil characteristics, aquifer data, and types of ecological receptors. Data collected at the national level include human exposure factors, ecological exposure factors, human health toxicity values, and ecological toxicity values.

Variability and Uncertainty Analysis. The 3MRA modeling system was designed specifically to support the analysis of uncertainty and variability in the risk outputs. The site-based approach provides a measure of intrasite variability in risk or hazard in terms of both spatial and temporal variables. With the current representative national data set, the representative sites provide a measure of nationwide intersite variability across industrial waste management sites; that is, the sites cover the range of different land-based WMU types and environmental settings. In addition, the 3MRA modeling system has been designed for a two-stage Monte Carlo analysis. For many inputs, distributions were developed to represent their variability.

2.2 Spatial and Temporal Scale

The 3MRA framework uses a site-based concept to address spatial issues. The data requirements for a multimedia modeling system that includes human and ecological receptors are considerable. Developing a realistic data set for modeling purposes requires consistency among all of the input data, such that the input data set represents a realistic, internally consistent environmental setting. To accomplish this, EPA adopted a site-based approach for the 3MRA modeling system, in which site-specific data, supplemented by regional and national data, are used to construct the environmental setting to be modeled. The inputs to the modeling system can be represented by distributions to reflect variability, and multiple distributions for selected variables can be developed to reflect uncertainty in those variables; however, the uncertainty and variability with respect to the modeling system inputs are all accounted for within the construct of a specific site. To conduct a national-scale assessment, multiple sites that either account for all relevant sites or statistically represent the population of such sites may be used as inputs to the 3MRA modeling system.

The 3MRA modeling system also manages time series data to address temporal issues. The multimedia, multipathway aspect of the modeling system results in exposures through the various pathways occurring at different times. Sometimes exposures can occur hundreds of years apart, especially when considering a ground water pathway versus an aboveground pathway. Thus, the ability to track contaminant concentrations and resulting exposures across all pathways on a common time scale is required. Although this time scale can vary depending on the needs of the analysis, a 1-year time step is used for outputs from all modules because the 3MRA modeling system was developed primarily for chronic exposures. Some of the modules use much shorter time steps to produce these annual outputs. For example, the Air Module uses an hourly time step, but provides annual averages as outputs.

2.2.1 Model Spatial Scale

The spatial scale for the 3MRA modeling system is defined in terms of the AOI surrounding a WMU or other source. The size of the AOI will depend on the goals of a particular application, as well as the ability to collect the needed data for the AOI. For most RCRA WMUs other than incinerators and other types of combustors, the AOI will be relatively constrained because of the operational nature of these sources. For the representative 201 sites in the modeling system, a distance of 2 km was selected for the AOI. If analysts or risk managers are interested in larger AOIs or in areas closer in to the WMU, site-based data sets can be created to provide additional spatial coverage or resolution for the risk results. By default, the 3MRA modeling system estimates risks for the following distance rings, as shown in Figure 2-4:

- Human risks are totaled within the 0 to 0.5 km, 0 to 1 km, and 0 to 2 km rings around the WMU.
- Ecological risks are totaled within the 0 to 1 km, 1 to 2 km, and 0 to 2 km rings around the WMU.

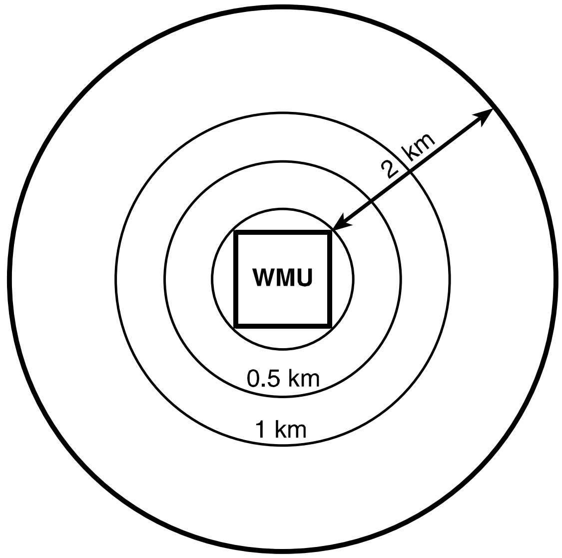

Figure 2-4. Current area of interest and concentric distance rings for human risk.

Within the AOI, two key spatial considerations are the location of watersheds and associated waterbodies, and the location of receptors (both human and ecological). These spatial considerations, in combination with other location-specific information, such as WMU characteristics, meteorology, hydrogeology, and land uses, constitute the environmental setting that is modeled. Figure 2-5 illustrates the delineation of watershed subbasins in the 3MRA modeling system. The watershed subbasins define the areas over which deposition rates and soil concentrations are averaged and assumed to be uniform. Although watershed size is determined mainly by the topography and hydrography at a site, the size of the subbasins within the watersheds is defined during watershed delineation. Currently, these are defined in the 3MRA modeling system so that there are generally about 10 to 12 watershed subbasins within the AOI at a site, with an average subbasin area within the AOI of about 1 million square meters. This provides the spatial resolution needed to map soil concentration gradients across the site while keeping the total number of watershed subbasins at a given site to a reasonable number.

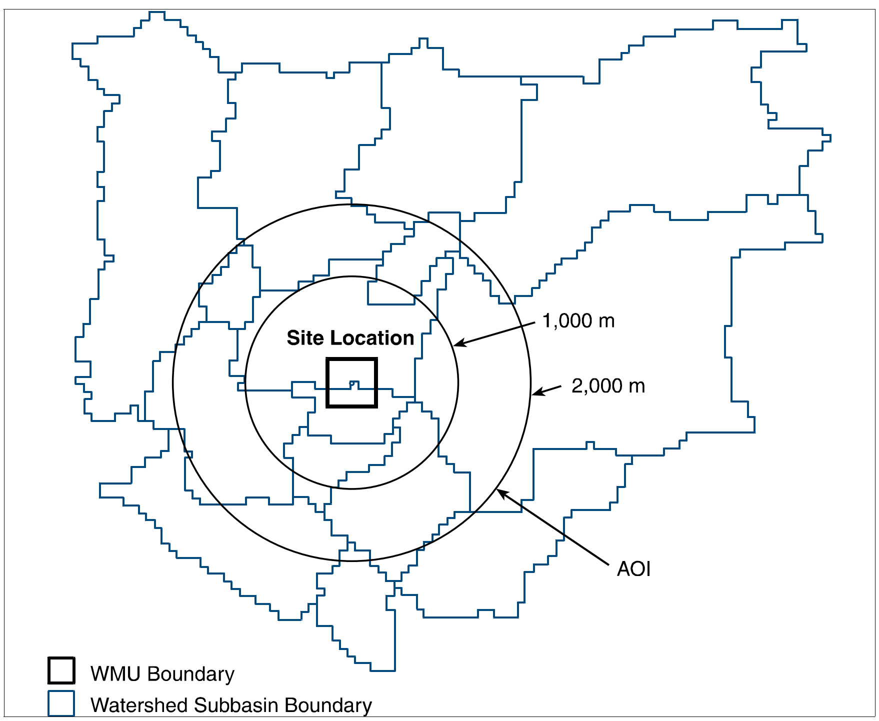

Figure 2-5. Example of watershed delineation for a typical site.

Figure 2-6 illustrates human receptor locations at an example site. Within the current representative national data set, human receptor locations at a site are defined by the centroids of Census blocks from the 1990 U.S. Census. If a distance ring crosses a Census block, then that block is divided into two components, each with a centroid, and the population associated with the Census block is apportioned to the two centroids based on the area of each of the two components. The density of receptor points varies with population density because Census blocks are sized by the population they contain.

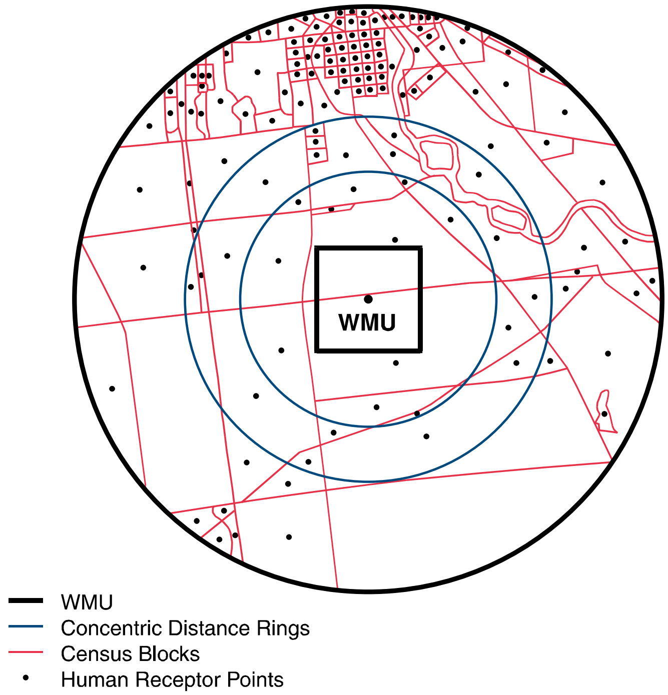

Figure 2-6. Example of human receptor placement for a typical site.

Ecological receptor locations are defined in terms of habitats and home range areas. Figure 2-7 presents a site example with five habitats delineated. The representative habitats are delineated for each site based on a variety of geographic information system (GIS) coverages of topography, land use, wetlands, and waterbodies. Within each habitat, receptor home ranges are placed at random. For each receptor, concentrations in contaminated media (soil or surface water), plants, and prey items are averaged over the home range. Various types of ecological receptors are placed within a home range, based on the available data for that particular receptor.

2.2.2 Model Temporal Scale

The 3MRA modeling system is designed to evaluate chronic exposures, with individual module results reported as a time series of annual average concentrations or fluxes. However, the time scale can be adjusted to be shorter or longer depending on the needs for a particular assessment. Individual modules may use input data on a shorter time scale to capture physical processes that vary within a year, and thus provide a more realistic estimate of the annual average. For example, annual average precipitation data do not provide details on storm events, which have a major effect on runoff and erosion processes that occur over a year. Therefore, the Land Application Unit (LAU) and Watershed Modules use daily precipitation data to more accurately estimate precipitation-driven runoff and erosion events. Similarly, for the Surface Impoundment Module, monthly temperature data can be used to capture temperature extremes across seasons, which can affect volatilization.

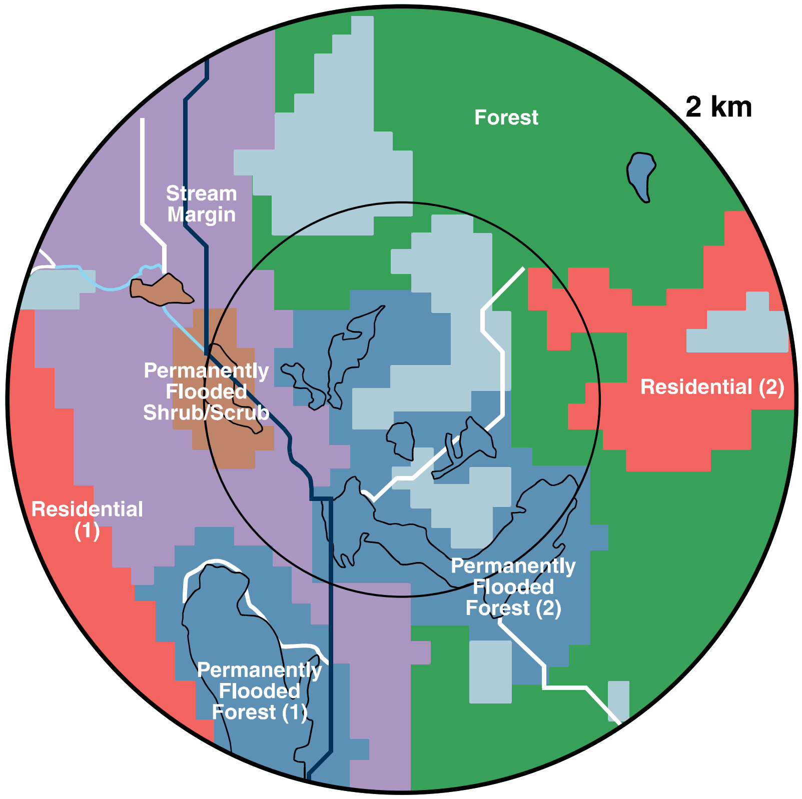

Figure 2-7. Example of representative ecological habitats delineated for typical site.

The time frame for estimating exposure and risk depends on a number of variables, including distance of receptors from the WMU, direction of receptors from the WMU, and physical-chemical properties affecting the transport and fate of contaminants in the receiving media. For most media (i.e., air, surface water, soil), the exposure and risk occur in the same time frame as the release from the WMU. For ground water, where the medium and chemical properties attenuate the transport process, the exposure and risk time frame can be hundreds to thousands of years after the release. The time frame, therefore, varies for each contaminant and environmental medium considered for each specific WMU. The 3MRA modeling system calculates concentrations for each pathway for each human receptor or habitat location until less than 1 percent of the maximum contaminant concentration in the medium is left in the medium, up to a maximum of 10,000 years. This is intended to capture the large majority of exposures from most contaminants that would be included in an analysis. A few metals and dioxin-like substances may move so slowly in the subsurface that exposures within the AOI may not be fully captured even in the 10,000-year time frame.

A given receptor is subject to exposures from various (but not necessarily all) pathways simultaneously, depending on whether there is a contaminant concentration in each pathway. The aggregate risk to any individual receptor is defined as the sum of the risks from each pathway over a given time period. In the 3MRA modeling system, all exposures are calculated on an annual basis. However, the risk can be based on an exposure duration defined for the analysis. All exposure durations greater than 1 year would be the average daily exposure (or average daily dose) over consecutive years that make up the duration. For example, if there were

100 years of exposure output and the exposure duration was defined as 10 years, there would be 91 10-year averages from which to calculate risk or hazard, beginning with years 1 to 10, 2 to 11, and 3 to 12 and ending with 91 to 100.

Given that the exposure in the different media can occur over significantly different time periods for each receptor location, aggregation of risk is performed for exposures that occur at the same time. For instance, exposures and risks due to contaminated air occurring during years 1 to 10 are not aggregated with exposures and risks due to contaminated ground water occurring during years 91 to 100. Figure 2-8 illustrates how risks during the same time periods are overlaid and aggregated across exposure pathways for a given receptor and contaminant. Risks are aggregated across different exposure routes (i.e., ingestion, inhalation) only after considering current EPA practices for combining exposures through different routes.

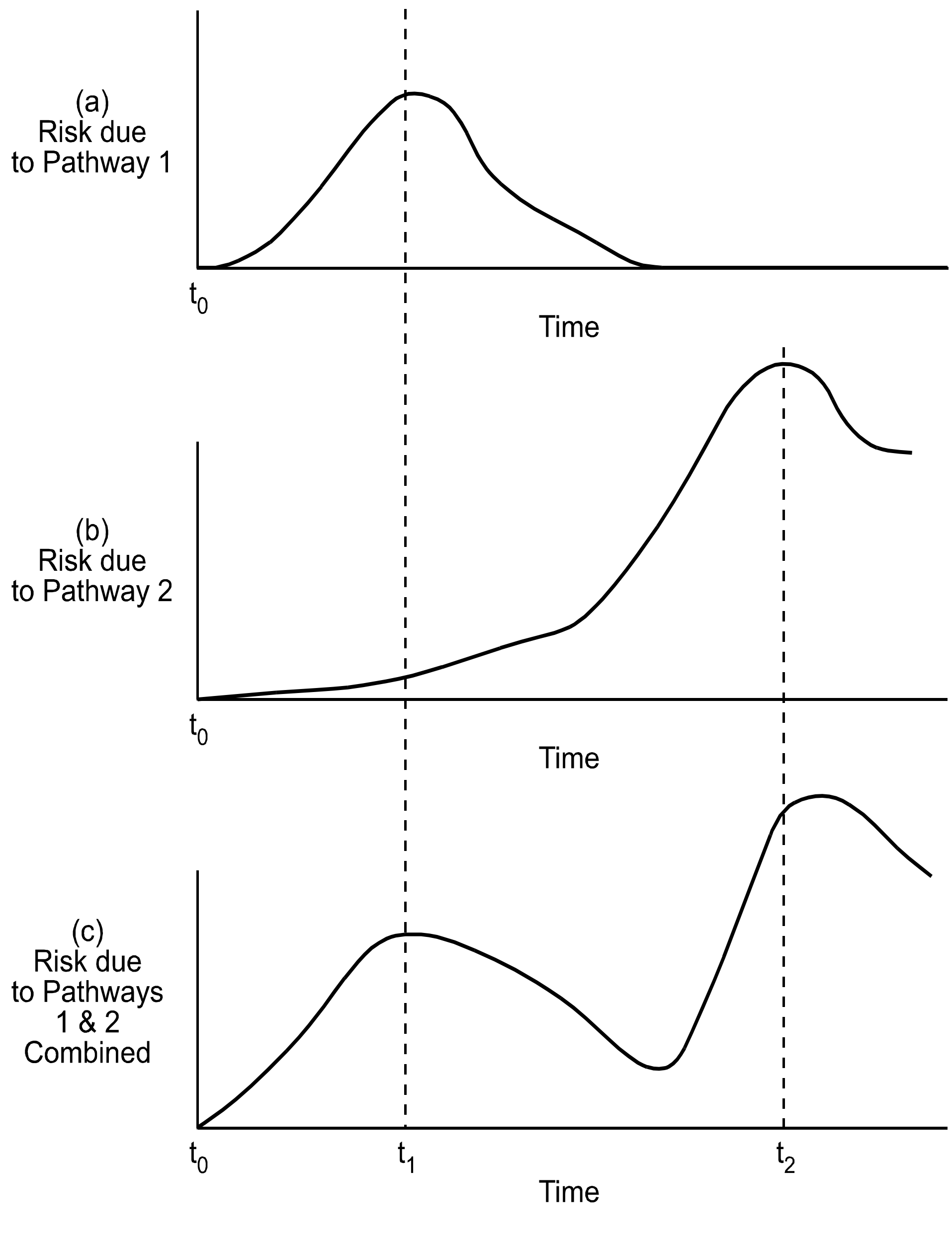

Figure 2-8. Illustration of concurrent time aggregation of risks.

Once risk estimates are calculated for each receptor at a location within the AOI, the critical time period during which the maximum risk and/or HQ occurs across the population for each receptor/cohort combination and for each exposure pathway and pathway aggregation is determined. The critical time period is determined in a similar fashion in the Ecological Risk Module for receptor types, groups, and distances. Types of outputs for the critical period include population-weighted risk or HQ; media-specific concentrations at receptor location; pathway-specific exposures for each receptor/cohort or indicator species at each receptor location; and various aggregations of media concentrations, pathways, or receptors.

2.3 Design of the 3MRA Modeling System Multipathway Modules

The design of the 3MRA modeling system takes into consideration how constituents are released to the environment, transported through various media, and result in exposure to humans and ecological receptors, and how these exposures are characterized in terms of risk measures. The major considerations in determining the design of the science modules in the 3MRA modeling system were

- Chemicals in the 3MRA database;
- Sources (WMUs);
- Transport media, fate processes, and intermedia contaminant fluxes;
- Food chain/food web components;
- Human exposure and risk; and
- Ecological exposure and risk measures.

2.3.1 Chemicals in the 3MRA Modeling System Database

Five categories used in the 3MRA modeling system to differentiate the behavior of contaminants in the environment:

- Organic chemicals
- Metals
- Mercury
- Dioxin-like chemicals
- Special chemicals (e.g., readily metabolizable)

The RCRA hazardous waste program covers more than 400 constituents. These constituents have a large range of physical-chemical properties, toxicological properties, and behavior in the environment. Some of these constituents are well characterized with respect to these properties and others are not. In developing the 3MRA modeling system, different approaches (e.g., algorithms) were required to appropriately model some groups of constituents versus others because of differences in properties. Five broad categories were established to differentiate the behavior of contaminants in the environment: organic chemicals, metals, mercury, dioxin-like chemicals, and special chemicals (such as those that readily metabolize or those that are completely soluble in water). Mercury and dioxin-like chemicals are specific subcategories for which EPA has developed special approaches for modeling release and fate and transport.

Table 2-1 shows the 46 constituents selected to develop and test the 3MRA modeling system. These constituents were selected to ensure coverage of the range of contaminant behavior in the environment. To facilitate this selection process, all RCRA-regulated

Table 2-1. Constituents in the 3MRA Database

3MRA Constituent Type	Representative Constituent Class	Constituent Name	CASRN
organic (1)	organonitrogen (1)	Acetonitrile	75-05-8
organic (1)	organonitrogen (1)	Acrylonitrile	107-13-1
organic (1)	organonitrogen (1)	Aniline	62-53-3
organic (1)	aromatic hydrocarbon (4)	Benzene	71-43-2
organic (1)	organosulfur (7)	Carbon disulfide	75-15-0
organic (1)	chlorinated aromatic (8)	Chlorobenzene	108-90-7
organic (1)	chlorinated hydrocarbon (9)	Chloroform	67-66-3
organic (1)	chlorinated pesticide (10)	2,4-Dichlorophenoxyacetic acid	94-75-7
organic (1)	brominated hydrocarbon (11)	Ethylene dibromide	106-93-4
organic (1)	misc. halogenated (12)	Hexachloro-1,3-butadiene	87-68-3
organic (1)	chlorinated pesticide (10)	Methoxychlor	72-43-5
organic (1)	carbon/hydrogen/oxygen (6)	Methyl ethyl ketone	78-93-3
organic (1)	carbon/hydrogen/oxygen (6)	Methyl methacrylate	80-62-6
organic (1)	chlorinated hydrocarbon (9)	Methylene chloride	75-09-2
organic (1)	organonitrogen (1)	Nitrobenzene	98-95-3
organic (1)	nonhalogenated phenolic (15)	Phenol	108-95-2
organic (1)	organonitrogen (1)	Pyridine	110-86-1
organic (1)	chlorinated hydrocarbon (9)	Tetrachloroethylene	127-18-4
organic (1)	carbamate group (17)	Thiram	137-26-8
organic (1)	aromatic hydrocarbon (4)	Toluene	108-88-3
organic (1)	chlorinated hydrocarbon (9)	Trichloroethylene	79-01-6
organic (1)	chlorinated hydrocarbon (9)	1,1,1-Trichloroethane	71-55-6
organic (1)	oxoanion metal (2)	Vanadium	7440-62-2
organic (1)	chlorinated hydrocarbon (9)	Vinyl chloride	75-01-4
metal (2)	oxoanion metal (2)	Antimony	7440-36-0
metal (2)	oxoanion metal (2)	Arsenic	7440-38-2
metal (2)	cationic metal (3)	Barium	7440-39-3
metal (2)	cationic metal (3)	Beryllium	7440-41-7
metal (2)	cationic metal (3)	Cadmium	7440-43-9
metal (2)	oxoanion metal (2)	Chromium (hexavalent)	18540-29-9
metal (2)	oxoanion metal (2)	Chromium (total)	7440-47-3
metal (2)	oxoanion metal (2)	Chromium (trivalent)	16065-83-1
metal (2)	cationic metal (3)	Lead	7439-92-1
metal (2)	cationic metal (3)	Nickel	7440-02-0
metal (2)	oxoanion metal (2)	Selenium	7782-49-2
metal (2)	cationic metal (3)	Silver	7440-22-4

(continued)

Table 2-1. (continued)

3MRA Constituent Type	Representative Constituent Class	Constituent Name	CASRN
metal (2)	oxoanion metal (2)	Thallium	7440-28-0
metal (2)	cationic metal (3)	Zinc	7440-66-6
mercury (3)	organometallic (13)	Methyl mercury	7439-97-6m
mercury (3)	cationic metal (3)	Mercury (elemental)	7439-97-6e
mercury (3)	cationic metal (3)	Mercury (total)	7439-97-6
dioxin-like (4)	dioxin/furan (16)	2,3,7,8-Tetrachlorodibenzo-p-dioxin	1746-01-6
special (5)	polynuclear aromatic (5)	Benzo[a]pyrene	50-32-8
special (5)	carbon/hydrogen/oxygen (6)	Bis-(2-ethylhexyl)phthalate	117-81-7
special (5)	polynuclear aromatic (5)	Dibenz[a,h]anthracene	53-70-3
special (5)	chlorinated phenol (14)	Pentachlorophenol	87-86-5

constituents were sorted into 17 representative constituent classes with similar physical-chemical properties. Only constituents with adequate constituent-specific toxicity and physical-chemical properties data were considered for model testing.

The specific properties used to establish these 17 constituent classes included

- Degree of aromaticity (the number and arrangement of benzene rings);
- Volatility;
- Presence of halogens, such as bromine and chlorine;
- Presence of other key elements, such as oxygen, nitrogen, sulfur, and/or phosphorus;
- Use (e.g., pesticides);
- Presence of organic functional groups, such as phenols and carbamates; and
- Similarities in ionic behavior (e.g., anionic and cationic metals).

Candidate constituents were selected from each of the 17 classes by considering toxicology, fate and transport properties, waste chemistry, and other considerations, such as

- Total number of RCRA constituents within a group,
- Range of expected toxicity of the constituents within a group,
- Availability of defensible physical-chemical property data and analytical methods,

- Differences in constituent structures within a group,
- Differences in degree or type of halogenation (chlorinated or brominated),
- The mix of isomers represented in the toxicity data,
- The potential for constituents to accumulate in food or prey,
- The degradation products required to understand the toxicity for a constituent,
- The significance of constituents to other EPA programs or their representativeness of a given class of chemicals (e.g., 2,3,7,8-TCDD for halogenated dioxins and furans), and
- The frequency or expectation of finding the constituent in many process waste streams.

The chemical properties were obtained through a combination of modeling, existing databases, and literature review. Many properties were modeled using SPARC (System Performs Automated Reasoning in Chemistry) and MINTEQA2. These models are described briefly below. Table 2-2 summarizes how the various chemical property values were obtained.

Table 2-2. Methodology and Data Sources for 3MRA Chemical Properties

Property	Methodology	References
Organic Chemicals		
Thermodynamic properties and partition coefficients[a]	SPARC-calculated values, adjusted by CPP for temperature and pH	U.S. EPA (2003) U.S. EPA (1999f)
Hydrolysis rates	Measured or estimated rate constants adjusted by CPP for temperature and pH	U.S. EPA (1996d) U.S. EPA (1999f)
Aerobic biodegradation rates	Measured values from literature, grouped by pH and temperature regimes	Aronson et al. (1999)
Anaerobic biodegradation rates	Measured values from literature, grouped by pH, temperature, and redox regimes	U.S. EPA (1999d)
Soil/water partition coefficients	Calculated from Kow by CPP	U.S. EPA (1999f)
Metals		
Partition coefficients (waste, soil, surface water, sediment)	Measured or estimated values, presented as national distributions	U.S. EPA (1999e)
Sorption isotherms	MINTEQA2 model	U.S. EPA (1998, 1999c)

[a] Molecular weight, density, volume, vapor pressure, boiling point, air and water diffusion coefficients, solubility, Henry's law constant, octanol/water partition coefficient (Kow), ionization coefficient.

SPARC. EPA developed the predictive modeling system SPARC to help meet the growing need for constituent-specific inputs for multimedia, multipathway, multireceptor risk assessment tools such as 3MRA. SPARC calculates a large number of physical and chemical parameters (such as solubility, vapor pressure, Henry's law constant, octanol/water partition coefficient, air diffusivity, water diffusivity, and ionization potential) from pollutant molecular structure and basic information about the environment (media, temperature, pressure, pH, etc.) For more information on SPARC, see U.S. EPA (2003).

MINTEQA2. The MINTEQA2 model was used to develop metal sorption isotherms that are contained within the 3MRA Vadose Zone and Aquifer Modules and used to provide the pH and concentration-adjusted soil/water partition coefficients needed to estimate sorption of metal contaminants in the subsurface. MINTEQA2 is an equilibrium metals speciation model that can be used to calculate the equilibrium composition of dilute aqueous solutions in the laboratory or in natural aqueous systems. The model can calculate the equilibrium mass distribution among dissolved species, adsorbed species, and multiple solid phases under a variety of conditions including a gas phase with constant partial pressure. For more information on MINTEQ, see U.S. EPA (1998) and U.S. EPA (1999c).

The 3MRA chemical database can be expanded as applications using this modeling system require and as constituent data become available.

2.3.2 Sources (WMUs)

The WMUs included in the 3MRA modeling system represent the major management practices where wastes are put into or on the land for recycling, recovery, reuse, treatment, or disposal. The various types of WMUs manage different types of industrial waste in different ways. For example,

- **Surface impoundments** are used to treat and dispose of liquid industrial waste and generate a sludge that requires further management;
- **Aerated tanks** are used to actively treat industrial liquid waste and generate a sludge requiring further management;
- **Landfills** are a common disposal site for many nonliquid industrial wastes;
- **Waste piles** are temporary storage areas on the ground for nonliquid industrial waste, such as ash or slag; and
- **LAUs** are used to reuse, treat, or dispose of industrial waste in liquid, semiliquid, or solid form. Some wastes are used as a soil amendment, which is a reuse practice; some wastes are applied to land for treatment through biological degradation; and some wastes are applied to land as a disposal method.

As illustrated in Table 2-3, each WMU source is modeled to conserve mass. That is, the WMU source boundaries are defined, and mass balance equations are derived to describe the various transport and loss mechanisms that affect the amount (mass or concentration) of

Table 2-3. WMU Types and Source Term Characteristics

WMU Type	Source Term Characteristics			
	Mass Balance	Partitioning	First-order Degradation	Finite Source
Surface Impoundment	✔	✔	✔	✔
Aerated Tank	✔	✔	✔	✔
Landfill	✔	✔	✔	✔
Waste Pile	✔	✔	✔	✔
LAU	✔	✔	✔	✔

constituent within the defined WMU. Each of the WMU source mass balances provides and accounts for partitioning of the constituent of interest among solids (sediment or soil particles), water, and air. First-order degradation terms are also included to account for constituent losses via hydrolysis and biodegradation.

Each of the WMUs included in the 3MRA modeling system is shown in Table 2-4, along with release mechanisms and directly affected media. Surface impoundments, landfills, waste piles, and LAUs were selected because they are the most likely destinations for industrial nonhazardous waste, according to an EPA industrial waste screening study (Westat, 1987). Aerated tanks were selected because hazardous waste management has been shifting from surface impoundments to aerated tanks since the 1987 study was conducted.

Table 2-4. WMU Types and Release Mechanisms Modeled

WMU Type	Release Mechanism (receiving medium)			
	Leaching (ground water)	Volatilization (air)	Windblown dust (air)	Runoff/erosion (soil and surface water)
Surface Impoundment	✔	✔		
Aerated Tank		✔		
Landfill	✔	✔	✔	
Waste Pile	✔	✔	✔	✔
LAU	✔	✔	✔	✔

As shown in Table 2-4, constituents enter four environmental media after direct release from a WMU: (1) air, where volatile and particulate releases disperse and are deposited onto soil, plants, and surface water; (2) soil, which receives runoff and eroded solids from WMUs; (3) surface water, which receives runoff and eroded solids from WMUs; and (4) ground water, which receives dissolved constituents leached from wastes in land-based units by infiltrating precipitation.

Certain characteristics were assumed for each of the five WMU types to complete the waste management scenario and to capture the variability in management practices. These characteristics drive the design and parameterization of the 3MRA source modules; they are described briefly below and in more detail in Sections 4 and 5 of this volume.

Surface Impoundment. The surface impoundment is used for the management of liquid wastes in an earthen basin. Constituents are released during the lifetime of the unit. The impoundment does not receive or contribute runoff to the watershed. No liner other than native soils is assumed to be present, and there is no cover to reduce volatile emissions. The unit is assumed to contain two well-mixed phases: liquid and sediment. The solids in the liquid waste settle to the bottom of the impoundment over time and fill pore space in the native soils. This attenuates the transport or leaching of constituents to underlying soils and ground water. In addition, a fraction of each surface impoundment is aerated, which enhances biodegradation and increases volatilization of some constituents. Each surface impoundment is assumed to operate for 50 years and then undergo clean closure—that is, all waste is removed from the unit.

The surface impoundment releases leachate to the unsaturated zone and volatile emissions to air. The surface impoundment is assumed to have a berm to prevent overrun; therefore, runoff is assumed not to occur. However, volatilized releases can be deposited on the surrounding soils, and erosion and runoff from these soils are modeled. The model accounts for hydrolysis, volatilization, and sorption, as well as settlement, resuspension, growth and decay of solids, and activated aerobic biodegradation (that is, a higher rate based on the amount of biomass present) in the liquid phase, and hydrolysis and anaerobic biodegradation in the sediment.

Aerated Tank. The aerated tank is used for the treatment of wastewaters. Only tanks the size of a drum or larger are modeled. Each aerated tank has a maximum lifetime of 20 years; therefore, the operating lifetime assumes a tank replacement every 20 years.

The aerated tanks are assumed not to fail or leak liquids to soil or surface runoff. As a result, the only release pathway for an aerated tank is volatile emissions to air. Constituent loss within the tank is simulated through hydrolysis and aerobic biodegradation.

Landfill. The design of the landfill assumes a series of vertical cells of equal volume that are filled sequentially, with each cell being filled in one year. The landfill model is based on the assumption that the constituent mass in the landfill cells may be linearly partitioned into aqueous, vapor, and solid phases. Releases from the landfill can occur over the active lifetime of the landfill (30 years) and continue until virtually all of the constituent mass is released, or, for immobile constituents, releases can continue for a specified number of years.

The landfill modeled in the 3MRA system assumes minimal controls and is constructed below grade. In particular, the unit has no liner, and the cover at closure is a layer of natural soil. The design of the landfill allows volatilization and particle emissions, as well as infiltration, which leaches constituents from the waste. The belowgrade design prevents runoff and erosion. In addition, anaerobic biodegradation and hydrolysis processes can degrade constituents within the landfill.

Waste Pile. The waste pile manages wastes in a storage pile on the ground (above grade). The wastes are removed and the pile site closed at the end of its active life (assumed to be 30 years). Constituents are released only during the operating lifetime of the pile. Waste pile height and area are set and held constant at each site based on the volume of waste stored. The waste in the pile is assumed to be completely removed annually and replaced with new waste.

The waste pile is assumed to be uncovered, so that constituents can be released through infiltration and leaching, volatilization, particle entrainment by wind, and erosion and runoff. In addition, constituents can be degraded by hydrolysis and aerobic degradation at the surface of the pile and by hydrolysis and anaerobic degradation within the pile. Although the waste pile design does not incorporate management controls, the pile is assumed to be located within a local watershed in such a way that there is no run-on of uncontaminated soil onto the pile.

For the waste pile, after runoff and erosion have occurred for some period of time, downslope land areas will be contaminated and their soil concentrations can approach the residual constituent concentrations in the waste pile itself. Thus, after extensive runoff and erosion from a waste pile, the entire downslope surface area can be considered a source and becomes an extended source area in the model construct. Consequently, a holistic modeling approach was taken with the waste pile to incorporate it into the watershed of which it is a part.

Land Application Unit. The LAU allows for the placement of wastes in an open field for degradation, treatment, or disposal. Some industrial wastes are applied to land for purposes of soil amendment, thus constituting a reuse of the waste. The waste is applied to the surface soil periodically and then tilled into the top layer of the soil during each of the assumed 40 years of operation. Releases from the LAU can occur over the active lifetime of the LAU and beyond, and continue until virtually all of the constituent mass is released or, for immobile constituents, for a specified number of years. Other than tilling into the soil and a crop cover, no management controls are present to limit releases from the LAU.

Constituents are released from the LAU by leaching to the unsaturated zone, volatilization and particulate emissions to the air, runoff of dissolved constituents, and erosion and runoff of particles. Constituent losses include hydrolysis and aerobic biodegradation. Also, LAUs are on the land surface and are an integral part of their respective watersheds. Consequently, they receive run-on and erosion from upslope land areas and affect downslope land areas through runoff and erosion.

For the LAU, after runoff and erosion have occurred for some time, downslope land areas will be contaminated, and their soil concentrations can approach the residual constituent concentrations in the LAU (or conceivably even exceed them, long after operation ceases). Thus, after extensive runoff and erosion from the LAU, the entire downslope surface area can be considered a source and becomes an extended source area in the model construct. Consequently, a holistic modeling approach was taken with LAU to incorporate it into the watershed of which it is a part.

2.3.3 Transport Media, Fate Processes, and Intermedia Contaminant Fluxes

The 3MRA modeling system has five main environmental transport modules: the Air Module, the Watershed Module, the Surface Water Module, the Vadose Zone Module, and the Aquifer Module. Four of these modules (Air, Surface Water, Vadose Zone, and Aquifer) are based on existing well-established, peer-reviewed regulatory models. These legacy models provide both state-of-the-art environmental transport and fate modeling and well-established acceptability through various peer-review cycles and applications that have undergone stringent scrutiny. They have been incorporated into the 3MRA modeling system through the use of pre- and postprocessors that integrate the specific input and output requirements of these legacy models into the 3MRA modeling system. Within the modules, all functionality of the original model is maintained, including degradation processes, transformation processes, and release processes. Each of the transport modules is summarized below.

3MRA Modeling System Transport Media, Fate Processes, and Intermedia Contaminant Fluxes

Transport Media
- Atmosphere
- Watershed
- Subsurface (vadose zone and aquifer)
- Surface water

Fate Processes
- Chemical/biological transformation (and associated products)
- Linear partitioning (water/air, water/soil, air/plant, water/biota)
- Nonlinear partitioning (metals in vadose zone)
- Chemical reactions/speciation (mercury in surface waters)

Intermedia Contaminant Fluxes
- Air ➔ Watershed/farm habitat soil (wet/dry deposition, vapor diffusion)
- Air ➔ Surface water (wet/dry deposition, vapor diffusion)
- Watershed soil ➔ Surface water (erosion, runoff)
- Surface water ➔ Sediment (sedimentation)
- Vadose zone ➔ Ground water (infiltration)
- Watershed soil ➔ Air (volatilization)
- Ground water ➔ Surface water

Air Module. The Air Module is based on the Industrial Source Complex Short-Term Model, Version 3 (ISCST3). ISCST3 uses a steady-state Gaussian plume modeling approach, which assumes that the plume of emissions follows a straight-line path in the direction of the mean wind flow, and the contaminant concentrations in the horizontal and the vertical directions (both orthogonal to the mean wind direction) follow a normal distribution. The model provides estimates of contaminant air concentration, dry deposition rates (particles only), and wet deposition rates (particles and vapors) for user-specified averaging periods (i.e., annually). It also has the capacity to model area sources or point sources. A point source is usually considered a source with a small area, such as a stack or vent. An area source is a source that covers some defined area that is large enough to have an impact on the dispersion across the source. The 3MRA modeling system sources are all area sources. There is also a regulatory definition of area source that refers to the quantity of emissions from a particular source. That definition does not apply to this discussion.

In the Air Module, ISCST3 is used to model the transport and diffusion of contaminants in the form of volatilized gases or fugitive dust emitted from area sources. The air concentrations are used to estimate bio-uptake from plants, and human exposures due to direct inhalation. The predicted deposition rates are used to determine contaminant loadings to watershed soils, terrestrial habitats, farms, and surface waters.

ISCST3 is used as legacy code in the 3MRA modeling system. That is, the model was left intact and the necessary interfacing to the modeling system is handled using pre- and postprocessors. Together, the EPA air quality model (ISCST3) and the pre- and postprocessing code that integrates ISCST3 into the 3MRA modeling system are referred to as the Air Module. The pre- and postprocessing code also provides additional functionality to support other 3MRA modeling system requirements. Additional detail on the Air Module can be found in Section 6 of this document.

Watershed Module. Contaminant mass can be released from a WMU in the form of volatile and particulate emissions. These emissions can then be transported and deposited onto the soils of nearby land areas as wet or dry deposition (estimated by the Air Module). Once deposited, a contaminant is then subject to fate and transport processes, such as runoff, erosion, and degradation. Contaminants in the soil of the AOI are available either for direct exposure to human or ecological receptors or for indirect exposure through uptake in a food web. The Watershed Module accounts for these fate and transport processes across the AOI.

The loss mechanisms for contaminants deposited onto soils are volatilization, leaching, and biological and/or contaminant degradation. The fate and transport mechanisms for contaminants that are deposited onto soils include runoff and erosion into adjacent waterbodies. Because the surface transport processes are hydrologically related, the land areas surrounding the WMU are disaggregated into watershed subbasins. A watershed subbasin can vary in size from a portion of a hillside to much larger areas encompassing regional stream or river networks. In all cases, a watershed subbasin is treated as a single, homogeneous area with respect to soil characteristics, runoff and erosion characteristics, and contaminant concentrations in soil. No spatial disaggregation below the watershed subbasin level is made; that is, no spatial contaminant concentration gradients are estimated within a given watershed subbasin. Additional detail on the Watershed Module can be found in Section 7 of this document.

Surface Water Module. The Surface Water Module has the ability to model streams, lakes, ponds, and wetlands at a site. Constituent mass released from a WMU can enter a nearby surface waterbody network through runoff and erosion directly from the WMU, atmospheric deposition to the water surface, runoff and erosion from adjoining watershed subbasins, and interception of contaminated ground water. The chemical is then subject to transport and transformation processes occurring within the waterbody network, resulting in variable chemical concentrations in the water column and underlying sediments. These chemical concentrations are the basis for direct exposure to human and ecological receptors and indirect exposure through uptake in the aquatic food web.

The Surface Water Module consists of a core model, Exposure Analysis Modeling System II (EXAMS II) (Burns, 1997; Burns et al., 1982), and the pre- and postprocessors for the model. The Surface Water Module estimates annual average total and dissolved chemical concentrations in the water column and sediments in stream reaches, ponds, and lakes. Transport and transfer processes modeled include advection, vertical diffusion, volatilization, deposition to the sediment bed, resuspension to the water column, and burial to deep sediments.

Transformation processes include hydrolysis and biodegradation as first-order reactions influenced by temperature and pH. Outputs from the Surface Water Module include water column and sediment concentrations that are used by the Aquatic Food Web Module and the Ecological Exposure Module.

EXAMS II is used as legacy code in the 3MRA modeling system. That is, the model is left intact and the necessary interfacing to the framework is handled using pre- and postprocessors. Together, the EXAMS II model and the pre- and postprocessing code that integrates EXAMS II into the 3MRA modeling system are referred to as the Surface Water Module. Additional detail on the Surface Water Module can be found in Section 8 of this document.

Subsurface (Vadose Zone and Aquifer) Modules. The Vadose Zone and Aquifer Modules consist of components from the EPA Composite Model for Leachate Migration with Transformation Products (EPACMTP) (U.S. EPA, 1996a,b,c) and pre- and postprocessors to provide compatibility with the 3MRA modeling system. The Subsurface Modules account for the fate and transport of constituents released from WMUs into the underlying unsaturated or vadose zone (soil) and saturated zone (surficial aquifer). The Subsurface Modules can consider the formation and transport of transformation products, the impact of ground water mounding on ground water velocity, finite source and continuous source scenarios, and metal transport.

The Vadose Zone Module consists of a one-dimensional (1-D) model that simulates infiltration and dissolved chemical transport through the unsaturated zone. The Vadose Zone Module accounts for sorption and attenuation of a chemical moving through the soils under the WMU. The Aquifer Module consists of a 3-D model that simulates transport through the saturated zone. The Aquifer Module consists of a 3-D ground water flow submodel and a 3-D contaminant transport submodel. The flow submodel accounts for the effects of leakage from a land disposal unit and regional recharge on the magnitude and direction of ground water flow. The contaminant transport submodel accounts for 3-D advection and dispersion and linear or nonlinear equilibrium sorption.

The Vadose Zone Module receives infiltration and solute mass fluxes from the source modules. The migration of chemicals in the vadose zone is terminated at the water table where the chemical fluxes, in the form of concentrations, are transferred to the Aquifer Module. The Aquifer Module also receives areal recharge from the Watershed Module. The Aquifer Module provides time-dependent, annual average chemical concentrations at receptor wells and annual average chemical fluxes at an intercepting stream, when present, in the AOI.

Both the Vadose Zone and Aquifer Modules can be described as legacy code in the 3MRA modeling system. The modules were left intact and the necessary interfacing to the framework is handled using pre- and postprocessors. The pre- and postprocessing code also provides additional functionality to support other 3MRA modeling system requirements. Additional detail on the Subsurface Modules can be found in Section 9 of this document.

2.3.4 Food Chain/Food Web Components

The 3MRA modeling system has three food web components: the Farm Food Chain Module, the Terrestrial Food Web Module, and the Aquatic Food Web Module. The processes in each component include the uptake by plants and animals of chemicals released from the sources and transported through the environmental media. The Farm Food Chain Module provides estimates of chemical concentrations in homegrown produce and crops, beef, and milk produced on a farm. The Terrestrial and Aquatic Food Web Modules provide estimates of chemical concentrations in plants and animals that are used to evaluate the hazards to ecological receptors. The Aquatic Food Web Module also provides estimates of chemical concentrations in fish that may be eaten by recreational fishers.

Food Chain/Food Web Fluxes

- Air ➔ Vegetation (particulate deposition; vapor diffusion)
- Farm/habitat/ garden soil ➔ Vegetation (root uptake, translocation, deposition)
- Vegetation, soil, surface water, ground water ➔ Animals (uptake)
- Surface water, sediment ➔ Aquatic organisms (uptake)

Farm Food Chain Module. The Farm Food Chain Module calculates the concentration of a chemical in homegrown produce (fruits and vegetables), farm crops for cattle (forage, grain, and silage), beef, and milk. The concentrations in homegrown produce, beef, and milk are inputs to the Human Exposure Module and are used to calculate the applied dose to human receptors who consume them. The modeling construct for the Farm Food Chain Module is based on recent and ongoing research conducted by EPA ORD and presented in *Methodology for Assessing Health Risks Associated with Multiple Exposure Pathways to Combustor Emissions* (U.S. EPA, 2000).

The Farm Food Chain Module is designed to predict the accumulation of a chemical in the edible parts of a plant from uptake of chemicals in soil and through transpiration and direct deposition of the chemical onto the plant surface. Concentrations are predicted for four main categories of food crops presumed to be eaten by humans: exposed fruits and vegetables (i.e., those without protective coverings, such as lettuce), protected vegetables (e.g., those with protective covering, such as corn), protected fruits (i.e., those in which the outer skins or rinds are not eaten, such as melons or bananas), and root vegetables (e.g., potatoes). In addition, the chemical concentrations in beef and milk are estimated from the biotransfer of chemical in feed (i.e., forage, grain, and silage), soil, and drinking water to beef and dairy cattle through ingestion.

The Farm Food Chain Module predicts the concentration of chemicals in produce grown by home gardeners and in food crops, beef, and milk produced on farms. The methodology for home gardeners uses point estimates of air and soil concentrations at the residential receptor location assigned to each Census block. In contrast, the methodology used for farms calculates an area-weighted average soil concentration for the farm and the average air concentration across the area of the farm. Thus, the predicted concentrations in farm food crops reflect the spatial average for the farm. Similarly, feed concentrations for cattle are derived using spatial averages. In predicting concentrations in beef and milk, the contribution from contaminated drinking water sources, such as farm ponds or wells on the farm, is also considered. However, irrigation of crops and home gardens is not modeled.

Because the behavior of each chemical is affected by its physical-chemical properties, the module takes into account whether the chemical is organic, metal, mercury, dioxin-like, or special. For most organic chemicals, the module calculates chemical-specific values for the biotransfer factors used in the various equations, including air-to-plant biotransfer factor, root concentration factor, and soil-to-plant biotransfer factor. For metals, dioxin-like chemicals, and special chemicals, the module generally uses empirical values from the literature for the various biotransfer factors, when available. If empirical values are not available for dioxin-like compounds and special chemicals, the biotransfer factors are calculated in the same way as for organics. Additional detail on the Farm Food Chain Module can be found in Section 10 of this document.

Terrestrial Food Web Module. The Terrestrial Food Web Module calculates chemical concentrations in soil, terrestrial plants, and various prey items consumed by ecological receptors, including earthworms, other soil invertebrates, and vertebrates. These concentrations are used as input to the Ecological Exposure Module to determine the applied dose to each receptor of interest (e.g., deer, kestrel). The module is designed to calculate spatially averaged soil concentrations in the top layer of soil (i.e., surficial soil), as well as deeper soil horizons (i.e., depth-averaged over approximately 20 cm). The spatial averages are defined by the home ranges and habitats that are delineated within the AOI at each site. Once the average soil concentrations are calculated, these values are multiplied by empirical bioconcentration factors (for animals) and biotransfer factors (for plants) to predict the tissue concentrations for items in the terrestrial food web.

The Terrestrial Food Web Module was designed to predict a range of concentrations in plants and prey items to which a given receptor may be exposed. The predator and various prey are represented in the site layout by allowing the respective home ranges to overlap. For plants and soil fauna, the Terrestrial Food Web Module estimates concentrations based on the spatially averaged soil and air concentrations across each home range. Receptors that ingest plants and soil invertebrates as part of their diet are presumed to forage only within that part of the home range that is contained within the AOI at a given site. Consequently, the home range defines the spatial scale for concentrations in soil, plants, and prey (both mobile and relatively immobile) to which a given receptor is exposed.

As with the Farm Food Chain Module, the Terrestrial Food Web Module modeling construct is based on recent and ongoing research conducted by EPA ORD and presented in *Methodology for Assessing Health Risks Associated with Multiple Exposure Pathways to Combustor Emissions* (U.S. EPA, 2000). The model can distinguish among different types of chemicals, using empirically derived algorithms for the various uptake factors for some chemicals and biouptake data from field or greenhouse studies for other chemicals. The Terrestrial Food Web Module accounts for uptake via root-to-plant translocation, air-to-plant transfer for volatile and semivolatile chemicals, and particle-bound deposition to edible plant surfaces.

To estimate the concentrations in other categories of terrestrial prey items (e.g., earthworms, small birds), the Terrestrial Food Web Module relies on soil-to-organism bioconcentration factors identified from empirical studies and/or generated using regression methods developed by the Oak Ridge National Laboratory (see, for example, Sample et al., 1998a,b). Additional detail on the Terrestrial Food Web Module can be found in Section 11 of this document.

Aquatic Food Web Module. The Aquatic Food Web Module calculates chemical concentrations in aquatic organisms that are consumed by humans (e.g., game fish), and in ecological receptors (e.g., emergent aquatic plants). These concentrations are used as input to the Human and Ecological Exposure Modules to determine the applied dose to receptors of interest. The module is designed to predict concentrations in aquatic organisms in both coldwater and warmwater aquatic habitats.

The underlying framework for the Aquatic Food Web Module is the development of representative freshwater habitats for warmwater and coldwater systems. Four basic types of freshwater systems were included for the two temperature categories: streams/rivers, permanently flooded wetlands, ponds, and lakes. Simple food webs were constructed for each of the eight freshwater habitats (four coldwater and four warmwater) that specify: (1) the predator-prey interactions, (2) the physical and biological characteristics of the species that are assigned to each habitat (e.g., size, lipid content), and (3) the dietary preferences for fish in trophic levels 3 (TL3) and 4 (TL4). For each freshwater habitat, the feeding guilds (i.e., groups of species that use environmental resources, such as food, in a similar way) for various types and sizes of fish were used to construct a simple food web and to map dietary preferences for organisms in each habitat (U.S. EPA, 1999a). The food web structure and species assignments are critical in determining concentrations of hydrophobic contaminants in aquatic organisms. Additional detail on the Aquatic Food Web Module can be found in Section 12 of this document.

To estimate the concentrations in aquatic prey items, the Aquatic Food Web Module relies on bioconcentration factors identified from empirical studies and/or generated using regression methods developed by the Oak Ridge National Laboratory (see, for example, Sample et al., 1998a,b).

2.3.5 Human Exposure and Risk

Within the 3MRA modeling system, human exposure and risk are estimated using two modules: one for exposure and one for risk. The Human Exposure Module provides an annual average dose (i.e., in mg/kg-d) and focuses on receptor types, exposure pathways, and routes. The Human Risk Module calculates risk or hazard and determines the number of people associated with these risk levels.

The conceptual approach used in developing the human exposure assessment and risk calculations within the 3MRA modeling system accounts for the major sources of variability in human exposures and risk. In particular, the approach considers variability through (1) a suite of human receptors that reflect different behaviors that lead to exposure; (2) a suite of age cohorts that reflect variability in exposure factors, such as body weight and consumption or intake rates across ages, as well as some behavior patterns such as showering; (3) the use of distributions for most exposure factors to account for the variability within a receptor/cohort category (e.g., adult resident, farmer child aged 12 to 18); (4) the location of receptors that reflect the spatial variability in contaminant concentration across an AOI, as well as the existence or nonexistence of some pathways, such as ground water being used as drinking water; (5) the variability in contaminant concentration in various pathways over time; and (6) the variability in the number of people exposed at various locations within an AOI. This site-based approach is used on multiple sites in a national assessment to address the variability both within an AOI for an individual site and across multiple sites.

3MRA Modeling System Human Receptors, Exposure Routes, and Risk Measures

Human Receptors*
- Resident
- Home gardener
- Dairy farmer
- Beef farmer
- Fisher**

* For each human receptor type, includes 5 age cohorts
** All other receptor types can also be fishers.

Age Cohorts
- Infants
- Child aged 1 to 5 years
- Child aged 6 to 11 years
- Child aged 12 to 19 years
- Adult (over 19 years)

Exposure Pathways
- Ingestion (plant, meat, milk, fish, water, soil, breast milk)
- Inhalation (gases, particulates)

Human Risk Measures
- Cancer (risk probability)
- Noncancer (hazard quotient)
- Population weighted risk distribution

Human Exposure Module. The Human Exposure Module calculates the applied dose (milligram of contaminant per kilogram of body weight) to human receptors from media and food concentrations calculated by other modules in the 3MRA modeling system. The applied dose is used for calculating carcinogenic risk for ingestion and inhalation exposure pathways and for calculating HQs for ingestion pathways. For inhalation pathways for noncarcinogens, the air concentration to which a receptor is exposed (expressed in milligrams of contaminant per cubic meter of air) is used. These calculations are performed for each receptor, cohort, exposure pathway, and year within each AOI.

Human Receptors. Human receptor types considered in the 3MRA modeling system are intended to cover the likely human receptor types evaluated at the national level. The 3MRA modeling system can model five different receptor types: residents, gardeners, beef farmers, dairy farmers, and fishers. Receptor types are used to differentiate behavior that leads to different profiles of exposure. For example, the exposure pathways evaluated for a resident include ingestion of soil and ground water and inhalation of airborne vapors and particulates. For home gardeners and farmers, contaminated foodstuffs are considered in addition to the pathways for the resident. Census and land use data are used to identify receptor types and populations that are potentially exposed within an AOI.

Age Cohorts. Each receptor type is divided into five age cohorts: infants, children aged 1 to 5 years, children aged 6 to 11 years, children aged 12 to 19 years, and adults. The only exposure pathway evaluated for the infant is the breast milk pathway, which is derived from the mother's exposure. The other age cohorts provide a way of estimating applied dose to different age groupings that exhibit different distributions of intake rates and body weights. For the 3MRA modeling system, statistical distributions were developed for these exposure factors for each age cohort using data available in the *Exposure Factors Handbook* (U.S. EPA 1997). The 3MRA modeling system samples from these distributions using a Monte Carlo approach.

Human Exposure Pathways. The approach used in human exposure assessment is to predict the type, timing, and magnitude of exposures that receptors may experience as a result of contact with the contaminants of potential concern. An exposure pathway describes the course that a contaminant takes from a source to an exposed individual. Exposures are evaluated for all potentially complete exposure pathways. An exposure pathway is complete when there is a route by which a human receptor takes up a constituent that was released from the source of concern. For example, if ground water is predicted to be contaminated, and there are no private wells located in the contaminated plume or the receptors across the AOI are connected to a municipal water supply that is not contaminated, the drinking water pathway (private ground water wells) and the shower inhalation pathway would not be complete.

Exposure routes include uptake mechanisms such as ingestion, dermal contact, and inhalation. When modeling human exposure, the exposure routes that are included in the Human Exposure Module include

- Ingestion of soil,
- Ingestion of contaminated ground water (private ground water wells only),
- Inhalation of contaminated shower air (private ground water wells only),
- Inhalation of volatile emissions,
- Inhalation of particulate emissions,
- Ingestion of homegrown produce (gardeners, farmers) and beef and milk produced on farms (farmers only),
- Ingestion of fish caught by recreational fishers, and
- Exposure of an infant through ingestion of breast milk from an exposed mother.

These routes define the exposure media to be modeled in the risk analysis (i.e., ground water, soil, air, vegetables, beef, milk, and fish). EPA considered the inclusion of the dermal route of exposure but decided that health benchmarks for dermal toxicity are not sufficiently developed at this time for use in analyses that could support regulatory decisions.

The exposure pathways and routes that can be evaluated for each receptor type are shown in Table 2-5. Residents can be exposed to contaminants in the air (inhalation) and in soil (incidental ingestion) and are assumed to be exposed to potentially contaminated ground water (inhalation and ingestion) if the house is not connected to a public water supply. Home gardeners are residents who also grow some portion of their fruits and vegetables. Farmers have the same exposure pathways as home gardeners with the additional exposure to either contaminated beef or milk (depending on the type of farms present at a site). Recreational fishers are any of the above receptors with the added pathway of ingestion of contaminated fish from local streams or lakes. Thus, some fraction of residents, home gardeners, and farmers are also fishers.

Table 2-5. Human Exposure Pathways by Receptor Type

Pathway	Resident	Resident Fisher	Home Gardener	Home Gardener Fisher	Beef Farmer	Beef Farmer Fisher	Dairy Farmer	Dairy Farmer Fisher
Air inhalation	✔	✔	✔	✔	✔	✔	✔	✔
Shower air inhalation	✔	✔	✔	✔	✔	✔	✔	✔
Soil ingestion	✔	✔	✔	✔	✔	✔	✔	✔
Water ingestion	✔	✔	✔	✔	✔	✔	✔	✔
Crop ingestion			✔	✔	✔	✔	✔	✔
Beef ingestion					✔	✔		
Milk ingestion							✔	✔
Fish ingestion		✔		✔		✔		✔

The Human Exposure Module aggregates exposures across exposure pathways and routes, when appropriate (e.g., daily doses of beef contaminated by uptake from forage, silage, grain, soil, and drinking water), and provides estimates of total exposure for the eight routes listed above.

Within the 3MRA modeling system, the evaluation of human exposure is accounted for by a contaminant's distribution across a site, a contaminant's concentration profile with time, and also the variability and uncertainty in exposure factors for each receptor type. The exposure for each receptor type is estimated at each receptor location within the study area to capture spatial variability, and for every year over the modeling time frame to capture temporal variability. Additional detail on the Human Exposure Module can be found in Section 13 of this document.

Human Risk Module. The Human Risk Module calculates the population-weighted risk or hazard for each receptor location, receptor type, age cohort, pathway, and exposure period. Hazard or risk distributions are developed across the site by doing these calculations at each receptor location, providing a measure of the spatial variability at a site. These same distributions are developed for each exposure period in the modeling time frame providing a measure of temporal variability. At each location, the number of people for each receptor type/cohort is matched with the risk or hazard estimates, providing a population weighting for each receptor location.

Once the risk or hazard calculations are complete, the model generates the risk distribution across a site for each exposure period in the modeling time frame. Given the resulting pathway-specific, receptor-type/cohort-specific, and location-specific risks, HQs, or margins of exposure (MOEs) (for the breast milk pathway), a time series of population-weighted risk data is generated for each. This can be thought of as a set of cumulative frequency histograms. The histogram for any given year is constructed as a series of risk bins defined by risk or HQ ranges. For any given year, the histogram contains the pathway-specific risk or HQ distribution of the number of people (corresponding to a given receptor type/cohort) across locations. Figure 2-9 shows an illustration of this concept of a cumulative frequency histogram

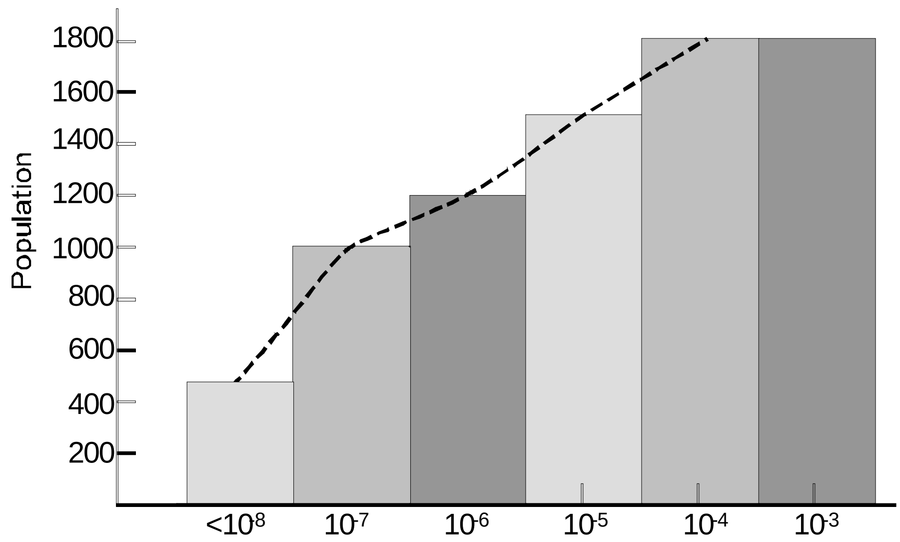

Figure 2-9. Example cumulative population risk histogram for the inhalation from ambient air pathway, single time period, single receptor.

of risk for one pathway, one receptor type, and one time period at one site. These types of histogram can be used to assess the variability of risk to the receptors across the AOI. The risk for this illustration ranges from below 10^{-8} for about 25 percent of the receptors within the AOI to 10^{-4} for about 15 percent of receptors within the AOI. Because the data sets are available for each pathway and each receptor type, a similar assessment can be made by looking at the contribution of each pathway to the total risk for a receptor type. When the time series for each data set is examined, the variability in risk from year to year can be assessed.

Given the resulting time series of pathway-specific and receptor-type/cohort-specific risk and/or HQ, the years during which the total risk or HQ across all receptors within the AOI is a maximum is determined. The exposure period with the highest total risk is termed the critical exposure period.

The distributions for the critical exposure period are the outputs for each site. For any given site, there is a distribution for each pathway independently, for all ingestion pathways together, for all inhalation pathways together, for the ground water-only pathways, and, where appropriate, for all ingestion and inhalation pathways together for each receptor type/cohort, all cohorts within a receptor type, and all receptor types together at various distances from the source and across the entire AOI. Additional detail on the Human Risk Module can be found in Section 14 of this document.

2.3.6 Ecological Receptors, Exposure Pathways, and Risk Measures

Ecological exposure and risk are estimated using two modules: one for exposure and one for risk. The Ecological Exposure Module focuses on receptors, habitats, and exposure pathways. The Ecological Risk Module estimates HQs for representative species.

Ecological Exposure Module. The Ecological Exposure Module calculates the applied dose (in mg/kg-d) to ecological receptors that are exposed to contaminants via ingestion of contaminated plants, prey, and media (i.e., soil, sediment, and surface water). However, just as in the Human Exposure Module, some exposures are expressed as media concentrations and are passed directly to the Ecological Risk Module to account for the way in which toxicity data are used. These dose estimates are then used as inputs to the Ecological Risk Module. The Ecological Exposure Module calculates the applied dose for each receptor placed within a terrestrial or freshwater aquatic habitat. Exposure is a function of (1) the habitat to which the receptor is assigned, (2) the spatial boundaries of the species' home range, (3) the food items (plants and prey) that are available in a particular home range, (4) the dietary preferences for food items that are available, and (5) the media concentrations in the receptor's home range. In essence, the module estimates an applied dose for birds, mammals, and selected herpetofauna that reflects the spatial and temporal characteristics of the exposure (i.e., exposure is tracked through time and space). The home range is the area within a habitat that is needed for each receptor.

The conceptual approach for the ecological exposure assessment within the 3MRA modeling system accounts for the major sources of variability in ecological exposures. In particular, the approach considers variability through (1) the development of representative habitats; (2) selection of receptors based on ecological region; (3) the recognition of opportunistic feeding and foraging behavior using probabilistic methods; (4) the creation of a dietary scheme specific to region, habitat, and receptor; and (5) the application of appropriate graphical tools to capture spatial variability in exposure. The underlying framework for the Ecological Exposure Module is based on a representative habitat scheme to increase the resolution of general terrestrial and freshwater systems.

Representative habitats currently delineated in the 3MRA modeling system include

Terrestrial Habitats
- Grasslands
- Shrub/scrub
- Forest
- Crop fields and pastures
- Residential

Waterbody Margin Habitats
- Rivers/streams
- Ponds
- Lakes

Wetland Margin Habitats
- Permanently or intermittently flooded grassland
- Permanently or intermittently flooded shrub/scrub
- Permanently or intermittently flooded forest

Depending on the type of habitat and contaminant-specific uptake and accumulation, animals may be exposed through the ingestion of plants (both aquatic and terrestrial), soil invertebrates, aquatic invertebrates, fish, terrestrial vertebrates, media, or any combination that is reflected by the dietary preferences of the particular species. For example, an omnivorous vertebrate that inhabits a freshwater stream corridor habitat may ingest fish, small terrestrial vertebrates found in the stream corridor, terrestrial and aquatic plants, surface water, and soil. The dietary preferences are independent of the contaminant type; therefore, contaminant concentrations in some food items may be near zero for contaminants that do not bioaccumulate. The dietary preferences for each receptor are supported by an extensive exposure factors database containing information on, for example, dietary habits and natural history for more than 50 representative species of interest. The module includes an innovative approach to characterizing the diet: a probabilistic algorithm that cycles through the database on minimum and maximum prey preferences to simulate dietary variability.

The Ecological Exposure Module focuses on two areas:

- Representative ecological receptors, and
- Potential and relevant exposure pathways.

Representative ecological receptors for the following groups of taxa were used to populate habitats:

Terrestrial Habitats
- Terrestrial plants
- Soil biota
- Reptiles
- Birds
- Mammals

Waterbody Margin Habitats
- Aquatic plants
- Aquatic invertebrates
- Benthos
- Fish
- Amphibians
- Birds
- Mammals

Wetland Margin Habitats
- Wetland plants
- Hydric soil-associated invertebrates
- Aquatic invertebrates
- Fish
- Amphibians
- Reptiles
- Birds
- Mammals

Representative Ecological Receptors. The ecological receptors considered in the 3MRA modeling system include the following:

- Terrestrial wildlife using habitats at or near WMUs; this wildlife can be directly exposed to waste or indirectly exposed by consuming plants, soil, or prey items that bioaccumulate constituents released from the WMU.

- Aquatic plants and other biota; these may be exposed to contaminants that are transported from a WMU to nearby aquatic habitats.

- Vascular plants and other terrestrial biota; these may be exposed to contaminants that are transported from the WMU to nearby terrestrial habitats (e.g., surficial soils).

Because all potentially affected species cannot be assessed in a national-level assessment, potentially affected plants and wildlife were organized into guilds of taxonomically and functionally related organisms (e.g., herbivorous birds, insectivorous birds, carnivorous mammals). Receptors were selected to represent each guild based on taxonomic relatedness, function in the ecosystem, and availability of wildlife exposure factors and toxicity data. For example, the American robin may be selected to represent insectivorous birds at a WMU because (1) it is a bird, (2) it eats insects and worms, (3) it is one of many thrushes observed at or near the WMU, and (4) wildlife exposure factors have been established (U.S. EPA, 1993).

Receptors were assigned to habitats within the AOI based on site-based and regional data. Representative ecological receptors were assigned to appropriate habitats based on documented foraging and feeding behavior and habitat usage.

Ecological Exposure Pathways. Table 2-6 lists the media and exposure routes that are evaluated by the Ecological Exposure Module. Direct contact with contaminants by terrestrial wildlife was not included in the model because dense undercoats or down effectively prevents contaminants from reaching the skin of wildlife species and significantly reduces the total surface area of exposed skin (Peterle, 1991; U.S. ACE, 1996). Also, results of exposure studies indicate that exposures due to dermal absorption are insignificant compared exposures due to ingestion for terrestrial wildlife (Peterle, 1991). Similarly, inhalation of volatile organic chemicals was not included in the model because concentrations of volatile chemicals released

Table 2-6. Ecological Exposure Routes Evaluated by Receptor Type

	Direct Contact			Ingestion		
Receptor	**Surface Water**	**Sediment**	**Soil**	**Surface Water**	**Soil**	**Food**
Plants						
Aquatic	✔	✔				
Terrestrial			✔			
Invertebrates						
Aquatic	✔	✔			✔	✔
Terrestrial			✔		✔	✔
Wildlife						
Fish	✔					
Amphibians	✔					
Reptiles				✔	✔	✔
Birds				✔	✔	✔
Mammals				✔	✔	✔

from soil to air are drastically reduced, even near the soil surface (U.S. ACE, 1996), and inhalation toxicity data for laboratory rats and mice show that volatile organic chemical concentrations in soils would have to be great to induce noncarcinogenic effects in wildlife (U.S. ACE, 1996). In addition, availability of inhalation benchmarks for ecological receptors is limited. Additional detail on the Ecological Exposure Module can be found in Section15of this document.

Ecological Risk Module. The Ecological Risk Module calculates HQs for a suite of ecological receptors assigned to habitats delineated at a site. These receptors fall into eight receptor groups: mammals, birds, herpetofauna, terrestrial plants, soil community, aquatic plants and algae, aquatic community, and benthic community. The spatial resolution of the Ecological Risk Module is largely determined by both the home ranges and the habitats delineated at each site. The habitats are intended to represent habitats across the United States that may be found at or near WMUs and that support wildlife receptors.

The habitat area is important in assessing risks to several receptor groups (e.g., benthic community); exposures and associated risks are considered across the entire habitat rather than for one or more home ranges. For example, contaminant concentrations to which the aquatic community is exposed are represented by a habitat-wide average that may include multiple stream reaches. The temporal resolution is based on annual average applied doses (for comparison with ecological benchmarks [EBs]) and media concentrations (for comparison with chemical stressor concentration limits [CSCLs]).

The 3MRA modeling system calculates HQs for all receptors assigned to the study site. The HQs are used in developing cumulative distributions of HQs. Each of the HQs calculated by the Ecological Risk Module has a series of attributes associated with it that allows ecological risks to be interpreted in a number of ways. For instance, distance from the source (i.e., 1 km, 1 to 2 km, or across the entire site) is important in understanding the spatial character of potential ecological risks. Other attributes considered relevant to ecological risks and regulatory decision making include the following:

- Habitat type (e.g., grassland, pond, permanently flooded forest),
- Habitat group (i.e., terrestrial, aquatic, and wetland),
- Receptor group (e.g., mammals, amphibians, soil community), and
- Trophic level (i.e., producers, TL1, TL2, TL3 top predators).

The maximum HQ across the site is also reported along with its ecological risk attributes.

In calculating receptor-specific HQs, the Ecological Risk Module does all of the necessary accounting to develop distributions based on the specific receptor and habitat groupings of interest. The Ecological Risk Module calculates HQs based on the EB or CSCL and the chemical exposure information, and provides summaries of ecological risk information from the simulation to determine when critical years with maximum HQs are experienced. Outputs are provided for all attributes associated with each receptor. Additional detail on the Ecological Risk Module can be found in Section 16 of this document.

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3.0 Spatial Aspects of Environmental Settings and Receptors

The 3MRA modeling system uses site-based modeling to provide the spatial distribution of contaminants and receptors in a specified area of interest (AOI) around each of the sources being modeled. To support this site-based strategy, a geographic information system (GIS) is used to provide a spatial frame of reference and associated attributes around each site or facility within the AOI. This section describes the spatial layout and the methods and data used to characterize the various spatial attributes of an AOI in the 3MRA modeling system. The spatial aspects of an AOI and the associated data collection process are discussed in greater detail in Volume II of this report, *Data Collection for the 3MRA Modeling System* (U.S. EPA, 2003).

3.1 Overview of Spatial Layout for Environmental Settings and Receptors

The spatial layout of an AOI used in the 3MRA modeling system consists of the source of constituent releases, the surrounding environment, and the location of receptors. The sources, such as a landfill, surface impoundment, industrial facility, or abandoned site, are stationary, and the AOI is defined by a distance from the source. This distance could be a few hundred meters up to 50 km. The 50 km limit is imposed by the limits of the air dispersion model (ISCST3) used in the 3MRA modeling system. The distance defining the AOI is wholly dependent on the areal extent needed for a specific analysis. For the sites and environmental settings currently in the 3MRA representative national data set, the AOI distance is set at 2 km from the boundary of the source.

3.1.1 Settings and Areas of Interests

The 3MRA modeling system incorporates a number of attributes that characterize the environmental setting of the AOI. A setting is defined as the source or sources of pollutant releases being modeled plus the characteristics of the environment in the AOI. The 3MRA modeling system uses more than 700 total variables to describe a site's environmental setting. The values for many of these variables are based on site-specific information. Others are based on regional or national data sets, when site-specific data are not available. A GIS was used to overlay layers of spatial data to construct a complete description of the environmental setting within the AOI.

3.1.2 Site Layout and Spatial Data Layers

Several key spatial data layers for each setting are shown in Figure 3-1. The layers are defined as follows:

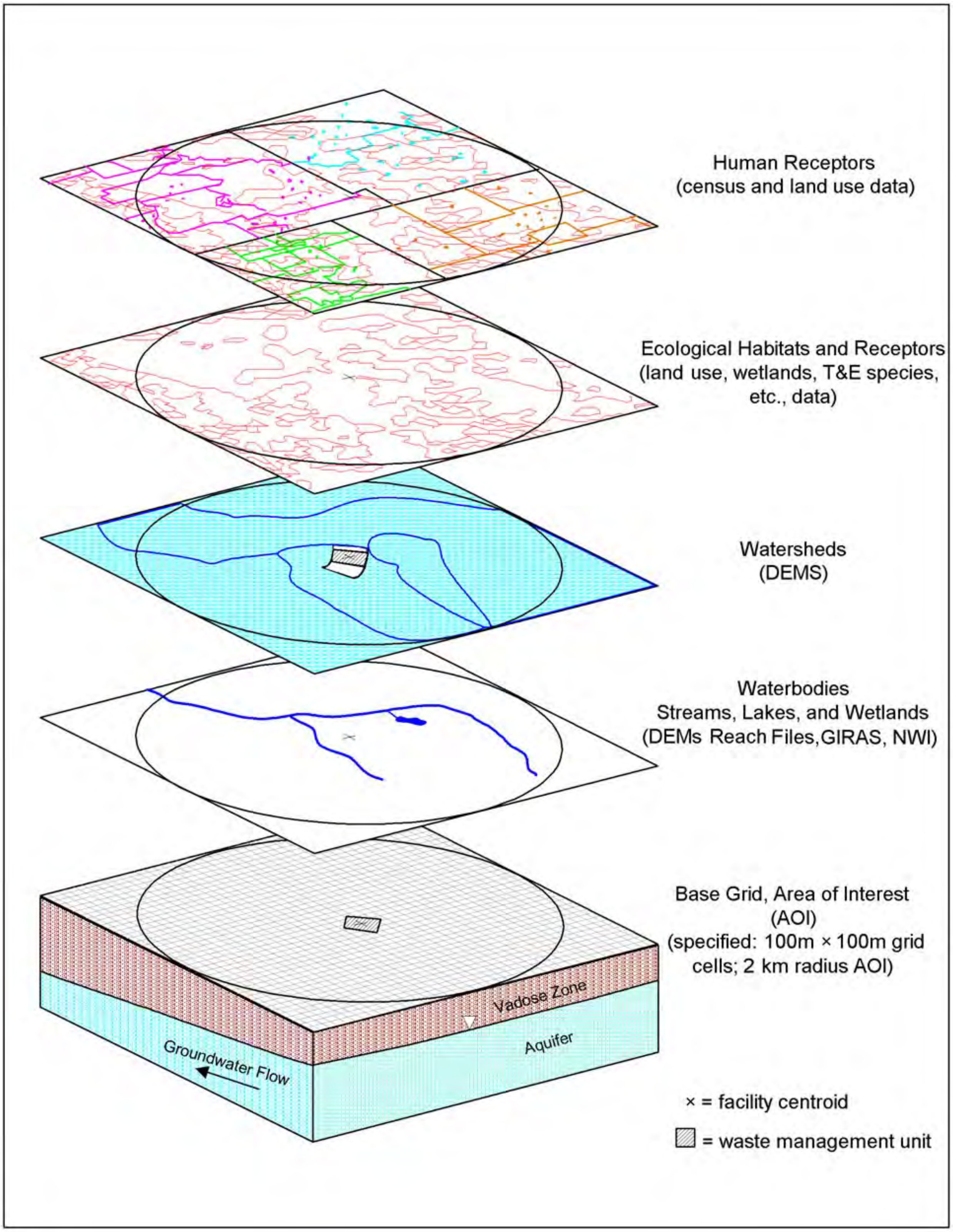


Figure 3-1. Site-based spatial overlays for 3MRA modeling system spatial framework (GIS analysis).

- **Human receptor points** are defined by U.S. Census block centroids for residents, and by randomly placed farms based on Census block group, agricultural census, and land use data for farmers (U.S. EPA, 2000b). These locations are used to calculate exposure concentrations in various media and to estimate risks to human receptors. A detailed explanation of how these data are used is provided in Volume II (U.S. EPA, 2003).

- **Ecological habitats and receptors** are defined by land use and other ecologically relevant data (U.S. EPA, 2002). These locations are used to calculate exposure concentrations in various media and to estimate risks to ecological receptors. A detailed explanation of how these data are used is provided in Volume II (U.S. EPA, 2000a).

- **Watersheds and watershed subbasins** are delineated either electronically, using digital elevation models (DEMs) of topography, or manually, based on Reach File 3 (RF3) stream networks (U.S. EPA, 2002). Watershed subbasins provide the spatial context and connectivity necessary to model chemical deposition, erosion, overland transport, and resultant soil concentrations in the Waste Pile, Land Application Unit (LAU), and Watershed Modules. Typically, a watershed is the entire drainage area for a particular stream network in the AOI, and is subdivided into subbasins. A watershed subbasin can vary in size from a portion of a hillside to much larger areas encompassing regional stream or river networks. In all cases, a subbasin is treated as a single, homogeneous area with respect to soil characteristics, runoff and erosion characteristics, and constituent concentrations in soil.

- **Waterbodies** (lakes, streams, and wetlands) are defined by DEMs, RF3 data, land use data contained in the Geographic Information Retrieval and Analysis System (GIRAS), and/or National Wetlands Inventory (NWI) (see U.S. EPA, 2002). Waterbodies provide the spatial context and connectivity necessary to model contaminant deposition, fate, transport, and the resulting water column and sediment concentrations in streams, lakes, and wetlands.

- **Subsurface features** include the surficial aquifer (the unconfined ground water source nearest the surface) and the vadose zone (the soil zone between the ground surface and the surficial aquifer). The vadose zone and surficial aquifer are assumed to directly underlie the source. The State Soil Geographic Data Base (STATSGO) provides vadose zone characteristics. Aquifer properties are obtained from regional data sets. Ground water flow is assumed to follow the topography of the land surface, as determined from the DEMs.

In the GIS, each of these spatial data layers is composed of 2-D polygons, except for streams, which are defined by 1-D vectors (stream reaches), and human receptor locations (residences and drinking water wells), which are defined as points. Because these polygon coverages could not be exported directly to the 3MRA modeling system, each spatial data layer was defined in terms of a base grid composed of 100 × 100 m cells, which roughly correspond to the minimum resolution of several site-specific data types (i.e., land use, topographic, and soil). This base grid (or *x*,*y* coordinate system) serves as the basis for defining receptor points at which the Air, Vadose Zone, and Aquifer Modules in the 3MRA modeling system produce constituent concentrations (and deposition rates for the Air Module) in terms of distance and direction from the constituent source. To provide the 3MRA modeling system with the data necessary to specify air points and well locations, spatial data are passed to the model using this site coordinate system for the following data layers:

- Watersheds,
- Waterbodies,
- Farms,
- Human receptor points,
- Wells (human receptor points with drinking water wells), and
- Ecological habitats.

The site coordinate system is described using metric *x*,*y* coordinates relative to the ground surface at the facility or site centroid (the georeference point). The 3MRA modeling system requires that the georeference point be specified using a latitude and longitude in the Universal Transverse Mercator (UTM) coordinate system. Using GIS, a coverage of spatial data (e.g., watersheds) is overlaid on the standard grid. If the centroid of a grid cell intersects a polygon, it receives the identifier for that polygon (see Figure 3-2).

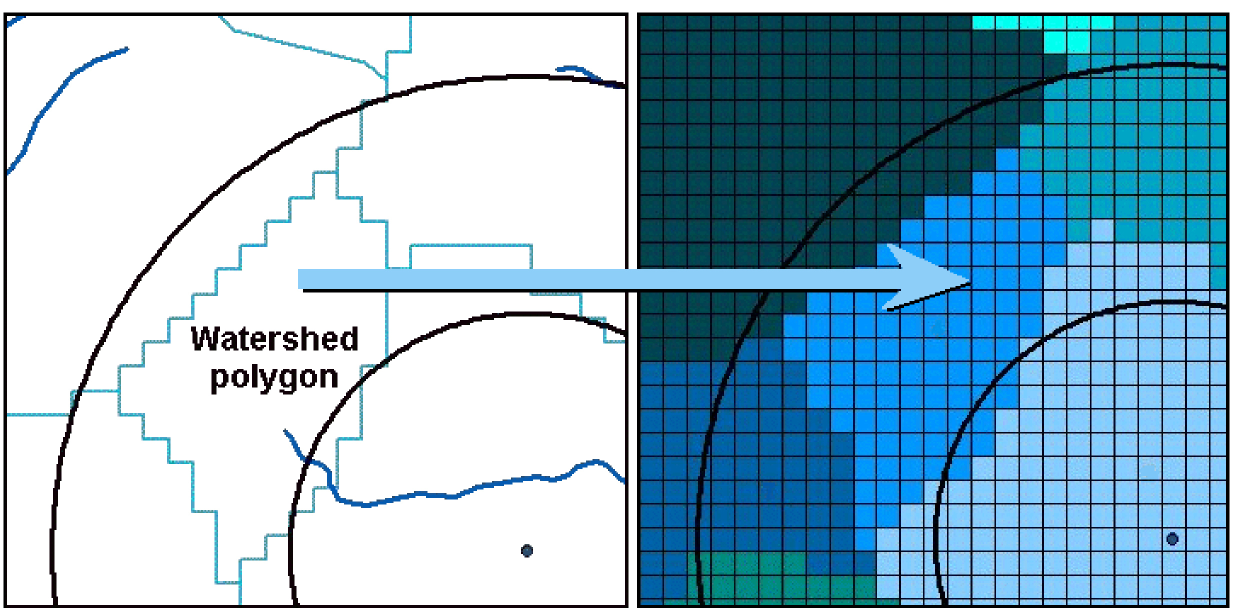


Figure 3-2. Example of transfer of polygons to 100 × 100 m template grid.

3.2 Watersheds and Waterbodies

To develop the watershed/waterbody layout, GIS programs were used to

- Compile available hydrologic, DEM land use, and wetlands data sets (see Figure 3-3);
- Extract site-specific data from these data sets; and
- Delineate the watershed subbasins, waterbodies, and local watersheds.

The watershed layout includes the watersheds that contribute flow to waterbodies within the AOI. Although the waterbody/watershed layout is important for several modules in the 3MRA modeling system, the layout primarily supports the surface water, regional watershed, and local watershed components. These components serve the following purposes:

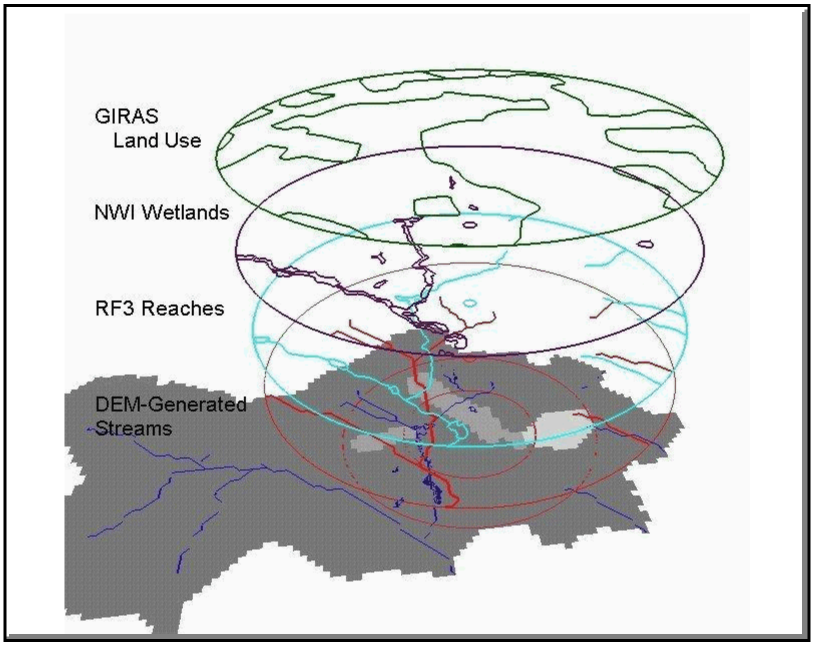


Figure 3-3. GIS data coverages for waterbody and watershed delineation.

- The Surface Water Module estimates water column and sediment chemical concentrations for use in the Aquatic Food Web and Ecological Exposure Modules.
- The Watershed Module estimates surficial (top 1 cm) and depth-averaged soil concentrations within the AOI for the Terrestrial Food Web, Farm Food Chain, and Human and Ecological Exposure Modules; generates hydrologic inputs (runoff, baseflow, soil loads) for the Surface Water Module; and generates annual average infiltration estimates for use as recharge for the Aquifer Module.
- The Waste Pile and LAU Modules estimate runoff, soil loads, and contaminant loadings for upgradient and downgradient subbasins. The modules simulate contaminant movement due to water erosion and overland transport from the LAU or waste pile to the nearest downslope waterbody.

Within the context of the 3MRA modeling system, certain assumptions and definitions were required to develop the procedures for delineating the watersheds, subbasins, and waterbodies to be modeled. Figure 3-4 illustrates the conceptual framework for the waterbody and watershed layout, which is detailed below.

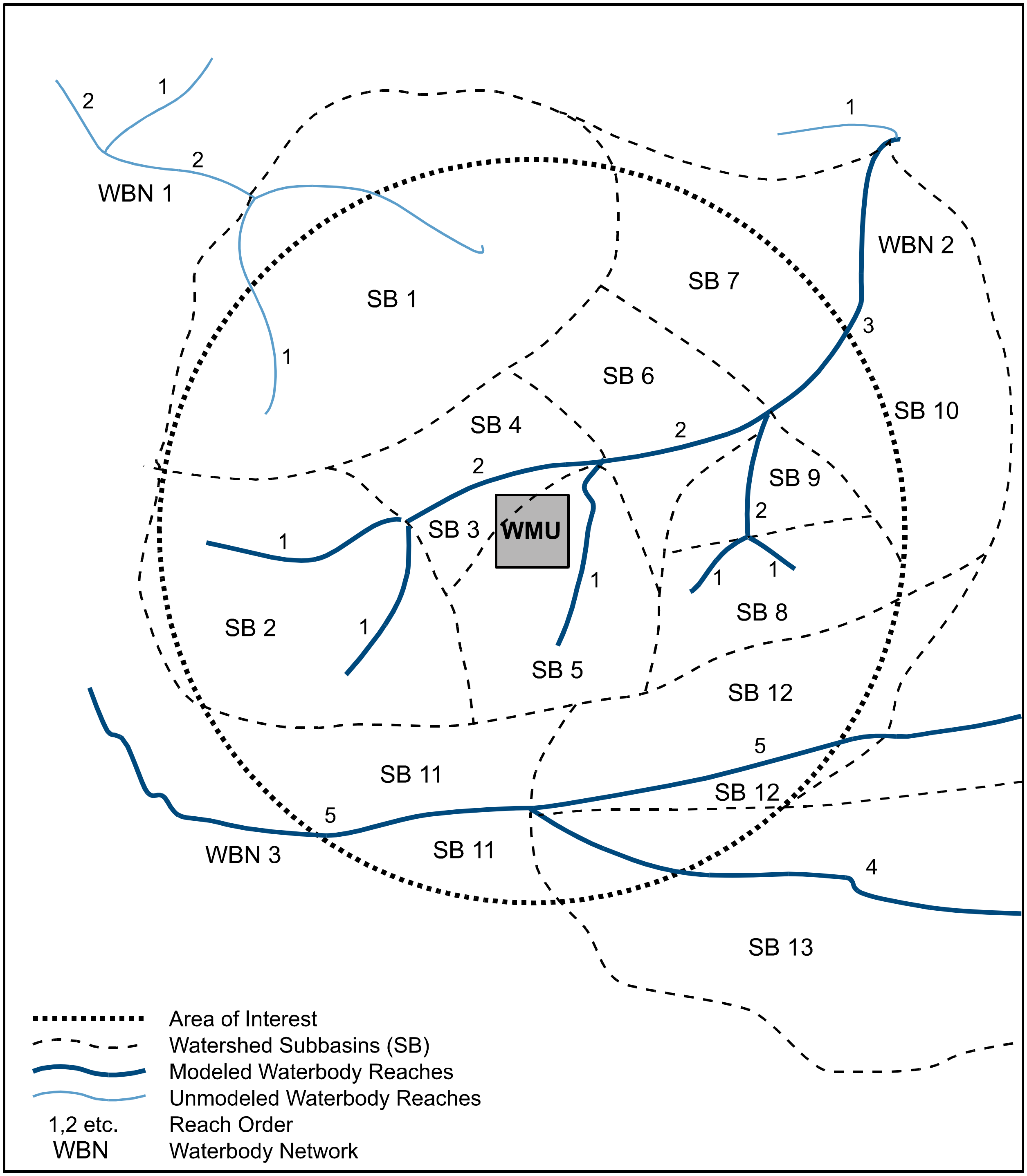


Figure 3-4. Regional watershed subbasin delineation.

3.2.1 Definitions for Watershed and Waterbody Layout

The 3MRA modeling system uses the following definitions for watersheds and waterbodies:

- A *waterbody reach* is a stream segment between tributaries, a lake (or pond),[1] a wetland, or an estuary. A waterbody reach is the basic modeling unit for the Surface Water Module.

- A *headwater reach* (or first-order reach) is a reach that flows into another reach downstream but has no upstream reaches draining into it.

- An *exiting reach* is a nonheadwater reach that flows out of the AOI.

- A *waterbody network* is a series of connected (possibly branching) reaches that defines a stream network (three waterbody networks are shown in Figure 3-4, labeled WBN 1, WBN2, and WBN3).

- A *regional watershed* is the entire drainage basin associated with an individual modeled waterbody network and can extend outside of the AOI, as necessary. Upstream watersheds are not delineated for any waterbody reaches greater than order 5[2] entering the AOI because the volume of water in this size stream or river would dilute the concentration of a constituent from a source. However, tributary lands for reaches greater than order 5 lying *within* the AOI are delineated as regional watershed subbasin(s) (see definition below) to simulate soil concentrations. Such subbasin delineations only include land within the AOI. The upstream boundary of any regional watershed is its natural, upstream boundary (i.e., its headwater basin, except, as noted above, for those greater than order 5).

- A *watershed subbasin* is a portion of the regional watershed. A regional watershed is composed of a set of nonoverlapping subbasins. The set of all subbasins for a given regional watershed will completely cover the regional watershed, except for watersheds greater than order 5. For example, in Figure 3-4, Subbasins 2 through 10 make up the regional watershed for Waterbody Network 2. Depending on the waterbody layout, one or more subbasins can drain into a reach, or a subbasin can drain into one or more reaches. The subbasin is the basic spatial modeling unit for the Watershed Module.

- A *local watershed* is a drainage area that contains a source, such as an LAU. A local watershed extends from the upslope drainage divide downslope to the first

[1] All lakes and ponds were designated as lakes in the waterbody layout data.

[2] As defined by the Strahler ordering system (Strahler, 1957).

defined drainage channel, lake, or pond. It is divided into subareas upslope and downslope of the source and the source itself. These subareas are the basic modeling units for the local watershed model.

3.2.2 Assumptions for Waterbodies and Regional Watersheds

The following assumptions were used to develop and collect data associated with the regional watershed and waterbody layout:

- The Surface Water Module models streams of order 1 through 5, ponds, lakes, and wetlands. All lakes, ponds, and wetlands connected to a waterbody network are modeled regardless of order. Only reaches lying completely or partially within the AOI are modeled.

- Streams of order 1 through 5, ponds, lakes, and wetlands within the AOI are modeled until they exit the AOI. Headwater reaches (order 1) that exit the AOI are not modeled.

- Order 3 and higher stream reaches, along with lakes, ponds, and certain wetlands, are assumed to be fishable (i.e., support fish populations suitable for recreational anglers). Order 1 and 2 stream reaches within the AOI that feed into an order 3 reach within the AOI contribute to the contaminant concentration in the order 3 stream reach.

- Surface waters in the waterbody network are subject to contaminant loads from indirect runoff (i.e., from aerial deposition of a contaminant on the watershed). Some are subject to both indirect and direct loads (i.e., direct runoff/erosion from the source).

3.2.3 Assumptions for Local Watersheds

Some sources, such as waste piles and LAUs, release pollutants directly through runoff and erosion. Assumptions associated with the local watershed component of these sources include the following:

- If the source overlaps a drainage divide, multiple local watersheds are modeled.

- The local watershed will contain at least two subareas—the source (or portion thereof) and a 30.5 m wide (100 ft wide) buffer between the source and the stream (see Figure 3-5). The local watershed may also include a third, upslope subarea from the drainage divide to the source.

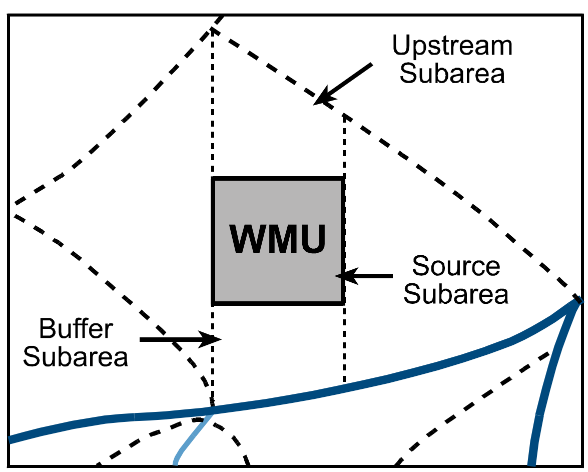


Figure 3-5. Local watershed delineation.

3.2.4 Watershed Soils

The 3MRA modeling system estimates surface soil concentrations of a contaminant in each of the watershed subbasins within the AOI. It also provides estimates of the vadose zone soil concentrations, but only under the source, not in the whole watershed. The depth of the surface soil is defined by the user in the 3MRA modeling system. Vadose zone soil extends from the ground surface to the water table. Depending on the location of the AOI, average (area- and depth-weighted) or predominant soil properties are derived for the soil depth zone of interest across each watershed subbasin or the waste management unit (WMU). All soil parameters are either site-specific or derived from site-specific data using national relationships and thus are considered site-based. Parameters derived from site-specific data include soil hydrologic properties derived from site-specific soil texture or hydrologic class, and other properties derived from a combination of soil texture or class and site-specific land use.

3.3 Human Receptors, Farms, and Wells

Human receptor locations, which include residences and farms, are one of the primary spatial data layers in the 3MRA modeling system. They enable human risk or hazard to be calculated spatially within the AOI where people are likely to be located. GIS data are used to locate these points and collect human receptor numbers and characteristics (e.g., receptor types, age cohorts). Using this information, the 3MRA modeling system can generate risk distributions around a source that are based on the population within the AOI. These data provide a level of specificity for a national-level assessment that accounts for how many people are affected and at what levels of risk.

As discussed in Section 2, human receptors are subdivided into groups that reflect differences in exposure patterns due to either location or behavior. Resident human receptor points are located and populated by 1990 Census block data (U.S. Bureau of the Census, 1987, 1992; U.S. EPA, 1995a,b). Farms are located and populated using Census block group boundaries, subdivided by farm land use, along with county-level agricultural census data.

Figure 3-6 illustrates many of the spatial data elements related to human receptors for a sample AOI. This AOI is relatively simple in terms of receptor population density. It includes 20 residential locations, one for each (populated) Census block, and one farm. Contaminant concentrations are estimated at the single point at the centroid of each Census block to evaluate the exposure and risk or hazard to residential receptors. For example, the soil concentrations are an areal average of the watershed subbasin soil concentrations within the Census block. The ambient air concentration is the average of air concentrations at points modeled within a Census block. The ground water concentration is evaluated at a well at the Census block centroid when some or all of the receptors in the Census block use private drinking water wells. The estimated risk or hazard quotient (HQ) for each receptor type and age cohort at a Census block centroid is assigned to all individuals in that Census block that are the same receptor type and age cohort. The farm is randomly located, but its location is constrained to be both within a Census block group that includes farming activity and also within an agricultural land use area within that block group. For this example, a single representative farm is placed within one block group to

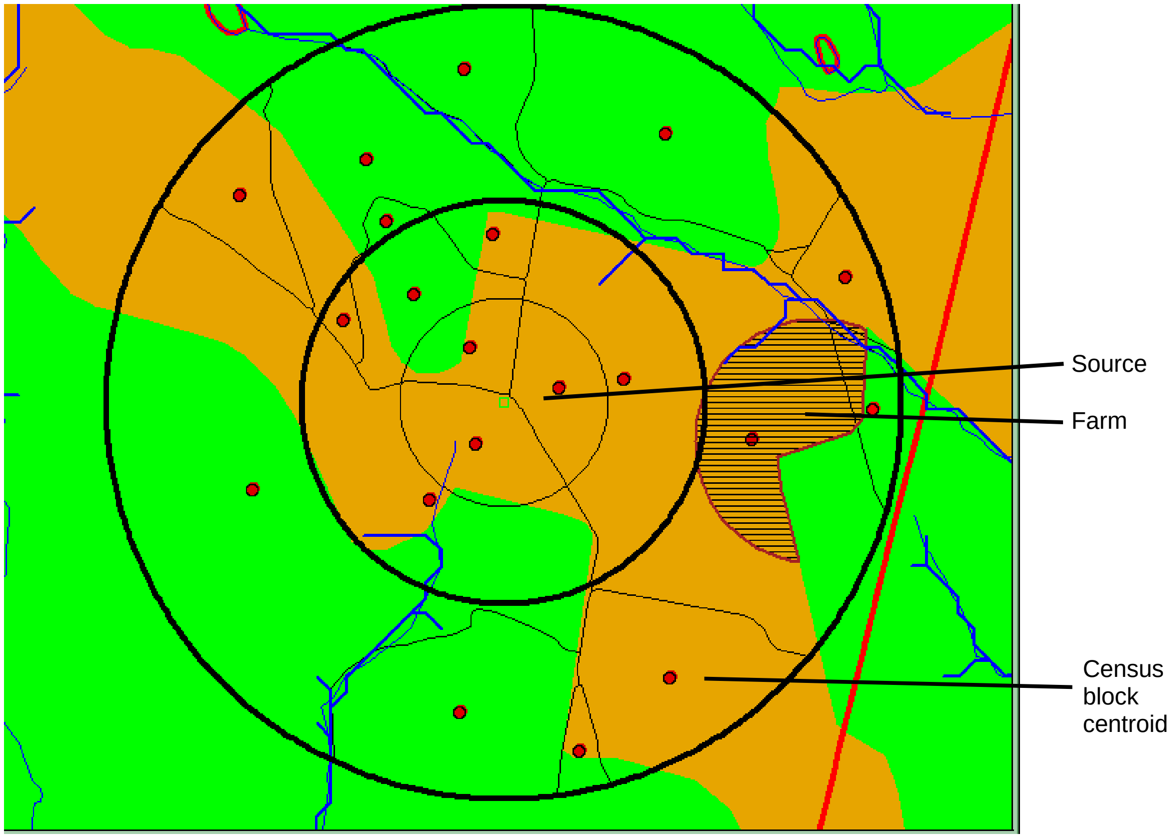


Figure 3-6. Example site layout for human receptors.

represent the exposures for all farms within that block group. The farm population by receptor type and age cohort in the block group is assigned the risk or HQ associated with the representative farm.

The following subsections summarize the steps used to place and process human receptor locations, farms, and wells. Details of these procedures are presented in U.S. EPA (2002). The total number of people within an AOI is divided into receptor types and age cohorts. These are further subdivided by whether or not households have a private drinking water well. Measures are taken with respect to numbers of people at a site in order to maintain the correct total population across the site.

3.3.1 Resident and Home Gardener Locations

Census block coverages are used to locate residents and home gardeners. A resident or home gardner receptor is placed at the centroid of every Census block within the AOI. The characteristics within each Census block coverage are linked to the other spatial data (regional watersheds, local watersheds, and distance rings) within the system. Area weightings are used to calculate the number of receptors/cohorts and the environmental characteristics within the AOI. For example, if Census blocks extend across distance rings, the block data are fractioned into each ring/block combination based on the relative areas in each.

3.3.2 Farm Locations

Farms within the AOI are located by overlaying the Census block groups with the land use coverages that identify areas with farm-type land use (i.e., crop and pastureland) in block groups that have farmers. A farm is placed in each of these block group/farm land use areas as follows:

- The median farm size for the county is determined based on a census of agriculture data (U.S. Bureau of the Census, 1987, 1992). These data include the median farm size and fractions of farms that are beef or dairy farms by county. The fraction of beef or dairy farms is calculated as the number of beef or dairy farms divided by the total number of farms in a county.

- If the block group/farm land use area is larger than the median farm size, then a random point inside the farm land use area of the block group is picked. A circle is created to represent the area of the farm. To keep the farm within the block group/farm land use area, the farm is clipped at the farm land use boundary. The radius of the circle is increased incrementally until a polygon is produced that approximately matches the median farm area and is completely contained within the farm land use area of the block group. If the farm land use area within a block group is smaller than the median farm size, then the entire area of farm land use in the block group is used as the farm.

- Once the farms are placed within the AOI, the fractions of beef farms and dairy farms are applied to the total number of farms to calculate the number of beef and dairy farmers in each block group with a farm.

3.3.3 Well Locations (Private Wells Only)

Census block group data include the number of households with private wells. These data are used to calculate the fraction of people with private water supplies within the AOI. This block group fraction is applied to block-level human receptor points within each block group to identify points with wells. Private wells are located at the centroid of each Census block that contains people on private wells. Wells are also located at the centroid of each farm. Every farm is assumed to be on a private well. Not all private wells are necessarily affected by the ground water contamination from the source, only those that are predicted to be within the plume of contamination given the characteristics at each site.

3.3.4 Recreational Fisher Locations

All receptor types may also be recreational fishers. The location of fishers is the same as for residents, gardeners, and farmers. For example, a recreational fisher may be a resident located at the centroid of a Census block. However, the fisher may catch fish in up to three streams within the AOI. If there are more than three fishable streams in the AOI, the location of the three streams in which a particular receptor fishes is randomly selected.

The number of recreational fishers in each category of receptor types is based on 1996 National Survey of Fishing, Hunting, Wildlife, and Recreation (NSFHWAR) data, which include the percentage of residences with recreational fishing licenses in urban, rural, and rural farm areas by state (U.S. DOI and U.S. DOC, 1997). For each state, recreational fisher percentages can be estimated using the NSFHWAR urban and rural population breakdowns and splitting the rural population into farm and nonfarm populations using farm/nonfarm population fractions calculated from 1990 U.S. Census data (U.S. Bureau of the Census, 1987, 1992). This provides state-by-state estimates of urban, rural-farm, and rural-nonfarm recreational fishers as a percentage of the total state population. These percentages are used to calculate the recreational fisher population within an AOI.

3.4 Habitats and Ecological Receptor Placement

Representative habitats were developed that can be used in the 3MRA modeling system to describe the biological conditions within an AOI. Receptors were assigned to habitats based on wildlife-habitat relationships documented in the literature. The habitat was chosen as the appropriate level of resolution for the spatial element of the ecological risk assessment used for a national-level assessment. In this context, the term habitat implies a level of detail and specificity that is meaningful for estimating exposure but that does not require extensive biological inventory or field investigation for identification or delineation.

Fourteen representative terrestrial, wetland, and margin habitats have been developed for use in the 3MRA modeling system. Table 3-1 presents an overview of the representative habitats. The representative habitats address areas inhabited by land-based receptors. In addition to terrestrial mammals, birds, and herpetofauna, terrestrial receptors also include some species that spend significant time in the water, such as the bullfrog and snapping turtle, and some that derive all or most of their food from the water, such as the osprey and muskrat. In order to fully assess exposure, the margin habitats are defined and delineated to include both the waterbody and its adjacent terrestrial areas, such as stream corridors and pond margins, in which all these receptors are potentially exposed. Section 15 of this document describes how the representative habitats were selected, characterized, and delineated.

3.4.1 Habitat Delineation

The GIS data layers needed for habitat delineation are

- Facility/site location,
- AOI,
- GIRAS land use,
- Delineated waterbodies—rivers, lakes, and wetlands,
- NWI wetlands,
- RF3 waterbodies,
- Managed areas,
- 2 m DEM contours, and
- Preprocessed habitat grid.

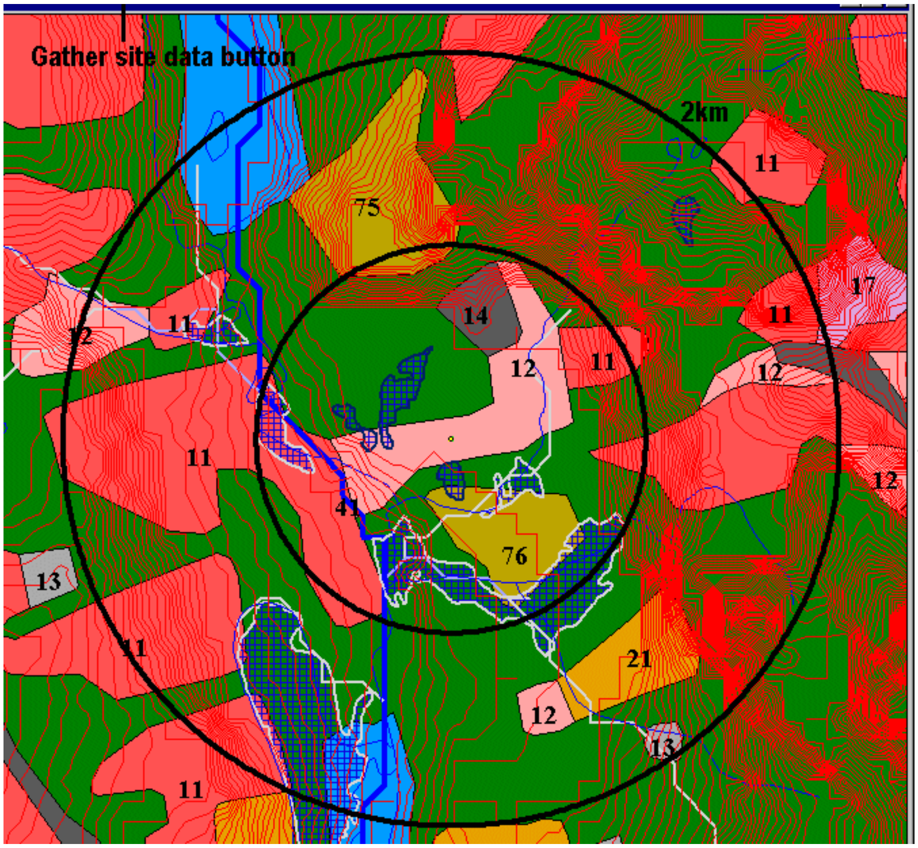
Table 3-1. Ecological Risk Assessment Representative Habitats for Terrestrial Receptors

Terrestrial Habitats
■ Grasslands
■ Shrub/scrub
■ Forests
■ Crop fields and pastures
■ Residential
Waterbody Margin Habitats
■ Rivers/streams
■ Lakes
■ Ponds
Wetland Margin Habitats
■ Intermittently flooded grasslands
■ Intermittently flooded shrub/scrub
■ Intermittently flooded forests
■ Permanently flooded grasslands
■ Permanently flooded shrub/scrub
■ Permanently flooded forests

Habitats are delineated by assigning each grid cell to one of the 14 representative habitats. A grid cell may also be designated as “no habitat” if it consists of highly developed or impervious areas (such as parking lots or train tracks) not likely to support wildlife. Figure 3-7 shows an example AOI with preprocessed habitats.

Terrestrial Habitat Delineation. To delineate the terrestrial habitats at each site, the representative habitat types were correlated with Anderson land use categories. The Anderson Land Use Classification, developed by the U.S. Geological Survey (USGS), assigns land use descriptors to areas that are distinguishable in satellite and other remotely sensed data. Digitized Anderson land use data are readily available through GIRAS and, therefore, provide a useful tool for locating habitats. GIRAS land use/land cover data delineate land use patterns wit respect to vegetation, human activity, and waterbodies. Although these data are 15 to 25 years old and therefore do not reflect current conditions in some locations, the GIRAS data set is the most complete and current national data set available. A full description of GIRAS data and methods for use in the 3MRA modeling system is presented in Volume II (U.S. EPA, 2003) of this report.

Delineation of Waterbody Margin Habitats. Some terrestrial receptor species depend on aquatic systems for some or all of their food. These exposure scenarios are assessed in the waterbody margin habitats. Stream corridor, pond, and lake margin habitats were delineated as being adjacent to the waterbodies generated as part of the waterbody and watershed layout data processing. Data sources for delineation of streams include DEM data (USGS, 1999), RF3 data



Land Use Code and Habitat

11 Residential

12 Commercial and services (nonhabitat)

13 Industrial (nonhabitat)

14 Transportation, communications, and utilities (included in largest adjacent habitat)

17 Other urban or builtup land (residential)

41 Deciduous forest land (forest)

21 Cropland and pastures (crop fields/pastures)

75 Strip mines, quarries, gravel pits (nonhabitat)

76 Transitional areas (nonhabitat)

Figure 3-7. Preprocessed habitat codes.

(U.S. EPA, 1994b), and GIRAS data (U.S. EPA, 1994a). Data sources for delineation of ponds and lakes include GIRAS and the NWI (U.S. FWS, 1998). Once the streams, ponds, and lakes were delineated for an AOI, their respective corridor and margin habitats were added.

There is no simple correlation between waterbody characteristics and wildlife distribution within corridors and margins. The vegetation, topography, and land use of the adjacent land, as well as the size, depth, flow, and aquatic food web of the waterbody, are but a few of the more prominent variables that affect wildlife use of waterbody margin habitats. Elevation contours were assumed to be the best indicators of stream corridors and pond and lake margins. Using DEM contour data, an attempt was made to determine a visual natural limit for the corridor or margin. Because waterbodies occur in the landscape along elevation contours, natural boundaries were frequently evident. If no contour-based boundaries were apparent, surrounding land use was used instead. For example, if GIRAS data indicated a forest buffer running parallel to a stream and a commercial or industrial area adjacent to the forest, the stream corridor would consist of the forest buffer. When neither contours nor land use indicated corridor or margin boundaries, a default minimum margin was delineated. Waterbody margin habitats at a single site were combined, or bridged, as were terrestrial habitat patches.

Delineation of Wetland Margin Habitats. National Wetlands Inventory (NWI) data provide the most complete and readily available digitized data on location and type of wetlands on a national scale (U.S. FWS, 1998). NWI data, however, have not yet been digitized for the entire United States. When digitized NWI data were not available for an AOI, wetland data from GIRAS were used.

Wetland identification and classification in the NWI are based on photo interpretation of remote imagery at a 1:24,000 scale. Each wetland polygon identified is classified and coded according to the Cowardin et al. (1979) classification system, which indicates substrate type, dominant vegetation, and hydrologic regime, as well as additional details regarding soil and water chemistry, where relevant. The GIRAS wetland data are on a 1:80,000 or smaller scale and distinguish only between wetlands dominated by woody vegetation and those not dominated by woody vegetation (Anderson et al., 1976). GIRAS data do not show wetlands smaller than 16 hectares (39.5 acres). Consequently, there is significant variation in the level of detail in the wetland data for sites with and without NWI coverage. Because the NWI provides high-quality data that readily distinguish the different wetland habitat types, it is the preferred source for wetland data.

Wetland Flood Regime. The 3MRA representative habitats include three intermittently flooded and three permanently flooded wetland types. The primary criterion for this division is the wetlands' ability to support fish populations. When NWI data were available, this distinction was made based on water regime modifiers in the NWI code. Based on the descriptions in Cowardin et al. (1979), a decision was made about whether each water regime would be expected to support an aquatic food web that includes TL3 or TL4 fish. Wetland types with flood regimes that indicate the presence of sufficient flooding to support fish populations are delineated as permanently flooded wetlands. Those wetlands in which flooding is infrequent or of short duration are delineated as intermittently flooded wetlands.

All GIRAS-identified wetlands are delineated as permanently flooded because GIRAS generally does not recognize wetland ecosystems at the drier end of the wetland flood regime continuum.

Treatment of Intermittently Flooded Wetlands. Wetlands designated as intermittently flooded were then further classified according to the representative wetland habitats as intermittently flooded grassland, shrub/scrub, or forest wetlands. These determinations also were made based on NWI codes.

For the 3MRA modeling system, the three intermittently flooded wetland habitat types were identified and designated, but their boundaries were not delineated. Many intermittent wetlands occur as small, scattered habitat patches and, even when combined with similar patches in an AOI, often do not constitute large enough areas for placement of home ranges. However, these areas are known to provide important, albeit highly fragmented, habitat. Therefore, during site layout data processing, intermittent wetlands were identified as inclusions within surrounding upland habitats (e.g., grassland, forest), and their unique receptor species are included in the data passed to the Ecological Exposure Module. In this manner, the receptors expected to occur in intermittent wetlands are included in the exposure assessment.

An exception was made to this approach for certain intermittent wetlands adjacent to streams of order 3 and higher. Water regimes for seasonally flooded and seasonally flooded/saturated imply seasonal flooding for extended time periods; in wetlands adjacent to fishable stream reaches (stream order 3 and higher), these water regimes could potentially support an aquatic food web. Therefore, these intermittent wetlands, whether grassland-, shrub/scrub-, or forest-dominated, were delineated when they occurred adjacent to streams of order 3 or higher. These habitats were treated in a manner identical to the permanently flooded wetlands, as described in the following section.

Delineation of Permanently Flooded Wetlands. Permanently flooded wetlands were identified at each site based on the NWI codes. When NWI data were not available, GIRAS data were used. GIRAS data classify wetlands as forested or nonforested and do not include any information on the flood regime. Because most national data sets generally apply the term wetlands to tidal and other aquatic habitats and do not recognize noninundated areas as wetlands, it is assumed that wetlands identified in GIRAS data fall within the permanently flooded wetland habitats. The forested GIRAS wetlands are delineated as permanently flooded forested wetlands; the nonforested GIRAS wetlands are delineated as permanently flooded grasslands. Although some of the wetlands included in the GIRAS data are undoubtedly dominated by shrub/scrub vegetation, the data do not allow this distinction to be made. In the absence of better data, the intermittently flooded grassland habitat is considered the most appropriate alternative.

Permanently flooded wetlands frequently occur in association with streams, rivers, lakes, and ponds. Thus, the potential arises for areas adjacent to waterbodies to include both wetlands and waterbody margin habitats. The most effective and straightforward approach to handling this situation appeared to be to default to the wetland habitat when wetlands and waterbody margin habitats overlapped. In fact, many wildlife receptors probably forage across both waterbody margin and wetland habitats, whereas other species show a preference for or tend to avoid the wetland habitat. Thus, wetland habitats were delineated whenever they were indicated, including within a waterbody margin. The wetlands occurring near waterbodies were not subsumed in the stream corridor or lake and pond margin habitat.

3.4.2 Assignment of Receptors to Habitats

Ecological risk is assessed for the wildlife receptors expected to be present within an AOI. A receptor is placed within the AOI if the AOI is located in the receptor’s geographic range and includes suitable habitat to support the receptor species.

The 3MRA modeling system was designed to support national analyses; therefore, the list of receptor species was developed to represent ecological regions throughout the contiguous United States. Within each AOI, the receptor species were assigned to represent all of the faunal classes, trophic levels, and feeding strategies that are typical of terrestrial and margin habitats, respectively. The simple food webs created for the terrestrial and margin habitats provided the context for receptor selection and were used to define the relationships between predators and prey. National applicability was achieved primarily by selecting species that are widely distributed throughout the contiguous United States, and then adding species to cover as many ecological regions as possible. Several criteria were established as part of the selection process, including geographical distribution, availability of data pertinent to exposure (e.g., body weight, dietary preferences), and representation of the faunal classes and functional niches represented in the terrestrial and margin food webs. In some instances, receptor species were added to represent food web components in regions where “common” species were not likely to be found.

For example, the mule deer was selected as a large herbivore representing regions where the white-tailed deer is not distributed. Similarly, the eastern cottontail rabbit and the black-tailed jackrabbit were both included in the grasslands receptor group in order to provide a small mammalian herbivore in temperate regions of the eastern United States and the drier western regions. Based on information describing Bailey's ecological regions (Bailey, 1994), receptor species were assigned to habitats within the AOI only if (1) the species was documented to occur in the ecological region containing the site, and (2) the species represented a component of the food web.

In assigning ecological receptors to habitats within a given AOI, it is important to recognize the implicit assumption with respect to species occurrence. Specifically, it is assumed that all receptor species occur in the representative habitats to which they are assigned regardless of the AOI's position in the landscape. That is, if a forest habitat is delineated at a site, all the species included in the terrestrial food web for the forest habitat that occur in that particular region are assumed to be present. Although the simple food webs represent the major trophic elements and feeding strategies that are likely present in a representative habitat, the food webs are simplifications of what may be very different structures. In fact, it is unlikely that all of the receptor species would be present, particularly those that are less adapted to human impacts and development. For example, the black bear is included in the forest habitat receptor group and is assessed at sites within its geographic range where forest habitat is indicated. However, the black bear requires tracts of land significantly larger than the AOI, and this pattern may not occur to an appreciable extent in areas where the land uses indicate an industrial base. As a result, the list of receptors assigned to each habitat delineated within the AOI does not reflect differences in species diversity related to habitat quality. Although small habitat patches were connected within the AOI to simulate typical habitat use by wildlife, "habitat bridging" reflected in the site layout files does not address the issue of habitat quality and species occurrence.

3.5 Summary

The delineation of habitats allows for the placement of receptor species within the AOI. A substantial amount of data processing is required, including the assignment of receptor species based on habitat and region, and the placement of the receptor species' home ranges at each site. These steps are described in detail in Volume II of this report (U.S. EPA, 2003) and in Section 15 of this document.

The spatial delineation of habitats throughout an AOI and the assignment of species to these habitats provides an overall approach for the national assessment of ecological risk. The variability in species affected by a contaminant release, both within a site and between sites, is captured by this approach.

3.6 References

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4.0 Wastewater Source Modules

4.1 Purpose and Scope

The Wastewater Source Modules simulate wastewater management. Two Wastewater Source Modules were developed for the 3MRA modeling system to represent the common management practices for industrial wastewaters. These management practices include flow equalization, storage, treatment (typically biological treatment or neutralization), and solids settling (clarification). The two waste management units (WMUs) for wastewaters are

- **Surface Impoundments**, which may be either aerated or quiescent and are used to treat, store, or dispose of many industrial wastewaters; and

- **Aerated Tanks**, which are aerated or mixed tanks used to treat or store many industrial wastewaters.

The two Wastewater Source Modules use a common set of algorithms with similar mass balance and transport equations. Both modules estimate volatile emissions to air. Leaching losses from the bottom of the unit are modeled only for surface impoundments; tanks are assumed to have impervious bottoms. Both modules simulate degradation and solids settling, although solids settling and accumulation are more significant for quiescent units.

The two Wastewater Source Modules were designed to provide estimates of annual average volatilization rates to air, which are used by the Air Module. In addition, the Surface Impoundment Module outputs annual average infiltration rates and leachate constituent flux rates, which are used by the Vadose Zone and Aquifer Modules, and annual average surface impoundment water concentrations, which are used by the Ecological Exposure Module to estimate exposure to wildlife receptors that may drink or consume organisms from the impoundment. Figure 4-1 shows the relationship and information flow between the Wastewater Source Modules and the 3MRA modeling system.

The Wastewater Source Modules have six major functions, as follows:

1. **Calculate constituent concentrations within the unit.** The Wastewater Source Modules use a mass-balance, temperature-adjusted approach to estimate constituent concentrations in the WMU. This approach considers constituent diffusion between wastewater and sediments, and constituent removal by volatilization, biodegradation, hydrolysis, partitioning to solids, solids settling, and, for surface impoundments only, infiltration through the bottom of the unit.

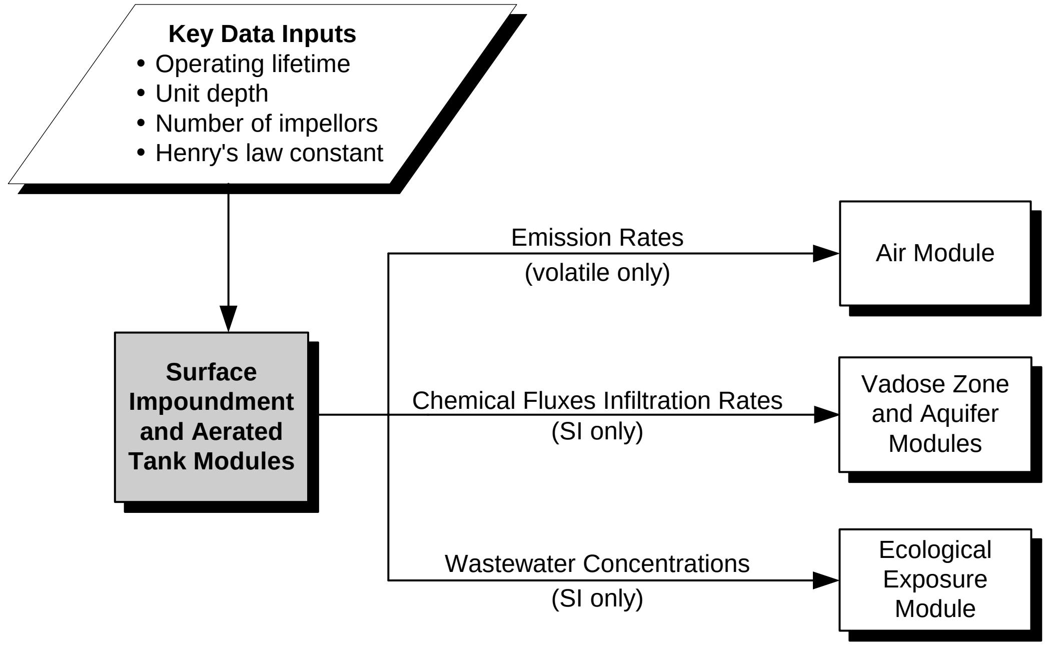

Figure 4-1. Information flow for the Wastewater Source Module in the 3MRA modeling system.

2. **Calculate solids concentrations within the unit.** The Wastewater Source Modules use a mass-balance approach to estimate solids concentrations in the WMU.

3. **Calculate volatile emission rates.** The modules calculate volatile emission rates for both aerated and quiescent surfaces.

4. **Estimate resuspension, sedimentation, and burial velocities within the units.**

5. **Estimate constituent release in leachate.** The Surface Impoundment Module calculates infiltration rates and constituent leachate flux rates for use in the Vadose Zone and Aquifer Modules. The Aerated Tank Module does not calculate leachate release, as tanks are assumed to have impervious bottoms.

6. **Adjust for temperature effects.** The modules account for the effect of temperature on air viscosity and density, water viscosity, chemical properties, and sediment biodegradation rates.

4.2 Conceptual Approach

This section first describes the two different WMUs and the processes modeled for them, and then describes the major model functions listed above.

4.2.1 Description of Waste Management Units

The Aerated Tank Module was developed to model tanks that are aerated or mixed; thus, it uses a well-mixed, steady-state mass balance solution. Tanks are assumed to have an impervious bottom; therefore, leaching is not modeled for tanks.

The Surface Impoundment Module was developed to model both aerated and quiescent impoundments using either a well-mixed, steady-state mass balance solution for well-mixed impoundments or a time-dependent mass balance solution for plug flow, batch, or disposal impoundments.

The following assumptions were made in developing both modules:

- The WMU is divided into three distinct compartments: impoundment liquid, unconsolidated sediment, and consolidated sediment. Each compartment has a fixed volume for a given monthly solution and volumes are readjusted to account for solids accumulation. The three-compartment model is described in more detail later in this section.

- Aerated tanks are well mixed; surface impoundments may be well mixed or have plug or batch flow. For well-mixed units, a steady-state mass balance solution is used. For plug or batch flow impoundments, a time-dependent mass balance solution is used.

- Volatilization in the liquid compartment follows first-order kinetics.

- Hydrolysis in both the liquid and sediment compartments follows first-order kinetics.

- Aerobic biodegradation in the liquid compartment follows first-order kinetics with respect to both constituent concentration and biomass concentration.

- Biomass growth rate follows Monod kinetics with respect to total biological oxygen demand loading.

- Solids settling follows first-order kinetics.

- Anaerobic biodegradation of constituent in the sediment compartments follows first-order kinetics.

- The biomass decay rate within the accumulating sediment compartment is first-order.

- There is no constituent in precipitation (i.e., rain or snow).

- Constituent partitioning among adsorbed solids, dissolved phases, and vapor phases is linear.

- A moisture-dependent infiltration rate is calculated using Darcy's law and Van Genuchten moisture relationships.

Figure 4-2 illustrates the model construct for both Wastewater Source Modules, which are divided into three primary compartments:

- The **liquid compartment** includes the influent and effluent wastewater streams and the aqueous liquid layer above the sediment at the bottom of the unit. Liquid enters the compartment via the influent waste stream and rainfall, and leaves the compartment via evaporation and the effluent waste stream. Processes modeled in this compartment include volatilization, aerobic biodegradation, constituent hydrolysis, biomass solids growth, and solids settling and resuspension.

- The **unconsolidated sediment compartment** is a layer of loose sediment immediately below the liquid compartment. Processes modeled in this layer include anaerobic biological degradation and burial of solids into the consolidated sediment compartment.

- The **consolidated sediment compartment** is a layer of compacted sediment at the bottom of the unit. Processes modeled in this compartment include anaerobic degradation and removal of solids by cleaning and dredging and by decomposition due to anaerobic digestion.

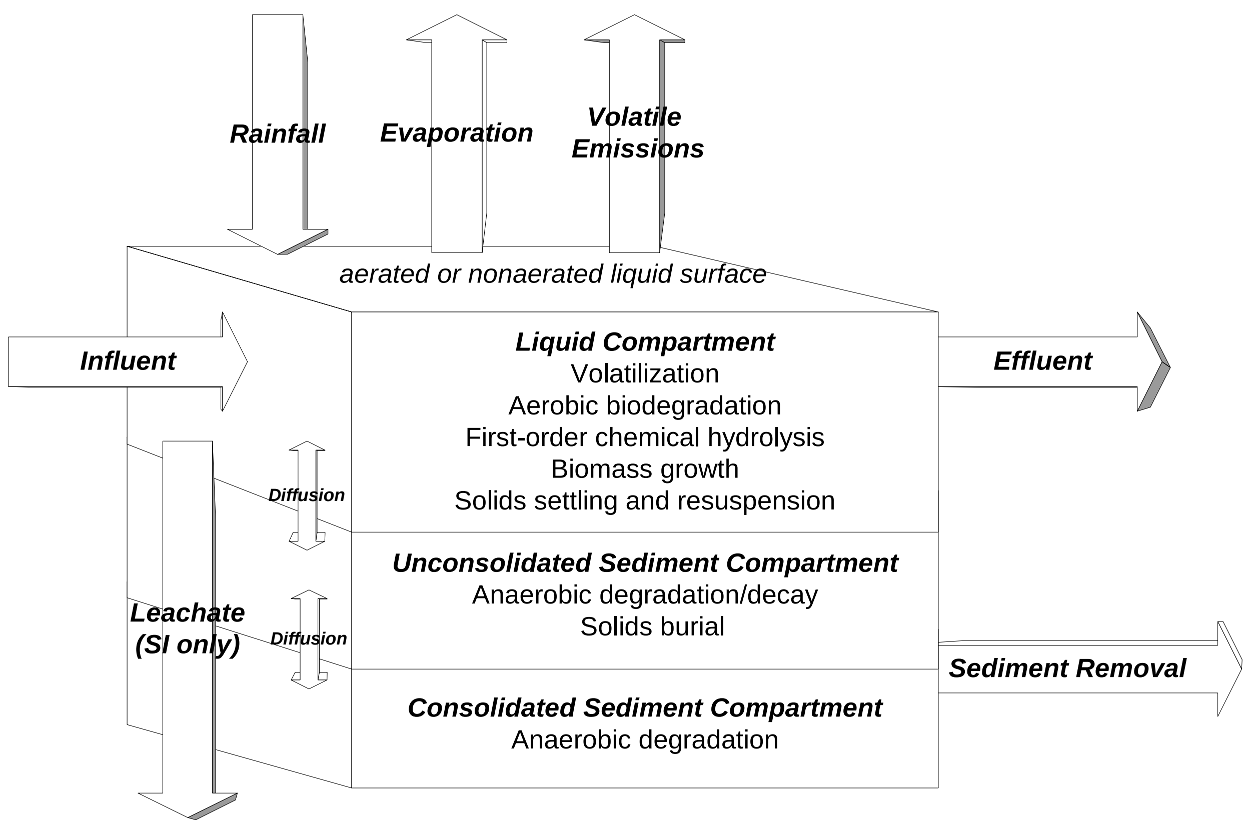

Figure 4-2. Conceptual model schematic for Wastewater Source Modules.

As shown in Figure 4-2, sediment is deposited and resuspended across the boundary between liquid and unconsolidated sediment, but only deposition (burial) occurs between the unconsolidated to consolidated sediment compartments. The modules also allow constituents to diffuse between adjacent compartments. In addition, the Surface Impoundment Module models leachate loss from the liquid layer through the sediment layers to the underlying soil.

For each compartment, the Wastewater Source Modules perform mass balances at time intervals small enough that the hydraulic retention time in the liquid compartment is not significantly affected by the solids settling and accumulation. In the liquid compartment, there is flow both in and out of the WMU, with constituent loss through volatilization, constituent decay (hydrolysis), aerobic biodegradation, and particle settling and burial (net sedimentation).

The following processes are modeled in the liquid compartment:

- Volatilization using a two-film model and mass transfer correlations from *Air Emissions Models for Waste and Wastewater* (U.S. EPA, 1994).

- Constituent loss due to hydrolysis and biodegradation using first-order rate constants for natural soil systems. These are adjusted to WMU conditions by assuming an effective system biomass concentration of 2.0×10^{-6} Mg/m.[3]

- Particle removal using particle settling velocities to estimate the projected sediment removal efficiency of the unit.

The modules estimate solids generation in the liquid compartment according the relationship of biological growth to the decomposition of organic chemicals (as biological oxygen demand) in the influent. These solids settle in the liquid to form the two underlying sediment compartments.

The following processes are modeled in the sediment compartments:

- Constituent loss due to hydrolysis and (anaerobic) biodegradation using first-order degradation rate constants. Hydrolysis and biodegradation rate constants are assumed to apply to the total constituent concentration (both dissolved and sorbed constituent) in each compartment.

- Constituent mixing between the compartments through constituent diffusion as well as particle sedimentation and resuspension.

- Solids destruction due to sludge digestion. It is assumed that solids decomposition is limited to the fraction of solids that are biologically active.

- Sediment compaction in the consolidated compartment, estimated by calculating the vertical effective stress across the consolidated sediment. Sediment compaction impacts the depth, volumetric water fraction, and the effective hydraulic conductivity of the consolidated sediment compartment.

Using the well-mixed assumption, the model assumes that the suspended solids concentration within the WMU is constant throughout the unit. However, some stratification of sediment is expected across the length and depth of the WMU so that the effective total suspended solids concentration within the unit is assumed to be a function of the WMU's total suspended solids removal efficiency rather than equal to the effluent total suspended solids concentration. The liquid (dissolved) phase constituent concentration within the unit, however, is assumed to be equal to the effluent dissolved phase concentration for the well-mixed model solution.

4.2.2 Calculate Constituent Concentrations within the Unit

The governing constituent mass balance equations used to calculate the constituent concentrations in the liquid and sediment compartments of the impoundment or tank are presented in the box starting on the next page. Detailed equations and solutions used by the Surface Impoundment and Aerated Tank Modules can be found in U.S. EPA (1999), including all assumptions and algorithms used in the modules.

The following sections describe the calculations for each of the three compartments, and the calculations for diffusion between the liquid and sediment compartments.

4.2.2.1 Liquid Compartment. The change in the constituent mass in the liquid compartment is a function of the constituent loss through volatilization, aerobic biodegradation, hydrolysis, and infiltration (denoting liquid flow of both liquid and entrained solids) into the sediment compartment. In addition, constituent is transported across the liquid/sediment compartment interface by constituent diffusion and by solids settling and resuspension. These processes are affected by whether the unit is aerated or nonaerated, and by constituent-specific properties.

For the time-dependent solution, the initial liquid compartment concentration is equal to the influent waste concentration.

4.2.2.2 Unconsolidated Sediment Compartment. The change in constituent mass within the unconsolidated sediment compartment is dependent on constituent infiltration from the liquid compartment (which includes entrained sediment), "filtered" leachate out the bottom of the compartment, and constituent loss through hydrolysis and anaerobic biodegradation. In addition, constituent is transported across the liquid/sediment compartment interface by constituent diffusion and by solids settling and resuspension. Constituent is transported across the unconsolidated/ consolidated compartment interface by constituent diffusion.

For aerated tanks, the initial sediment depth is zero. For surface impoundments, the initial sediment depth is 20 cm (unconsolidated). When the simulation begins, this initial sediment mass is partitioned equally into the unconsolidated and consolidated sediment compartments. The sediment mass is fixed for each month. At the end of the month, the gross mass of sediment accumulated during the month is calculated and the mass of sediment that is decomposed is calculated, resulting in a net mass of accumulated sediment. This net mass of accumulated sediment is partitioned equally into the unconsolidated and consolidated sediment compartments prior to simulation of the next month.

Aerated Tank and Surface Impoundment Constituent Mass Balance Equations
(variables listed on next page)

Liquid Compartment

Time-dependent constituent mass balance (surface impoundments only):

$$\frac{-\partial\left(V_1\,C_{tot,1}\right)}{\partial t} = Q_{leach}C_{tot,1} + f_{d,1}K_{OL}\,A\,C_{tot,1} + V_1\left(k_{bm}k_{ba}[TSS]_1 + k_{hyd}\right)\,C_{tot,1}$$
$$+\,v_{sed}\,A\;\;f_{p,1} - v_{res}\,A\;\;f_{p,2}\;\;C_{tot,2} - v_{diff\,12}\,A\;\;\theta_{liq,2}\;\left(f_{d,2}\;\;C_{tot,2} - f_{d,1}\;\;C_{tot,1}\right)$$

Well-mixed model solution:

$$Q_{\inf l}C_{tit,\inf l} - Q_{out}C_{tot,out} = Q_{leach}C_{tot,1} + f_{d,1}K_{OL}\,A\,C_{tot,1} + V_1\left(k_{bm}k_{ba}[TSS]_1 + k_{hyd}\right)\,C_{tot,1}$$
$$+\,v_{sed}\,A\;\;f_{p,1} - v_{res}\,A\;\;f_{p,2}\;\;C_{tot,2} - v_{diff\,12}\,A\;\;\theta_{liq,2}\;\left(f_{d,2}\;\;C_{tot,2} - f_{d,1}\;\;C_{tot,1}\right)$$

Unconsolidated Sediment Compartment

Time-dependent constituent mass balance (surface impoundments only):

$$\frac{-\partial(V_2\;C_{tot,2})}{\partial t} = Q_{leach}\,f_{d,2}\;C_{tot,2} - Q_{leach}\;C_{tot,1} + (k_{bs} + k_{hyd})\;V_2\;C_{tot,2}$$
$$+\;(v_{res} + v_b)\;A\;f_{p,2}\;C_{tot,2} - v_{sed}\;A\;f_{p,1}\;C_{tot,1}$$
$$+\;v_{diff12}\;A\;\theta_{liq,2}\;(f_{d,2}\;C_{tot,2} - f_{d,1}\;C_{tot,1})$$
$$+\;v_{diff23}\;A\;\theta_{liq,3}\;(f_{d,2}\;C_{tot,2} - f_{d,3}\;C_{tot,3})$$

For the well-mixed model solution, the left-hand side of the time-dependent mass balance equation is set to zero.

Consolidated Sediment Compartment

Time-dependent constituent mass balance (surface impoundments only):

$$\frac{-\partial(V_3\;C_{tot,3})}{\partial t} = Q_{leach}\,f_{d,3}\;C_{tot,3} - Q_{leach}\,f_{d,2}\;C_{tot,2} + (k_{bs} + k_{hyd})\;V_3\;C_{tot,3}$$
$$+\;v_{diff23}\;A\;\theta_{liq,3}\;(f_{d,3}\;C_{tot,3} - f_{d,2}\;C_{tot,2})$$

For the well-mixed model solution, the left-hand side of the time-dependent mass balance equation is set to zero.

(continued)

Constituent mass balance equation variables
(x denotes the compartment: 1 = liquid, 2 = unconsolidated sediment, 3 = consolidated sediment)

$C_{tot,\,x}$	=	total constituent concentration in compartment x (mg/L = g/m^3)
$C_{tot,\,infl}$	=	total constituent concentration in influent (mg/L = g/m^3)
$C_{tot,\,out}$	=	total constituent concentration in effluent (mg/L = g/m^3)
Q_{leach}	=	leachate flow rate from WMU (m^3/s)
Q_{infl}	=	influent flow rate into WMU (m^3/s)
Q_{out}	=	effluent flow rate out of WMU (m^3/s)
K_{OL}	=	overall volatilization mass transfer coefficient (m/s)
A	=	total surface area of WMU (m^2)
V_1	=	volume of liquid compartment in WMU= d_1 A (m^3)
V_2	=	volume of unconsolidated sediment compartment in WMU (m^3)
V_3	=	volume of consolidated sediment compartment in WMU (m^3)
d_1	=	depth of liquid compartment (m)
k_{hyd}	=	hydrolysis rate (1/s)
k_{bm}	=	complex first order biodegradation rate constant (m^3 /Mg-s)
k_{ba}	=	ratio of biologically active solids to the total solids concentration (i.e., k_{ba} = $[MLVSS]_1/[TSS]_1$)
k_{bs}	=	anaerobic biodegradation decay rate of constituent (1/s)
$[TSS]_x$	=	concentration of total suspended solids (TSS) in compartment x (g/cm^3 = Mg/m^3)
$[MLVSS]_1$	=	concentration of biomass as mixed liquor volatile suspended solids (MLVSS) liquid compartment (g/cm^3 = Mg/m^3)
ν_{sed}	=	solids settling or sedimentation velocity (m/s)
ν_{res}	=	solids resuspension velocity (m/s)
ν_b	=	solids burial velocity (m/s)
ν_{diff12}	=	mass transfer coefficient between liquid compartment (1) and unconsolidated sediment compartment (2) (m/s)
ν_{diff23}	=	mass transfer coefficient between unconsolidated sediment compartment (2) and consolidated sediment compartment (3) (m/s)
$f_{d,x}$	=	dissolved constituent fraction in compartment x: $$f_{d,x} = \frac{C_{liq,x}}{C_{tot,x}} = \frac{1}{\left(\theta_{liq,x} + k_{ds}\left[TSS\right]_x\right)}$$
$f_{p,x}$	=	particulate constituent fraction in compartment x: $$f_{p,x} = \frac{C_{sol,x}\left[TSS\right]_x}{C_{tot,x}} = \frac{k_{ds}\left[TSS\right]_x}{\left(\theta_{liq,x} + k_{ds}\left[TSS\right]_x\right)}$$
$\theta_{liq,x}$	=	volumetric liquid fraction of compartment x (m^3/m^3)
$C_{liq,x}$	=	liquid-phase constituent concentration in compartment x (mg/L = g/m^3)
$C_{sol,x}$	=	solid-phase constituent concentration in compartment x (mg/kg = g/Mg)
k_{ds}	=	solid-water partition coefficient (m^3/Mg) = $K_{oc}\times f_{oc}$ for organics
K_{oc}	=	soil-water partitioning (m^3/Mg)
f_{oc}	=	fraction organic carbon in the waste (mass fraction).

The initial sediment compartment concentrations are calculated by assuming the sediment compartments' void space is filled with waste at the influent concentration and that the initial sediment particles do not contain any contaminant. For each subsequent month, the initial sediment compartment constituent concentration (for either sediment compartment) is estimated based on the previous month's sediment compartment concentration. For a plug-flow unit, it is estimated as the log mean average sediment compartment concentration across the unit from the previous month. For batch units, it is the final sediment compartment concentration at the end of the previous month.

4.2.2.3 <u>Consolidated Sediment Compartment</u>. The change in the constituent mass within the consolidated sediment compartment is a function of "filtered" leachate flow from the unconsolidated sediment compartment and out the bottom of the unit, and constituent loss through hydrolysis and biodegradation. In addition, constituent is transported across the unconsolidated/consolidated compartment interface by constituent diffusion.

4.2.2.4 <u>Diffusion between Liquid and Sediment</u>. The models estimate effective diffusion velocity between the liquid and sediment compartments using the following two-resistance model based on the liquid phase mass transfer coefficient for quiescent surfaces and the porosity of the sediment compartment:

$$v_{diff} = \left(\frac{1}{k_{l,q}} + \frac{1}{k_{eff,2}} \right)^{-1} \tag{4-1}$$

where

v_{diff} = effective diffusion velocity between liquid and sediments
$k_{l,q}$ = liquid phase mass transfer coefficient for quiescent surface areas as calculated in Section 4.2.3 (m/s)
$k_{eff,2}$ = effective liquid mass transfer coefficient in sediment compartment (m/s).

To determine the effective liquid mass transfer coefficient in the sediment compartment, the models first calculate the effective liquid diffusion rate from the porosity of the sediment layer using the Millington-Quirk tortuosity model (Millington and Quirk, 1961):

$$D_{eff,2} = \theta_{liq,2}^{\frac{4}{3}} \times D_{i,l} \tag{4-2}$$

where

$D_{eff,2}$ = effective liquid diffusion rate (cm^2/s)
$\theta_{liq,2}$ = volumetric porosity (assumed to be liquid filled) of sediment compartment = $1 - [TSS]_2 / \rho_{TSS}$, where ρ_{TSS} = density of total suspended solids
$D_{i,l}$ = diffusivity in liquid (water) (cm^2/s).

In most cases, the sediment accumulating at the bottom of the WMU will be more of a viscous sludge layer than a rigid mass of particles. Therefore, the top layer of sediment is expected to be affected by the bulk currents within the WMU (caused by wind shear, aeration, or mixing) similar to the liquid phase mass transfer coefficient for quiescent surfaces. The liquid phase quiescent mass transfer coefficient is primarily a function of the liquid diffusivity raised to the two-thirds power; therefore, the models estimate the effective liquid mass transfer coefficient from the liquid compartment to the sediment layer as follows:

$$k_{eff,2} = k_{l,q} \left(\frac{D_{eff,2}}{D_{i,1}} \right)^{\frac{2}{3}} = k_{l,q} \; \theta_{liq,2}^{\frac{8}{9}} \tag{4-3}$$

4.2.3 Calculate Solids Concentrations within the Unit

The time-dependent solution for estimating total solids concentration within the unit is based on the assumption of constant total suspended solids (TSS) concentration. In reality, the solids concentration varies, either increasing as a result of biomass production from the consumption of organic material in the waste stream or decreasing as a result of solids settling. The solids mass balance for the liquid compartment is

$$\frac{-\,\partial(V_1[TSS]_1)}{\partial t} = r_{BOD}\lambda \; k_{ba}[TSS]_1 V_1 + v_{res}[TSS]_2 A - Q_{leach}[TSS]_1 - v_{sed} A [TSS]_1 \tag{4-4}$$

where

$[TSS]_1$	=	concentration of total suspended solids in liquid compartment ($g/cm^3 = Mg/m^3$)
V_1	=	volume of liquid compartment (m^3)
r_{BOD}	=	normalized biodegradation rate of BOD_5 (g-BOD/g-biomass/sec)
λ	=	biomass yield (g-biomass (dry basis)/g-BOD consumed)
k_{ba}	=	ratio of biologically active solids to the total solids concentration (i.e., $k_{ba} = [MLVSS]_1/[TSS]_1$)
v_{res}	=	solids resuspension velocity (m/s)
$[TSS]_2$	=	concentration of total suspended solids in unconsolidated sediment compartment ($g/cm^3 = Mg/m^3$)
A	=	total surface area of WMU (m^2)
Q_{leach}	=	leachate flow rate from WMU (m^3/s)
v_{sed}	=	solids settling or sedimentation velocity (m/s).

The normalized BOD_5 biodegradation rate is estimated using the Monod equation as follows:

$$r_{BOD} = \frac{K_{bmax} \; C_{BOD}}{(K_{b2} + C_{BOD})} \tag{4-5}$$

where

$C_{BOD,infl}$ = BOD_5 concentration in the influent to the WMU (g/cm^3 or Mg/m^3)

$C_{BOD,1}$ = BOD_5 concentration in the liquid compartment (g/cm^3 or Mg/m^3)

K_{bmax} = maximum BOD_5 biodegradation rate (g-BOD/g-biomass/sec or Mg/Mg/sec)
= $6.94 \times 10^{-6} \times T_{corr}$ (the value of 6.94×10^{-6} comes from the maximum rate of 0.6 g-BOD/g-biomass/hr ÷ 86,400 sec/hr)

K_{b2} = half-saturation constant = 0.00005 (g/cm^3 or Mg/m^3)

T_{corr} = temperature correction factor for biodegradation rate constants (see Section 4.2.7.5).

The models estimate the maximum BOD_5 degradation rate constant based on a typical design value for F/M (a 0.6 food-to-biomass ratio) for activated sludge systems based on values in Eckenfelder et al. (circa 1984) and Hermann and Jeris (1992). The model uses a typical half-saturation rate constant (K_{b2}) of 50 mg/L (0.00005 g/cm^3) selected from values reported in the literature (Tabak et al., 1989; Gaudy and Kincannon, 1977; Goldsmith and Balderson, 1989; Rozich et al., 1985) .

The integration period time steps in the overall time-dependent solution depend on how quickly the total suspended solids and BOD_5 concentrations vary. The model selects an initial time step based on the initial concentrations of TSS and BOD_5 so that BOD_5 concentrations are effectively constant. The TSS concentration is calculated at the end of the first time step and compared to the starting total suspended solids concentration; if it changes by more than a factor of 5, additional time steps are added until the starting and ending TSS concentrations differ by less than a factor of 5. This allows the model to determine an effective average TSS concentration across a given time step that can be used as a constant value in the constituent mass balance solution equations.

For well-mixed systems, the model calculates the effluent TSS concentration from the predicted solids removal efficiency of the unit (see Section 4.2.5) and the predicted BOD_5 removal efficiency. The TSS mass balance for the liquid compartment can be written as

$$[TSS]_{out} = \frac{Q_{infl}}{Q_{out}} \left(1 - \epsilon_{TSS}\right) \left([TSS]_{infl} + \lambda\, \epsilon_{BOD}\, C_{BOD,infl}\right) \qquad (4\text{-}6)$$

where

ϵ_{TSS} = total suspended solids mass removal efficiency in WMU (unitless)
λ = biomass yield (g-biomass (dry basis)/g-BOD)
ϵ_{BOD} = biological oxygen demand removal efficiency of WMU (unitless)
$C_{BOD,infl}$ = biological oxygen demand of influent (Mg/m^3)
$[TSS]_{infl}$ = concentration of total suspended solids in the influent (g/cm^3 = Mg/m^3).

$[TSS]_{out}$	=	concentration of total suspended solids in the effluent ($g/cm^3 = Mg/m^3$).
Q_{infl}	=	influent flow rate into WMU (m^3/s)
Q_{out}	=	effluent flow rate out of WMU (m^3/s).

Although the liquid compartment is assumed to be well mixed, solids removal and growth are not expected to be instantaneous. To account for gradients of TSS concentration along the WMU length and depth, the model estimates the effective TSS concentration within the WMU as the log-mean average between the influent and effluent total suspended solids concentrations (based on first-order sedimentation). Given the influent and effluent total suspended solids concentrations, the effective (mean) total suspended solids concentration in the liquid compartment is

$$[TSS]_1 = \exp\left[\frac{(\ln[TSS]_{infl}) + \ln([TSS]_{out})}{2}\right] \tag{4-7}$$

The model calculates BOD_5 removal efficiency based on the BOD_5 biodegradation rate using the same biodegradation rate model used for the time-dependent solution. Because the model uses BOD_5 degradation rate primarily to determine the production rate of biological solids within the impoundment, it neglects decreases in BOD_5 concentrations due to dilution by precipitation (i.e., the influent flow rate is assumed to be equal to the effluent flow rate plus the leachate flow rate). With this simplification, the BOD_5 removal efficiency can be written as follows:

$$\epsilon_{BOD} = \frac{r_{BOD}\ k_{ba}\ [TSS]_1\ V_1}{Q_{infl}\ C_{BOD,infl}} \tag{4-8}$$

Because BOD_5 removal efficiency depends on the effective TSS concentration, the model calculates the effective TSS concentration using iterative calculations between estimating the BOD_5 removal efficiency and the effective TSS concentration.

4.2.4 Calculate Volatile Emission Rates

The Wastewater Source Modules use an overall mass transfer coefficient that determines the rate of volatilization based on a two-resistance model: a liquid-phase mass transfer resistance and a gas-phase mass transfer resistance. The liquid- and gas-phase mass transfer resistances for turbulent surfaces are very different from those for quiescent (laminar flow) surfaces. Therefore, the overall mass transfer coefficient is a composite of the coefficients for the turbulent surface area and the quiescent surface area. The overall coefficient is based on an area-weighted average, as follows:

$$K_{OL} = \frac{K_{OL,t}\ A_t + K_{OL,q}\ A_q}{A} \tag{4-9}$$

where

K_{OL}	=	overall mass transfer coefficient for the WMU (m/s)
$K_{OL,t}$	=	overall mass transfer coefficient for turbulent surface areas (m/s)
A_t	=	turbulent surface area = f_{aer} A(m^2)
f_{aer}	=	fraction of total surface area affected by aeration (unitless)
A	=	total surface area (m^2)
$K_{OL,q}$	=	overall mass transfer coefficient for quiescent surface areas (m/s)
A_q	=	quiescent surface area (m^2) = (1-f_{aer}) × A (Note: A_t + A_q must equal A).

The overall mass transfer coefficient for turbulent surface areas based on the two-resistance model is

$$K_{OL,t} = \left(\frac{1}{k_{l,t}} + \frac{1}{H' \; k_{g,t}} \right)^{-1} \tag{4-10}$$

where

$k_{l,t}$	=	liquid-phase mass transfer coefficient for turbulent surface areas (m/s)
H'	=	dimensionless Henry's law constant = H/RT_H
$k_{g,t}$	=	gas-phase mass transfer coefficient for turbulent surface areas (m/s).

Similarly, the overall mass transfer coefficient for quiescent surface areas is

$$K_{OL,q} = \left(\frac{1}{k_{l,q}} + \frac{1}{H' \; k_{g,q}} \right)^{-1} \tag{4-11}$$

where

$k_{l,q}$	=	liquid-phase mass transfer coefficient for quiescent surface areas (m/s)
$k_{g,q}$	=	gas-phase mass transfer coefficient for quiescent surface areas (m/s).

The mass transfer correlations used to estimate the liquid- and gas-phase mass transfer coefficients for turbulent and quiescent surfaces are the same as those used in the WATER8 and CHEMDAT8 emission models developed by EPA. Basic equations are provided in U.S. EPA (1999) with a more detailed treatment in Chapter 5 of the CHEMDAT8 model documentation (U.S. EPA, 1994).

4.2.5 Estimate Resuspension, Sedimentation, and Burial Velocities

To solve the constituent and sediment mass balance equations, the Wastewater Source Modules must estimate the transfer rate of the sediment (and its associated constituent content) between the liquid and sediment compartments in the WMU. Key parameters in this calculation include

- Sedimentation velocity, which establishes the rate at which particles in the liquid compartment enter the unconsolidated sediment compartment;

- Resuspension velocity, which establishes the rate at which particles in the unconsolidated sediment compartment enter the liquid compartment; and

- Burial rate, which is the net accumulation rate of sediment in the two sediment compartments.

Sediment movement between the liquid and sediment compartment should vary primarily with the dimensions and flow characteristics of the WMU, and with the relative surface area affected by turbulent mixing. The general approach used to estimate the various sediment transport rates is to first estimate the suspended solids mass removal efficiency of the WMU. Given this removal efficiency, the Wastewater Source Modules estimate the resuspension, sedimentation, and burial velocities based on the characteristics of the mean-sized particles in the WMU.

4.2.5.1 <u>Estimate Design Sediment Removal Efficiency</u>. The WMU quiescent surface area and flow rate are used to calculate the vertical (or “upflow”) velocity of the impoundment as follows:

$$v_{upflow} = \frac{Q_{infl}}{A_q} \tag{4-12}$$

where

ν_{upflow} = upflow velocity (m/s)
Q_{infl} = influent flow rate into WMU (m^3/s)
A_q = quiescent surface area (m^2).

The upflow velocity is assumed to act on the liquid compartment to cause an upward flux of particles. The model estimates sediment removal efficiency in the WMU from WMU flow rate, surface area (i.e., the upflow velocity), and particle size distribution characteristics (mean particle size and relative standard deviation) by considering the terminal settling velocity of the particles. Particles with a terminal settling velocity greater than the upflow velocity settle within the WMU, while particles with a terminal velocity less than the upflow velocity remain suspended and are entrained in the effluent.

The model assumes that suspended solids are spherical when calculating the terminal velocity (or critical particle diameter) and the mass-to-volume ratio of the particles. The model calculates the terminal velocity of a sphere using Stoke’s Law (see U.S. EPA 2001), and determines the particle diameter that has a terminal velocity equal to the upflow velocity. The mass sediment removal efficiency of the WMU is then calculated from the particle size distribution (model input parameters assuming lognormal distribution) and the mass of particles of a given diameter (based on spherical particles). The lognormal distribution density function is

$$\phi(d_{part}) = \frac{1}{d_{part}\ \sigma\ (2\pi)^{½}} \times \exp\left(\frac{-[\ln(d_{part}/d_{mean})]^2}{2\sigma^2}\right) \tag{4-13}$$

where

$\phi(d_{part})$ = distribution density function for sediment particles
d_{part} = particle diameter (cm)
σ = standard deviation of $\ln(d_{part})$
d_{mean} = geometric mean particle diameter = exp[mean of $\ln(d_{part})$] (cm).

The mass of the particle is proportional to its volume; therefore, the density function is evaluated based on the cube of the particle diameter (assuming spherical particles) using the following factor:

$$WtFactor_{part} = \frac{\pi}{6}\ d_{part}^3 \tag{4-14}$$

where

$WtFactor_{part}$ = particle weighting factor (cm^3)
d_{part} = particle diameter (cm).

Note that it is assumed that all particles have the same density, regardless of size. Therefore, the design volumetric solids removal efficiency equals the design mass solids removal efficiency, which is calculated as

$$\epsilon_{TSS,o} = \frac{\int_{d_{crit}}^{+\infty} [\phi(d_{part}) \times WtFactor_{part}]}{\int_{0}^{+\infty} [\phi(d_{part}) \times WtFactor_{part}]} \tag{4-15}$$

where

$\epsilon_{TSS,o}$ = design mass solids removal efficiency of WMU (mass fraction).

Because of the solution algorithm selected, the equations become unsteady as $\epsilon_{TSS,o}$ approaches 1. To prevent taking the logarithm of zero, the design mass solids removal efficiency is capped at 99.9 percent.

4.2.5.2 <u>Estimate Resuspension, Sedimentation, and Burial Velocities for Time-dependent Model Solution</u>. The time-dependent model uses the design removal efficiency to set a target effluent concentration. The sedimentation velocity is calculated as the terminal velocity for the mean particle size diameter, which is an input to the model, using Stoke's Law (see U.S. EPA, 2001). The resuspension velocity is set so that the total mass of sediment resuspended will equal the mass settling when the average TSS concentration in the liquid compartment equals the target effluent concentration. The burial rate is then calculated for each individual time step based on the difference in the sedimentation and resuspension rates at the average liquid compartment TSS concentration for that time step. The equations for the resuspension, sedimentation, and burial velocities are

$$v_{sed} = 100 \times v_{part,mean} \tag{4-16}$$

$$v_{res} = v_{sed} \frac{[TSS]_{target}}{[TSS]_2} \tag{4-17}$$

$$v_b = \left(\frac{Q_{leach}}{A} + v_{sed} \right) \frac{[TSS]_{ave}}{[TSS]_2} - v_{res} \tag{4-18}$$

where

$v_{part,mean}$	=	particle settling velocity of a mean-diameter particle (cm/s)
$[TSS]_{target}$	=	target effluent TSS concentration = $[TSS]_{infl}$ (1- $\epsilon_{TSS,o}$) ($g/cm^3 = Mg/m^3$)
$[TSS]_{ave}$	=	average TSS concentration in the liquid compartment for a given time interval ($g/cm^3 = Mg/m^3$)
v_b	=	solids burial velocity (m/s).

As constructed, the time-dependent solution assumes the mass rate of sediment resuspension will equal the mass rate of sediment settling at the target or design effluent TSS concentration. The rate at which the target TSS concentration is reached is dependent on the particle characteristics as well as the growth rate of biomass (i.e., the BOD_5 consumption rate). The actual effluent TSS concentration predicted by the model may not reach the target TSS concentration at very low hydraulic residence times or where significant quantities of biosolids are produced. As sediment accumulates in the WMU, the corresponding change in the hydraulic residence time may also affect the predicted effluent TSS concentration.

4.2.5.3 Estimate Resuspension, Sedimentation, and Burial Velocities for Well Mixed Model Solution. In the well-mixed model, mass balance consideration of the sediment requires that the suspended solids burial (or accumulation) rate be determined from the predicted sediment removal efficiency. As constructed, the design sediment removal efficiency is independent of WMU depth, and therefore does not change as sediment accumulates in the WMU. This will generally be true for large depths, but for shallower depths, the increased lateral flow rates tend to cause "short-circuiting" flow patterns, which decrease the sediment removal efficiency of the WMU. To take this phenomenon into account, it is assumed that the sediment removal efficiency remains constant at the design efficiency (i.e., $\epsilon_{TSS} = \epsilon_{TSS,o}$) at liquid depths of 1.2 meters (4 feet) or more based on design considerations of settling chambers. As the liquid depth becomes less than 1.2 meters, it is assumed that the sediment removal efficiency will decrease as a function of the liquid retention time. A first-order sedimentation rate constant is estimated based on the design sediment removal rate and the WMU retention time at a liquid depth of 1.2 meters. This first-order sedimentation rate constant is calculated as

$$k_{sed} = \frac{-\ln\left(1 - \epsilon_{TSS,o}\right)}{\dfrac{(1.2\ m)\ A}{Q_{infl}}} \tag{4-19}$$

where

k_{sed} = apparent first-order sedimentation rate at a liquid depth of 1.2 meters (1/s).

For liquid depths less than 1.2 meters, the removal efficiency is estimated using this first-order sedimentation rate constant and the hydraulic retention time as

$$\epsilon_{TSS} = 1 - e^{\left(\frac{-k_{sed}\ d_1\ A}{Q_{infl}}\right)} \tag{4-20}$$

where

ϵ_{TSS} = predicted mass sediment removal efficiency of the WMU as sediment accumulates (mass fraction)
d_1 = depth of liquid compartment.

The predicted mass sediment removal efficiency is assumed to apply equally to influent sediment and sediment generated within the unit. The net rate of sediment transfer or burial from the liquid compartment to the sediment compartment can be calculated based on a mass balance of sediment in the liquid compartment, which can be rearranged to calculate the burial velocity (defined in terms of the sediment concentration in the sediment compartment) as follows:

$$v_b = \frac{Q_{infl}\ \left([TSS]_{infl} + \lambda\ \epsilon_{BOD}\ C_{BOD}\right)\ \left(1 - \epsilon_{TSS}\right)}{A\ [TSS]_2} \tag{4-21}$$

The resuspension velocity acts on the sediment compartment, and it is assumed to affect the same upward flux of sediment as the upflow velocity. Therefore, the resuspension velocity can be calculated from the upflow velocity and the relative concentrations of particles in the liquid and sediment compartments as follows:

$$v_{res} = v_{upflow} \frac{[TSS_1]}{[TSS_2]} \tag{4-22}$$

The sedimentation rate is calculated from the mass balance of sediment in the sediment compartment (Equation 4-4), which can be rearranged as follows.

$$v_{sed} = (v_{res} + v_b) \frac{[TSS]_2}{[TSS]_1} - \frac{Q_{leach}}{A} \tag{4-23}$$

where $[TSS]_1$ is calculated from $[TSS]_{infl}$ and $[TSS]_{out}$ using Equation 4-7.

4.2.5.4 <u>Estimate Sediment Decomposition</u>. The burial rate is the total sediment accumulation rate for the time step. To account for the reduction in solids typically associated with anaerobic digestion, a sediment decomposition rate (or sludge digestion rate) is included in the burial (accumulation) compartment. If the entire sediment compartment included this anaerobic digestion term, a more rigorous accounting of the biological (organic) versus inert solids would be required, but, ultimately, the sediment compartment will reach a steady state (i.e., biomass growth equals biomass decay). By including it only in the burial (accumulation) compartment, sediment reduction (which includes a constituent reduction associated with the sediment) by digestion can be included without significantly complicating the model. The net accumulation of sediment over a time step is estimated as

$$\Delta d_2 = v_b \, \Delta t \, [1 - k_{ba} (1 - e^{-k_{dec} \Delta t})] \tag{4-24}$$

where

Δd_2	=	change in depth of the unconsolidated sediment compartment (m)
Δt	=	time step (s)
k_{ba}	=	ratio of biologically active solids to the total solids concentration - assumed to be the same ratio as present in the liquid compartment
k_{dec}	=	anaerobic digestion/decay rate of the organic sediment (1/sec).

Prior to the next time step calculations, Δd_2 is added to d_2 (and subtracted from d_1). Additionally, the total amount of sediment in the tank or impoundment and the total time since the last cleaning or dredging action is compared to the input cleaning and dredging parameter (i.e., the fraction of the WMU that can be filled with sediment before the WMU is cleaned or dredged). The module will also automatically run the dredge subroutine in the event that the sediment settling for the next time step (based on the sediment settling for the current time step) would completely fill the WMU. The removed sediment and the contaminant associated with the removed sediment is recorded; this removal acts as a sink for the overall system.

4.2.6 Estimate Constituent Release in Leachate (Surface Impoundments only)

The Surface Impoundment Module estimates leachate infiltration rate from liquid depth and from the hydraulic conductivities and thicknesses of three layers: the sediment compartment, a clogged soil layer, and the underlying native soil. This procedure follows the method outlined in EPA's Composite Module for Leachate Migration with Transformation Products (EPACMTP) background document (U.S. EPA, 1996), except that the liquid depth is known and there is a sediment layer between the impoundment liquid and the underlying soil layer.

The Surface Impoundment Module treats the unconsolidated sediment layer as free liquid to calculate the pressure head on the consolidated sediment layer and underlying soil. The model calculates the infiltration rate in an iterative manner. It makes an initial estimate of the infiltration rate, calculates the associated pressure profile in the underlying soil, and compares the calculated pressure head at the ground water surface with the boundary condition (i.e., pressure head of zero). Based on this comparison, the model revises the infiltration rate estimate and iterates until the boundary conditions are met.

Based on Darcy's law, the leaching (infiltration) rate for a given soil sublayer is:

$$I_n = K_{s,n} k_{rw,n} \left(\frac{\psi_n - \psi_{n+1}}{d_{s,n}} + 1 \right) \tag{4-25}$$

where

I_n = infiltration rate (m/d)
$K_{s,n}$ = hydraulic conductivity of the n^{th} soil sublayer (m/d)
$k_{rw,n}$ = relative permeability of the n^{th} soil sublayer (unitless)
Ψ_n = pressure head at top of the n^{th} soil sublayer (m)
Ψ_{n+1} = pressure head at base of the n^{th} soil sublayer (m)
$d_{s,n}$ = depth of the n^{th} soil sublayer (m).

The relative permeability is a function of the effective saturation and can be expressed by soil class parameters using relationships developed by Van Genuchten (1980) as follows:

$$if\ \psi \geq 0 \qquad k_{rw,n} = 1 \tag{4-26a}$$

$$if\ \psi < 0 \qquad k_{rw,n} = \frac{(1 - (-\alpha_n\psi_n)^{\beta_n - 1}[1 + (-\alpha_n\psi_n)^{\beta_n}]^{-\gamma_n})^2}{(1 + (-\alpha_n\psi_n)^{\beta_n}]^{\gamma_n/2}} \tag{4-26b}$$

where

α_n = soil retention model parameter alpha for n^{th} soil sublayer (1/m)
β_n = soil retention model parameter beta for n^{th} soil sublayer (unitless)
γ_n = soil retention model parameter gamma for n^{th} soil sublayer = $1 - 1/\beta_n$ (unitless).

Solution methods for these equations can be found in the EPACMTP background document (U.S. EPA, 1996). As shown in Figure 4-3, the Surface Impoundment Module applies these equations to the sediment layer, a liner or soil layer clogged with sediment particles immediately below the surface impoundment, and another soil layer under that. The sediment layer is assumed to be saturated and modeled as a single layer using Darcy's law. The underlying soil layers are partially saturated and are modeled with five sublayers using the Van Genuchten relationships and algorithms developed for EPACMTP.

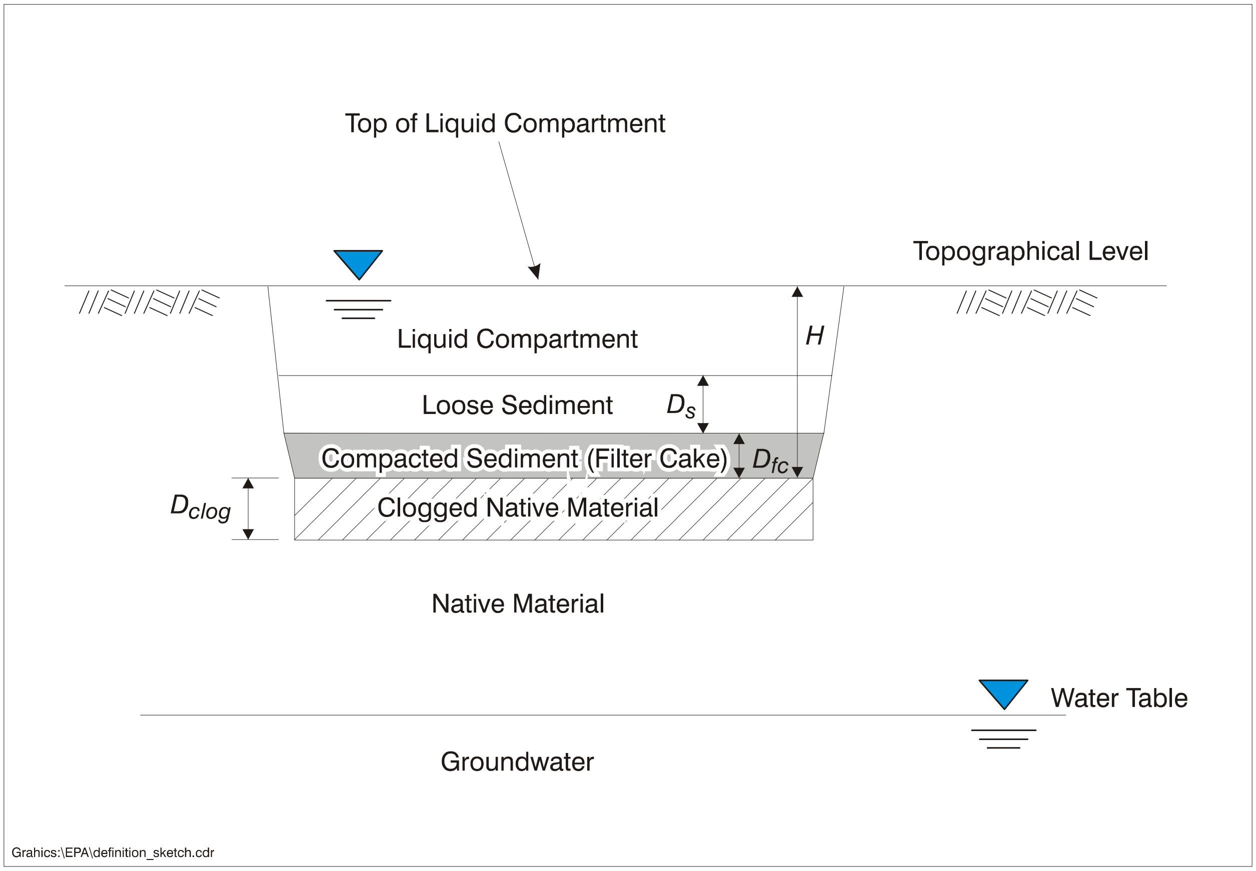

Figure 4-3. Surface impoundment cross-section showing sediment and soil layers modeled by the Surface Impoundment Module infiltration rate algorithms.

4.2.6.1 __Leachate Infiltration Rate__. The Surface Impoundment Module calculates a leachate infiltration rate through three compartments: consolidated sediment, clogged soil, and subsoil. Initially, the consolidated sediment is saturated and the soil layers are unsaturated. The model divides each of the soil sublayers into three sublayers. The unconsolidated sediment layer is loose (fluid) so that the effective pressure head for the consolidated sediment layer is simply the liquid depth plus the depth of the unconsolidated sediment. The model assumes that the system is at steady state; therefore, a water balance dictates that the infiltration rate is the same for all compartments and compartment sublayers. Assuming the pressure head at the ground water interface is zero, the general solution for the infiltration rate is shown in Equation 4-27.

$$I = \frac{(d_1 + D_s + \sum d_{s,n})}{\Sigma\left(\frac{d_{s,n}}{K_{s,n}k_{rw,n}}\right)} \tag{4-27}$$

where

I	=	leachate infiltration rate (m/d)
$K_{s,n}$	=	saturated hydraulic conductivity of the n^{th} soil sublayer (m/d)
$k_{rw,n}$	=	relative permeability of the n^{th} soil sublayer (unitless)
$d_{s,n}$	=	thickness of the n^{th} soil sublayer (m).

The relative permeabilities (k_{rw}) of the clogged soil and native soil sublayers are a function of whether the previous sublayer is saturated. The model uses an interative, steady-state method to solve this equation across the sediment and soil compartments, as described in U.S. EPA (1999). The infiltration rate is then set equal to the lowest of the calculated rates across the three compartments.

The Surface Impoundment Module calculates the volumetric leachate flow rate from the calculated infiltration rate as follows:

$$Q_{leach} = \frac{I \times A}{24 \times 3600} \tag{4-28}$$

where

Q_{leach}	=	leachate infiltration flow rate (m^3/s)
A	=	surface impoundment area (m^2).

Leachate flow rates and leachate contaminant concentrations (calculated as the liquid contaminant concentration in the consolidated sediment compartment) are output from the Surface Impoundment Module as a time series of annual-average values.

4.2.6.2 <u>Hydraulic Conductivity of Consolidated Sediment</u>. As sediment accumulates at the bottom of the impoundment, the weight of the liquid and upper sediments tends to compress (or consolidate) the lower sediments. This consolidated sediment acts as a filter cake, and its hydraulic conductivity may be much lower than the unconsolidated sediment. The Surface Impoundment Module assumes that the sediment compartment is at pseudo-steady-state and that all sediment layer thicknesses are nearly stationary and approximately constant. The Surface Impoundment Module sets the initial sediment depth at 20 cm to account for sediment and compaction created during the excavation of the impoundment.

The Surface Impoundment Module simulates the effective stress of the overlying liquid and unconsolidated sediment on the compacted sediment layer, and the effects of this stress on the porosity and hydraulic conductivity of the filter cake. The filter cake thickness is capped at one-half of the total sediment depth and is assumed to have a minimum thickness of 100 cm. Equations for the algorithms used to estimate sediment consolidation and the resulting hydraulic conductivity of the compacted sediment layer can be found in U.S. EPA (1999).

4.2.6.3 Hydraulic Conductivity of Clogged Native Material. The Surface Impoundment Module simulates the effect of sediment particles that enter and clog the soil layer immediately underlying the surface impoundment by assuming that the saturated hydraulic conductivity of the clogged soil zone is one-tenth of the hydraulic conductivity of the native soil. The model also constrains the hydraulic conductivity of the clogged layer to be less than or equal the underlying soil layer and greater than or equal to the consolidated sediment filter cake. Based on observed penetration depths of up to about 0.45 m, the depth of the clogged layer is fixed at 0.5 m.

4.2.6.4 Limitations on Maximum Infiltration Rate. If the surface impoundment infiltration rate exceeds the rate at which the aquifer can transport ground water, the ground water level will rise into the vadose zone, and the assumption of zero pressure head at the base of the vadose zone would be violated. This ground water "mounding" will reduce the effective infiltration rate to a maximum infiltration rate. The model estimates this maximum rate as one that does not cause the ground water mound to rise to the bottom elevation of the surface impoundment unit, using the following equation (U.S. EPA, 1999; HydroGeoLogic, 1999):

$$I_{Max} \leq \frac{2K_{aqsat}D_{aqsat}(D_{vadose} - H)}{R_0^2 \ln\frac{R_\infty}{R_0}} \tag{4-29}$$

where

I_{Max}	=	maximum allowable infiltration rate (m/d)
K_{aqsat}	=	hydraulic conductivity of the aquifer (m/d)
D_{aqsat}	=	aquifer thickness (m)
D_{vadose}	=	vadose zone thickness (m)
H	=	the effective head in the WMU (see Figure 4-3)
R_0	=	equivalent source radius (m)
R_∞	=	length between the center of the source and the downgradient boundary where the boundary location has no perceptible effects on the heads near the source (m).

Under certain conditions of high soil hydraulic conductivity and long residence time in the surface impoundment, the leachate flow rate may exceed the influent flow rate. Rather than reiterating the infiltration rate calculation with liquid depth as a variable, the leachate rate is limited to 99 percent of the influent flow rate. This limit is based on a volumetric balance on the WMU and an assumption that the effluent flow is never less than 1 percent of the influent flow.

4.2.6.5 Surface Impoundment Effluent Flow Rate. The Surface Impoundment Module calculates effluent flow based on a volumetric water balance on the WMU:

$$Q_{out} = Q_{infl} - Q_{leach} + A \, (P_{rain} - P_{evap}) \tag{4-30}$$

where

Q_{out}	=	effluent flow rate (m^3/s)
Q_{infl}	=	influent flow rate (m^3/s)
Q_{leach}	=	leachate infiltration rate (m^3/s)
A	=	surface impoundment surface area (m^2)
P_{rain}	=	precipitation rate (m/s)
P_{evap}	=	evaporation rate (m/s).

Under certain conditions of influent flow rate, impoundment dimensions, infiltration, precipitation, and evaporation, there may be months that Equation 4-30 predicts a negative or zero effluent rate, which would violate the pseudo-steady-state assumption. Therefore, if Equation 4-30 produces an effluent flow rate of less than 1 percent of the influent flow rate, the effluent flow rate is calculated as

$$Q_{out} = 0.01 \; Q_{infl} \tag{4-31}$$

When the infiltration rate is capped at 99 percent of the influent rate, the model uses this equation only if the evaporation rate exceeds the precipitation rate. Depending on the various rates, Equation 4-31 can be triggered even if the infiltration rate is not capped at 99 percent of the influent rate. The Surface Impoundment Module uses Equation 4-31 to limit (or cap) the evaporation rate to prevent a zero or negative effluent rate.

4.2.7 Adjust for Temperature Effects

Temperature can affect a number of the inputs used by the Wastewater Source Modules, including air density and diffusivity, biodegradation rate, liquid viscosity, and Henry's law constant. Some of the equations employed by the modules already include a temperature correction factor. For example, the liquid-phase, turbulent surface mass transfer coefficient includes a temperature correction term of 1.024^{T-20}. The modules use the ambient air temperature to adjust the air-side properties (air diffusivity, air density, etc.). Liquid-side properties (liquid diffusivity, liquid viscosity, etc.) are adjusted using the wastewater temperature within the tank or surface impoundment.

4.2.7.1 Estimating Temperature in the Waste Management Unit. The Wastewater Source Modules use a simplified energy balance around the aerated tank or surface impoundment to estimate the liquid temperature in the WMU from the liquid temperature of the influent, the monthly ambient air temperature, and the liquid residence time in the WMU. The model uses coefficients for free and forced convective heat transfer coefficients for both water and air (Kreith and Black, 1980) to estimate the average overall heat transfer coefficient. The model assumes that there is forced convection on the air side (windspeed greater than 0 m/s), free convection on the quiescent liquid side, and forced convection on the turbulent liquid side. Assumptions and equations used to estimate tank and surface impoundment temperature are detailed in U.S. EPA (1999).

The surface impoundment and aerated tank temperature estimates do not take into account the heat of fusion from ice formation, and can yield liquid temperatures below 0°C. When this happens, the models set the liquid temperature to 0.1°C and estimate the amount of ice formed based on the specific heat capacity and density of water and ice. The models translate the additional heat loss in taking the water from 0°C to below 0 into a mass of ice formation, and estimate the volume or depth of ice formed using the following equation:

$$d_{ice} = \frac{A \times d_{wmu} \times (-T_l)}{80\ (0.9)\ A} \tag{4-32}$$

where

d_{ice} = depth of ice layer formed (m)
A = area of the unit (m^2)
T_l = temperature of liquid (C).

This equation tends to overestimate ice formation because convective heat transfer from the surrounding soil is not included in the heat balance. Also, although a small amount of ice formation will not significantly impact the emission estimates and other parameters estimated by the module, if a solid crust of ice forms over the entire impoundment for a prolonged period of time, the model would overstate the potential for volatile emissions because it does not consider volatilization through an ice layer. Therefore, when the depth of the projected ice layer is 10 cm or more for 3 consecutive months, the model generates a warning message that significant ice formation is projected.

4.2.7.2 Temperature Effects on Air-Side Properties. Air-side properties include density and viscosity. The model estimates air density at a given temperature using the ideal gas law (U.S. EPA, 1999). Because the viscosity of air is only slightly affected by temperatures in the temperature range of interest, with a range from 1.75×10^{-4} to 2.17×10^{-4} g/cm-s from 0°C to 100°C (Kreith and Black, 1980), it is not adjusted for temperature in the Wastewater Source Modules.

4.2.7.3 Temperature Effects on Liquid-Side Properties. The density of water is basically insensitive to temperature and no temperature adjustments are used in the Wastewater Source Modules. The viscosity of water varies by more than a factor of 5 over the temperature range of interest (0°C to 100°C). This temperature dependency is important not only for mass transport, but also for its effect on the solids settling rate (terminal velocity) at lower Reynolds numbers. Using the data from Kreith and Black (1980), the modules use the a correlation developed using a log-log least squares linear regression to adjust the viscosity of water (U.S.

EPA, 1999). The values for the viscosity of water calculated from this equation agree well with the values reported by Liley and Gambill (1973) for temperatures between 0°C and 100°C.

4.2.7.4 <u>Temperature Effects on Constituent-specific Properties</u>. Air diffusivity, water diffusivity, Henry's law constant, and aerobic and anaerobic biodegradation rates are constituent-specific properties used by the Wastewater Source Modules to estimate constituent volatilization, degradation, and release to air and, for the surface impoundment, through leachate. In the 3MRA modeling system, these properties are supplied by the Chemical Properties Processor (CPP). Because the Wastewater Source Modules operate on monthly time steps, they call the CPP for these properties according to the temperature in the unit at the beginning of each month. Details on the temperature correction routines used by the CPP can be found in the documentation for the 3MRA CPP (Pacific Northwest National Laboratory, 1998).

4.2.7.5 <u>Temperature Effects on BOD and Sediment Biodegradation Rates</u>. The Wastewater Source Modules assume that the BOD and sediment decay rates (k_b, k_{dec}) are relatively unaffected by temperature over a reasonably wide range of temperature. At temperatures above 50°C and at temperatures near freezing, the decay rate is assumed to drop rapidly. The modules incorporate a simple temperature correction factor for these decay rates based on these assumptions: between 7°C and 40°C, the biodegradation rate temperature correction factor is assumed to be 1 (i.e., no correction). If temperatures fall below 3°C or above 60°C, the model stops biodegradation of BOD and the sediment mass by setting the temperature correction factor to 0. A linear extrapolation is used to determine the temperature correction factor between 3°C and 7°C and between 40°C and 60°C.

4.3 Module Discussion

4.3.1 Strengths and Advantages

The Wastewater Source Modules are based on sound engineering principles and algorithms, many of which have been tested and peer reviewed. Some of the strengths and advantages of these modules include the following:

- **Volatilization model.** The volatilization component of the Wastewater Source Modules employs the mass transfer correlations recommended by the Office of Air Quality Planning and Standards (OAQPS) as developed though the CHEMDAT and WATER series of models. These correlations have been developed, tested, and validated through many years of EPA research and public and industry use.

- **Leachate infiltration model.** The leachate infiltration flow rate component of the Surface Impoundment Module was adapted from the equations and algorithms developed by the Office of Research and Development (ORD) in the EPACMTP model, which has undergone peer review by SAB. The infiltration rate component also considers the effect of sediment consolidation within the impoundment and sediment impregnation (or clogging) of the underlying soil layer on the infiltration rate. These components were recommended by ORD based on their on-going research and development efforts.

- **Solids mass balance.** The modules account for sorption using correlations for Kd and K_{oc} developed jointly by OSW and ORD. The modules also perform an explicit mass balance on the solids within the unit, including production of new solids through biomass growth, transport of solids between compartments due to infiltration (for the Surface Impoundment Module), sedimentation, resuspension and burial, and decay of sediment due to anaerobic digestion. Therefore, the modules evaluate total, dissolved, and sorbed phase concentrations needed for various potential exposure mechanisms.

- **Loss mechanisms.** The modules account for a variety of loss mechanisms, including volatilization, sorption, hydrolysis, biodegradation, and infiltration (for the Surface Impoundment Module). Furthermore, the modules account for changes in the rates of these loss mechanisms due to monthly changes in ambient temperature, the predicted effect of these ambient temperature changes on the average wastewater temperature, and monthly evaporation and precipitation rates. Therefore, the modules can elucidate differences in constituent fate through seasonal variations. Additionally, the modules account for the accumulation and consolidation of sediment within the units, and thus can be used to predict the transient nature of the constituent fate as a unit fills with sediment (thereby reducing the hydraulic retention time and perhaps the sediment removal efficiency).

- **Site-specific data.** The modules have a numerous input variables that allow modeling of site-specific units using detailed site-specific information. Without this site-specific data, the modules can be used with Monte-Carlo-derived inputs to derive a reasonable expectation of the expected fate of the constituent and the extremes of the potential exposure concentrations. The modules employ predominantly analytical solutions so that numerous module runs, such as those needed when performing a Monte Carlo analysis, can be completed in a short time frame.

- **Plug-flow and batch operations.** The Surface Impoundment Module provides solution algorithms for modeling plug-flow and batch operations in addition to the well-mixed solution algorithm. The Aerated Tank Module was designed specifically for aerated or mixed tanks, and therefore assumes a well-mixed liquid compartment. However, non-aerated, quiescent tanks that operate in a plug-flow or batch mode can be effectively be modeled using the Surface Impoundment Module by setting a very low sediment hydraulic conductivity (driving the infiltration rate essentially to zero).

4.3.2 Uncertainty and Limitations

The most significant uncertainties and limitations of the Wastewater Source Modules include the following:

- **Applicable only to dilute aqueous wastes.** By using a simple biodegradation rate model together with Henry's law partitioning coefficients, the modules are most applicable to dilute aqueous wastes. High constituent concentrations can reduce or inhibit biodegradation of toxic constituents. Also, if constituents exceed solubility limits and free phase is present, Henry's law can overestimate volatilization (the modules output a warning when this occurs).

- **Reaction byproducts not considered.** Daughter products from hydrolysis and biological degradation of the constituents are not addressed by the modules (i.e., volatile emissions or leaching are not considered for reaction intermediates or end products).

- **No oxygen balance.** The modules assume that the liquid compartment is aerobic and contains adequate oxygen for the degradation of influent carbon (as BOD). However, high influent carbon loadings for certain low-aeration surface impoundments can result in anoxic (low oxygen) conditions that can limit biological degradation. In such cases, the modules will tend to overestimate constituent and BOD removal through biodegradation and sorption, and underestimate volatile emissions and leachate flux.

- **No delineation of organic solids.** The modules maintain suspended solids characteristics in the influent throughout the simulation. However, as influent carbon loading converts to biomass, the characteristics of the suspended solids within the WMU could change. For example, the fraction organic carbon, average particle density, and fraction of biologically active solids can change dramatically from influent solid values within a biologically active unit. Such changes can affect the solids balance (e.g., the sedimentation rate) as well as the partitioning of organic chemicals. This limitation creates uncertainty in the output variables for WMUs with high biomass generation rates relative to the influent solids loading. For units with relatively high solids loading rates and low biomass generation rates, changes in suspended solid particle characteristics due to biological growth would be limited.

- **No spatial variations in sediment depths.** For all units (well-mixed, plug-flow or batch; aerated or nonaerated), the modules assume that sediment compartment depths are uniform throughout the impoundment. For plug-flow systems, it is likely that the sediment will accumulate fastest near the influent point. For aerated units, sediment depths directly beneath high-speed aerators are expected to be less than in less agitated parts of the WMU. Because including areas of different sediment depth would greatly complicate the model construct and solution, the even distribution of sediment is a reasonable simplifying assumption. However, the modules could overestimate leachate flux for plug-flow units that have significant sediment accumulation near the influent (i.e., lower infiltration rates where the concentrations are the highest), and could underestimate leachate flux for units that have low sediment accumulation areas below mechanical aerators.

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5.0 Land-based Source Modules

5.1 Purpose and Scope

The Land-based Source Modules simulate partitioning and emission of constituents from land-based waste management units (WMUs). Three Land-based Source Modules were developed for the 3MRA modeling system to represent the major management practices where wastes are put into or on the land for recycling, recovery, reuse, treatment, or disposal. These modules simulate waste management practices in the following types of WMUs:

- **Landfills,** which are a common disposal unit for many nonliquid industrial wastes;

- **Waste piles,** which are temporary storage areas on the ground for nonliquid industrial waste, such as ash or slag; and

- **Land application units (LAUs),** which are used to reuse, treat, or dispose of industrial waste in liquid, semiliquid, or solid form. Some wastes are used as a soil amendment, which is a reuse practice; some wastes are applied to land for treatment through biological degradation; and some wastes are applied to land as a disposal method.

The three Land-based Source Modules were designed to provide estimates of annual average constituent concentrations in surface soil and constituent mass emission rates to air, surface water, and ground water, and to maintain mass balance between the source and all release routes. The emission rate estimates are then used in the 3MRA modeling system, which links source modules with environmental fate and transport modules. Figure 5-1 shows the relationship and information flow among the Land-based Source Modules in the 3MRA modeling system.

Each of the three Land-based Source Modules provides some similar and some different features in terms of the ways constituents of concern can be released to the environment. All three modules have the potential to release constituents to the air by volatilization or particle entrainment, and to the subsurface and ground water by leaching. Waste piles, because they are elevated and have more surface area exposed, are assumed to have a greater potential for air emissions than the other two source types. Waste piles and LAUs have the additional release mechanism of erosion and runoff to the surrounding watershed and the nearest stream or other waterbody. Landfills are assumed to be below grade and, thus, do not release constituents through erosion or runoff.

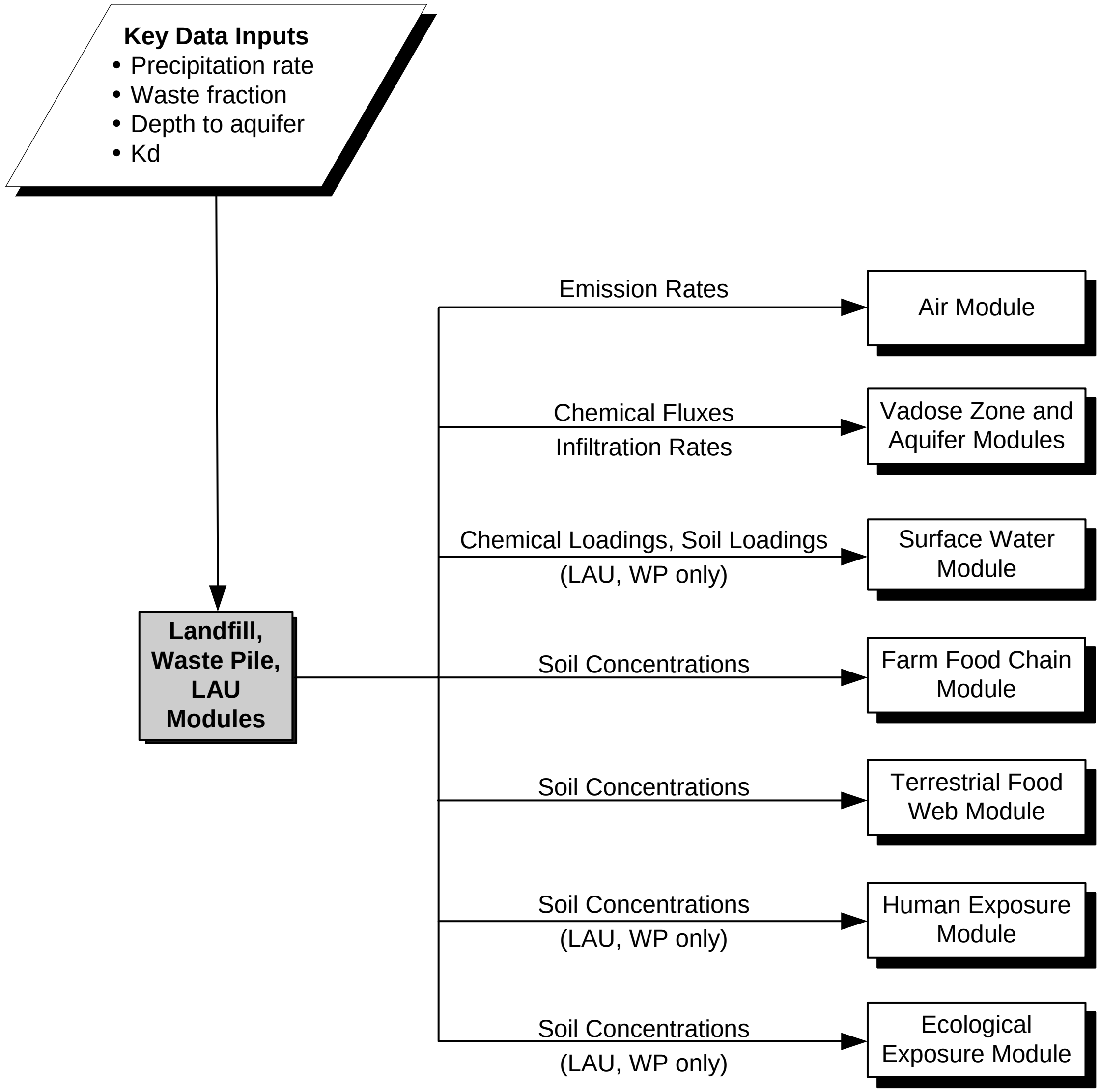

Figure 5-1. Information flow for the Land-based Source Modules in the 3MRA modeling system.

The Land-based Source Modules contain the following three models:

1. **The Generic Soil Column Model (GSCM)** was developed to describe the constituent fate and transport in a porous medium, such as soil or waste. It provides the vertical concentration profile in the soil/waste column at various times. The GSCM provides the following outputs:

   – Annual average constituent concentrations in surficial soil (top 1 cm), as well as depth-averaged concentrations over greater depths in the WMU

   – Annual average constituent concentrations in surficial soils, as well as depth-averaged concentrations over greater depth in the buffer zone located downslope from the WMU (LAU and waste pile only)

- Annual average constituent fluxes in leachate draining from the WMU into the vadose zone

- Annual average leachate flow rates (infiltration rates) draining from the WMU into the vadose zone

- Annual average constituent fluxes volatilizing from the WMU surface into the atmosphere.

2. **The Local Watershed Model** (for LAUs and waste piles) is based on mass balances of solids and constituents in the runoff and the top layer of the soil column modeled by the GSCM. The "local watershed" comprises the land area between the waterbody and the top of the hillside containing the WMU and may include an upslope area, the WMU, and a downslope or buffer area between the WMU and the waterbody. The Local Watershed Model provides the following outputs:

   - Annual average constituent loadings in surface runoff and erosion that enter the nearest surface waterbody downslope of the WMU

   - Annual average eroded soil loadings in surface water runoff that enter the nearest surface waterbody downslope of the WMU

   - Annual average runoff that enters the nearest surface waterbody downslope of the WMU.

3. **The Particulate Emissions Model** was designed to provide estimates of the annual average emission rate of constituent mass adsorbed to particulate matter less than 30 μm in diameter. The release mechanisms considered differ for each WMU, but may include wind erosion, vehicular activity, unloading operations, tilling, and spreading/compacting operations. The Particulate Emissions Model provides the following outputs:

   - Annual average particulate constituent fluxes emitted into the atmosphere from the WMU surface sorbed to soil particles via wind erosion or other surface disturbances

   - Annual average particulate fluxes (the particles themselves) emitted into the atmosphere from the WMU surface via wind erosion or other surface disturbances.

   - Particle size distributions for airborne soil particles.

Figure 5-2 illustrates how these three models interact within the Land-based Source Modules. The GSCM models vertical movement of constituents within and from the waste/soil (with the exception of particulate emissions, which are modeled by the Particulate Emissions Model). The Local Watershed Model simulates lateral movement and mass balance between waste/soil and runoff. In addition to the constituent fluxes indicated in the figure, various media fluxes are also calculated (e.g., leachate flux, by the GSCM; eroded soil load, by the Local Watershed Model; and particle size distribution, by the Particulate Emissions Model). The three models interact to maintain mass balance between the compartments. The GSCM calculates the depth-averaged soil concentration in the buffer zone for several additional layers below the surficial layer only in order to maintain mass balance between the surficial layer and the deeper layers. The deeper layers represent a sink, and the soil concentrations in those layers are not used. Similarly, the GSCM calculates volatilization from the buffer zone only to maintain mass balance—the volatilization flux from the buffer zone is not used in the Air Module. The omission of volatilization and leachate flux from the buffer zone from subsequent modeling results in an underestimation of air and ground water concentrations. However, these emissions likely have a less significant impact on the results than do direct releases from the WMU.

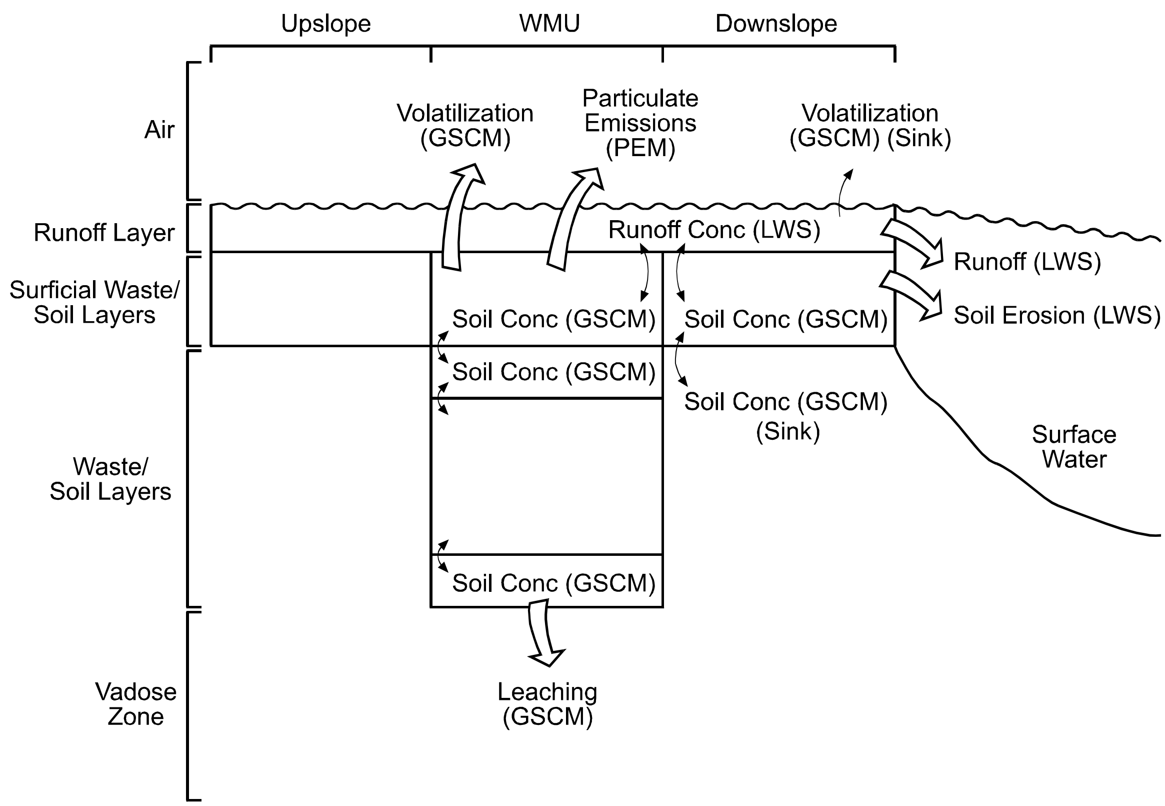

Figure 5-2. Interaction of the models that form the Land-based Source Modules.

5.2 Conceptual Approach

The physical processes modeled differ among the three land-based sources. As a result, not all of the models listed above are used for all three of the WMUs. This section first describes the three different WMUs and the processes modeled for each and then describes the three models that are contained in the Land-based Source Modules in greater detail.

5.2.1 Description of WMUs

Landfills. Figure 5-3 shows the physical processes being modeled in the 3MRA modeling system for landfill operations. The Landfill Module was developed to approximate the effects of the gradual filling of active landfills. The landfill is divided into vertical cells of equal volume running from the site surface to the bottom of the landfill, each sized so that they require 1 year to fill. Waste mass is added gradually, forming layers of waste. After 1 year, the cell is full and the waste may be covered with a clean soil cover (this is optional). This process is repeated with the next cell and continues until the landfill reaches maximum capacity. The landfill can be sized to operate for a specified number of years (such as a 20-year operational life). Releases of a constituent to the air or ground water are modeled for up to 200 years or until the concentration in the landfill is 1 percent of the maximum concentration during the operating life of the landfill.

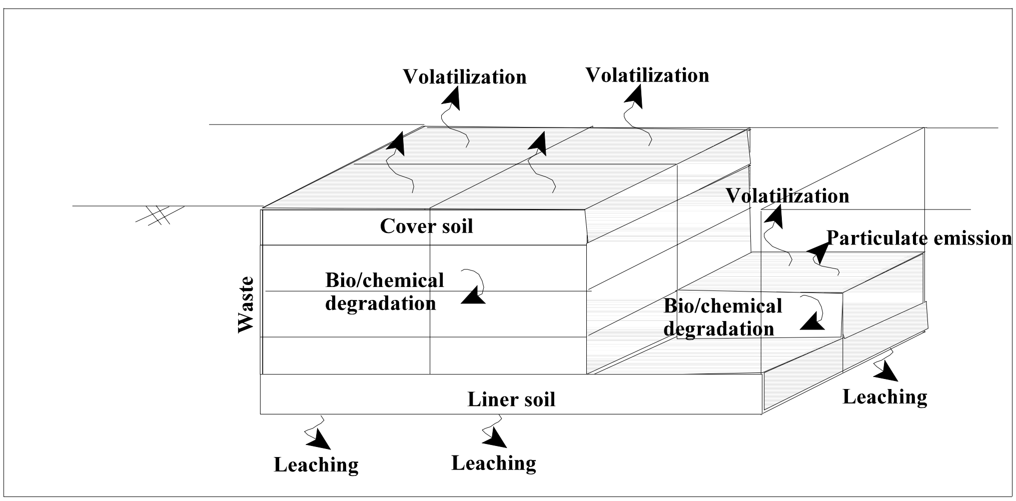

Figure 5-3. Illustration of landfill (shown with six cells and three waste layers).

The following assumptions were made in developing the Landfill Module:

- The empty landfill is approximated as an excavated volumetric rectangle. The volume is assumed to be completely below grade; therefore, it is assumed that no constituent mass is lost due to runoff or erosion.

- Each landfill cell is approximated as a soil column consisting of three homogeneous zones: soil cover, landfill waste, and subsoil. Each zone is modeled as a homogeneous porous medium whose properties are uniform in space and time within the zone. The soil cover zone and subsoil zones are optional.

- The concentration of constituents is constant in the incoming waste, and waste is added at a constant rate. The concentration of constituents in the waste can be adjusted to account for other wastes entering the landfill that do not contain the constituent of interest.

- There is no lateral transport of constituents between cells.

- Waste is added to the landfill cell in layers. A waste layer, for the purposes of the model, is simply a zone within which initial concentrations are assumed to be uniform. Waste layers are conceptualized as being formed over time by the dumping of loads of waste next to one another in the landfill cell until eventually a waste layer of uniform depth is formed. At this point, a new layer is started.

- At the start of the landfill cell simulation, one waste layer is assumed to be present. After each time period, another waste layer is laid down until the landfill cell is full. At that time, it may be covered with clean soil or left exposed.

- The first-order constituent and biological loss processes modeled for the entire landfill (including cover soil, waste, and liner material) are anaerobic biodegradation and hydrolysis.

- The first-order loss rate due to particulate emissions from an active landfill cell includes losses due to wind erosion, vehicular activity, and spreading and compacting. Only losses due to wind erosion are modeled from inactive landfill cells.

Waste Piles. Figure 5-4 shows the physical processes modeled in the 3MRA modeling system for waste piles, which may manage ash, slag, or other similar types of waste. The waste pile has a constant height and constant area equal to the footprint of the waste pile. At the start, the waste pile is filled to capacity. After each period of time, such as 1 year, the entire waste pile is removed and instantaneously refreshed (i.e., replaced with fresh waste). In reality, waste is added and removed from a waste pile incrementally; the assumption that the waste is instantaneously refreshed is made to simplify modeling. Because the waste surface is not refreshed between placements of new waste, the Waste Pile Module may underestimate volatile losses relative to a model where waste is built up gradually. This is unlikely to affect emission estimates for nonvolatile constituents.

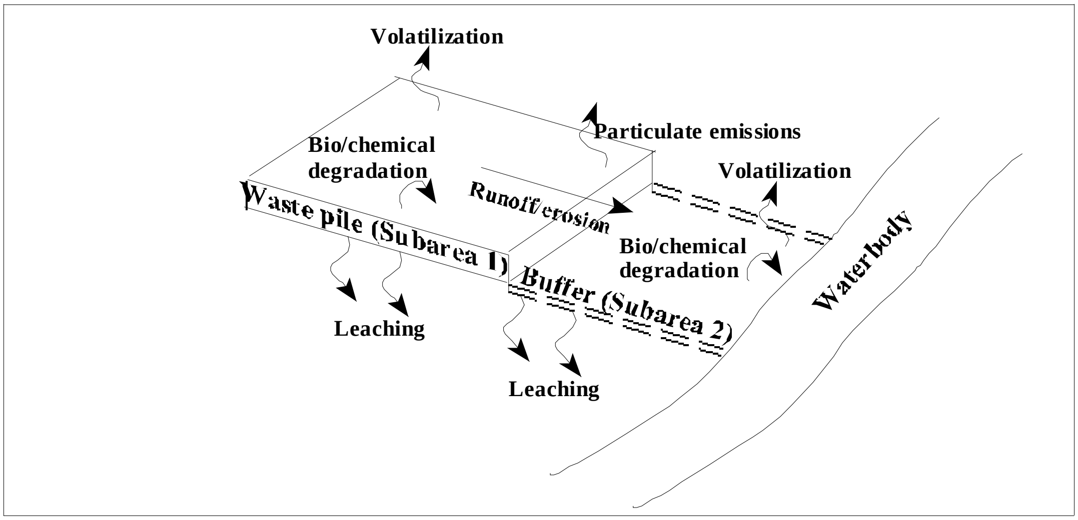

Figure 5-4. Illustration of a waste pile in local watershed.

The following assumptions were made in developing the Waste Pile Module:

- The waste pile site is used for a finite number of years, after which it is assumed that the final waste pile is removed and clean soil is placed on the surface where the waste pile was.

- The first-order constituent and biological loss processes modeled for the waste pile are aerobic biodegradation and hydrolysis in the surface waste layer, and anaerobic biodegradation and hydrolysis in all subsurface layers of the waste pile.

- The first-order loss rate due to particulate emissions from an active waste pile is applied only to the surface layer and includes losses due to wind erosion, vehicular activity, and spreading and compacting operations from an active waste pile.

- For purposes of modeling runoff and erosion processes, the following assumptions were made:

  – The waste pile is conceptualized as having side slopes (from which increased runoff and erosion would occur) that are insignificant (i.e., the sloped surface area is small relative to the surface area of the top of the waste pile).

  – The top surface of the waste pile has the same slope as the average slope of the local watershed.

  – No run-on occurs to the waste pile subarea from upslope subareas in the local watershed.

– Following removal of the waste pile, there is no subsequent runoff/erosion transport pathway from the waste pile subarea of the local watershed downslope to the surface water, only from the buffer subarea. That is, there is assumed to be no remaining surficial contamination in the subarea that contained the waste pile because clean soil is assumed to be placed there after waste pile removal.

■ The topmost layer of waste serves as the "soil" compartment in the runoff and erosion algorithms used in the local watershed model. For the purposes of applying these algorithms, it is assumed that the surface layer is 0.01 m deep.

Land Application Unit. Figure 5-5 shows the physical processes being modeled in the 3MRA system for LAUs. Such units are used to manage liquid, semi-solid, and solid wastes. Certain types of waste can be used to amend agricultural soils, thus constituting a reuse of the waste.

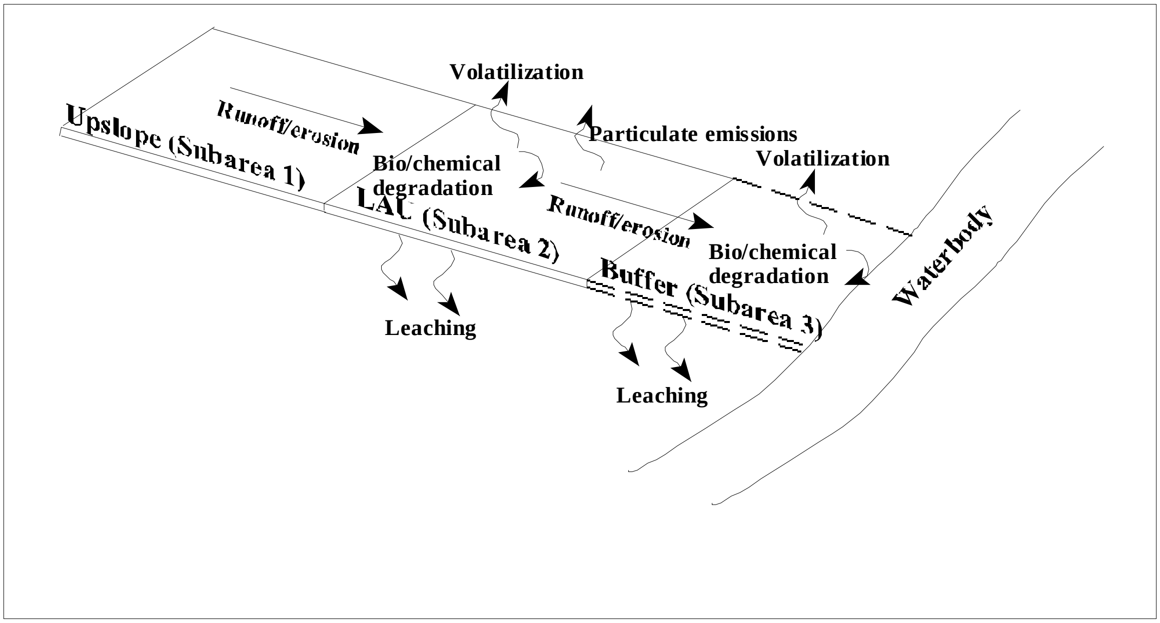

Figure 5-5. Illustration of LAU in local watershed.

The following assumptions were made in developing the LAU Module:

■ Waste is applied to the soil surface periodically at even intervals (e.g., quarterly) and then tilled into the soil to a specified depth, such as 0.2 m.

■ The till zone is completely mixed upon each application of waste to soil.

■ The modeled surface layer consists of one homogeneous zone—the till zone—which consists of a soil/waste mixture. The till zone properties can be estimated as the depth-weighted average of the soil and waste properties according to the depth of soil and waste in the till zone.

- The water contained in the wet waste added to the LAU increases the annual average infiltration rate.

- The constituent mass is concentrated in the solids portion of the waste and is re-partitioned among the solid, aqueous, and gas phases in the soil column.

- Waste applications do not result in significant buildup of the soil surface, nor does erosion significantly degrade the soil surface (i.e., the vertical distance from the site surface). As a result, there is no naturally occurring limit to the modeled concentration of constituents in the soil other than the limit for nonaqueous-phase liquids (NAPLs). As a result, the modeled concentration in the till zone could exceed the concentration in the waste. This is physically possible for highly immobile constituents if the waste matrix is organic and decomposes, leaving behind the constituent to concentrate over multiple applications.

- The LAU is operated for a specified number of years (e.g., 40 years).

- The first-order constituent and biological loss processes in the till zone include aerobic biodegradation and hydrolysis.

- The first-order loss rate due to particulate emissions from an active LAU is applied to the surface layer of the till zone only and includes losses due to wind erosion, vehicular activity on the surface of the LAU, and tilling operations. The particulate emission loss rate from an inactive LAU includes wind erosion only.

- The topmost waste/soil layer serves as the soil compartment in the runoff and erosion algorithms used in the Local Watershed Model. For the purposes of applying these algorithms, it is assumed that the surficial soil layer is 0.01 m deep.

5.2.2 The Generic Soil Column Model

The GSCM was developed to describe the dynamics of constituent mass fate and transport within land-based WMUs and near-surface soils in watersheds. Figure 5-6 illustrates the processes modeled by the GSCM. The GSCM solves the 1-D partial differential equation describing constituent fate and transport in a porous medium in space and time. This solution represents the vertical concentration profile in the soil/waste column at various times, where the soil/waste column can represent either the top portion of the vadose zone of a land application site (LAU Module), the waste pile (Waste Pile Module), a landfill cell (Landfill Module), or the top portion of the vadose zone of an entire watershed subbasin (Watershed Module). In addition to describing vertical constituent concentration gradients in the soil column, the GSCM is coupled with a runoff/erosion model (either the Local Watershed Model, described in Section 5.2.3, or the Watershed Module, described in Section 7) that describes constituent

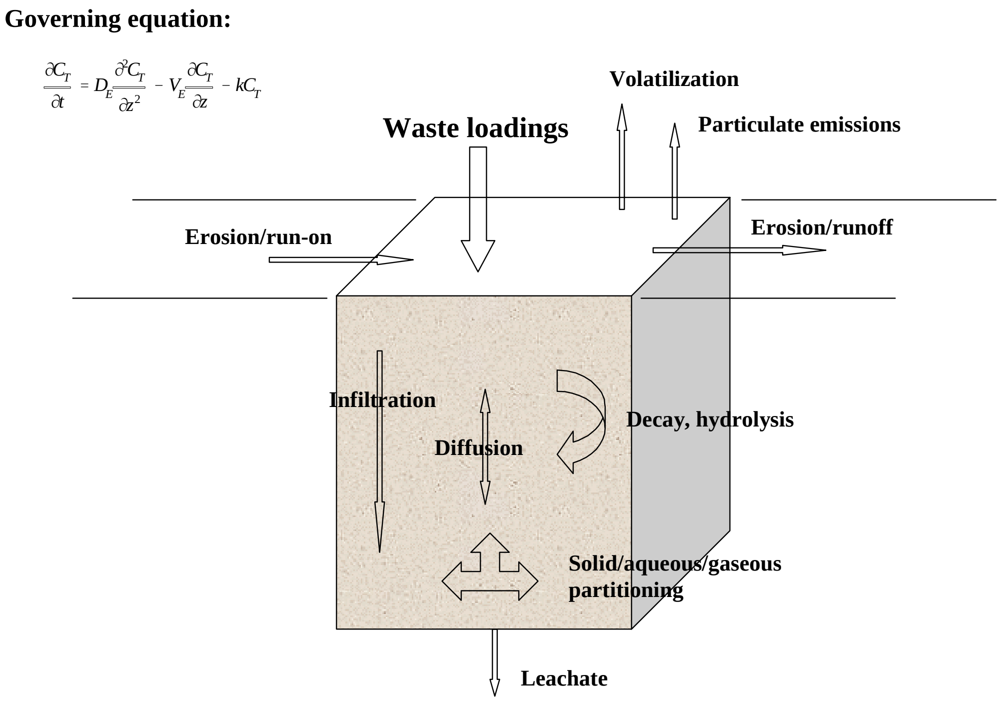

Figure 5-6. Conceptual diagram of the Generic Soil Column Model (GSCM).

concentration in surface runoff across the soil column's surface and interactions between this runoff compartment and the soil compartment.

The GSCM was originally designed for the 3MRA modeling system to make some modest improvements to the well-known Jury model (Jury et al., 1983, 1990), which had been used historically to simulate constituent fate and transport in land-based sources. Similar to the GSCM, the Jury model describes vertical concentration gradients over time of total (sorbed, dissolved, and gaseous) constituent concentration. The Jury model was limited for 3MRA purposes for two reasons. First, it assumes an infinitely deep soil column. The 3MRA Land-based Source Modules output leachate fluxes to the Vadose Zone Module, so the infinite depth assumption is inappropriate. Second, the Jury model's initial conditions cannot accommodate the vertical gradient resulting from periodic waste loadings, such as from repeated land application of waste, over all or a portion of the soil column depth. The governing equations for the GSCM are similar to those used by Jury et al. (1983, 1990) and Shan and Stephens (1995), but have been applied in a manner that avoids these limitations. Use of the GSCM allows

- Constituent mass balance;
- Waste additions and removals to simulate active facilities; and

- Joint estimation of

  – Volatilization of gas-phase constituent mass from the surface to the air,

  – Leaching of aqueous-phase constituent mass by advection or diffusion from the bottom of the WMU or vadose zone,

  – Concentrations of constituents within the soil or waste column, and

  – First-order losses, which can include hydrolysis and anaerobic and aerobic biodegradation.

The following assumptions were made in the development of the GSCM for use in all of the Land-based Source Modules:

- The constituent partitions to three phases: adsorbed (solid), dissolved (liquid), and gaseous (as in Jury et al., 1983, 1990). The governing equation is

$$C_T = \rho_b\, C_s + \theta_w\, C_L + \theta_a\, C_G \tag{5-1}$$

where

C_T = total constituent concentration in soil (g/m^3 of soil)
ρ_b = soil dry bulk density (g/cm^3)
C_s = adsorbed-phase constituent concentration in soil (µg/g of dry soil)
θ_w = soil volumetric water content (m^3 soil water/m^3 soil)
C_L = aqueous-phase constituent concentration in soil (g/m^3 of soil water)
θ_a = soil volumetric air content (m^3 soil air/m^3 soil)
C_G = gas-phase constituent concentration in soil (g/m^3 of soil air).

- The constituent undergoes reversible, linear equilibrium partitioning between the adsorbed and dissolved phases (as in Jury et al., 1983, 1990):

$$C_s = K_d \times C_L \tag{5-2}$$

where

K_d = linear equilibrium partitioning coefficient (cm^3/g).

For organic constituents,

$$K_d = f_{oc} \times K_{oc} \tag{5-3}$$

where

f_{oc} = organic carbon fraction in soil or waste (unitless)
K_{oc} = equilibrium partition coefficient, normalized to organic carbon (cm^3/g).

Alternatively, K_d can be specified as an input parameter for inorganic constituents.

It is implicit in this linear equilibrium partitioning assumption that the sorptive capacity of the soil column solids is considered to be infinite with respect to the total mass of constituent over the duration of the simulation, i.e., the soil column sorptive capacity does not become exhausted.

- Constituents in the dissolved and gaseous phases are assumed to be in equilibrium and to follow Henry's law (as in Jury et al., 1983, 1990):

$$C_G = H' \times C_L \tag{5-4}$$

where

H′ = dimensionless Henry's law coefficient.

The Henry's law coefficient is adjusted for seasonal variations in temperature.

- The total constituent concentration in soil can also be expressed in units of mass of constituent per mass of dry soil (µg/g) by dividing by the soil dry bulk density.

- Using the linear equilibrium approximations in Equations 5-2 through 5-4, C_T can be expressed in terms of C_L, C_S, or C_G as follows:

$$C_T = K_{TL} \times C_L = \frac{K_{TL}}{K_d} \times C_S = \frac{K_{TL}}{H'} \times C_G \tag{5-5}$$

where

K_{TL} = dimensionless equilibrium distribution coefficient between the total and aqueous-phase constituent concentrations in soil.

- The total water flux or infiltration rate is constant in space and time (as in Jury et al., 1983, 1990) and greater than or equal to zero. It is specified as an annual average.

- Material in the soil column (including bulk waste) can be approximated as unconsolidated homogeneous porous media (i.e., one whose basic properties, which include dry bulk density, fraction organic carbon, soil volumetric water content, soil volumetric air content, and total soil porosity, are uniform in space). Waste/soil properties are specified as annual average values.

- Constituent mass may be lost from the soil column through one or more first-order loss processes.

- The total constituent flux is the sum of the vapor flux and the flux of the dissolved solute (as in Jury et al., 1983, 1990).

- The constituent is transported in one dimension through the soil column (as in Jury et al., 1983, 1990).

- The effective diffusivity in soil may be calculated by the model of Millington and Quirk (1961) (as in Jury et al., 1983, 1990).

- The modeled spatial domain of the soil column remains constant in volume and fixed in space with respect to a vertical reference, such as the water table.

Under the above assumptions, the governing mass fate and transport equation for the GSCM can be written as follows:

$$\frac{\partial C_T}{\partial t} = D_E \frac{\partial^2 C_T}{\partial z^2} - V_E \frac{\partial C_T}{\partial z} - kC_T \qquad (5\text{-}6)$$

where

t	=	time
k	=	total first-order loss rate (1/d)
D_E	=	effective diffusivity in soil (m^2/d)
z	=	verticle direction (m), depth
V_E	=	effective solute convection velocity (m/d).

D_E can be considered to be the algebraic sum of the effective gaseous and water diffusion coefficients in soil. V_E is equal to the water flux corrected for the constituent partitioning to the solid phase (m/d).

A solution of the complete convective-diffusive-decay concentration model (Equation 5-6) was undertaken to evaluate, in a soil/waste column,

- Total constituent concentration as a function of time and depth below the surface, and

- Constituent mass fluxes across the upper and lower boundaries of the soil column.

A quasi-analytical approach was developed that allows for relative computational speed and significantly reduces concern about numerical diffusion and lack of stability associated with fully numerical solution techniques. The tradeoff is a loss of ability to evaluate short-term trends in concentration and diffusive flux profiles. The method was developed to allow estimation of long-term (i.e., annual average) constituent concentration profiles and mass fluxes.

The solution for the simplified case, in which the soil column consists of one homogeneous zone whose properties are uniform in space and time, is described below. Adaptations of the solution technique to account for variations from this simplified case (e.g., more than one homogeneous zone, as for a landfill with cover soil zone atop the waste zone) are described in the module-specific sections of the 1999 background documents at www.epa.gov/epaoswer/hazwaste/id/hwirwste/risk.htm.

The quasi-analytical approach is a step-wise solution of the three components of the governing equation (Equation 5-6) on the same grid. That is, the following equations are solved individually and then added (under the principle of superposition for linear differential equations) to yield the complete solution:

$$\frac{\partial C_T}{\partial t} = D_E \frac{\partial^2 C_T}{\partial z^2} \tag{5-7}$$

$$\frac{\partial C_T}{\partial t} = -V_E \frac{\partial C_T}{\partial z} \tag{5-8}$$

$$\frac{\partial C_T}{\partial t} = -k\,C_T \tag{5-9}$$

Equations 5-7 through 5-9 each have an analytical solution that can be combined to obtain a pure diffusion solution that moves with velocity V_E through the porous medium (Jost, 1960). The solution of the general differential equation then has the form of the solution of the diffusive portion with its time dependence, translating in space with velocity V_E, and decaying exponentially with time.

The following boundary conditions are assumed:

- Zero concentration is assumed at the upper boundary of the soil column.[1] This is consistent with the assumption that the air is a sink for volatilized constituent mass, but requires the approximate method for estimating the mass flux of volatile emissions.

- At the lower boundary of the soil column, the flexibility exists with this solution technique to specify a value between 0 and 1 for the ratio of the total constituent concentration in the soil directly below the modeled soil column and in the soil column. A ratio of 1 corresponds to a zero gradient boundary condition. A ratio of 0 corresponds to a zero concentration boundary condition. For the 3MRA modeling system, a boundary condition of 0 was assumed.

[1] The $C_T = 0$ "boundary condition" is not a boundary condition in the usual sense of it determining the functional form of the resulting analytical solution to the underlying diffusion differential equation (Equation 5-7). That solution is based on an implicit boundary condition of $C_T = 0$ at +/- infinity, similar to heat transfer in an infinitely long plate. The $C_T = 0$ boundary condition is here intended to reflect the fact that there is no *back*-diffusion from the overlying air column into the soil column, i.e., the concentration in the overlying air column is assumed to be zero so that no back-diffusion results.

5.2.3 Local Watershed Model

The Local Watershed Model is a holistic model used in the Waste Pile and LAU Modules to incorporate them into the watershed of which they are a part. These WMUs are on the land surface, so they are integral land area components of their respective watersheds and, consequently, are not only affected by runoff and erosion from upslope land areas, but also affect downslope land areas through runoff and erosion. After some period of operation during which runoff and erosion have occurred from these WMUs, the downslope land areas will have been contaminated and their surface concentrations could approach the residual constituent concentrations in the WMU itself (or conceivably even exceed them long after the WMU ceases operation). Thus, after extensive runoff and erosion from a WMU, the entire downslope surface area can be considered a source, and it becomes important to consider these extended source areas in the model.

As stated earlier, the watershed that includes the LAU or waste pile is termed here the “local watershed” (see Figures 5-4 and 5-5). A local watershed is defined as the drainage area that contains the WMU (or a portion thereof) in the lateral direction (perpendicular to runoff flow) and the area in which runoff occurs as overland flow (sheet flow) only. Thus, a local watershed extends downslope only to the point that runoff flows and eroded soil loads would enter a well-defined drainage channel (e.g., a stream, lake, or some other waterbody). The sheet-flow-only restriction is based on the assumption that any subareas downslope of the WMU subarea are subject to constituent contamination from the WMU through overland runoff and soil erosion.

The local watershed is conceptualized as a 2-D, two-medium system. The dimensions are longitudinal (i.e., downslope or in the direction of runoff flow), and vertical (i.e., through the soil column). The media are the soil column and, during runoff events, the overlying runoff water column. The local watershed is assumed to be made up (in the longitudinal direction) of an arbitrary number of land subareas[2] that may have differing surface or subsurface characteristics (e.g., land uses, soil properties, and constituent concentrations).

The Local Watershed Model comprises the following three components:

- A hydrology model,
- A soil erosion model, and
- A constituent fate and transport model.

These are described below.

Hydrology Model. Hydrologic modeling is performed to simulate watershed runoff and ground water recharge (termed here “infiltration”). The hydrology model is based on a daily soil water balance performed for the root zone of the soil column as shown in Figure 5-7.

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[2] The conceptual approach allows for an arbitrary number of subareas, but in the implementation of the 3MRA modeling system, that number has been limited to a maximum of 3: upslope, WMU, and downslope/buffer.

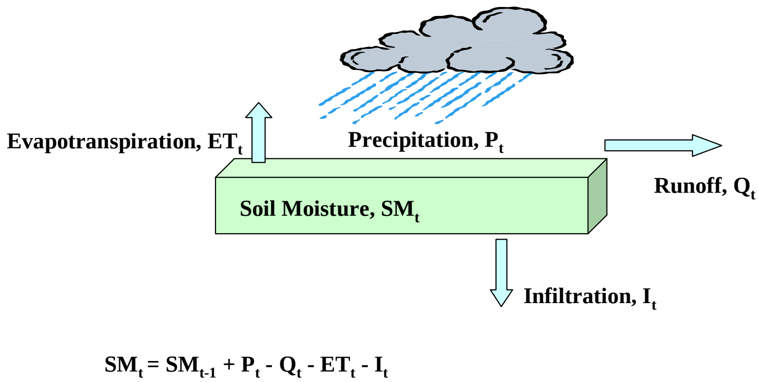

Figure 5-7. Daily water balance model.

At the end of a given day, the soil moisture in the root zone of an arbitrary watershed subarea is given as

$$SM_t = SM_{t-1} + P_t - Q_t - ET_t - I_t \tag{5-10}$$

where

SM_t	=	soil moisture in root zone at end of day *t* for subarea *i* (cm)
SM_{t-1}	=	soil moisture in root zone at end of previous day for subarea *i* (cm)
P_t	=	total precipitation on day *t* (cm)
Q_t	=	net of storm runoff on day *t* leaving the subarea and storm runoff entering the subarea (cm)
ET_t	=	evapotranspiration from root zone on day *t* for subarea *i* (cm)
I_t	=	infiltration (ground water recharge) on day *t* for subarea *i* (cm) .

Precipitation is undifferentiated between rainfall and frozen precipitation; that is, frozen precipitation is treated as rainfall.

The daily runoff estimate is based on the Soil Conservation Service's (SCS's) widely used "curve number" procedure (USDA, 1986) and is a function of land use and current and antecedent precipitation. Land use is considered empirically by application of the curve numbers, which are catalogued by land use or cover type (e.g., woods, meadow, impervious surfaces), treatment or practice (e.g., contoured, terraced), hydrologic condition, and hydrologic soil group.

Curve numbers are typically presented in the literature assuming average antecedent moisture conditions (AMC II), but can be adjusted for drier (AMC I) or wetter (AMC III) conditions (Chow et al., 1988). These three categories are used in the hydrology model; a distinction is made within them between the dormant season and the growing season. The growing season is assumed to be June through August (Julian Day 152 to 243) throughout the country. These adjustments have the effect of increasing runoff under wet antecedent conditions and decreasing runoff under dry antecedent conditions, relative to average conditions.

Potential evapotranspiration (PET) is the demand for soil moisture from evaporation and plant transpiration. When soil moisture is abundant, actual evapotranspiration equals the PET. When soil moisture is limiting, actual evapotranspiration will be less than PET. The extent to which it is less under limiting conditions has been expressed as a function of PET, available soil water, and available soil water capacity (Dunne and Leopold, 1978).

The more theoretically based models for daily evapotranspiration (e.g., the Penman-Monteith equation [Monteith, 1965]) rely on the availability of significant daily meteorological data, including temperature gradient between surface and air, solar radiation, windspeed, and relative humidity. For 3MRA, it is assumed that all of these variables will not be readily available for all applications. Therefore, a less data-demanding model, the Hargreaves equation (Shuttleworth, 1985), is used. The Hargreaves method, which is primarily temperature-based, has been shown to provide reasonable estimates of evaporation (Jensen, et al., 1990)—presumably because it also includes an implicit link to solar radiation through its latitude parameter (Shuttleworth, 1993).

Soil moisture in excess of the soil's field capacity, if not used to satisfy evapotranspiration, is available for gravity drainage from the root zone as infiltration to subroot zones (Dunne and Leopold, 1978). The resulting infiltration rate will, however, be limited by the root zone soil's saturated hydraulic conductivity.

In the event that infiltration is limited by the saturated hydraulic conductivity (rather than evapotranspiration), the hydrology model includes a mechanism to increase the previously calculated runoff volume by the amount of excess soil moisture (i.e., the soil water volume in excess of the field capacity). This adjustment is made to preserve water balance and is based on the assumption that the runoff curve number method, which is only loosely sensitive to soil moisture (through the antecedent precipitation adjustment) has admitted more water into the soil column than can be accommodated by evapotranspiration, infiltration, and/or increased soil moisture. After the runoff is increased by this excess, the evapotranspiration, infiltration, and soil moisture are updated to reflect this modification and preserve the water balance.

Soil Erosion Model. The soil erosion model used in the Local Watershed Model is based on the Universal Soil Loss Equation (USLE), an empirical methodology (see, e.g., Wischmeier and Smith, 1978) derived from measured soil losses at small, experimental field-scale plots throughout the United States. The USLE predicts sheet and rill erosion from hillsides upslope of defined drainage channels, such as streams. It does not predict streambank erosion. The eroded soil flux from a hillside area averaged over a specific time period is predicted by the USLE as the product of the six variables discussed below.

The rainfall factor (*R*, 1/time) accounts for the erosive (kinetic) energy of falling raindrops, which is essentially measured by rainfall intensity. The kinetic energy of an individual storm multiplied by its maximum 30-minute intensity is sometimes called the erosivity index factor. Rainfall factors have been compiled throughout the United States on a long-term annual average basis. These rainfall factors were developed by cumulating individual storm erosivity index factors.

The soil erodibility factor (K, kg/m^2) is an experimentally determined property that is a function of soil type, including particle size distribution, organic content, structure, and profile. Soil erodibility factor values are reported by soil type in the literature.

The cropping management factor (C, unitless) varies between 0 and 1. It accounts for the type of cover (e.g., sod, grass type, fallow) on the soil. The cropping management factor is used to correct the USLE prediction relative to the cover type for which the experimentally determined soil erodibility values were measured (fallow).

The practice factor (P, unitless) accounts for the effect of erosion control practices, e.g., contouring or terracing. The practice factor is never negative, but could be greater than 1 if land practices actually encourage erosion relative to the original experimental plots on which soil erodibility factors were measured.

The combined length-slope factor (LS, unitless) is given by U.S. EPA (1985b). The length-slope factor increases with increasing flow distance because runoff quantity and erosive energy generally increase with downslope distance. LS increases with slope because runoff velocity generally increases with slope.

The sediment delivery ratio (Sd, unitless) estimates the fraction of on-site eroded soil that reaches a particular downslope or downstream location in the subbasin (Shen and Julien, 1993). The sediment delivery ratio is used here to account for deposition of eroded soil from the local watershed in ditches, gullies, or other depressions before reaching a downslope stream or waterbody. Vanoni (1975) developed the relationship between the sediment delivery ratio and the watershed drainage area.

The USLE is implemented within the Local Watershed Model on a storm event basis, i.e., the “modified” USLE (MUSLE) is used. This implementation requires determining a rainfall factor value for each daily storm event that specifies the erosivity of that individual storm.

Constituent Fate and Transport Model. Constituent and suspended solids concentrations in storm-event runoff are simulated using a two compartment model for the soil surface layer: a runoff compartment and the soil compartment. Figure 5-8 presents the conceptual runoff /erosion model showing the two compartments and the fate and transport processes considered. Hydrolysis, volatilization, and biodegradation processes are not simulated in the runoff compartment. These processes are continuously simulated in the surface layer of the soil column by the GSCM. The percentage of time that runoff is actually occurring is sufficiently short that any additional constituent losses in the runoff water due to these processes should be minimal.

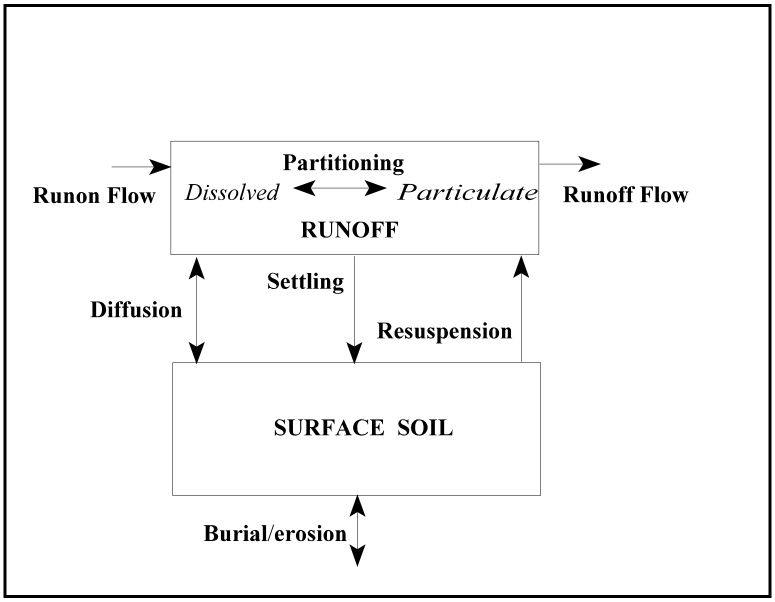

Figure 5-8. Runoff/erosion conceptual model.

Runoff Compartment. The runoff compartment model is based on mass balances of solids and constituent in the runoff water and the top soil column layer. A simplifying assumption is made that solids and constituent concentrations in the runoff are at instantaneous steady-state equilibrium during each individual runoff event, but can vary among runoff events (i.e., a quasi-dynamic approach is used). While assumption of instantaneous, steady-state equilibrium for each storm event is not strictly accurate, it was deemed appropriate for the following reasons:

- Data will not generally be available at the temporal scale to accurately track within-storm event conditions (e.g., rainfall hyetographs).

- The actual time to steady-state may not differ significantly from the 1 day or less implicitly assumed here because of the anticipated relatively small surface areas of the watershed subareas and the associated relatively small runoff volumes. A sensitivity analysis was performed, using a dynamic form of the runoff compartment model, which suggested relatively little difference in calculated soil concentrations as a function of the steady-state versus dynamic assumption.

- To the extent that the actual time to steady-state is greater than 1 day, the model is biased toward overestimating downslope concentrations and waterbody loads.

A steady-state mass balance of solids in the runoff, i.e., suspended solids from erosion, is used:

$$0 = Q'_{i-1}\, m_{l,i-1} - Q'_i\, m_{1,i} - vs_i\, A_i\, m_{1,i} + vr_i\, A_i\, m_2 \qquad (5\text{-}11)$$

where

Q'_{i-1}	=	total run-on flow volume (water plus solids) from subarea *i*-1 (m^3/s) (see Equation 5-12)
Q'_i	=	total runoff flow volume (water plus solids) leaving subarea *i* (m^3/s) (see Equation 5-12)
$m_{1,i-l}$	=	solids concentration in runon from subarea *i*-l (g/m^3)
$m_{1,i}$	=	solids concentration in runoff from subarea *i* (suspended solids) (g/m^3)
m_2	=	solids concentration in the top soil column layer of subarea *i* (g/m^3)
vs_i	=	settling velocity (m/s)
vr_i	=	resuspension velocity (m/s)
A_i	=	surface area of subarea *i* (m^2).

$$Q'_i = Q_i + \frac{CSL_i}{\rho} \tag{5-12}$$

where

Q_i	=	runoff flow leaving subarea *i* (m^3/s)
CSL_i	=	cumulative soil load leaving subarea *i* (g/s)
ρ	=	particle density (g/m^3) (i.e., 2.65 g/m^3).

The first term in Equation 5-11 is the flux of soil across the upslope interface of subarea *i*. The second term is the flux of soil across the downslope interface. The third term is an internal sink of soil due to settling, and the fourth term is an internal source due to resuspension.

Solids mass transport from or to the soil compartment within any given watershed subarea is assumed to occur only in a vertical direction, i.e., no downgradient advection of the top soil column layer itself is considered. The downslope mass transport of soil occurs because of vertical erosion or resuspension of soil followed by advective transport of the soil in the runoff water as suspended solids. The transport is described in terms of three parameters: settling, resuspension, and burial/erosion velocities. Under the assumption of no advective transport of the soil column layer, the steady-state mass balance equation for the surficial soil layer is

$$0 = vs_i \ m_{1,i} \ A_i - vr_i \ m_{2,i} \ A_i - vb_i \ m_{2,i} \ A_i \tag{5-13}$$

where

vb_i = burial/erosion velocity (m/s).

The first term of Equation 5-13 is a source of soil mass to the surficial soil column layer due to settling from the overlying runoff water. The second term is a sink from resuspension. The third term is either a source or a sink depending on the sign of the burial/erosion velocity as described subsequently.

A steady-state mass balance of constituents in the runoff is achieved, as illustrated in Figure 5-8. The governing equation for this mass balance is as follows:

$$0 = Q'_{i-1}c_{1,i-1} - Q'_i c_{1,i} - vs_i A_i Fp_{1,i} c_{1,i} + vr_i A_i Fp_{2,i} Er_i c_{2,i} + vd_i A_i \left(\frac{Fd_{2,i}}{\Phi_2} c_{2,i} - \frac{Fd_{1,i}}{\Phi_{1,i}} c_{1,i}\right) \tag{5-14}$$

where

$c_{1,i}$	=	total constituent concentration (particulate + dissolved) in runoff in subarea *i* (g/m^3)
$c_{2,i}$	=	total constituent concentration in soil (g/m^3)
$V_{1,i}$	=	subarea-specific (not cumulative) runoff volume for subarea *i* (m^3)
$Fp_{j,i}$	=	fraction of constituent sorbed to particulate in runoff in soil compartment *j* (unitless)
$Fd_{j,i}$	=	fraction of constituent sorbed to dissolved in runoff in soil compartment *j* (1-$Fp_{j,i}$)
vd_i	=	diffusive exchange velocity (m/s)
Er_i	=	enrichment ratio (unitless)
$\Phi_{1,i}$	=	porosity of the runoff (unitless)

Note that ϕ_2 is equivalent to porosity (η) in the GSCM.

Equation 5-14 can be used to express $c_{1,i}$ as a function of $c_{1,i-1}$ and $c_{2,i}$, as shown in Equation 5-15:

$$c_{1,i} = \frac{Q'_{i-1}c_{1,i-1} + [vr_i A_i Fp_{2,i} Er_i + vd_i A_i (Fd_{2,i}/\Phi_2)]c_{2,i}}{Q'_i + vs_i A_i Fp_{1,i} + vd_i A_i (Fd_{1,i}/\Phi_{1,i})} \tag{5-15}$$

where $c_{2,i}$ is determined by the GSCM, as described in Section 5.2.2.

An enrichment ratio is used to account for preferential erosion of finer soil particles—with higher specific surface areas and more sorbed constituent per unit area—as rainfall intensity decreases. That is, large (highly erosive) runoff events may result in average eroded soil particle sizes and associated sorbed constituent loads that do not differ much from the average sizes/loads in the surficial soil column layer. However, less intense runoff events will erode the finer materials, and resulting constituent loads could be significantly higher than represented by the average soil concentration. U.S. EPA (1985b) gives the storm-event-specific enrichment ratio as a power function of sediment discharge flux.

Soil Compartment. From the discussion of the GSCM, the governing differential equation for the surface soil layer of subarea *i* is

$$\frac{\partial C_{2,i}}{\partial t} = D_E \frac{\partial^2 C_{2,i}}{\partial z^2} - V_E \frac{\partial C_{2,i}}{\partial z} - \Sigma k_j C_{2,t} + ss_i \tag{5-16}$$

where k_j represents first-order rate constant due to not including runoff/erosion processes, i.e., biological decay and hydrolysis and wind/mechanical action. The last term, ss_i, is a source/sink term representing the net effect of runoff and erosion processes on the concentration in the surface soil. The term ss_i comprises the following terms respectively: a source of constituent due to settling from the overlying runoff water, a sink of constituent due to resuspension, and a source or sink (depending on the relative values of $C_{1,i}$ *and* $C_{2,i}$) due to constituent diffusion from/to the runoff.

Referring back to the governing equation for the surface soil column layer, the first two-component equations remain the same, while the third is revised to

$$\frac{\partial C_{2,i}}{\partial t} = - k' C_{2,i} + ld_{i-1} \tag{5-17}$$

With this equation, the constituent mass balance is maintained in the GSCM with the inclusion of erosion and runoff.

5.2.4 Particulate Emissions Model

The Land-based Source Modules have been designed to provide estimates of the annual average, area-normalized emission rate of constituent mass adsorbed to particulate matter less than 30 µm in diameter (PM_{30}), as well as annual average particle size distribution information in the form of the mass fractions of the total particulate emissions in four aerodynamic particle size categories—30 to 15 µm, 15 to 10 µm, 10 to 2.5 µm, and <2.5 µm.

A variety of release mechanisms are considered. The release mechanisms considered differ for each WMU and may include wind erosion, vehicular activity, unloading operations, tilling, and spreading and compacting operations. Table 5-1 summarizes the mechanisms considered for each WMU.

Table 5-1. Summary of Mechanisms of Release of Particulate Emissions for Each WMU

	WMU Type[a]						
	LAU		LF		WP		
Mechanism of Release	**Active**	**Inact.**	**Active**	**Inact.**[b]	**Active**	**Inact.**	**Algorithm Reference**
Wind erosion from open area	✓	✓	✓	✓			Cowherd et al. (1985)
Wind erosion from waste pile					✓	✓	U.S. EPA (1985a)
Vehicular activity	✓		✓		✓		U.S. EPA (1995a)
Unloading			✓		✓		U.S. EPA (1995a)
Spreading and compacting or tilling	✓		✓		✓		U.S. EPA (1985a)

[a] Active = Operating WMU. Inact. = Inactive WMU where no additional constituent mass is being added.
[b] Inactive (full) and uncovered landfill cell. Assume no emissions from a covered landfill cell.

This section describes the algorithms and assumptions used to estimate the following for each mechanism of release:

- Annual average PM_{30} emission rate due to each mechanism considered for each WMU, and
- Annual particle size range mass fractions, i.e., the mass fractions of PM_{30} in the aerodynamic particle size categories identified above.

For each WMU, the following are estimated:

- The total annual average PM_{30} emission rate due to all release mechanisms,
- The annual average emission rate of constituent sorbed to PM_{30},
- Annual average particle size range mass fractions of the total annual average PM_{30} emission rate, and
- Annual average first-order loss rate from the soil surface due to constituent mass losses caused by particulate emissions.

Wind Erosion from Open Fields. The algorithm for the estimation of PM_{30} emissions due to wind erosion from an open field is based on the procedure developed by Cowherd et al. (1985). The annual average emission rate of PM_{30} per unit area of the contaminated surface is estimated in the LAU and Landfill Modules.

To account for differing degrees of vegetation, surface roughness height, and frequency of disturbances per month in active versus inactive WMUs, different values are assigned to these parameters according to whether the WMU is active or inactive.

The methodology of Cowherd et al. (1985) was originally developed to estimate the emission rate of PM_{10}. Emission rates for PM_{30} can be approximated from those for PM_{10} with knowledge of the ratio between PM_{30} and PM_{10} emissions for wind erosion. According to Cowherd (1998), a good first approximation of this ratio is provided by the particle size multiplier information presented in U.S. EPA (1995) for wind erosion from open fields, where PM_{30}/PM_{10} is equal to 2. Therefore, a factor of 2 has been incorporated into Cowherd et al.'s (1985) equations for PM_{10} emission to allow estimation of PM_{30} emissions.

Wind Erosion from Waste Piles. The equation used in the Waste Pile Module to estimate the annual average PM_{30} emission rate per unit area of contaminated surface due to wind erosion is an adaptation of the empirical equation developed for total suspended particulate matter (TSP) from active sand and gravel waste piles presented in U.S. EPA (1985a, referred to here as AP-42; see Equation 3, p. 11.2.3-5). TSP is defined as those particulates measured by a high-volume sampler, and the effective cutoff commonly assigned to standard high-volume samplers is 30 μm (U.S. EPA, 1985a). A dust-control efficiency factor is added.

A more recent version of AP-42 (U.S. EPA, 1995) recommends the use of an event-based algorithm for estimating wind emissions from a waste pile. The updated algorithm was evaluated for use in 3MRA, but it was determined that it requires detailed site-specific information, typically not available for site-based analyses. The algorithm used in the Land-based Source Modules will tend to overestimate emissions relative to the event-based algorithm (Meyers, 1998).

Vehicular Activity. To estimate the PM_{30} emissions from vehicular travel on the surface of the WMU, an algorithm was derived from an empirical equation presented in U.S. EPA (1995; Equation 1, p. 13.2.2-1) for the kilograms of size-specific particulate emissions emitted per vehicle per kilometer traveled on unpaved roads. (In the Land-based Source Modules, the EPA parameter “fraction of waste on unpaved roads” is equal to 1 because travel is on the surface of the WMU.) EPA's equation has been adapted for the Land-based Source Modules to provide emissions normalized to the contaminated surface area and to account for the control of emissions with a dust-control efficiency factor.

Unloading Operations. The equation for estimating the PM_{30} emission rate due to unloading operations at waste piles and landfills was adapted from U.S. EPA (1995, Equation 1, p. 13.2.4-3). The EPA equation was adapted by multiplying it by the average annual loading rate, normalizing the emissions for the contaminated surface area, and applying the particle size multiplier for <30 μm.

Spreading and Compacting or Tilling Operations. The equation for estimating the rate of PM_{30} emissions due to spreading and compacting or tilling operations was adapted from an equation in U.S. EPA (1985a, Equation 1, p. 11.2.2-1) that was developed for estimating emissions due to agricultural tilling in units of kilograms of particulate emissions per hectare per tilling (or spreading and compacting) event. The particle size multiplier for <30 μm is applied in the Land-based Source Modules.

5.3 Module Discussion

5.3.1 Strengths and Advantages

Relative to other models that might have been candidates for the functionality provided by the Land-based Source Modules, the strengths and advantages of the modules include the following:

- **The GSCM extends the functionality of the Jury model.** The GSCM was developed to make modest improvements to known limitations of the Jury-type approach for the 3MRA modeling system. Specifically, the GSCM simulates changes over time in vertical soil constituent concentration profiles given (1) periodic waste applications that result in non-uniform vertical concentration profiles (this addresses the Jury limitation of uniform initial condition); and (2) a finite vertical boundary is formed by an underlying vadose zone so that the lower boundary depth is not infinite (this addresses the Jury limitation of a zero concentration boundary condition at the lower soil column boundary. (A zero concentration *gradient* boundary condition is assumed instead.)

- **The "local watershed" algorithm gives the LAU and Waste Pile Modules spatial resolution appropriate for the 3MRA modeling system exposure scenarios.** The conceptual site models for both the LAU and waste pile WMUs assume that both WMUs are adjoined by a buffer area (non-WMU land area). (The landfill is not assumed to have off-site runoff or erosion, so it is not considered part of a "local watershed".) The buffer area is situated between the WMU and the nearest surface waterbody. Overland flow runoff is assumed to occur from the WMU, across the buffer area, and into the nearest surface waterbody. This set of land areas (WMU, buffer, and a possible third land area upslope of the WMU, which contributes stormwater runoff and eroded soil, but not constituent loads, to the WMU area) located on the "hillside" adjacent to a surface waterbody constitutes the "local" watershed. Because constituent concentrations over time will be different in the buffer area than in the WMU area, and receptors can be located in the buffer area, it was important for the purposes of the 3MRA modeling system to have this spatial resolution. Prior to developing the Land-based Source Modules, which contain this "local watershed" functionality, a review of candidate watershed/vadose zone models was performed, and none were found to have this spatial resolution. For example, EPA's PRZM model, widely used for pesticide contamination studies in agricultural fields, is "field-scale" only (i.e. it considers the agricultural field only and does not consider a downslope, adjacent buffer area). On the other end of the spectrum, the HSPF watershed model is "watershed-scale"(i.e., it considers entire drainage basins), so that what might be happening on a two-land-area hillside in that larger drainage area is considered by its resolution. Thus, the "hillside-scale" of the local watershed algorithm is intermediate between these two extremes and needed to be custom-developed.

- **A carefully selected hybrid of empirical and mechanistic algorithms and approaches are used to maximize overall suitability for the purposes of the 3MRA modeling system.** From a model development perspective, the 3MRA framework presented an interesting challenge with regard to selection of appropriate levels of mechanistic detail. As a general rule, the more scientifically based (mechanistic) the model, the greater potential that model has to provide realistic simulations under a variety of environmental conditions. However, that potential carries with it the price of including many parameters that must be estimated from site-specific data before these highly parameterized models reach their potential. This essentially requires site-specific model calibration. In contrast, the 3MRA modeling system is applied to many sites with limited site-specific data. Thus, site-specific model calibration was not an option and the modeling approach needed to be based on a judicious balance of theoretical and empirical methods. For example, the GSCM is based on theoretical, mass balance principles, but involves parameters that are readily available (e.g., soil properties, constituent properties), or can be assigned from national distributions. Other algorithms (e.g., the Universal Soil Loss Equation for eroded soil losses, and the "curve number" method for estimating stormwater runoff) are fundamentally empirical, yet are widely used and acknowledged to provide reasonable predictions for planning-level applications.

5.3.2 Uncertainty and Limitations

The major limitations of the Land-based Source Modules and their implications are summarized below, by model.

Limitations of the GSCM. The GSCM was originally designed for the 3MRA modeling system to make some modest improvements to the well-known Jury model, which had been used historically to simulate constituent fate and transport in land-based sources. The GSCM has the following limitations:

- **Series solution of diffusion/decay/advection processes is problematic for some constituents.** The GSCM solves the mass balance differential equation sequentially in time in the following order: diffusive transport, internal decay, advective transport. This approach is justified conceptually on the basis of linearity and the principle of superposition; however, the approach can lead to simulation errors when relatively large time steps are taken. The magnitude of the errors would depend on the relative loss rates associated with the three processes. For example, if the first-order loss rate due to degradation were high and the calculation time step were sufficiently large, then leachate flux and the mass of constituent remaining for the advection phase would be underestimated.

- **Leachate flux postprocessing algorithm is artificial.** The leachate flux postprocessing algorithm postprocesses the constituent mass in the leachate over the total period of the simulation and attempts to allocate it over the individual years in the simulation in a reasonable manner. This postprocessing is needed for persistent constituents (highly sorbed, low decay) because of the combination of choice of the GSCM state variable (total concentration) and the diffusion/decay/advection series solution methodology. Patterned after the Jury model, the GSCM's state variable is total (sorbed + dissolved + gaseous) constituent concentration. The "effective solute convection velocity," V_E, is not the actual leachate infiltration velocity, but rather (because total constituent is the state variable, not dissolved) is corrected to account for sorption. When that sorption is high (high K_d), the V_E is small and can be much smaller (slower velocity) than the true infiltration rate. "Convective transport," i.e., advection, of the constituent from one computational layer in the GSCM down to the next (or from the bottom layer out as leachate) is performed only when the constituent has traversed the depth of a computational layer (typically 1 cm) as measured by V_E, not by the actual infiltration velocity. For very small values of V_E, this travel time can become excessive, and vertical movement due to advective transport may not occur within a year; indeed, it may not occur for the duration of the simulation. In reality, dissolved constituent would be more or less continuously moving downward due to advective transport in the infiltration. The postprocessing algorithm attempts to mitigate this limitation.

- **Treatment of different zones with appropriate boundary conditions is not possible.** The GSCM solution methodology does not allow a rigorous treatment of scenarios where different zones with different sorption characteristics exist in the soil column. For example, if the landfill soil column had a liner with less sorptive capacity than the waste zone, then the liner's sorptive capacity would be exhausted relatively quickly relative to the waste zone's. At this point, since dissolved constituent drains from the waste zone into the liner, there would be no further reduction by sorption, and a boundary condition should be applied at this waste/liner interface that requires a zero concentration gradient for dissolved constituent. The GSCM cannot accommodate such a boundary condition (or an analogous waste/cover boundary condition), superficially because the GSCM is simulating total (not dissolved) constituent, but more fundamentally because the underlying solution methodology is simply not designed to deal with internal boundary conditions. This limitation has been addressed by imposing the relevant boundary condition, but it is not a mathematically rigorous treatment of the physical scenario. A more elegant solution would be to include nonlinear sorption kinetics in the algorithm, so that the relative differences in sorption among different zones are intrinsically accounted for, rather than imposed by boundary conditions.

- **Simulation of volatilization out of the surface of soil column is artificial.** The GSCM's simulation of diffusive transport over the depth of the soil column transports total constituent in accordance with the constituent's water diffusivity (corrected for sorption). This diffusion is treated as diffusion in an infinitely deep column, so that *total* constituent flux is effectively diffused across the surface of the soil column. The amount of constituent mass that is not in the gaseous phase, but has been (erroneously) diffused across the surface, is then calculated and "replaced" back into the top computational layer to better reflect reality. It is unclear to what extent this overall treatment biases the volatile emissions themselves, but the re-introduction of the escaping, nongaseous constituent into the surficial soil layer is artificial and could lead to an erroneous estimate of the vertical concentration gradient.

Limitations of the Local Watershed Model. The Local Watershed Model has the following limitations:

- **Burial/erosion introduces minor mass balance error.** The burial/erosion mechanism introduces a minor mass balance error into the model. The model for surface soil/runoff fate and transport is based on a conceptual model originally developed for use in a stream/sediment application where the sediment compartment location relative to a reference point below the surface can move vertically ("float") as burial and erosion occur. In that moving frame of reference, burial/erosion of a constituent does not introduce a mass balance error. However, in this application, the frame of reference is not allowed to float, but is fixed by the elevation of the lower boundary (e.g., top of the vadose zone). Thus, if sorbed constituent is eroded from the surface computational cell, that surface cell, which is vertically fixed, must have a "source" that is internal to the modeled soil column to compensate for this sink, or its internal mass balance is not maintained. The magnitude of this mass balance error is equal to the mass of eroded soil from the surface over the duration of the simulation multiplied by its average sorbed constituent concentration. In most cases, this error as a percentage of the total constituent mass in the modeled WMU will be quite small, and that has been confirmed in multiple executions of the model.

- **Potential hot spots in the local watershed buffer subarea are diluted.** The local watershed buffer is considered as a single, homogeneous subarea by the model to facilitate watershed delineation and decrease run time. This will result in spatial averaging of concentrations in the buffer.

- **Sheet flow is assumed across buffer subarea.** The conceptual Local Watershed Model construct assumes that runoff and erosion occur as sheet flow from the WMU across the buffer subarea to the waterbody. This implicitly assumes no short-circuiting of constituent loads directly from the WMU to the downslope waterbody, such as might actually occur in runoff/erosion-created ditches or swales. To the extent that such short-circuiting might occur, the model will underestimate waterbody constituent loadings. Mass balance is maintained; consequently, any such underestimation would come at the expense of overestimating soil concentrations in the (bypassed) buffer zone.

- **Hydrological responses are limited by the available record.** The hydrology model uses available historical records of meteorological data. When the number of years simulated exceeds the number of years in the record, the record is repeated. Thus, unusual hydrological events (e.g., major storms) are limited to those actually observed. Events not observed, but possible nonetheless, that could result in increased source releases or media transport will not be included in the simulation.

Limitations of the Particulate Emissions Model. The particulate emissions models were incorporated without changes from the sources discussed (mostly AP-42) and have the same limitations as those models, as described in the source documentation.

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6.0 Air Module

6.1 Purpose and Scope

The Air Module estimates the annual average air concentration of dispersed constituents and the annual deposition rates of vapors and particles at various receptor points in the area of interest (AOI). This module simulates the transport and diffusion of constituents in the form of volatilized gases or fugitive dust emitted from area sources into the air. The predicted air concentrations are used to estimate biouptake into plants, and human exposures due to direct inhalation. The predicted deposition rates are used to determine constituent loadings to farm crops and soils, watershed soils, and surface waterbodies. Figure 6-1 shows the relationship and information flow between the Air Module and the 3MRA modeling system.

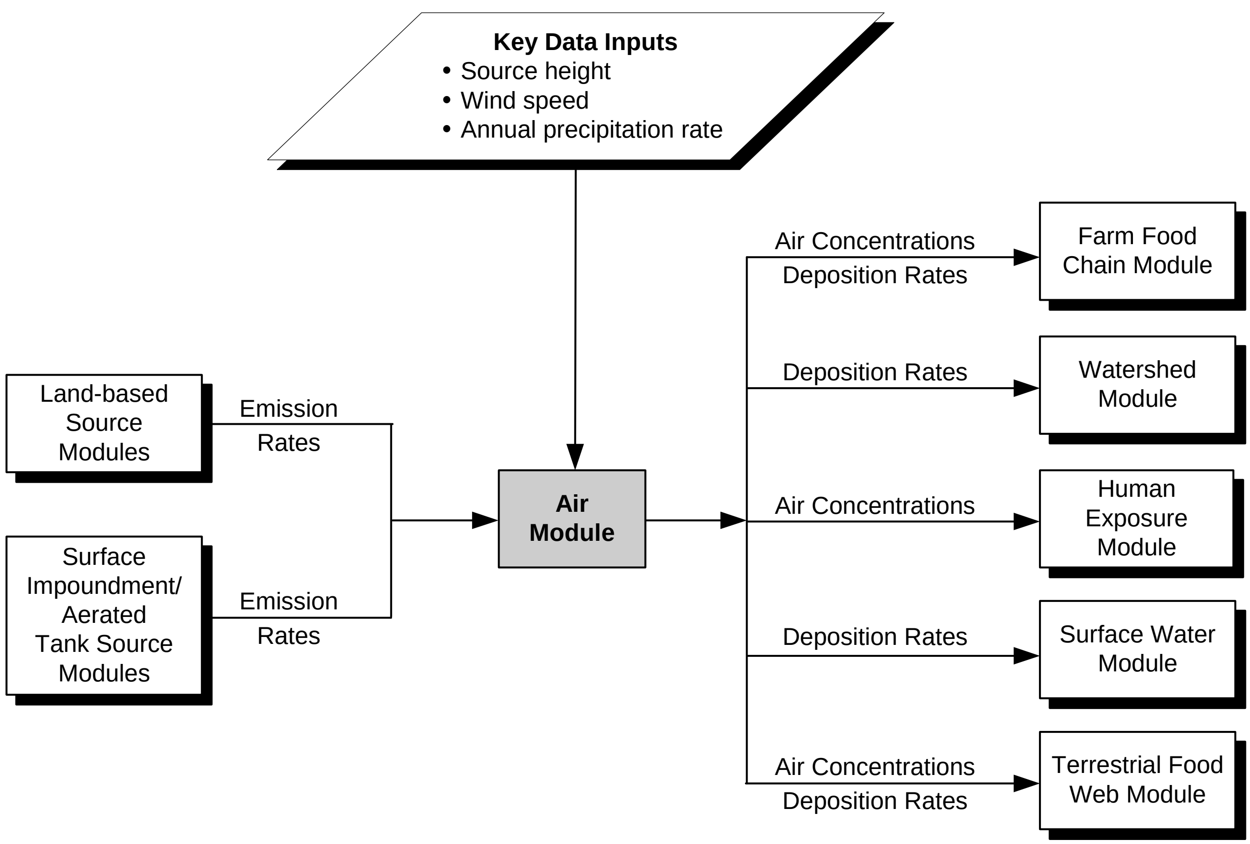

Figure 6-1. Information flow for the Air Module in the 3MRA modeling system.

The Air Module performs four major functions, as follows:

1. **Characterize source-specific parameters.** For each AOI, the Air Module characterizes emission sources in terms of waste management unit (WMU) dimensions, emission rate, and the site-layout. As an option, the module can calculate the source-specific, long-term average particulate mass fraction distribution from the outputs of the Land-based Source Modules (Landfill, Waste Pile, or Land Application Unit [LAU] Module).

2. **Calculate receptor locations (polar grid or site specific).** The Air Module provides the option to model directly to all site-specific output coordinates needed by the 3MRA modeling system or to model to a set of polar coordinates and then use a two-dimensional (2-D) cubic spline method to interpolate from the polar set to the larger set of interest. The spline interpolation can be used to reduce the ISCST3 run time.

3. **Calculate receptor-specific contaminant concentration and deposition rates.** The Air Module calculates annual average air concentration and deposition rates for each receptor location specified. Concentrations and deposition rates calculated include

   – Air concentration of vapors,
   – Air concentration of particles,
   – Wet deposition rate for vapors and particles, and
   – Dry deposition rate for particles.

4. **Calculate constituent-specific annual average concentrations and deposition rates.** The Air Module converts the receptor-specific concentrations and deposition rates based on unit emission rates (e.g., 1 $g/m^2{-}s$) to constituent-specific estimates by multiplying the values by the constituent-specific emission rate for each year.

ISCST3 may be implemented by the 3MRA modeling system during a run, or it may be run outside of the 3MRA modeling system and the results called by the Air Module.

6.2 Conceptual Approach

Figure 6-2 illustrates the dispersion and subsequent deposition of vapors and particles from a WMU into the environment. Constituents are released to the air as vapors (by volatilization) or sorbed to particulates (by wind erosion and mechanical disturbances) and move through the air to locations around the AOI. The constituents can then deposit on soil and plant surfaces. The Air Module is used to estimate air concentrations and deposition rates for each contaminant/site/WMU combination.

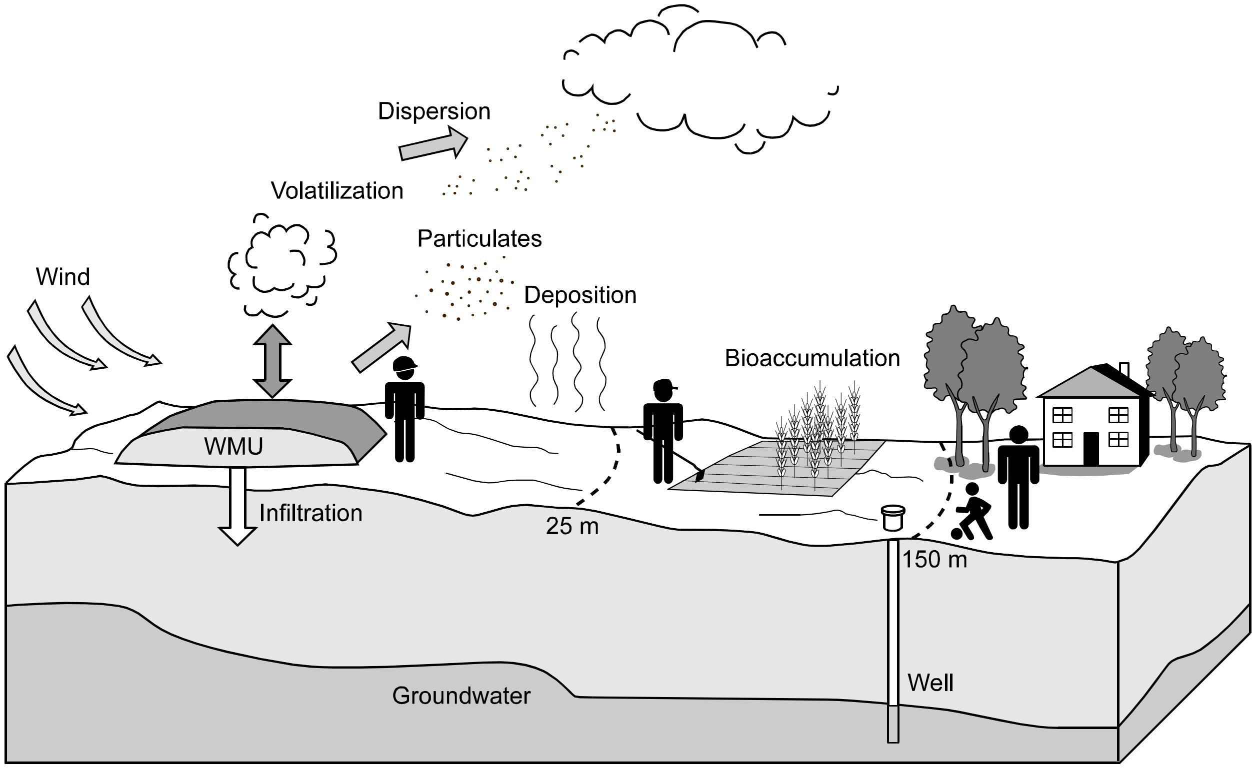

Figure 6-2. Conceptual diagram of dispersion and deposition.

The Air Module is based on EPA's legacy model Industrial Source Complex Short-Term Model, Version 3 (ISCST3), which has been used extensively by EPA in regulatory applications. ISCST3 was extended for use in the Air Module through the development of pre- and postprocessors that serve as an interface between the ISCST3 model and the rest of the 3MRA modeling system. The preprocessor reads initial settings from files generated by the 3MRA modeling system. Next, the preprocessor determines if the core model needs to be run or if results from previous runs can be reused. If the ISCST3 model needs to be run, the preprocessor builds an input file that controls the ISCST3 run. The ISCST3 model is run using unit constituent emissions to produce receptor-specific annual average air concentrations and deposition rates normalized to emission rate. The postprocessor converts these normalized concentration and deposition estimates to constituent-specific annual averages by multiplying the values by the constituent-specific emission rates for the WMU for each year. The resulting annual averages are written to files that are read by other modules. A summary of the ISCST3 model—the science and the computational core of the Air Module—is presented in the accompanying text box. The full capabilities of the ISCST3 model are explained in U.S. EPA (1995).

Summary of ISCST3

ISCST3 (U.S. EPA, 1999a,b), a recommended dispersion model in EPA's *Guideline on Air Quality Models* (U.S. EPA, 1999c), is a steady-state, Gaussian plume dispersion model. A steady-state model is one in which the model inputs and outputs are constant with respect to time. The term "Gaussian plume" refers to the kind of mathematical solution used to solve the air dispersion equations. It essentially means that the constituent concentration is dispersed within the plume laterally and vertically according to a Gaussian distribution, which is similar to a normal distribution. These assumptions and solutions hold for each hour modeled. The results for each hour are then processed to provide values for different averaging times depending on the user's needs (e.g., annual average).

ISCST3 is capable of simulating dispersion of pollutants from a variety of sources, including point, area, volume, and line sources. ISCST3 can account for both long- and short-term air concentration of particles and vapor, and wet and dry deposition of particles and vapor. In addition to deposition, wet and dry plume depletion can be selected to account for removal of matter by deposition processes and to maintain mass balance. Receptor locations can be specified in polar or cartesian arrays or can be set to discrete points as needed. Flat or rolling terrain can be modeled, but only flat terrain can be used for area sources. ISCST3 considers effects of the environmental setting on dispersion by allowing the user to set urban or rural dispersion parameters. Dispersion around the centerline of the constituent plume is estimated by empirically derived dispersion coefficients that account for horizontal and vertical movement of constituents. In addition, constituent movement from the atmosphere to the ground is also modeled to account for deposition processes driven by gravitational settling and removal by precipitation.

For area sources, the general equation for ground-level concentration at a receptor is given by a double integral in the upwind (x) and crosswind (y) directions:

$$\chi = \frac{Q_A K}{2\pi u_s} \int_x \frac{VD}{\sigma_y \sigma_z} \left(\int_y \exp\left[-0.5 \left(\frac{y}{\sigma_y} \right)^2 \right] dy \right) dx \tag{6-1}$$

where

χ	=	concentration at x, y (mass per volume)
Q_A	=	area source emission rate (mass per unit area per unit time)
K	=	scaling coefficient to convert calculated concentrations to desired units (default value of 1×10^6 for Q in g/s and concentration in μg/m^3)
x	=	distance from source in wind direction (positive is downwind) (m)
y	=	distance from centerline of plume across the wind direction (m)
z	=	vertical distance from ground (m)
V	=	vertical term accounting for vertical distribution of the plume (includes effects of source elevation, receptor elevation, plume rise, limited mixing in the vertical direction, and gravitational settling and dry deposition of particulates)
D	=	decay term accounting for pollutant removal by physical or chemical processes
σ_y,σ_z	=	standard deviation of lateral and vertical concentration distribution (m)
u_s	=	mean wind speed at release height (m/s).

Key meteorological data required for the ISCST3 model include

- **Wind Direction:** Determines the direction of the greatest impacts.
- **Windspeed:** Affects ground-level air concentration. The lower the windspeed, the higher the concentration.
- **Stability Class:** Affects rate of lateral and vertical diffusion. The more unstable the air, the greater the diffusion.
- **Mixing Height:** Determines the height to which constituents can be diffused vertically.
- **Deposition Parameters:** Additional site-specific meteorological parameters are required to make estimates of dry and/or wet deposition. These parameters vary based on land-use types and seasons.

6.2.1 Characterize Source-specific Parameters

For each site, ISCST3 input files can be created that contain source-specific information such as WMU type, dimensions, location, and source variables for settling, removal, and deposition. Values for most of these data are obtained from other modules or the 3MRA modeling system databases.

The WMU types modeled by the 3MRA modeling system include surface impoundments, aerated tanks, landfills, waste piles, and LAUs. All of these unit types except for aerated tanks and waste piles are modeled as ground-level area sources. Aerated tanks and waste piles are modeled as elevated area sources.

For the 201 sites in the representative national data set, the location and the size of the sources is determined from the 3MRA modeling system site-based database. The sources for these data are the 1985 Screening Survey for Industrial Subtitle D Establishments (Westat, 1987) and the 1986 Survey of Hazardous Waste Treatment, Storage, and Disposal Facilities (U.S. EPA, 1987). Neither of these data sources includes data on height; therefore, the waste pile heights were calculated based on volume of waste and area of the pile, and tank height was randomly selected. These heights are stored in the representative national data set.

When modeling particulates, ISCST3 requires a long-term average particle size distribution. As an option, the Air Module can calculate the source-specific mass fraction distribution. The Land-based Source Modules (i.e., the Landfill, Waste Pile, and LAU Modules) generate a time series of particle size distributions (i.e., one distribution per year), as well as a time series of particulate mass fluxes for particles less than or equal to 30 µm in diameter. From these inputs, the Air Module estimates the long-term average particle size distribution.

Other source-specific information included in the ISCST3 input file include specification of rural versus urban setting and meteorological dataset information.

Rural vs. Urban. The rural vs. urban setting in ISCST3 allows the user to account for differences between rural and urban environments. In urban environments, the built environment (e.g., buildings, roads, and parking lots) alters the dispersion character of the atmosphere, particularly at night as a result of building-induced turbulence and reduced nighttime cooling. Thus, there is greater nighttime mixing of constituents in urban areas compared with rural areas. For the purposes of ISCST3 modeling, the urban classification applies mainly to large cities; even small cities and suburban areas are more appropriately classified as rural.

Meteorological data. The 3MRA modeling system database contains meteorological data collected at regional meteorological stations. Each of the 201 sites contained in the 3MRA modeling system representative national data set was assigned to the nearest station with similar weather conditions and adequate data. In making these assignments, EPA considered all available data from 218 meteorological stations across the United States to find the best data for each site. Each of the meteorological data sets included in the 3MRA modeling system database contains a minimum of 10 years of data.

6.2.2 Calculate Receptor Locations (polar-grid or site-specific)

The user has the option of specifying receptor locations for all cartesian coordinates as requested by the 3MRA modeling system, or for a set of polar grid coordinates. If the second option is chosen, a two-dimensional cubic spline method (discussed below) is applied to interpolate from the polar grid coordinates to the larger receptor location set requested by the 3MRA modeling system.

The run time for ISCST3 can be long, particularly for large area sources with many receptor locations. As an option to minimize run time of ISCST3 in the Air Module, a 2-D cubic spline interpolation algorithm was developed. Under this option, run-time savings can be achieved by using ISCST3 to model directly to a smaller number of points than the total number of receptor locations for the site. In general, the points making up the spline data set are not a subset of the points representing the receptor locations; rather, they are uniformly distributed across the AOI. A requirement of the spline algorithm is that the data provided to the spline must be "complete" in the sense that, no matter how the underlying 2-D grid is configured (e.g., cartesian, polar), data must be provided at each intersection of the two dimensions. A polar grid is used as the basis for the spline because, under the "complete" data set requirement of the spline algorithm, a polar grid can achieve a higher resolution in the region of steepest output gradients (near the WMU) with fewer total points than can a cartesian grid.

The spline method can only interpolate values. Consequently, value estimates cannot be made for any points located outside of the outermost grid circle.

6.2.3 Calculate Receptor-specific Concentration and Deposition Rates

Both vapors and particles are modeled for landfills, waste piles, and LAUs; only vapors are modeled for surface impoundments and aerated tanks. The type of model output selected depends on the WMU being modeled. Concentration, dry deposition rate, and wet deposition rate are calculated for landfills, waste piles, and LAUs. Only concentration and wet deposition rate are calculated for surface impoundments and aerated tanks because the pollutants are only emitted in vapor phase. Dry deposition of vapors is not modeled in the Air Module. Consequently, dry deposition of vapors is calculated as part of the Farm Food Chain Module (see Section 10.2.1) and Terrestrial Food Web Module (see Section 11.2.1). These modules estimate the dry deposition rate of vapors based on the vapor-phase concentration in air and a vapor-phase dry deposition velocity (default value of 1 cm/s).

The Air Module calculates the following concentrations and deposition rates:

- *Air concentration of particles and vapors.* ISCST3 estimates air concentrations of particles and vapors, accounting for downwind movement of the plume. It also accounts for dispersion of vapors and particles around the centerline of the plume as the plume travels in a downwind direction. Removal of constituent mass from the plume occurs as a result of wet and dry deposition.

- *Wet deposition of particles and vapors.* Wet deposition is the loss of material from a plume onto a surface as a result of precipitation. A scavenging ratio approach is used to model the deposition of gases and particles through wet removal. The amount of material removed from the plume by wet deposition is a function of the scavenging rate coefficient, which is based on particle size (U.S. EPA, 1995) for particles. For vapors, ISCTS3 is typically run with constituent-specific scavenging coefficients. These scavenging coefficients are read in by the Air Module. When multiple constituents are run, this method proportionately increases the number of runs that need to be made. As an alternative to applying constituent-specific coefficients, a single set of gas scavenging coefficients stored in the representative national data set can be applied. These coefficients are based on approximating the gases as very small particles.

- *Dry deposition of particles*. Dry deposition refers to the loss of material from a plume that has been deposited onto a surface (e.g., ground, vegetation) as a result of processes such as gravitational settling, turbulent diffusion, and molecular diffusion. Dry deposition of particles is calculated as the product of air concentration and dry deposition velocity.

To reduce the computational burden on the 3MRA modeling system, several new features were added to ISCST3 to create the Air Module. A complete description of the technical algorithms for these features is provided in U.S. EPA (1999b). Operational instructions are provided in U.S. EPA (1999a).

- **Revised Plume Depletion Scheme.** The version of ISCST3 distributed by EPA's Office of Air Quality Planning and Standards contains the Horst (1983) plume depletion algorithm. This algorithm was found to be computationally intensive. A new plume depletion and settling algorithm developed by Venkatram (1998) was implemented for the Air Module, resulting in a faster, more robust approach. This approach is based on depleting material in a surface-based internal boundary layer that grows with distance from the source. In conjunction with this change, the deposition velocity algorithm was also modified by removing the inertial impaction term. The inclusion of this term appears to provide deposition velocity estimates that are too high for some particle sizes.

- **Sampled Chronological Input Model (SCIM).** To reduce model run time, an option was added to the Air Module to sample the long-term meteorological record at regular, user-specified intervals and scale the model results at the end of the run to produce annual average estimates. This method is called the Sampled Chronological Input Model (SCIM). An advantage of this method is that it uses hourly meteorological data that maintain the serial correlation between wet deposition rate and concentration. The user specifies two sampling intervals. Using the first interval, the meteorological data are sampled, ignoring any recorded precipitation, and the air concentration and dry deposition rate are calculated for each receptor location of interest. The second interval specifies the sampling rate for the hours of meteorological data during which precipitation was recorded. This latter sampling rate is used to determine the air concentration, dry deposition rate, and wet deposition rate. The estimates from these separate schemes are combined at the end of the model run using a weighted average based on the number of hours sampled in each interval.

- **Output by Particle Size.** The Human Risk Module requires air concentrations for pollutants with particle sizes $\leq 10\ \mu m$ to calculate inhalation risks. ISCST3 does not output concentrations by particle size. Therefore, an option was added to output concentration and deposition rate by particle size.

6.2.4 Calculate Constituent-specific Annual Average Concentrations and Deposition Rates

To reduce the number of required runs, the ISCST3 model is executed using unit emissions. The output from the model is normalized receptor-specific annual average concentrations and deposition rates. These estimates are converted to constituent-specific annual average concentrations and deposition rates by multiplying the values by the yearly emissions provided as output from the source modules.

6.3 Module Discussion

6.3.1 Strengths and Advantages

The Air Module has the following strengths and advantages:

- **ISCST concentration algorithms.** The concentration algorithms have been extensively reviewed and evaluated. The model has a long history of use by the EPA for fine-scale modeling and has been promulgated in the Guideline on Air Quality Modeling (U.S. EPA, 1999c).

- **ISCST particle deposition algorithms.** The particle deposition algorithms were selected for inclusion based on an extensive comparison against other algorithms and field data.

- **10-year period of record for the meteorological data.** The 10-year period of record ensures that all conditions typical of a meteorological station are modeled. EPA guidance for air quality applications requires only 5 years of off-site meteorological data to establish representativeness.

- **Urban or rural conditions.** The Air Module is appropriate for urban or rural dispersion conditions, depending on the location of the source.

- **Flexible receptor locations.** Receptors may be placed anywhere in the area of interest.

- **Options for use of meteorological data.** The Air Module provides the flexibility to either model all hours of meteorological data or use SCIM to model only selected hours if runtime is a constraint, which decreases runtime considerably.

6.3.2 Uncertainty and Limitations

The Air Module includes the following limitations or uncertainties:

- **Meteorological data gaps.** A data quality review uncovered missing data within the various meteorological data time series required for the 3MRA modeling system. Programs were written to automatically find and correct these data gaps, within the technical constraints established by Atkinson and Lee (1992). Missing data were typically less than 2 percent of all hours, so this correction should have little effect on the results.

- **Lack of data on shapes and heights of WMU.** Currently, none of the national-level data sets contain data on the shape, height, or orientation of the sources. In the absence of these data, units can be characterized as a square, rectangle, or 20-sided polygon shaped to approximate a circle.

- **Wet deposition of vapors.** Modeling of wet deposition of vapors requires constituent-specific scavenging coefficients and separate runs for each constituent. When modeling multiple constituents, the number of runs, as well as, the availability of constituent-specific data may be an issue. As an alternative, a set of gas scavenging coefficients that can be used for all constituents is stored in the representative national data set. These coefficients are based on approximating the gases as very small particles. This approach may lead to underprediction of wet deposition for some gases and overprediction of others depending on the Henry's law coefficient for the gas.

- **Dry deposition of vapors.** The Air Module is not designed to allow modeling of dry deposition of vapors using ISCST3. Modeling of dry deposition of vapors requires input of several contaminant-specific variables, which would require separate runs for each chemical. Alternatively, dry deposition of vapors is estimated in the Farm Food Chain Module (see Section 10.2.1) based on vapor-phase concentration in air and a default vapor-phase dry deposition velocity. Because dry deposition is calculated external to ISCST3, the plume is not depleted within the model. However, the impact to the modeling results is not expected to be significant.

6.4 References

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7.0 Watershed Module

7.1 Purpose and Scope

The Watershed Module estimates contaminant concentrations in soil resulting from aerial deposition of contaminants throughout the area of interest (AOI) around each modeled site and the resulting contaminant loadings to surface waterbodies from runoff and erosion. It also estimates some hydrological inputs for the Surface Water Module (flows, eroded soil loads) and the Vadose Zone and Aquifer Modules (infiltration rates). Figure 7-1 shows the relationship and information flow between the Watershed Module and the 3MRA modeling system .

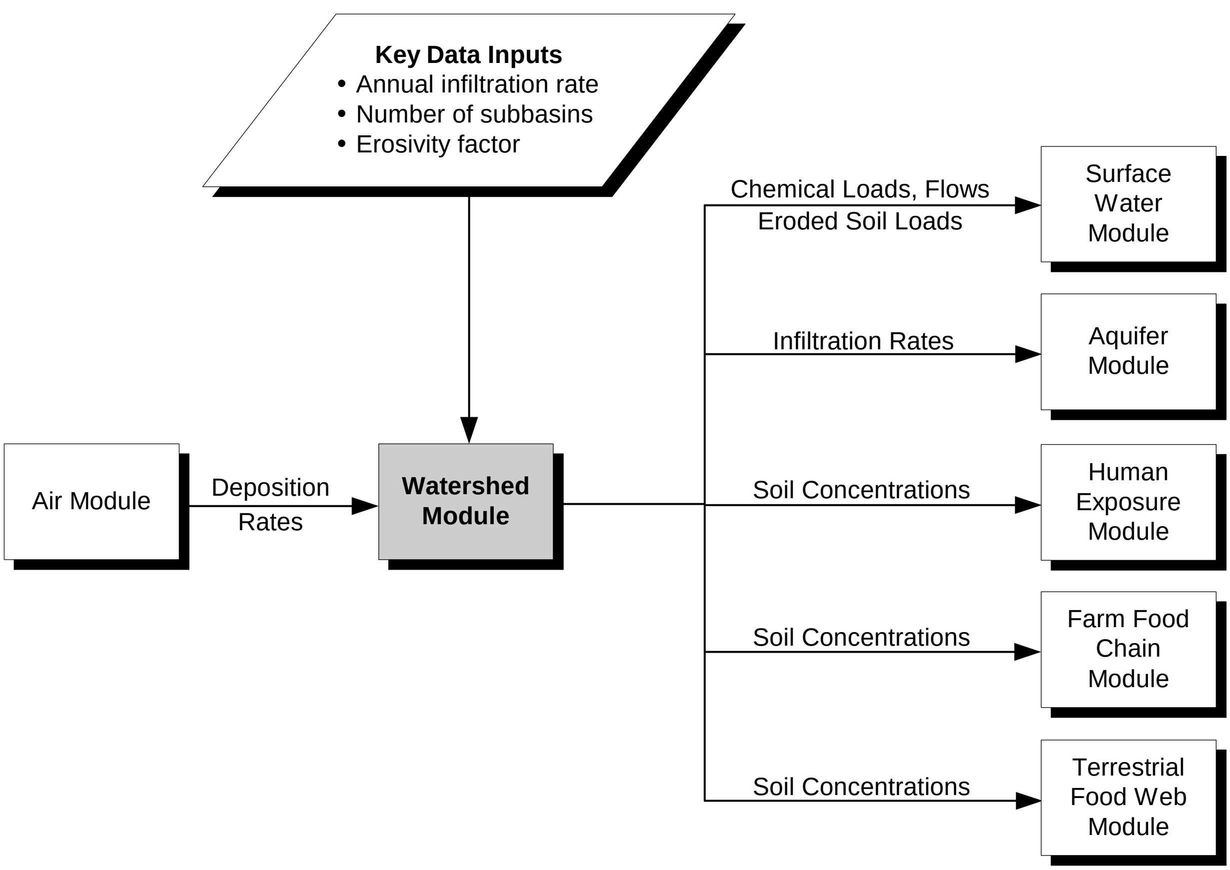

Figure 7-1. Information flow for the Watershed Module in the 3MRA modeling system.

The Watershed Module has two major functions:

1. **Calculates soil contaminant concentrations and surface water loadings.** The Watershed Module predicts the fate and transport of contaminants in the watershed subbasins to estimate soil concentration in the regional watershed and contaminant loadings to surface water from runoff and erosion.

2. **Calculates hydrological inputs.** The Watershed Module calculates several hydrological inputs for the Surface Water Module and the Aquifer Module.

7.2 Conceptual Approach

The AOI around a site consists of a set of contiguous watershed subbasins. Excluding surface waterbody areas, all land surfaces within the AOI fall within one of the watershed subbasins. Depending on the watershed delineation criteria, the number of watershed subbasins constituting an AOI varies; the typical range is 6 to 25 for the current site-based data set. Figure 7-2 shows a sample AOI that consists of all or part of 12 watersheds.

The watersheds are delineated so that surface runoff in one subbasin is modeled independently of surface runoff in other subbasins; therefore, the 3MRA modeling system assumes no contaminant transport occurs from one watershed subbasin to another via erosion or runoff, or via wind suspension or volatilization and subsequent deposition. Thus, the Watershed Module models each of the watershed subbasins at a site independently of the other subbasins at that site.

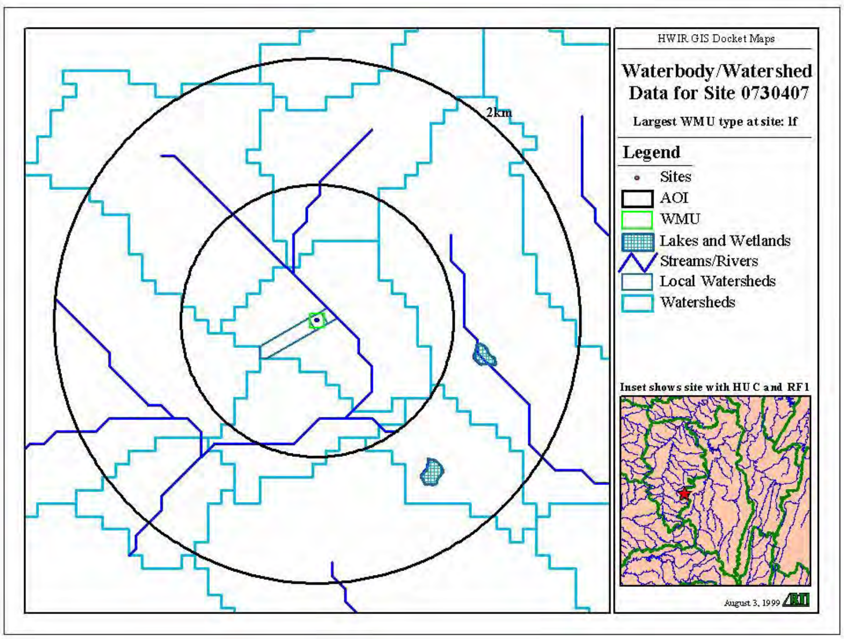

Figure 7-2. Illustration of watersheds within an AOI.

The Watershed Module simulates contaminant fate and transport related only to loads that result *indirectly* from the WMU (i.e., through the process of wind erosion or other particulate suspension or volatilization from the WMU into the atmosphere and through subsequent deposition from the atmosphere onto the surrounding regional watersheds). Contaminant loads to a waterbody resulting from *direct* runoff and erosion from a WMU are simulated by the LAU or Waste Pile Module. Watershed soil concentrations and waterbody concentrations are a function of both direct and indirect contaminant loads. Thus, if a receptor is located in a buffer area between a WMU and the downslope waterbody (i.e., in the WMU's local watershed), the total soil concentration to which the receptor is exposed is the aerial deposition-related concentration estimated by the Watershed Module plus the WMU runoff and erosion-related concentration estimated by the relevant source module.

There is significant overlap between the models used in the Watershed Module and those used in the Land-based Source Modules described in Section 5. This section discusses only aspects of the Watershed Module that are different from the Land-based Source Modules.

7.2.1 Calculate Soil Contaminant Concentrations and Surface Water Loadings

The Watershed Module is based on conceptual and mathematical models similar to those used in the LAU and Waste Pile Modules, such as the Generic Soil Column Model (GSCM) algorithm described in Section 5. The GSCM provides the solution to a one-dimensional (1-D), partial differential equation that describes the spatial and temporal distribution of contaminants in a porous medium subject to advective/dispersive transport and first-order losses. The governing equation is

$$\frac{\partial C_2}{\partial t} = D_E \frac{\partial^2 C_2}{\partial z^2} - V_E \frac{\partial C_2}{\partial z} - k' C_2 + ld_{dep} \qquad (7\text{-}1)$$

where

C_2 = total contaminant concentration in the soil layer (g/m^3)

D_E = vertical diffusion coefficient (m^2/d)

V_E = infiltration rate (m/d)

z = vertical distance (m)

ld_{dep} = annual average wet plus dry deposition contaminant mass loading rate (g/m^3−d)

k' = lumped first-order decay rate (d^{-1})—equal to the sum of the hydrolysis loss rate, the aerobic biodegradation loss rate, and two additional first-order rate constants that quantify the rainfall runoff and erosional loss processes.

As described in Section 5.0, Equation 7-1 is disaggregated into three component equations—diffusion, convection, and first-order losses, each solved individually on the soil column's numerical grid. For the Watershed Module, while the first two component equations remain the same as in the GSCM, the third is revised to Equation 5-17 (Section 5.2.3), with watershed parameter ld_{dep} replacing the local watershed subarea run-on load, ld_{i-1}. The solution to Equation 5-17 is the same as that described for the LAU or Waste Pile Module, with the same substitutions noted above.

After C_2 in the surface layer of the soil column is determined at the end of a given time step, $\bar{C}_1$, the contaminant concentration in the runoff water, averaged over the time step, is determined using Equation 5-15 (Section 5.2.3), where all the parameters are annual averages determined from annual average runoff flow and cumulative soil load (or mass of eroded soil). The time-step-averaged contaminant concentration in the soil compartment, $\bar{C}_2$ in Equation 5-15, is calculated using the following equation:

$$\bar{C}_2 = \frac{1}{T}\int_0^T C_2\,dt = \begin{cases} \dfrac{C_2^0}{k'T}\left(1 - \exp(-k'T)\right) + \dfrac{ld_{dep}}{k'^2T}\left(k'T - 1 + \exp(-k'T)\right) & k' > 0 \\ C_2^0 + \dfrac{ld_{dep}T}{2} & k' = 0 \end{cases} \tag{7-2}$$

where T is the averaging time period, which is the same as the computational time step here, and C_2^0 is the contaminant concentration in the soil compartment at the start of the averaging period (which is the same as at the end of the previous time step).

At the end of each year's simulation, annual average C_1 (g/m^3) is determined and multiplied by the annual average runoff rate (m^3/d) to determine the annual average contaminant mass load to the waterbody due to runoff and erosion (g/d).

In the Watershed Module, the depth of the soil column is a user-specified input, set at a default of 5 cm. Each soil column layer is 1 cm thick. The surficial soil column layer (top 1 cm) is linked to the runoff compartment during runoff events using the local watershed/soil column algorithm described in Section 5. Similar to the presentation in Section 5, the runoff water during a runoff event is considered as Compartment 1 and the surficial soil column layer as Compartment 2 of the two-compartment conceptual model for the watershed/soil column algorithm. The total (particulate-sorbed plus dissolved) contaminant concentration in the watershed runoff is coupled to the total concentration in the soil layer.

In the subsurface layers of the soil column, the contaminant mass fate and transport governing equation is also given by Equation 7-1; however, the total first-order loss rate (k′) is equal to the sum of the input first-order loss rates due to hydrolysis and anaerobic biodegradation only.

The solution technique for Equation 7-1 is identical to that used for the LAU, Waste Pile, and Landfill Modules and is described completely in Section 5. The concentration in the surface layer of the soil column is determined at the end of each time step by solving Equation 7-1; its average value over the time step is calculated by integrating over the time step. The contaminant concentration in the runoff water averaged over the time step is calculated as a function of the surface soil concentration averaged over the time step, as described in Section 5. At the end of each year's simulation, the annual average contaminant concentration in the runoff water is determined and multiplied by the annual average runoff rate to determine the annual average contaminant mass load to the waterbody due to runoff and erosion.

7.2.2 Calculate Hydrological and Soil Erosion Inputs

Streamflow. The Watershed Module uses the identical hydrology submodel described in detail in Section 5 to estimate stormwater runoff and ground water infiltration. The hydrology submodel is applied to entire, individual watersheds where no further spatial disaggregation occurs.

Streamflows are assumed to be made up of both stormwater runoff and baseflow. Baseflow is streamflow occurring during nonrunoff periods and is derived from ground water discharge to streams or interflow (shallow infiltration flowing parallel to the ground surface). For a given stream reach, baseflow can vary seasonally, or even near-continuously, as ground water levels and/or interflow varies and can be estimated for a given time period by analyzing runoff hydrographs that include runoff as well as pre- and post-runoff flows. For the purposes of the 3MRA modeling system, within-year variability in baseflows is not estimated. Rather, a single estimate is sought that reasonably characterizes annual average baseflow conditioned on stream reach order (or tributary drainage area), year, and hydrologic region.

The single flow statistic that best represents annual average baseflow for a given region, reach order, and year is an important issue. The widely available annual average streamflow would, in general, tend to overestimate baseflow. Conversely, the common-low flow statistic, 7Q10 (the minimum 7-day average flow expected to occur within a 10-year return period, i.e., at least once in 10 years), would tend to underestimate baseflow. Therefore, the 30Q2 low flow, i.e., the minimum 30-day average flow occurring, on average, at least once every other year, was selected as a reasonable estimate of annual average baseflow for any given year.

The Watershed Module estimates the 30Q2 flow based on the area of the regional watershed, using a regression equation developed for each of the 18 hydrologic unit codes (HUCs) in the United States. The general equation is

$$Q = a \times WSA^b \tag{7-3}$$

where

Q	=	30Q2 baseflow
a	=	HUC-specific regression parameter
WSA	=	watershed area
b	=	HUC-specific regression parameter.

Watershed Slope for Soil Erosion. The Watershed Module uses the modified Universal Soil Loss Equation (MUSLE), as described in Section 5, to predict soil erosion from entire watersheds. To do so, the slope parameter of the length-slope factor presented in Section 5 must reflect subbasin average conditions, rather than local watershed conditions as in the source modules application. Sheet-flow slope for a subbasin is not the slope of the stream network draining the watershed; rather, it is the average slope of the (essentially infinite) individual sheet-flow paths that form the land surfaces of that subbasin. As presented in Williams and Berndt

(1977), the watershed-subbasin-average slope is estimated from the following equation, which was first proposed by Horton (1914):

$$S = \frac{Z\ (LC_{25} + LC_{50} + LC_{75})}{4A} \tag{7-4}$$

where

S	=	watershed-subbasin-average slope (percent)
Z	=	difference in the subbasin's maximum and minimum elevations (m)
A	=	total surface area of the subbasin (m^2)
LC_{25}	=	total length of the contour line at the 25th percentile of Z (m)
LC_{50}	=	total length of the contour line at the 50th percentile of Z (m)
LC_{75}	=	total length of the contour line at the 75^{h} percentile of Z (m).

7.3 Module Discussion

7.3.1 Strengths and Advantages

The Watershed Module has two overall objectives: (1) to simulate contaminant concentrations in soils surrounding WMUs over time, and (2) to generate needed hydrological inputs and contaminant loads required by other modules. Relative to other known models that might have been candidates to satisfy these two objectives, the strengths and advantages of the Watershed Module include the following:

- **Appropriate level of spatial resolution.** Time-varying soil concentrations surrounding the WMU arise only as a result of deposition of airborne particles and vapors originating from the WMU. The threshold issue for designing the spatial resolution for the Watershed Module was, how fine did this resolution need to be, given such competing objectives such as minimizing runtime and data collection? Although it could reasonably be expected that soil concentrations resulting from atmospheric deposition are, on average, quite low, it is also possible that land areas in relative proximity to the WMU might have loadings of some concern. Therefore, the balancing act for spatial resolution was to not have so much spatial resolution as to unduly affect runtimes and data collection, but to also avoid as much as possible the "diluting-out" of contaminant hot spots in close proximity to the WMU. This balance led to the approach of assuming that each watershed subbasin being modeled has uniform spatial concentration (a "completely-mixed" approach), which tends to "dilute" hot spots, but having the ability to delineate "watershed subbasins" such that those in close proximity to the WMU are relatively small, so that hot spots are reasonably well represented, with larger (and presumably less important from a contamination standpoint) subbasins resulting at greater distances. Thus, although each watershed subbasin is modeled identically, the spatial resolution is highly flexible. Within a watershed subbasin, concentrations at any time are uniform, but they can vary among watershed subbasins. This level of spatial resolution control by the model user has proven to be very advantageous in modeling a wide variety of different sites.

- **Ability to generate needed hydrological inputs for other modules.** In addition to having the functionality to simulate different time-varying soil concentrations (and depth profiles) within each watershed subbasin (to be used for subsequent exposure calculations), the Watershed Module also generates hydrological inputs and contaminant loadings for other modules over a variety of meteorological and environmental conditions representing the continental United States. For the Surface Water Module, subbasin-specific stormwater runoff, eroded soil loads, stream baseflow (dry weather streamflow), and contaminant loadings are generated as time series outputs. Of particular note, baseflows are estimated in addition to stormwater runoff, so that the total streamflow (not just the surface runoff portion of streamflow) is available to the Surface Water Module. The baseflows were estimated by fitting 18 different regression models to USGS low-flow streamflow data, each model being specific to one of 18 Hydrologic Units (HUCs) in the conterminous United States. This ability to simulate region-specific baseflow is extremely important as a Surface Water Module input.

- **Consistency between the GSCM and hydrology algorithms used for Land-based Source Modules.** The algorithms used to estimate watershed subbasin soil concentrations over time and depth (the GSCM), as well as the hydrological algorithms (soil erosion, stormwater runoff) are identical to those used in the Land-based Source Modules. This commonality provided not only economies with respect to model development and ease-of-use, but, more importantly, it ensures that underlying assumptions are consistent across these modules.

7.3.2 Uncertainty and Limitations

The Watershed Module includes the following limitations or uncertainties:

- **GSCM limitations.** As mentioned previously, the GSCM is the computational engine for the Watershed Module. Accordingly, all uncertainties or limitations inherent to the GSCM, and described for the LAU, Waste Pile, and Landfill Modules (Section 5), also apply to the Watershed Module.

- **Spatial dilution of intra-watershed hot spots.** Because each watershed subbasin is assumed to be uniform with respect to contaminant concentrations in soil, hot spots resulting from nonuniform aerial deposition within a watershed subbasin will not be detected.

- **30Q2 equivalent to baseflow.** There is uncertainty in the baseflow estimates with regard to whether the module uses the correct low flow statistic (e.g., 30Q2), and in representing the variability in the baseflow for any given watershed. For a given watershed, the 30Q2 estimate of constant baseflow is a point estimate, generated from a regression model. Thus, the same baseflow will always be estimated for a given hydrologic region and watershed size.

- **Sheetflow runoff assumption**. The MUSLE application to estimate eroded solids loads and associated sorbed contaminant loads in each watershed subbasin assumes that sheet-flow runoff and erosion apply across the entire area. Watersheds are delineated, in part, to support that assumption, but it is unlikely that true sheetflow runoff occurs over all portions of all watershed subbasins. To the extent that channelized flow occurs and can “short-circuit” contaminant loads directly to adjacent waterbodies without first traversing downslope land surfaces, the estimated contaminant loadings to waterbodies may be underestimated by the Watershed Module. Conversely, this assumption would tend to overestimate average soil concentrations across the watershed subbasin, leading to overestimated soil exposures.

7.4 References

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8.0 Surface Water Module

8.1 Purpose and Scope

The Surface Water Module simulates contaminant concentrations in surface waterbodies throughout the area of interest (AOI) around each site modeled. Inputs to the Surface Water Module include contaminant loadings from direct air deposition onto surface waters, contaminant loadings from runoff and soil erosion from land areas associated with sources (LAU and waste piles only), contaminant loadings from contaminated ground water plumes that are intercepted by surface waters, contaminant loadings in runoff and eroded soil from the watersheds in the AOI, and hydrological inputs (flows, soil loads) from the watersheds. Surface Water Module outputs include water column and sediment contaminant concentrations, which are then used by the Aquatic Food Chain Module, Farm Food Chain Module, and Ecological Exposure Module. All inputs and outputs are annual average time series. Figure 8-1 shows the relationship and information flow between the Surface Water Module and the 3MRA modeling system.

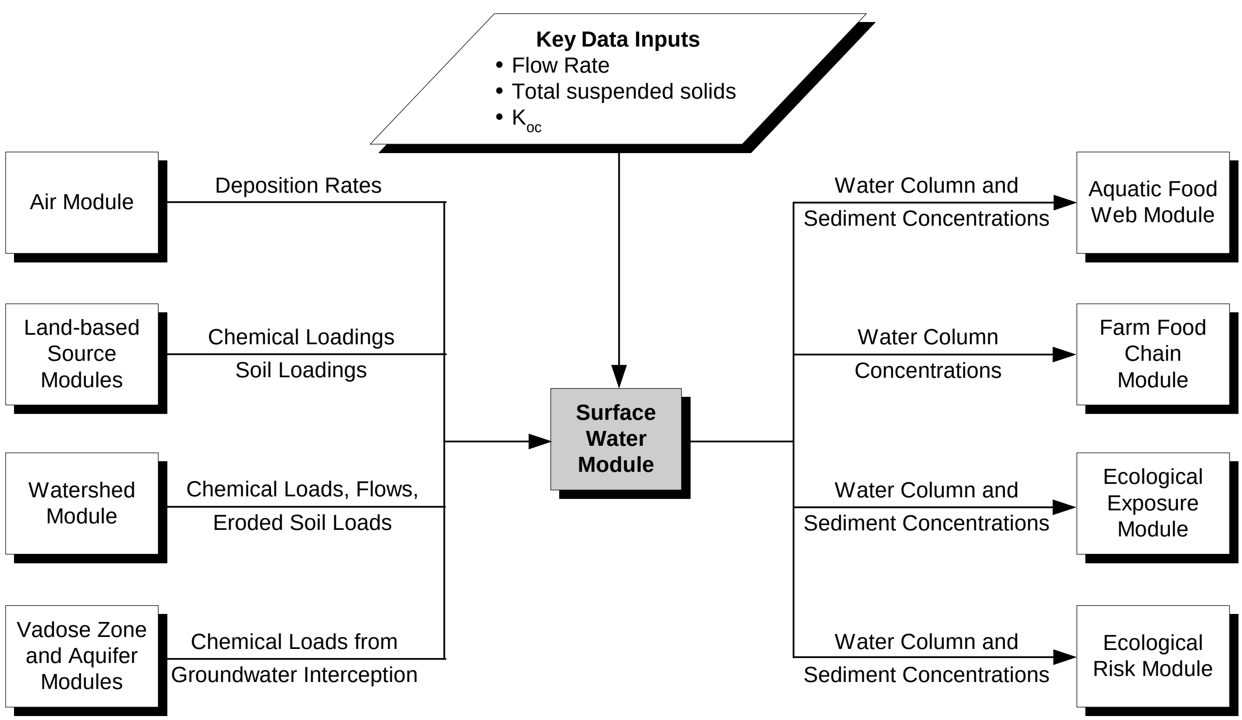

Figure 8-1. Information flow for the Surface Water Module in the 3MRA modeling system.

The major tasks performed by the Surface Water Module to simulate contaminant concentrations throughout the surface waterbody network in the AOI are as follows:

1. **Constructs waterbody network.** For each AOI, the Surface Water Module identifies the streams, wetlands, ponds, lakes, and bay reaches to be modeled, and assigns lengths, areas, depths, and volumes to them.

2. **Routes hydraulic flow and solids through the waterbody network.** The Surface Water Module conducts water and solids balances for each waterbody in each year of the simulation.

3. **Constructs and solves the mass balance equations describing contaminant fate and transport throughout the waterbody network.** The Surface Water Module calculates the contaminant concentration in each waterbody for each year. Outputs include total water column concentration and dissolved concentration.

8.2 Conceptual Approach

The Surface Water Module is based on the EPA legacy model Exposure Analysis Modeling System (EXAMS II), which has been thoroughly verified and validated in numerous applications (see Volume III of this report). EXAMS II was extended for use in the Surface Water Module primarily through development of the pre- and postprocessor, EXAMS IO. EXAMS IO is the interface between EXAMS II and the rest of the 3MRA modeling system. It reads data from other 3MRA modules and databases, builds the EXAMS input files describing the waterbody environment and chemical properties, builds a command file that specifies the contaminant loading history, and controls the EXAMS simulation. EXAMS IO passes control to EXAMS II, which conducts the simulation and produces intermediate results files. EXAMS IO then processes the intermediate files and passes the output data back to the proper 3MRA modeling system database. The complete system consisting of EXAMS II and EXAMS IO is denoted in this discussion as the Surface Water Module. A summary of EXAMS II—the science and computational core of the Surface Water Module—is presented in the accompanying text box. The full capabilities of EXAMS II are explained in Burns et al. (1982) and Burns (1997).

8.2.1 Construct Waterbody Network

Each Monte Carlo iteration of the 3MRA system represents a unique site covering the AOI around a waste management unit (WMU). This AOI is defined by a radius of 2 km in the sample 3MRA data set. One or more contiguous waterbody networks may be located within each AOI. For each of the 201 sites currently contained in the 3MRA modeling system database, waterbody networks were defined using the EPA Reach File system supplemented with digital elevation modeling. Each waterbody network is divided into reaches, which are characterized and stored in the site database.

Summary of EXAMS II

EXAMS II combines contaminant loadings, transport, and transformations into a set of differential equations using the law of conservation of mass as an accounting principle. It accounts for all the contaminant mass entering and leaving a system as the algebraic sum of external loadings, transport processes that export the compound from the system, and transformation processes within the system that convert the contaminant to daughter products. (Transformation from parent to daughter products is implemented only for mercury in the Surface Water Module.)

EXAMS II represents each waterbody by a set of segments or distinct zones in the system. The program is based on a series of mass balances for the segments that give rise to a single differential equation for each segment. Working from the individual transport and transformation process equations, EXAMS II compiles an overall equation for the net rate of change of contaminant concentration in each segment. The resulting system of differential equations describes the mass balance for the entire system, which is then solved by the method of lines.

EXAMS II includes process models of the physical, chemical, and biological phenomena governing the transport and fate of compounds. Each of the unit process equations used to compute the transformation kinetics of contaminants accounts for the interactions between the chemistry of a compound and the environmental forces that shape its behavior in aquatic systems. This “second-order” or “system-independent” approach allows one to study the fundamental chemistry of compounds in the laboratory and then, based on independent studies of the levels of driving forces in aquatic systems, evaluate the probable behavior of the compound in systems that have never been exposed to it. Most of the process equations are based on standard theoretical constructs or accepted empirical relationships. The user can specify reaction pathways for the production of transformation products of concern, whose further fate and transport can then be simultaneously simulated by EXAMS II.

EXAMS II contains process modules for several chemical reactions. Equilibrium reactions are used for sorption and ionization. Kinetic reactions are used for volatilization, hydrolysis (acid, base, and neutral), biodegradation (water column and sediments), photolysis, oxidation, and reduction. (The Surface Water Module does not consider photolysis or oxidation/reduction.) EXAMS II uses these modules as determined by the input chemical properties. EXAMS II has been designed to accept standard water quality parameters and system characteristics that are commonly measured by limnologists throughout the world and contaminant data sets conventionally measured or required by EPA regulatory procedures.

The contaminant fate algorithms in EXAMS II model sorption to suspended solids, biotic solids, and sediment solids, but EXAMS II does not simulate a solids balance. Solids concentrations are specified as input data. The effects of settling and resuspension on contaminant fate are accounted for in a bulk sediment-water exchange term.

EXAMS II incorporates a few major assumptions. The model was designed to evaluate the consequences of longer-term, primarily time-averaged contaminant loadings that ultimately result in trace-level contamination of aquatic systems. EXAMS II generates a steady-state, average flow field (long-term or monthly) for the ecosystem. The program cannot therefore fully evaluate the transient, high concentrations that arise from chemical spills, although spills under average hydrological conditions can be studied. An assumption of trace-level contaminant concentrations was used to design the process equations. The contaminant is assumed not to radically change the environmental variables that drive its transformations. EXAMS II uses linear sorption isotherms and second-order (rather than Michaelis-Menten-Monod) expressions for biotransformation kinetics, which are known to be valid for low concentrations of pollutants. Sorption is treated as a thermodynamic or constitutive property of each compartment in the system; that is, sorption-desorption kinetics are assumed to be rapid compared with other processes. Although this assumption may be violated by extensively sorbed contaminants, these contaminants tend to be captured by benthic sediments, where their release to the water column is controlled by benthic exchange processes.

From the 3MRA modeling system site database, stream reaches are assigned lengths, and wetland, pond, lake, and bay reaches are assigned surface areas. Site-specific reach depths and volumes were not available. Instead, depths were assigned to water column and sediment compartments for wetlands, ponds, lakes, and bays based on national distributions. The Surface Water Module calculates volumes from the site-specific surface areas and the estimated depths. For stream reaches, the Surface Water Module calculates widths and depths using discharge coefficients based on empirical observations of stream hydrogeometric relationships (Leopold and Maddox, 1953) and then calculates volumes using site-specific reach lengths.

8.2.2 Route Hydraulic Flow Through the Waterbody Network

Water Balance. The Surface Water Module conducts a water balance for each simulated year, routing incoming flows from the headwater reaches down through the network and out of the exiting reach. For a given reach in a given year, the outflow is the sum of all the inflows minus evaporation, as shown in the water balance equation:

$$Q_{r,t} = \sum_{i=1}^{NumTrib_r} QTrib_{i,r,t} + \sum_{i=1}^{NumWS_r} QRunoff_{i,r,t} \times WSFrac_{i,r} + QBase_r + \frac{\left(R_t - E_t\right) \times AS_r}{100 \times 365} \quad (8\text{-}1)$$

where

$Q_{r,t}$	=	average outflow from reach *r* during year *t* (m^3/d)
$NumTrib_r$	=	number of upstream reaches flowing into reach *r*
$QTrib_{i,r,t}$	=	average upstream outflow from reach *i*, tributary to reach *r* during year *t* (m^3/d)
$NumWS_r$	=	number of adjacent watersheds flowing into reach *r*
$QRunoff_{i,r,t}$	=	annual average surface runoff inflow from watershed *i* immediately adjacent to reach *r* during year *t* (m^3/d)
$WSFrac_{i,r}$	=	area fraction of adjacent watershed *i* contributing to reach *r*
$QBase_r$	=	average constant baseflow flow to reach *r* (m^3/d)
R_t	=	precipitation for year *t* (cm/yr)
E_t	=	evaporation for year *t* (cm/yr)
AS_r	=	surface area for reach *r* under baseflow conditions (m^2).

The Surface Water Module also checks and ensures positive flow in every reach for every year. If evaporation exceeds the total inflow to a reach for a year, the evaporation in that reach is reduced to a level supporting an outflow equivalent to precipitation of 0.5 mm/yr over the entire contributing watershed.

Long-term average baseflow from each adjacent watershed is calculated in the Watershed Module using empirical correlations based on watershed surface area and regional location (see Section 7). The input data map these watersheds to the appropriate reaches and also specify the fraction of each contributing watershed's baseflow that should be associated with each reach. Total reach baseflow is then calculated by the fraction-weighted sum over all contributing watersheds. This baseflow estimate is derived from observed stream flow data and accounts for both seepage inflow and evaporation loss during dry periods.

This water balance is conducted for each simulated year on a reach-specific basis by EXAMS IO. The resulting internal flows are used to calculate stream hydrogeometry and the waterbody solids balance. Information on stream widths and depths and all hydrologic inflows is used to recalculate the water balance on a compartment-specific basis for use in the chemical advection calculations.

Solids Balance. EXAMS IO conducts a solids balance for each simulated year, routing incoming solids from the headwater reaches down through the network and out of the exiting reach. The total suspended solids (TSS) in a reach may originate in erosion from adjacent watershed subbasins, from internal bank erosion, or from internal production of biotic solids. Solids deposition and burial are not simulated in this version of the Surface Water Module; consequently, TSS is modeled as a conservative substance. Therefore, for a given reach in a given year, the average solids flux per year through the reach is the sum of all the solids mass inflows, including upstream loads, loads delivered from adjacent surface erosion, and loads generated internally to the reach:

$$TSS_{r,t} = \sum_{i=1}^{NumTrib_r} QTrib_{i,r,t} \times TSSTrib_{i,r,t} + \sum_{i=1}^{NumWS_r} TSSRunoff_{i,r,t} \times WSFrac_{i,r} + Q_{r,t} \times \left[SolBnk_r + \frac{Phyt(TI_r)}{FocBio} \right] \quad (8\text{-}2)$$

where

$TSS_{r,t}$	=	average solids load exiting reach *r* in year *t* (g/day)
$QTrib_{i,r,t}$	=	average upstream outflow from reach *i*, tributary to reach *r* during year *t* (m^3/d)
$TSSTrib_{i,r,t}$	=	TSS concentration exiting upstream reach *i*, tributary to reach *r*, in year *t* (g/m^3)
$TSSRunoff_{i,r,t}$	=	solids load delivered from erosion of watershed *r*, immediately adjacent to reach *r*, in year *t* (g/day)
$SolBnk_r$	=	TSS concentration from incremental bank erosion in reach *r* (g/m^3)
Phyt (TI_r)	=	biotic carbon concentration for trophic index for reach *r* (g-C/m^3)
TI_r	=	trophic index for reach *r* (unitless)
FocBio	=	organic carbon fraction of biotic solids (g-C/g-solids)

The first right-hand term in this equation accounts for upstream tributary solids loads. The second term accounts for adjacent watershed erosion loads. This erosion load is generated by the Watershed Module and already incorporates an area-dependent sediment delivery ratio. The third term accounts for bank erosion loading, where the TSS concentration from incremental bank erosion is the resulting incremental concentration in reach *r*. This is set to 5 g/m^3 for stream reaches and 1 g/m^3 otherwise. The final term accounts for internal biotic production of solids. The trophic index is related to the net biotic production.

8.2.3 Construct and Solve the Mass Balance Equations Describing Contaminant Fate and Transport throughout the Waterbody Network

The primary function of the Surface Water Module is to simulate the spatial and temporal contaminant concentrations for both fate within the individual waterbody segments (or compartments) and transport among these compartments that make up the overall waterbody network.

Contaminant fate and transport are calculated on a compartment-specific basis, by accounting for a set of transport and transformation processes applied within a mass balance framework. The Surface Water Module is based on a series of mass balances, which give rise to a single differential equation for each compartment. Working from the transport and transformation process equations, the Surface Water Module generates an equation for the net rate of change of contaminant concentration in each compartment. The resulting system of differential equations describes a mass balance for the entire waterbody network. These equations have the following general form:

$$V \frac{dC}{dt} = L_e + L_i - V \times K \times C \tag{8-3}$$

where

V	=	volume of water in the compartment (m^3)
C	=	total contaminant concentration (mg/L)
L_e	=	total external loading to the compartment (g/h)
L_i	=	total internal loading to the compartment resulting from contaminated flows among system compartments and the formation of chemical reaction products (g/h)
K	=	overall pseudo-first-order loss constant that expresses the combined effect of transport and transformation processes (i.e., advection, dispersion, volatilization, biodegradation) that decrease contaminant concentration (h^{-1})

The Surface Water Module provides for a total of five kinds of compartment loadings:

- Point source or stream-borne loadings,
- Nonpoint source loadings,
- Contaminated ground water seepage,
- Precipitation washout from the atmosphere, and
- Spray drift (or miscellaneous) loading.

The inputs include monthly and annual average loadings to all appropriate compartments.

The Surface Water Module evaluates the status of individual loadings entering the compartments to prevent conditions outside of its numerical operating range. For example, the Surface Water Module makes no provision for crystallization of the compound from solution, nor for dissolution of a compound from a solid or liquid phase. In addition, reaction nonlinearities potentially present at high contaminant concentrations are not incorporated into the model. The Surface Water Module calculates the dissolved contaminant concentration within each input flow, using receiving compartment pH and temperature to account for ionization and using entrained sediment loads to account for sorption. If the dissolved concentration for any of the species is greater than half its solubility, then the simulation is not used by the 3MRA modeling system.

The system of mass balance differential equations is solved numerically, using the method of lines algorithm. The solution to the system of mass balance equations yields the distribution of contaminant concentration throughout the surface waterbody network (water column and sediments) over time. Three temporal options are available for EXAMS II:

- Steady-state (no change over time),
- Quasi-dynamic with time-constant environmental data, and
- Quasi-dynamic with monthly-varying environmental data.

The Surface Water Module implements the quasi-dynamic mode with time-constant environmental data. In this mode, the dynamic simulations consider annually-varying flows and loadings but assume other environmental conditions are at their long-term average values.

8.3 Module Discussion

8.3.1 Strengths and Advantages

The Surface Water Module has the following strengths and advantages:

- **Use of a proven chemical fate model.** The Surface Water Module uses EXAMS to simulate chemical fate for the scenarios presented by the 3MRA framework. EXAMS is an EPA legacy model that has been used for two decades to assess the fate of organic chemicals in aquatic systems. It has been used routinely to evaluate pesticide exposure scenarios for the Office of Pesticide Programs. During its extensive application history, EXAMS has been tested in a wide variety of waterbodies for a wide range of organic chemicals. The code has been verified, and its predictions have been confirmed in several site-specific evaluations published in the technical literature.

- **Applicable to a variety of conditions.** EXAMS is a robust model that solves the governing differential equations effectively under a variety of conditions.

8.3.2 Uncertainty and Limitations

The following uncertainties and limitations are inherent in the Surface Water Module:

- **Limited temporal resolution of numerical solution.** The current 3MRA modeling system methodology calls for repeated, relatively long simulations (e.g., 200 or more years). Accordingly, run time must be relatively fast, which precludes small time steps that might otherwise give insight into contaminant behavior on a finer temporal scale.

- **Limited resolution of loadings and flows.** Use of annual-average loadings and flows rather than daily loadings and flows leads to calculated annual-average concentrations that are biased high, depending on the correlation between flow and loading at a particular site. This bias is somewhat mitigated for reactive and volatile contaminants where the loss rate is proportional to the concentration.

- **Limited site-specific data.** Use of national distributions rather than site-specific environmental data could cause calculated concentrations to be low or high at a given location, with no known general bias.

- **Simplistic solids balance.** The simple solids balance overestimates suspended solids concentrations slightly in streams and more significantly in ponds, wetlands, and lakes. Calculated total water column contaminant concentrations will be high, whereas the dissolved contaminant fraction will be low. The net result for dissolved water column contaminant concentrations, which are used for fish exposure, is not expected to be biased significantly high or low.

- **Procedure for preventing drying of surface water reaches.** For sites that experience periodic drying (due to limited inflows), a small default positive flow equivalent to 0.5 mm per year of direct precipitation onto the waterbody surface is maintained to keep the model functioning. This procedure conducts contaminant loads downstream within a remnant waterbody reach rather than within runoff over a dry bed or subsurface flow within the bed. Although the mass balance is maintained, the contaminant and solids concentrations will tend to be elevated within the remnant reach. These elevated concentrations are probably realistic for years in which evaporation exceeds all hydrologic inflows.

8.4 References

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Burns, L.A. 1997. *Exposure Analysis Modeling System (EXAMS II): User's Guide for Version 2.97.5.* U.S. Environmental Protection Agency, Athens, GA. EPA-600/R-97/047.

Leopold, L.B. and T. Maddox. 1953. *The Hydraulic Geometry of Stream Channels and Some Physiographic Implications.* Professional Paper 252, U.S. Geological Survey, Washington, DC.

9.0 Vadose Zone and Aquifer Modules

9.1 Purpose and Scope

The Vadose Zone and Aquifer Modules simulate the subsurface movement of contaminants in leachate from surface impoundments, landfills, land application units (LAUs), and waste piles to downgradient drinking water wells and waterbodies. The modules are not used for aerated tanks, because tanks are assumed not to leak. Detailed information on the Vadose Zone and Aquifer Modules can be found in the technical background document (U.S. EPA, 1999e). Figure 9-1 shows the relationship and information flow between the Vadose Zone and Aquifer Modules and the 3MRA modeling system.

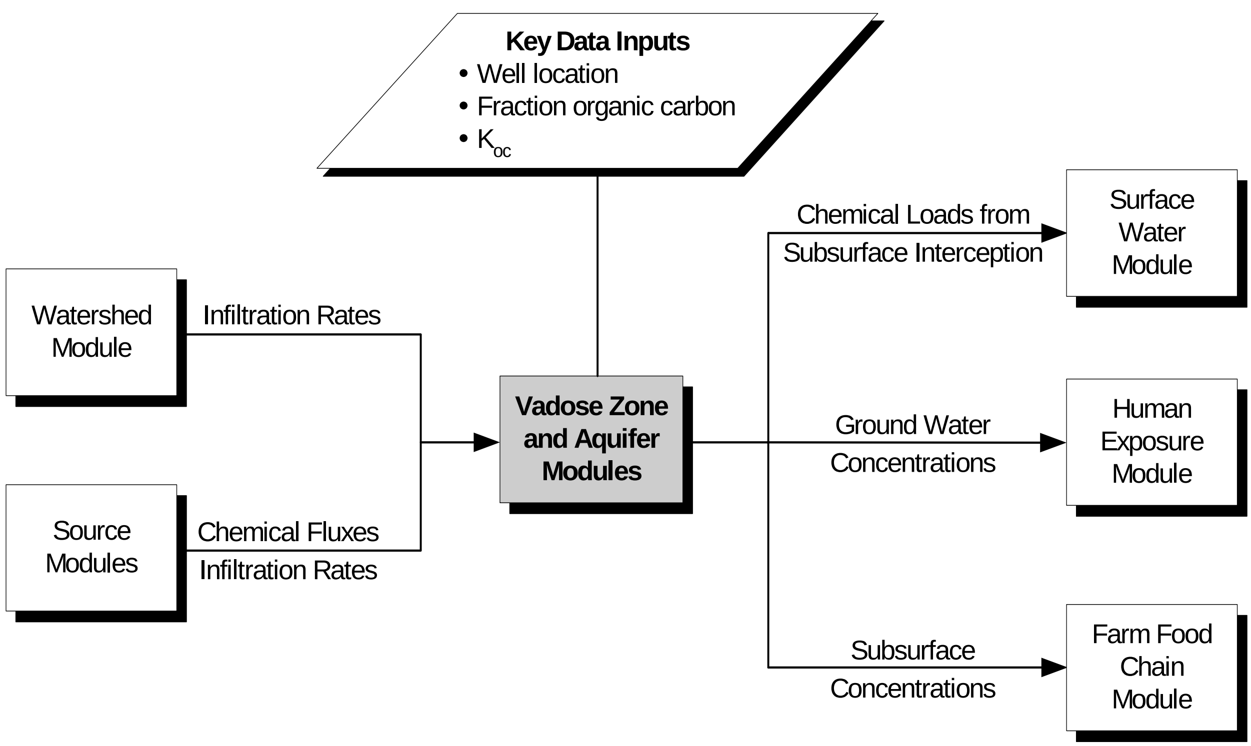

Figure 9-1. Information flow for the Vadose Zone and Aquifer Modules in the 3MRA modeling system.

The Vadose Zone and Aquifer Modules simulate the fate and transport of dissolved contaminants from a point of release at the base of a WMU, through the underlying soil, and through a surficial aquifer (or ground water source). Module outputs include ground water contaminant concentrations in wells, which are used by the Human Exposure Module to estimate exposures through drinking water and showering, and by the Farm Food Chain Module to estimate contaminant concentrations in beef and milk from farm well use; and contaminant fluxes into waterbodies, which are used by the Surface Water Module, along with contaminant fluxes from atmospheric deposition and overland flow, to estimate contaminant concentrations in streams, lakes, and wetlands.

The Vadose Zone and Aquifer Modules are used by the 3MRA modeling system only if there are wells or downgradient streams, lakes, or wetlands at a site. Waterbodies are downgradient if they are in the direction of ground water flow away from the WMU.

The Vadose Zone and Aquifer Modules perform the following functions:

1. **Model vadose zone flow and transport**. The one-dimensional (1-D) Vadose Zone Module simulates infiltration and dissolved contaminant transport, by advection and dispersion, leaching from the bottom of a WMU through the soil above the water table (i.e., the vadose zone) to estimate the contaminant and water flux to the underlying ground water.

2. **Model ground water flow and transport**. The pseudo-3-D Aquifer Module simulates ground water flow and contaminant transport, by advection and dispersion, from the base of the vadose zone to estimate contaminant concentrations in drinking water wells and contaminant discharge fluxes to intercepted waterbodies.

3. **Model subsurface chemical reactions.** Both the Vadose Zone and Aquifer Modules simulate sorption to soil or aquifer materials and biological and chemical degradation, which can reduce contaminant concentrations as they move through soil and ground water. In cases where degradation of a contaminant yields other contaminants that are of concern, the modules can account for the formation and transport of up to six different daughter and granddaughter degradation products. For metals, the modules use sorption isotherms that allow adjustment of sorption behavior to account for varying metal concentrations and geochemical conditions.

The Vadose Zone and Aquifer Modules in the 3MRA modeling system were extracted from EPA's Composite Model for Leachate Migration with Transformation Products (EPACMTP) (U.S. EPA, 1996 a,b,c; 1997). EPACMTP is used in EPA regulatory efforts by OSW and has been subject to extensive peer review and public comment. EPACMTP is a tool used routinely to predict potential ground water pathway exposure at a downstream receptor well for regulatory development purposes.

9.2 Conceptual Approach

Figure 9-2 illustrates how the Vadose Zone and Aquifer Modules work together to calculate receptor well contaminant concentrations and contaminant fluxes to downgradient waterbodies. The Vadose Zone Module and the Aquifer Module are described below, followed by a discussion of the chemical reaction modeling that operates in both modules. Additional

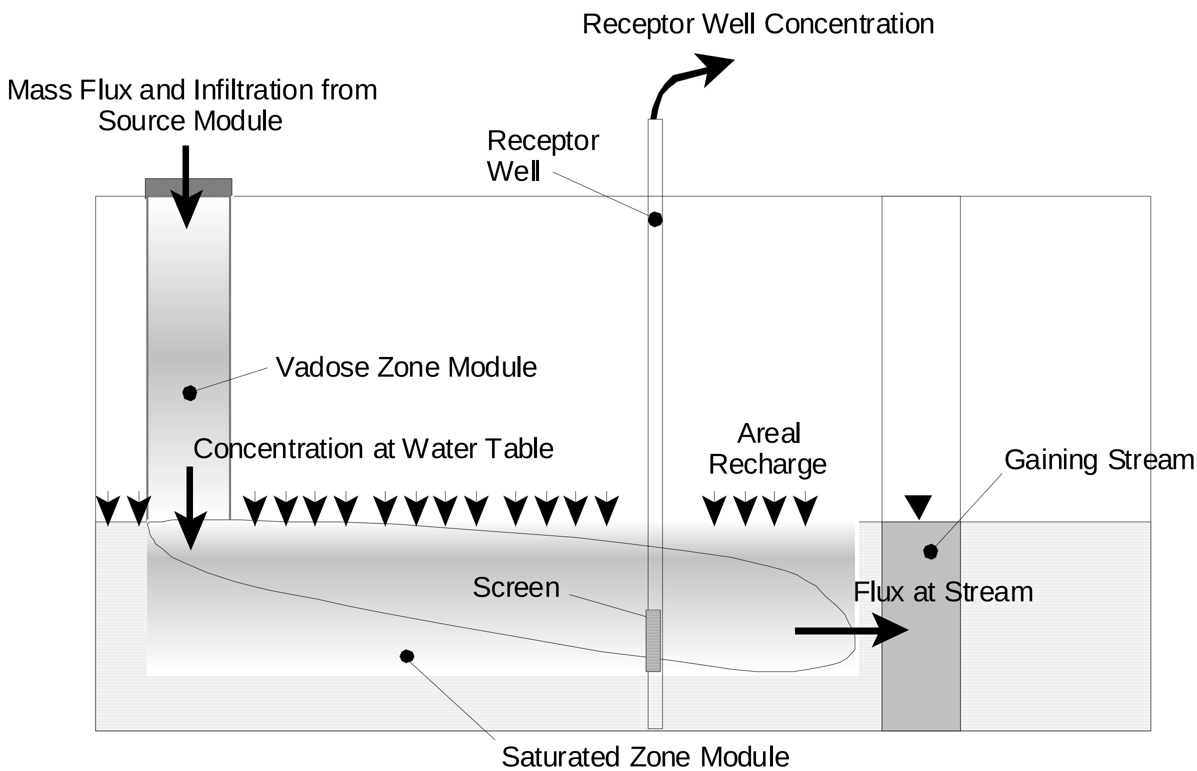

Figure 9-2. Conceptual diagram of Vadose Zone and Aquifer Modules.

details on the Vadose Zone and Aquifer Modules, including governing equations and detailed input/output specifications, can be found in the 3MRA modeling system background documents (U.S. EPA, 1999a-e).

9.2.1 Vadose Zone Module

The Vadose Zone Module simulates the flow and transport of contaminants in leachate from the upper boundary of the vadose zone at the base of the WMU source to the lower vadose zone boundary at the water table. The Vadose Zone Module estimates contaminant flux to the Aquifer Module given leachate flows and fluxes from the 3MRA source modules. Its outputs are the long-term, steady-state infiltration rate and the annual time series of contaminant concentrations that are used as inputs to the Aquifer Module.

Vadose Zone Flow. The Vadose Zone Module performs 1-D analytical and numerical solutions for water flow and contaminant transport in unsaturated soil underlying a WMU source. The model assumes that flow in the vadose zone is at steady-state and flows vertically from underneath the source toward the water table.[1] The Vadose Zone Module receives a time series of annual average infiltration rates and contaminant mass fluxes from the source modules.

[1] Because water flow in the vadose zone is predominantly gravity driven, the vertical flow component accounts for most of the fluid flux between the WMU source and the water table.

Because the Vadose Zone Module assumes steady-state flow, it calculates a long-term average infiltration rate from the full series of annual average infiltration rates. At the same time, the model calculates an annual average time series of leachate concentrations, while conserving contaminant mass with respect to the full series of contaminant mass fluxes output from the source modules.

The vadose zone flow model assumes that the unsaturated soil is a uniform porous medium and that the infiltrating flow rate (*I*) is governed by Darcy's law (Freeze and Cherry, 1979):

$$I = -K_S\, k_{rw}\, (d\psi/dz - 1) \qquad (9\text{-}1)$$

where

ψ	=	pressure head (cm)
z	=	depth in the soil column (positive downward) (cm)
K_S	=	saturated hydraulic conductivity (cm/h)
k_{rw}	=	relative permeability (unitless).

Solution of this equation for unsaturated soil conditions requires stipulation of the relationships between

- The relative permeability and the volumetric water content of the porous medium and

- The volumetric water content and the pressure head.

The Vadose Zone Module assumes that these relationships follow the equations given by Van Genuchten (1980). The governing equations for these relationships are given in U.S. EPA (1999e). Solution methods for these equations can be found in the EPACMTP background document (U.S. EPA, 1996a-c).

Vadose Zone Transport. The Vadose Zone Module simulates transient[2] contaminant transport through the unsaturated zone by advection and dispersion. The model assumes that the unsaturated zone is initially free of contamination and that contaminants migrate downward along with the leachate flow from the WMU. Chemical reactions modeled include single- or multiple-chemical chain decay reactions, and linear or nonlinear sorption. The chemical reaction modeling is described in Section 9.2.3.

The vadose zone transport model simulates the 1-D transport of contaminants through the soil column using the advection-dispersion equation of Huyakorn and Pinder (1983). This equation estimates contaminant degradation using a first-order decay constant that calculates mass fractions of both parent and daughter compounds. The effect of equilibrium sorption of a species is expressed by a retardation coefficient that can simulate sorption using either linear or nonlinear (Freundlich) sorption isotherms. The governing equations for the vadose zone

[2] Contaminant transport can vary year to year as leachate concentration changes.

transport model can be found in U.S. EPA (1999e). Solution methods for these equations, which are semianalytical for organic chemicals and analytical for nonlinear metals, are described in detail in the EPACMTP background documents (U.S.EPA, 1996a-c).

Key assumptions of the Vadose Zone Module

- Contaminants are released from a square WMU source; there is no contaminant flux outside the WMU
- Flow and transport are steady-state and are 1-D, with year-to-year changes in leachate concentration from the source.
- Flow is vertical with no horizontal component.
- The soil is initially free of contamination.
- Soil is a uniform porous medium with uniform properties; there are no soil layers or preferential pathways such as fractures or soil macropores.
- Contaminant transport is by advection and dispersion only; there is no facilitated transport by colloids or nonaqueous-phase liquid (NAPL).
- Contaminants are in the aqueous and sorbed phases only; there are no mass transfer processes between phases other than adsorption onto soil particles.
- There is no volatilization from the unsaturated zone (i.e., no gas-phase release or transport).
- NAPLs (e.g., oil) are not present.

Vadose Zone Module Assumptions, Inputs, and Outputs. Key assumptions for the Vadose Zone Module are summarized in the text box. Vadose Zone Module inputs include

- Contaminant fluxes and infiltration rates from the 3MRA source modules;
- Soil hydrologic properties (saturated hydraulic conductivity, total porosity, residual water content, and Van Genuchten water-retention parameters);
- Soil bulk density, pH, temperature, and organic matter content;
- Vadose zone thickness (depth to ground water); and
- Contaminant-specific degradation and sorption variables (see Section 9.2.3).

The module produces a time series of contaminant concentrations in the infiltrate passed to the aquifer, a long-term average infiltration rate, and the duration of the WMU source of contamination, all of which are used as input to the Aquifer Module. Detailed specifications of these inputs and outputs can be found in U.S. EPA (1999e).

9.2.2 Aquifer Module

The Aquifer Module simulates 1-D ground water flow and pseudo-3-D contaminant transport to calculate ground water contaminant concentrations at downgradient[3] drinking water wells and intercepting surface waterbodies (i.e., streams, rivers, lakes, and wetlands). The Aquifer Module estimates these concentrations and fluxes given infiltration flow rate and contaminant concentrations from the Vadose Zone Module. Its primary outputs include time series of annual average contaminant concentrations at each downgradient receptor well (used by

[3] "Downgradient" refers to features (wells, waterbodies) that are downstream from the WMU with respect to the ground water flowpath.

the Human Exposure and Farm Food Chain Modules) and a time series of contaminant fluxes at any downgradient intercepting waterbody (used by the Surface Water Module).

Ground Water Flow. The Aquifer Module simulates ground water flow in a surficial (unconfined) aquifer of constant thickness using a 1-D ground water flow submodel and a pseudo-3-D transport submodel. The model accounts for the effects of infiltration from the WMU and regional recharge downgradient of the WMU on the magnitude and direction of ground water flow.[4] The concept accounts for regional flow in the horizontal direction driven by a regional hydrologic gradient, along with vertical disturbance of this regional flow by water infiltrating to the surficial aquifer from the overlying vadose zone and WMU.

The Aquifer Module uses a 1-D, steady-state solution for predicting hydraulic head and Darcy velocities that assumes that the aquifer is composed of a uniform porous media, and that ground water flow is governed by Darcy's law (Bear, 1972). This solution begins with the classic 3-D governing equation for ground water flow that is used in EPACMTP:

$$K_x (\partial^2H/\partial x^2) + K_y (\partial^2H/\partial y^2) + K_z(\partial^2H/\partial z^2) = 0 \tag{9-2}$$

where

H	=	hydraulic head (cm)
K_x	=	hydraulic conductivity in the longitudinal direction (cm/hr)
K_y	=	hydraulic conductivity in the horizontal-transverse direction (cm/hr)
K_z	=	hydraulic conductivity in the vertical direction (cm/hr).

The 1-D flow equation used in the Aquifer Module is derived by setting the transverse hydraulic conductivity, K_y, equal to zero and invoking the Dupuit- Forchheimer assumption (Bear, 1972), in conjunction with vertical boundary conditions to account for infiltration and recharge. Details on this flow equation, its derivation, and its solution can be found in U.S. EPA (1999e).

The ground water flow model accounts for ground water mounding beneath the WMU by allowing different recharge rates underneath and outside the source area. Within the model, ground water mounding beneath the source is represented in the flow system by increased hydraulic head values at the top of the aquifer. This approach is reasonable so long as the height of the mound is small relative to the thickness of the saturated zone.

Ground Water Transport. The Aquifer Module simulates contaminant transport through ground water by use of a pseudo-3-D advection and dispersion submodel. The model assumes that ground water is initially free of contamination; that is, initial aquifer contaminant concentration and concentration gradients along the downstream and upstream boundaries are set to zero. Contaminants enter the aquifer only from the vadose zone immediately underneath the

[4] Recharge is provided by the 3MRA Watershed Module as a time series of annual average recharge rates with units of m/d. Because the aquifer model requires a steady-state recharge rate with units of m/yr, the model calculates an effective long-term recharge rate as the average of the time series recharge rates received from the Watershed Module.

WMU, which is modeled as a square, horizontal plane source. Chemical reactions modeled include single- or multiple-chemical chain decay reactions, and linear or nonlinear sorption. The chemical reaction modeling is described in Section 9.2.3.

The Aquifer Module simulates the advective-dispersive transport of dissolved contaminants in one dimension with dispersion in the other two dimensions added analytically (pseudo-3-D). The model is based on the 3-D advection-dispersion governing equation from Huyakorn and Pinder (1983), with a simplified solution to model advection in one dimension (horizontally) and dispersion in three dimensions. Although this may not be as accurate as a fully 3-D solution, it is much more computationally efficient and therefore more suitable for the large-scale Monte Carlo simulations for which the 3MRA modeling system was designed.

The ground water transport equation estimates contaminant degradation using a first-order decay constant that calculates mass fractions of both parent and daughter compounds. The effect of equilibrium sorption of a species is expressed by a retardation coefficient, R, that can simulate sorption using linear sorption isotherms.[5] Governing equations for the ground water transport equation can be found in U.S. EPA (1999e), which also describes the solution methods applied to develop the Aquifer Module.

Aquifer Fractures and Heterogeneity. The basic Aquifer Module described above assumes that aquifers are homogeneous and isotropic, and it therefore does not simulate flow and transport under the fractured or heterogenous subsurface conditions that are common across the United States. Because modeling fractures and heterogeneity is complex, EPA developed an indirect approach to address fracture flow, which tends to increase the rate of migration compared to nonfractured settings, and heterogeneity, which can cause a plume to break into fingers of higher concentration ground water that can increase the concentration at a well.

To address fracture flow conditions, the Aquifer Module uses an equivalent porous media (EPM) approach, which applies uniform porous media analogues to a fractured aquifer flow field. The approach involves developing “fracture multiplier” distributions for several classes of fractured hydrogeologic settings. For a particular Monte Carlo realization at a fractured site, the 3MRA modeling system selects a multiplier from the appropriate distribution and applies it to increase the porous media hydraulic conductivity selected for the site. Additional details on the development and implementation of this approach can be found in U.S. EPA (1999d,e).

To incorporate effects of heterogeneity in aquifers, EPA first assessed the effects of heterogeneity on receptor well concentrations (U.S. EPA, 1999b). Based on this study, EPA developed an algorithm and database to estimate the effects of heterogeneity on well concentrations output by the Aquifer Module. For a particular Monte Carlo realization, the 3MRA modeling system selects input values from the database and applies the algorithm to adjust the well concentration outputs. Additional details on the development and application of this method can be found in U.S. EPA (1999b,e).

[5] The aquifer model differs from the vadose zone model in that it does not allow the use of nonlinear Freundlich sorption isotherms.

Aquifer Module Assumptions, Inputs, and Outputs. Key assumptions for the Aquifer Module are summarized in the text box. Aquifer Module inputs include

- WMU length, width, and location;
- A steady-state, long-term annual infiltration rate and a time series of annual average contaminant concentrations from the Vadose Zone Module;
- A time series of annual-average recharge rates from the Watershed Module;
- Aquifer properties, including saturated hydraulic conductivity, regional gradient, thickness, pH, organic carbon content, and temperature;
- The ratio of horizontal to vertical hydraulic conductivity;
- Longitudinal, horizontal-transverse, and vertical dispersivities;
- Receptor well locations and screen depths below the water table surface;
- Location coordinates for downgradient intercepting waterbodies;
- Direction of regional ground water flow, measured clockwise from due north;
- Contaminant-specific degradation and sorption variables (see Section 9.2.3); and
- Termination criteria for ending the model simulation, including the maximum simulation time.

Key assumptions of the Aquifer Module

- The aquifer is unconfined and subject to recharge from the ground surface.
- The upper aquifer boundary is the water table; the lower aquifer boundary is an impermeable confining layer.
- The aquifer has a uniform (regional) hydraulic gradient and a constant saturated thickness.
- Aquifer materials are uniform porous media that are isotropic except for hydraulic conductivity, which can vary between the vertical and horizontal directions.
- The downgradient intercepting waterbody does not fully alter the ground water flow pattern.
- The ground water is initially free of contamination.
- Contaminants are released from a square WMU source; there is no contaminant flux outside the WMU.
- Recharge of contaminant-free water occurs from the watershed outside of the WMU.
- Ground water flow is steady-state, but contaminant concentrations can vary year to year.
- Preferential flow through fractures and aquifer heterogeneities is addressed by adjusting aquifer hydraulic conductivity or well concentrations.
- The model does not apply to solution openings in karst limestone.
- Contaminants are transported by advection and dispersion only; there is no facilitated transport by colloids or NAPL.
- Contaminants are in the aqueous and sorbed phases only; there are no mass transfer processes between phases other than adsorption onto soil particles.
- There is no volatilization from the water table (i.e., no gas-phase release or transport).
- NAPLs (e.g., oil, halogenated solvents) are not present.

The Aquifer Module outputs time series of annual average ground water contaminant concentrations at downgradient receptor wells within the ground water plume, which are used by the Human Exposure and Farm Food Chain Modules. It also outputs a time series of contaminant fluxes into a downgradient intercepting waterbody that is read by the Surface Water Module. Detailed specifications of aquifer model inputs and outputs can be found in U.S. EPA (1999e).

9.2.3 Chemical Reaction Modeling

The Vadose Zone and Aquifer Modules include code to simulate sorption to soil or aquifer materials and biological and chemical degradation, processes that can reduce contaminant concentrations as they move through soil and ground water. The Vadose Zone and Aquifer Modules use very similar methods to model these processes. The modeling approach differs between organic chemicals, which can degrade but follow simple sorption relationships, and metals, which do not degrade but require a more complicated approach to model sorption.

Organic Chemical Reactions. For organic chemicals, the Vadose Zone and Aquifer Modules simulate decay reactions for single compounds or multiple-compound chains, along with sorption to solid soil and aquifer components. Degradation reactions addressed include both chemical and biological transformation processes, with all transformation reactions represented by first-order decay processes. The models account for chemical and biological transformations by combining first-order degradation rates derived for chemical hydrolysis and biological degradation.

The transport of organic chemicals is influenced in part by hydrolysis. Chemical hydrolysis is modeled using acid-catalyzed, base-catalyzed, and neutral contaminant-specific hydrolysis rates (i.e., the Arrhenius equation). These rate constants are all influenced by ground water temperature, while the acid- and base-catalyzed rate constants are also influenced by pH. If chemical degradation by-products are hazardous and their contaminant-specific parameters are known, they can also be modeled in the simulation as part of a decay chain.

To model biological degradation, the Vadose Zone Module uses aerobic biodegradation rates and the Aquifer Module uses anaerobic rates. Both modules combine biological and chemical degradation rates to determine an overall decay rate. As a result, the modules cannot explicitly consider the separate effects of hydrolysis and biodegradation.

The Vadose Zone and Aquifer Modules take into account adsorption behavior of organic chemicals by calculating a retardation factor based on a contaminant-specific partition coefficient (K_d). To develop K_d, the modules apply a distribution coefficient normalized for organic carbon (K_{oc}), in conjunction with a media-specific fractional organic carbon content to obtain the soil (or aquifer material)/water partition coefficient (K_d). The use of K_{oc} to estimate sorption is appropriate for organic compounds that tend to sorb preferentially on the natural organic matter in the soil or aquifer. Although the Aquifer Module is limited to using linear K_d values, the Vadose Zone Module can use nonlinear Freundlich sorption coefficients if these are available for the organic compounds being modeled.

Additional details on modeling organic chemical reactions within the Vadose Zone and Aquifer Modules can be found in U.S. EPA (1996a,b; 1999e).

Organic Chemical Reaction Model Assumptions and Inputs. Key assumptions for the organic chemical reaction models used in the Vadose Zone and Aquifer Modules are summarized in the text box. Chemical-specific inputs include

- Acid-catalyzed, base-catalyzed, and neutral hydrolysis rates, specified for a default constant reference temperature of 25°C;
- First-order aerobic (vadose zone) and anaerobic (aquifer) biodegradation rates; and
- Partition coefficient normalized for soil organic carbon (K_{oc}).

Key assumptions and limitations of vadose zone and aquifer organic chemical reaction models

- The sorption of contaminants onto soil or aquifer solids occurs instantaneously and is entirely reversible.
- Sorption of organic compounds is linear in the saturated zone but can be nonlinear in the vadose zone.
- The model cannot explicitly consider the separate effects of multiple degradation processes.
- Biodegradation is aerobic in the unsaturated zone and anaerobic in the saturated zone.
- All transformation reactions are adequately represented by first-order decay processes.

When a multichemical simulation is desired for parent and daughter compounds, the necessary chemical-specific parameters must be repeated for all species in the decay chain.

Metal Sorption Processes. Site geochemistry and metal concentration are important determinants of the fate and transport of metals in the subsurface. Varying geochemical conditions result in the large variability in metal sorption behavior observed from site to site. Metals that sorb readily at low concentrations usually show much lower sorption at higher concentrations. To help capture this effect, the Vadose Zone and Aquifer Modules use sorption isotherms generated by the MINTEQA2 metal speciation model (Allison et al., 1991) that are nonlinear with respect to metal concentration. To represent nationwide variability in geochemistry, these concentration-dependent partition coefficients have been developed for various combinations of key geochemical parameters (MINTEQA2 master variables) known to affect metal sorption, including the pH, iron oxide content, and organic matter content of the subsurface environment, and the leachate organic matter content.

MINTEQA2 generates concentration-dependent effective partition coefficients (K_d values) for various combinations of the key geochemical parameters that are sampled during the Monte Carlo runs. Based on the selection of these parameters for each Monte Carlo realization, the Vadose Zone and Aquifer Modules call the appropriate sorption isotherm for use during transport modeling. The set of sorption isotherms included in the 3MRA modeling system has two subsets of isotherms for each metal: one representing vadose zone conditions and the other representing aquifer conditions.

In the Vadose Zone Module, metal partition coefficients are adjusted using these isotherms to represent the general increase in K_d that is expected to occur as contaminant concentrations decrease along the transport path. This procedure is described in the background document for the modeling of metals transport (U.S. EPA, 1996c). Because it is limited to linear sorption processes, the Aquifer Module selects a single K_d value from the isotherm that corresponds to the maximum ground water concentration under the source. This is appropriate because by the time the contamination reaches the Aquifer, metal concentration is usually low enough that a linear isotherm (i.e., a single K_d value not dependent on metal concentration) is appropriate.

The MINTEQA2 sorption isotherms implemented in the Vadose Zone and Aquifer Modules are a significant improvement over those used in previous EPACMTP versions in that they address a greater number of metals, including antimony, arsenic (+3 and +5 species), barium, beryllium, cadmium, chromium (+3 and +6), copper, lead, mercury, nickel, selenium, silver, thallium, vanadium, and zinc. This was made possible by improvements in MINTEQA2's thermodynamic, iron oxide sorption, and organic matter sorption databases (U.S. EPA, 1999a).

Additional details on the MINTEQA2 modeling approach, including model formulation, assumptions, and limitations, can be found in Allison et al. (1991) and U.S. EPA (1996c, 1999a, e).

9.3 Module Discussion

9.3.1 Strengths and Advantages

The Vadose Zone Module simulates the migration of constituents from the bottom of land-based WMUs to the water table, and the Aquifer Module simulates the migration of constituents in the saturated zone from the water table immediately below the WMU to the downgradient receptor wells. Both modules have been used in regulations and have been thoroughly tested and verified and validated as part of the integrated groundwater model EPACMTP. Strengths of the two modules include the following:

Strengths and advantages for the Vadose Zone Module are listed below:

- **Widely used, verified, validated, state-of-the-science approach.** The Vadose Zone Module is based on a state-of-the-science approach and is part of EPACMTP, which has been tested, verified, and validated. EPACMTP has been used to support regulations and has undergone various peer and public reviews, including reviews by the SAB.

- **Computational efficiency.** The solution procedure used in the Vadose Zone Module is computationally very efficient and stable and, therefore, ideally suited for use in Monte Carlo frameworks.

- **Use of nonlinear metal isotherms.** The module can handle nonlinear metal isotherms while maintaining computationally efficiency because of the solution technique used in the module.

- **Simulation of multiple transformation products.** The module can simulate the transformation of waste constituents into multiple transformation products (up to seven chain members).

Strengths and advantages for the Aquifer Module are listed below:

- **Widely used, verified, validated, state-of-the-science approach.** The Aquifer Module is based on a state-of-the-science approach and is part of EPACMTP, which has been tested, verified, and validated. EPACMTP has been used to support regulations and underwent various peer and public reviews, including reviews by the SAB.

- **Computational efficiency.** The module uses a special solution technique for the boundary value problem that makes it computationally very efficient and stable. The module is ideally suited for use in Monte Carlo framework in which numerous simulations are required using a minimum amount of computational time.

- **Complete transient response at multiple well locations.** The module provides complete transient response at multiple receptor well locations.

- **Simulation of multiple transformation products.** The module can simulate the transformation of waste constituents into multiple transformation products (up to seven chain members).

9.3.2 Uncertainty and Limitations

Limitations and uncertainties for the Vadose Zone Module are listed below:

- Transient effects of the flow are not considered (i.e., year-to-year variability in infiltration is not considered).

- Multiphase flow and transport are not modeled; NAPL flow and transport are not modeled.

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- Volatilization and vapor-phase diffusion are not modeled.

- Preferential flow due to fractures or heterogeneity in the vadose zone is not considered.

- Clay lenses or other potential flow and transport barriers in the vadose zone are not considered.

- Decay is limited to first-order reactions; lag time for decay is not considered.

- The transport domain in the saturated zone is kept constant. Effects due to mounding caused by infiltration from WMUs are not considered. These effects would decrease the depth of the flow and transport domain in the vadose zone.

Limitations and uncertainties for the Aquifer Module are listed below:

- Transient effects of ground water flow, recharge, and infiltration are not considered.
- Spatially varied recharge is not considered.
- Source geometry is limited to an idealized square, with two opposite sides parallel to the flow direction.
- Multiphase flow and transport are not modeled. NAPL flow and transport are not modeled.
- Contaminant contribution to the saturated zone via vapor-phase diffusion above the water table is not modeled.
- Karst conditions are not modeled.
- Decay is limited to first order. Lag time for decay is not considered.
- The presence of different hydrogeologic zones in the flow and transport domain is not considered.
- The transport domain in the saturated zone is kept constant. Effects due to significant mounding caused by infiltration from WMUs are not considered.
- Domain geometry is limited to the idealized rectangular shape. Other geometries are not considered.
- Only gaining streams, with axes normal to the ground water flow direction, are permitted. Effects of streams on the flow field are not considered.
- Only receptor wells with small extraction rates are considered. Effects of well extraction on the ground water flow field are not considered.
- There are many sources of uncertainty associated with the distribution coefficients generated by the metal speciation model. These can be categorized as uncertainty arising from model input parameters, uncertainty in database equilibrium constants, and uncertainty due to application of the model. Additional details can be found in the MINTEQA2 background documents (Allison et al., 1991; U.S. EPA, 1991a, c).

9.4 References

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10.0 Farm Food Chain Module

10.1 Purpose and Scope

The Farm Food Chain Module predicts the accumulation of contaminants in the edible parts of plants through the uptake of constituents from soil and the deposition of vapor-phase and particle-bound constituents from the air. Concentrations are predicted for fruits and vegetables that are grown above ground, as well as for root vegetables. In addition, the module predicts the annual average contaminant concentrations in beef and milk products from cattle raised on farms. The concentrations in produce, beef, and milk are used as inputs to the Human Exposure Module to calculate the applied daily dose to human receptors that consume fruits and vegetables from home gardens or consume produce, beef, or milk produced on a farm. Figure 10-1 shows the relationship and information flow between the Farm Food Chain Module and the 3MRA modeling system.

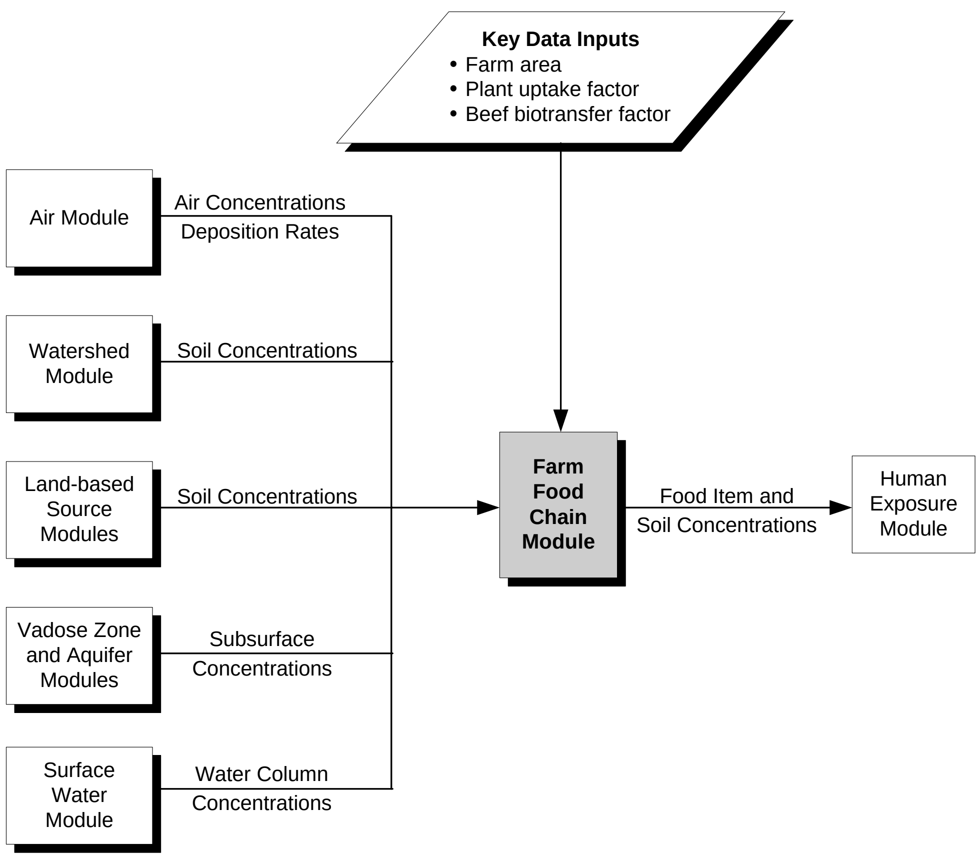

Figure 10-1. Information flow for the Farm Food Chain Module in the 3MRA modeling system.

The modeling construct for the Farm Food Chain Module is based on recent and ongoing research conducted by EPA ORD and presented in *Methodology for Assessing Health Risks Associated with Multiple Pathways of Exposure to Combustor Emissions* (U.S. EPA, 1998).

The Farm Food Chain Module performs the following four functions:

1. **Calculates contaminant concentrations in plants due to contaminants in air.** The Farm Food Chain Module calculates the contaminant concentration in plants due to the deposition of particle-bound and vapor-phase contaminants onto fruits, vegetables, and feed crops that grow above the ground.

2. **Calculates contaminant concentrations in plants due to contaminants in soil.** The Farm Food Chain Module calculates the contaminant concentration in plants due to uptake and translocation of contaminants in the soil into the edible parts of fruits, vegetables, and feed crops that grow above the ground.

3. **Calculates total contaminant concentrations in plants.** The Farm Food Chain Module sums plant concentrations across relevant mechanisms, including direct deposition of particle-bound contaminants, vapor-phase uptake, and translocation of contaminants from soil into the edible parts of plants.

4. **Calculates contaminant concentrations in beef and milk.** The Farm Food Chain Module calculates exposures to beef and dairy cattle through ingestion of contaminated forage, feed crops, soil, and drinking water, and the resulting beef and milk concentrations.

For each year in the simulation, the module predicts point estimates and spatially averaged contaminant concentrations within the area of interest (AOI). Point estimate concentrations are used to evaluate exposures of residential home gardeners that grow and eat fruits and/or vegetables within the AOI. The point estimates reflect the concentrations at locations of residential receptors that are used to represent the populations in various Census tracts throughout the AOI. The spatially averaged concentrations are used to evaluate exposures of farmers that raise and eat their own produce, beef, or milk products. The spatial averages reflect the concentrations at the farm boundaries delineated in the site layout that defines all of the characteristics of the AOI.

For both farmers and residential receptors, the Farm Food Chain Module predicts a time series of annual average concentrations of contaminants in fruits and vegetables.[1] With the exception of root vegetables, the categories of fruits and vegetables are designated as either "exposed" or "protected." For the farm food chain modeling framework, the term "exposed" means that the contaminant concentrations in plants are calculated based on uptake and accumulation from aerial deposition, uptake during transpiration, and uptake from the soil. The

[1] Botanically, "fruit" refers to any part that develops from the flower (i.e., reproductive parts) and includes what are considered fruits as well as vegetables in terms of grocery items. For example, tomatoes, corn, and beans all develop from the flower and are thus all considered fruits in botanical terms. In the context of this discussion, the conventional grocery item terminology is used.

term “protected” means that the outer covering of the fruit or vegetable is not edible and serves as a barrier to contaminant transfer from air; therefore, the contaminant concentration in plants for protected fruits and vegetables is attributed only to the uptake of contaminants from the soil through the root system and subsequent translocation into the edible portions of the plant. Each of these categories is explained in further detail below.

- **Exposed vegetables.** The term “exposed vegetables” refers primarily to fruiting plants and edible vegetative parts of plants that are exposed directly to contaminant loadings from the air. For example, cucumbers, tomatoes, peppers, and green beans are included in the category of exposed vegetables, although all of these develop from flowers and are thus considered fruits in botanical terms.

- **Protected vegetables.** This category of vegetables includes plants with the edible part protected from airborne contaminants by a nonedible covering. For example, peas and lima beans are considered protected vegetables because their outer pod is not eaten, so the plant load in the edible portion is exclusively due to soil-to-plant transfer of contaminants. Aerial deposition is not included in calculating contaminant concentrations in protected vegetables.

- **Exposed fruits.** Exposed fruits include produce commonly referred to as “fruits” that have an edible outer skin. This category includes a number of tree fruits (e.g., apples, pears) and various vine fruits (e.g., strawberries, grapes). The plant concentration is a function of both the soil concentration and the concentration of contaminants in the air.

- **Protected fruits.** Protected fruits have limited relevance in many exposure scenarios because they include fruits with a fairly narrow range of growing climates. Although a variety of melons can be grown throughout much of the contiguous United States, other protected fruits, such as bananas, pineapples, and citrus fruit, are found only in the warmest regions of the country, such as Florida. As with protected vegetables, the concentration in the edible portion is exclusively due to soil-to-plant transfer of contaminants.

- **Root vegetables.** This category cuts across a number of botanical categories, including leaves, stems, and roots. The common theme for this category is that all of these vegetables grow below the soil surface. Consequently, a host of vegetables, such as potatoes (stem), onions (leaf), and carrots (root), are lumped together in this category. As with protected fruits and vegetables, the concentration in root vegetables is predicted based on the contaminant concentrations in soil.

In addition to calculating the concentrations in produce, the Farm Food Chain Module also generates a time series of annual average contaminant concentrations in beef and milk for each farm within the AOI. The beef and milk concentrations are calculated for beef and dairy cattle, respectively, and include three exposure pathways for cattle: (1) the consumption of contaminated feed crops (i.e., forage, grain, and silage), (2) incidental ingestion of contaminated soil while foraging, and (3) ingestion of contaminated drinking water from streams, ponds, or wells on the farm.

10.2 Conceptual Approach

As shown in Figure 10-2, contaminants may be released from waste management units (WMUs) into the air and subsequently deposit on soil and plants through wet and dry deposition. Plants may accumulate contaminants directly from the air through the deposition of both particle-bound and vapor-phase contaminants. In addition, this deposition may increase the contaminant concentrations in the soil over time, and plants may take up contaminants from the soil through the root system. Contaminants may be translocated as part of the transpiration stream to fruits and edible vegetative parts, or they may sorb directly to the outer skin of root vegetables. The Farm Food Chain Module performs the calculations for each of these uptake and accumulation mechanisms. The total contaminant concentration in plants will depend both on the category of plant being modeled (i.e., exposed versus protected) and the properties of the contaminant of concern. For example, the uptake and accumulation of dioxin-like chemicals from soil to edible parts of plants has been shown to be negligible because these chemicals are strongly sorbed to the organic fraction of soil. Because the behavior of each contaminant is, to a large degree, determined by its chemical properties, the Farm Food Chain Module uses both empirical data and regression algorithms to predict the uptake and accumulation of contaminants from environmental media into edible plant tissue.

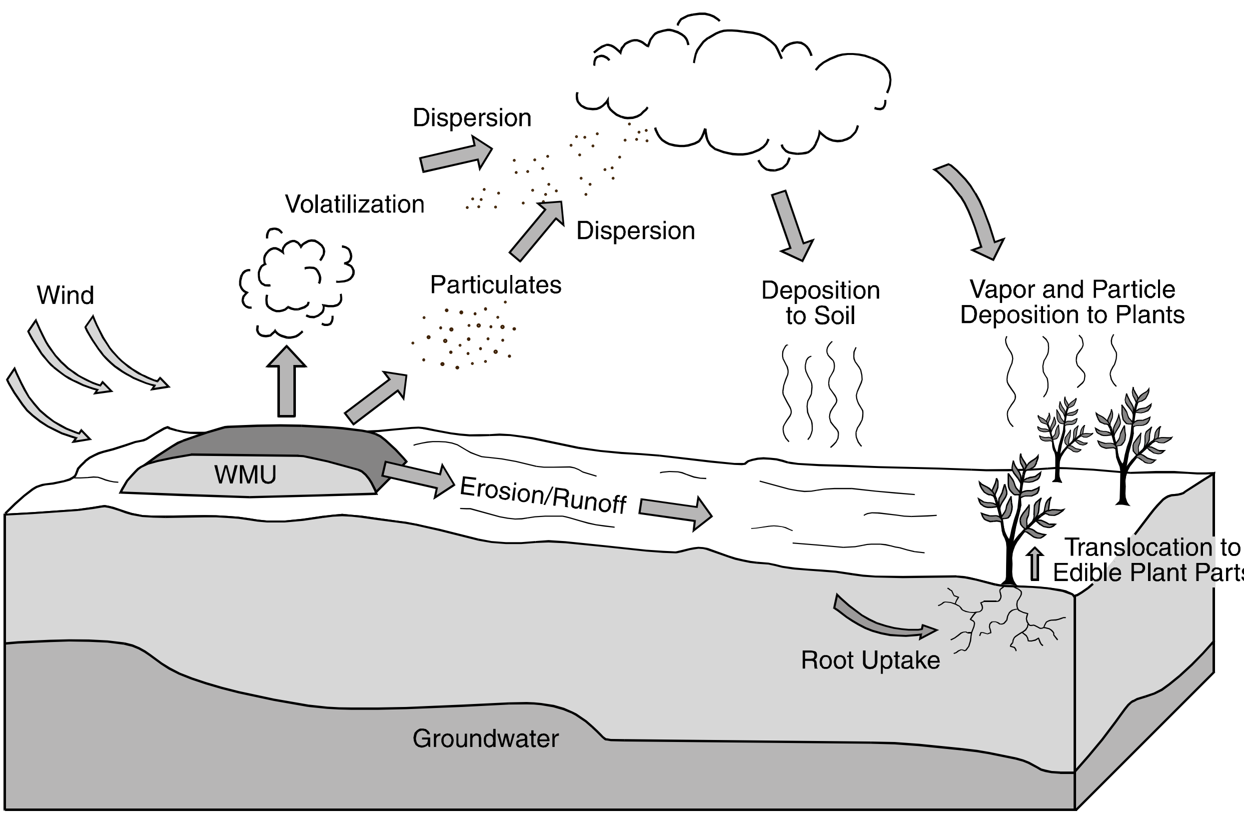

Figure 10-2. Release, exposure, and uptake mechanisms of contaminants in plants.

10.2.1 Calculate Contaminant Concentrations in Plants due to Contaminants in Air

Contaminants in the air may accumulate on the surface of plants by two mechanisms: particle-bound deposition and vapor-phase deposition. The algorithms used to simulate these processes and predict contaminant concentrations in exposed fruits, exposed vegetables, forage, and silage due to contaminants in air are described below. These processes are not considered for protected fruits, protected vegetables, or root vegetables.

The total plant load from air may be driven by either particle or vapor deposition, depending on the contaminant of concern. For example, metals released from a land application unit (LAU) can be deposited onto plant surfaces through particle-bound but not vapor-phase deposition. Therefore, the Farm Food Chain Module must include both of these mechanisms in calculating the contaminant load to plants from air. The deposition rates are calculated by the Air Module for the entire AOI; the areal average deposition rates for farms and average point estimates for residential receptor locations are provided by year.

Throughout this subsection, the subscript *i* refers to the following plant types:

- Exposed vegetables,
- Exposed fruits,
- Forage,
- Grains, and
- Silage.

Plant Concentrations from Deposition of Particle-Bound Contaminants. The plant concentration due to deposition of particle-bound contaminants in the air is calculated as a function of wet and dry deposition rates as follows:

$$C_{plant-dep^i} = \frac{1000 \times 365 \left(ParDryDep + ParWetDep \times FracAdhere^i \right) \times FracInt^i \times \left(1.0 - \exp^{\left(-kpPar^i \times LengthEx \right)} \right)}{Yield^i \times kpPar^i} \quad (10\text{-}1)$$

where

$C_{plant\_dep}^{i}$	=	concentration in plant *i* due to particulate deposition (mg/kg DW)
1000	=	units conversion factor (1,000 mg/g)
365	=	units conversion factor (365 d/yr)
ParDryDep	=	average dry deposition rate of particulates (g/m^2-d)
ParWetDep	=	average wet deposition rate of particulates (g/m^2-d)
$FracAdhere^i$	=	fraction of wet deposition that adheres to plant *i* (unitless)
$FracInt^i$	=	interception fraction for plant *i* (unitless)
$kpPar^i$	=	surface loss of particle-bound contaminant for plant *i* (1/yr)
$LengthEx^i$	=	length of plant exposure during the growing season for plant *i* (yr)
$Yield^i$	=	yield or standing crop biomass for plant *i* (kg DW/m^2).

The contaminants that reach the plant through dry deposition are assumed to remain on the plant surface until weathering occurs. In contrast, only a fraction of the contaminant from wet deposition remains on the plant's surface; the rest washes off immediately. This is reflected by the variable for the fraction of wet deposition that adheres to a plant's surface.

Not all airborne particles will contact a plant's edible surface; some will fall to the ground, others will fall on other surfaces that will undergo weathering processes, such as wind removal, water removal, and growth dilution, and most will end up in the soil or runoff. The interception fraction represents the fraction of airborne contaminants that make contact with the surface of the plant. The plant's surface can also lose contaminants due to weathering, and the aggregate losses from the plant surface are represented by the surface loss coefficient.

The duration of exposure for the plant is often determined by the growing season and is represented by the length of exposure of the plant to contaminants in the air. For instance, the exposure duration for a tomato begins when the flower begins to ripen and lasts until the tomato is harvested.

Finally, the plant biomass or yield represents the amount of standing crop assumed for a farm or residential garden, as appropriate.

Plant Concentrations from Deposition of Vapor-Phase Contaminants. As with particle-bound contaminants, vapor-phase deposition to plant surfaces occurs through wet and dry deposition processes. The mechanisms for wet and dry deposition are, to a large degree, dependent on the properties of the contaminant. Plant concentrations due to deposition of vapor-phase contaminants can be calculated two ways. For many organic chemicals, the form of the equation is essentially the same as the equation used to predict plant concentrations from particle-bound contaminants, as follows:

$$C_{plant-vap^i} = \frac{1000 \times \left(VapDryDep + VapWetDep \times FracAdhere^i \times 365\right) \times FracInt^i \left(1 - e^{\left(-kpVap \times LengthEx^i\right)}\right)}{Yield^i \times kpVap^i} \quad (10\text{-}2)$$

where

$C_{plant\_vap}{}^{i}$	=	concentration in plant *i* due to vapor deposition (mg/kg DW)
1000	=	units conversion factor (1,000 mg/g)
365	=	units conversion factor (365 d/yr)
VapDryDep	=	average dry deposition rate of vapor-phase contaminants (g/m^2-yr)
VapWetDep	=	average wet deposition rate of vapor-phase contaminants (g/m^2-d)
$FracAdhere^i$	=	fraction of wet deposition that adheres to plant *i* (unitless)
$FracInt^i$	=	interception fraction for plant *i* (unitless)
$kpVap^i$	=	surface loss of vapor-phase contaminant for plant *i* (1/yr)
$LengthEx^i$	=	length of plant exposure during the growing season for plant *i* (yr)
$Yield^i$	=	yield or standing crop biomass for plant *i* (kg DW/m^2).

The variables have the same function as previously described for the particulate-phase equation above.

For organic chemicals with a high affinity for lipid tissue, the mechanism of uptake and accumulation of vapor-phase organics for plants is not accurately described by the term "deposition." Evidence has shown that these compounds can be essentially stripped from the air simply by coming in contact with vegetation. An alternative model for the dry deposition of vapor-phase organic compounds is termed the "transfer" approach. This model describes the plant concentration in terms of air-to-plant transfer rather than physical deposition on plant surfaces. Further, EPA has demonstrated that wet deposition of vapor-phase lipophilic compounds can be considered negligible. Consequently, the plant concentration for lipophilic organic chemicals (functionally defined as those contaminants with an octanol-water partition coefficient greater than or equal to 100,000) is calculated as follows:

$$C_{plant\_vap^i} = \frac{Cair_{vapor} \times BTF_{air-plant^i} \times ECF_{exposed^i}}{1000 \times \rho_{air}} \tag{10-3}$$

where

$C_{plant\_vap^i}$	=	concentration in plant *i* due to vapor transfer into plant (mg/kg DW)
C_{air_vapor}	=	vapor-phase concentration in air (µg/m^3)
$BTF_{air\_plant^i}$	=	contaminant-specific air-to-plant biotransfer factor for plant *i* ([µg/g DW]/[µg/g air])
$ECF_{exposed^i}$	=	empirical correction factor to convert the air-to-plant biotransfer factor derived for leafy vegetation to a value for plant *i* (unitless)
1,000	=	units conversion factor (L/m^3)
ρ_{air}	=	density of air (constant at 1.19 g/L).

The dry deposition rate of vapor-phase, organic chemicals is not currently calculated by the Air Module. Rather, it is estimated from the vapor-phase concentration in air using the methods presented in U.S. EPA (1998), as follows:

$$VapDryDep = 0.31536 \times CvAve \times VapDdv \tag{10-4}$$

where

VapDryDep	=	average dry deposition rate of vapors (g/m^2-yr)
0.31536	=	units conversion factor ((m/yr)/(cm/s) and (g/µg))
CvAve	=	vapor-phase concentration in air (µg/m^3)
VapDdv	=	vapor phase dry deposition velocity - default value of 1 (cm/s).

The air-to-plant biotransfer factor describes the relationship between the contaminant concentration in exposed plant parts and the vapor-phase contaminant concentration in air. It may be calculated from chemical-physical properties (as it is for most organic chemicals), or it may be derived using empirical data in the open literature or EPA sources. The empirical correction factor (ECF) for each plant category *i* reflects the fact that experiments have shown that lipophilic, persistent organics such as polycyclic aromatic hydrocarbons (PAHs) and dioxins tend not to translocate from the outer surfaces of plants to inner plant parts. Without this adjustment, the biotransfer factor would be inappropriately high (i.e., it would overpredict the contaminant concentration in the inner parts of the plant). The ECF may be further adjusted to account for washing and peeling of fruits and vegetables that reduce contaminant residues in the outer skin. However, the contaminant-specific data set available to the 3MRA modeling system does not currently include washing and peeling losses in the empirical adjustment factor for aboveground plants. For many organic chemicals, the correction factor is simply set to 1, indicating that the contaminant is efficiently translocated to the inner plant parts. See Volume II of this report for further details.

10.2.2 Calculate Contaminant Concentrations in Plants due to Contaminants in Soil

Contaminant releases to soil from WMUs may occur through aerial deposition, as well as through erosion and runoff mechanisms. Over time, the contaminant concentration in soil may increase, with a resultant increase in the contaminant concentrations in the edible parts of plants. Depending on the properties of a given contaminant and the physiology of exposed vegetation, plants may take up aqueous-phase contaminants in the soil through the roots and subsequently translocate them to edible plant parts. The plant concentration is a function of the contaminant concentration in the root zone soil and the soil-to-plant bioconcentration factor, as follows:

$$C_{plant\_soil}{}^{i} = C_{soil\_RZ} \times BCF_{soil-plant}{}^{i} \tag{10-5}$$

where

$C_{plant\_soil^i}$	=	plant concentration uptake from soil and translocation (mg/kg DW)
C_{soil_RZ}	=	depth-averaged soil concentration in the root zone (µg/g soil)
$BCF_{soil\text{-}plant^i}$	=	contaminant-specific soil-to-plant bioconcentration factor ([µg/g DW]/[µg/g soil]).

The root zone soil concentration is area averaged for farms. The soil-to-plant bioconcentration factor quantifies the potential for a contaminant in soil to be taken up by the roots and translocated to edible plant parts. As with the air-to-plant biotransfer factor, the soil-to-plant bioconcentration factor may be calculated from chemical-physical properties for most organic chemicals, or derived from empirical data for metals (see Volume II for more details). Equation 10-5 is appropriate for metals for all categories of plants (e.g., exposed fruits, protected vegetables, forage) and for organic chemicals for all categories of plants except root vegetables. However, for certain types of chemicals, such as dioxins, studies have shown that this pathway for uptake and accumulation of contaminants in plants is negligible; that is, the soil concentration does not correlate with the contaminant concentration in plant tissue. For dioxin-like chemicals, the concentration in exposed fruits and vegetables is exclusively a function of air-to-plant transfer and particle deposition.

For root vegetables, the mechanism by which plants take up organic chemicals from the soil and accumulate them in edible tissue is not adequately described by Equation 10-5. Experiments suggest that contaminant concentrations found in the outer parts of the vegetable, or skin, are due more to sorption than to passive uptake via transpiration water (U.S. EPA, 1998). Although water-soluble contaminants may be distributed more or less uniformly throughout the entire plant, lipophilic contaminants may be bound almost exclusively to the outer portion of the root vegetable. The relationship between soil pore water and plant concentrations is expressed as a root concentration factor, which is used to predict the contaminant concentration in root vegetables as follows:

$$C_{Root\_Veg} = \frac{C_{soil\_RZ} \times RootCF \times ECF_{RootVeg}}{K_{d_{soil}}} \tag{10-6}$$

where

C_{Root_Veg}	=	contaminant concentration in root vegetables (mg/kg WW)
C_{soil_RZ}	=	depth-averaged soil concentration in the root zone (µg/g soil)
RootCF	=	root concentration factor ([µg/g WW plant]/[µg/mL soil water])
$ECF_{RootVeg}$	=	empirical correction factor for root vegetables (unitless)
K_{d_soil}	=	soil-water partition coefficient (mL/g).

The soil-to-plant bioconcentration factor (in Equation 10-5) is replaced with the root concentration factor, which quantifies the potential for organic chemicals in soil pore water to accumulate in root vegetables. An empirical correction factor is included to adjust for differences between the experimental data and the application in the 3MRA modeling system. For example, much of the experimental data on the root concentration factor is based on whole barley roots, and a contaminant-specific correction factor is needed to adjust for the volume differences between barley roots and bulky root vegetables. The empirical correction factor may be further adjusted for peeling, cooking, or cleaning, which can all reduce the contaminant concentration. However, the chemical-specific data set available to the 3MRA modeling system does not currently include these losses in the empirical adjustment factor for root vegetables.

10.2.3 Calculate Total Contaminant Concentrations in Plants

The Farm Food Chain Module calculates the total contaminant concentration in plants for exposed fruits, exposed vegetables, forage, and silage by summing the concentrations for all potential exposure pathways for plants. This summation step is not needed for protected fruits and vegetables or root vegetables, because they take up contaminants only from the soil. For all aboveground produce, the total contaminant concentrations in plants are converted from dry weight (DW) to wet weight (WW) by adjusting for the moisture content. These WW concentrations (also referred to as whole weight or fresh weight) are required by the Human Exposure Module. This conversion is not needed for root vegetables, because the concentration is calculated in WW directly.

10.2.4 Calculate Contaminant Concentrations in Beef and Milk

The Farm Food Chain Module uses the predicted soil, plant (i.e., forage, grains, and silage), and drinking water concentrations to predict the contaminant concentrations in beef and milk for animals raised on beef and dairy farms within the AOI. The exposure scenario is based on the assumption that beef and dairy cattle consume some fraction of forage, feed grain, and silage that is grown on the farm. Furthermore, cattle are presumed to ingest surface water from stream reaches or ponds that are delineated on the farm; where no such waterbody exists, ground water is assumed to be the drinking water source for cattle. The incidental ingestion of soil refers only to the surficial soil (i.e., the top 1 cm of soil) that may be ingested during normal grazing by cattle on untilled soil. All of the exposure concentrations reflect the spatial averaging across each individual farm modeled within the AOI and a consistent temporal scale; that is, the time series of input concentrations for plants, soil, and water are matched for each year in the simulation.

The contaminant concentrations in beef or milk are calculated as a function of the biotransfer of contaminants from plants, soil, and water into beef and milk as follows:

$$C_{beef/milk} = BTF_{feed} \times \left[\sum_i \left(C_i \times CR_i \times f_i \right) + \left(C_{soil} \times CR_{soil} \times FracBio_{soil} \right) \right] + BTF_{water} \times \left(C_{water}{}^{d} \times IR_{water} \right) \quad (10\text{-}7)$$

where

$C_{beef/milk}$	=	contaminant concentration in beef or milk (mg/kg WW)
BTF_{feed}	=	contaminant biotransfer factor from feed into beef or milk (d/kg tissue WW)
C_i	=	contaminant concentration in plant type *i* (forage, grain, or silage) grown on farm (mg/kg DW)
CR_i	=	consumption rate of plant type *i* for beef or dairy cattle (kg plant DW/d)
f_i	=	fraction of plant type *i* grown on farm (unitless)
C_{soil}	=	contaminant concentration in surficial soil at farm (mg/g soil)
CR_{soil}	=	quantity of contaminated soil eaten by beef or dairy cattle (kg soil/d)
$FracBio_{soil}$	=	fraction of contaminant in soil bioavailable relative to vegetation (unitless)
BTF_{water}	=	contaminant biotransfer factor from water into beef or milk (d/kg tissue WW)
$C_{water}{}^{d}$	=	dissolved contaminant concentration in drinking water (mg/L)
IR_{water}	=	drinking water ingestion rate for beef or dairy cattle (L/d).

Biotransfer from the soil (the second term in brackets in Equation 10-7) is adjusted by the fraction of bioavailable contaminant in the soil to account for differences in absorption efficiency when contaminant is ingested through plant matter versus through contaminated soil. For some contaminants, vegetation is a more efficient vehicle for biotransfer because the breakdown and digestion of plant matter releases a higher fraction of contaminant available for absorption in the gut. Data on the biotransfer of dissolved contaminants in water into beef and milk were not identified. Nevertheless, because of the potential importance of this pathway, the biotransfer factor for water is set equal to the biotransfer factor for feed for beef and milk, respectively.

The values for biotransfer factors, consumption rates, and fraction contaminated food items differ for beef and dairy cattle. For example, dairy cattle tend to have a much higher water ingestion rate to support lactation; therefore, the ingestion rate of water for dairy cattle is higher than the value for beef cattle.

10.3 Module Discussion

10.3.1 Strengths and Advantages

The Farm Food Chain Module was developed to (1) predict contaminant concentrations in produce and feed crops from multiple routes of plant uptake, and (2) estimate concentrations in beef and milk for cattle raised on farms within the AOI. Relative to other approaches that were considered for the 3MRA modeling system (e.g., fugacity-based models such as PlantX), the strengths and advantages of the Farm Food Chain Module include the following:

- **Widely used and reviewed approach.** The Farm Food Chain Module was based on the science, algorithms, and data developed by EPA's ORD to assess indirect exposures to contaminated produce, beef, and milk. This approach has been widely used by the EPA regions as well as other EPA program offices, and represents a considerable investment by EPA in indirect exposure assessment. Consequently, the approach has been reviewed in a variety of different applications and corresponds well with the categories of produce (e.g., protected vs. unprotected vegetables) that are needed to assess human health exposures.

- **Simulates major plant uptake processes for contaminants in air and soil.** The contaminant concentration in plants is a function of all relevant pathways given the type of plant and the chemical properties of the contaminant. The concentration is summed across uptake for soil as well as uptake from the air, as appropriate, for each contaminant/plant combination. Because some of the WMUs release contaminants into the air, it is crucial that the model accounts for the contaminant mass taken up directly from the air and deposited on the soil from the air and taken up via the roots.

- **Calculates point estimates for home gardeners and areal average estimates for farmers.** The Farm Food Chain Module can calculate the contaminant concentrations both in plants at points specifically designated in the site layout file (i.e., home gardeners) as well as spatially averaged across farm areas within the AOI. The averaging functions in the module include calculating contaminant concentrations in soil that includes both the regional and local watersheds, and air concentrations of particulates and vapor in the same spatial area and time frame. Therefore, the contribution to the total plant concentration from air and soil reflects the spatial character of the plant exposure through time such that the plant loads from soil and from air can be meaningfully added together.

- **Data requirements and module performance consistent with the overall design of the 3MRA modeling system.** The data requirements for the Farm Food Chain Module are relatively limited with respect to plant physiology. As a result, a wide variety of chemical contaminants may be evaluated without requiring inputs for individual plant species included under the categories presented in Section 10.1 (e.g., exposed vs. protected vegetables), or for specific compartments within the plants. Similarly, the module is computationally efficient so that national-scale analyses involving time-varying concentration profiles in air, surface water, ground water (for ingestion by cattle), and soil are feasible in a reasonable run time. Because the system simulations may span many years, and because the system was designed to support full Monte Carlo implementation, it was critical that the module be able to produce results within a time frame that would be useful to the user.

10.3.2 Uncertainty and Limitations

The technical approach underlying the Farm Food Chain Module includes a number of assumptions, uncertainties, and limitations. Limitations include the following:

- **System constraints limit each simulation to a single WMU and contaminant**. The 3MRA modeling system was designed to simulate a single contaminant for a single WMU. Consequently, multiple plant loadings of a contaminant from different WMUs (either within or outside of the AOI) are not considered. This assumption may result in an underprediction of the contaminant concentrations in produce, beef, and milk.

- **Farm Food Chain Module is driven by empirical data.** The Farm Food Chain Module uses empirically derived regression algorithms, as well as empirical data on uptake and accumulation, to predict contaminant concentrations in plants, beef, and milk. Although other models that were evaluated have several advantages over empirical models (for example, the compartmental model developed by Trapp and MacFarlane, 1995), many of these models were considered too data intensive or too computationally demanding for use in a modeling tool intended to support national-level analyses. The reliance on empirical data and models limits the model's ability to predict contaminant concentrations outside of the narrow range for which data are available.

- **Resuspension and redeposition of particle-bound contaminants are not considered.** Plant concentrations are a function of the deposition of the contaminants that have been emitted from the WMU. The plant concentration due to particle deposition does not include resuspension and redeposition of contaminants bound to soil particles. Resuspension and redeposition can occur due to tillage, wind erosion, vehicular resuspension, and rainsplash and can increase the contaminant concentration in exposed fruits, exposed vegetables, forage, and silage.

- **Contaminant concentrations in beef and milk consider only the ingestion pathway.** The beef and milk calculations consider only ingestion pathways for plant matter, surficial soil, and drinking water; exposures from the inhalation or dermal contact are not included in the calculations. For some contaminants, this limitation may result in an underestimate of the contaminant concentrations in beef and milk.

10.4 References

Trapp, S. and J. Craig Mc Farlane, 1995. Plant Contamination: Modeling and Simulation of Organic Chemical Processes. CRC Press, Boca Raton.

U.S. EPA (Environmental Protection Agency). 1997. *Parameter Guidance Document.* EPA, National Center for Environmental Assessment, NCEA

U.S. EPA (Environmental Protection Agency). 1998. *Methodology for Assessing Health Risks Associated with Multiple Exposure Pathways to Combustor Emissions*. EPA 600/R-98/137. National Center for Environmental Assessment, Cincinnati, OH. December.

11.0 Terrestrial Food Web Module

11.1 Purpose and Scope

The Terrestrial Food Web Module calculates the annual average contaminant concentrations in terrestrial plants and prey (such as earthworms or small mammals) eaten by wildlife. In addition, the module calculates spatially-averaged soil concentrations in two soil horizons— surficial soil and root zone soil—for the receptor home ranges placed within each of the habitats delineated within the area of interest (AOI). These concentrations are used as input to the Ecological Exposure Module to calculate the applied dose to receptors of interest, and the root zone soil concentration is also used by the Ecological Risk Module to predict risks to terrestrial plants and soil communities. Figure 11-1 shows the relationship and information flow between the Terrestrial Food Web Module and the 3MRA modeling system.

For each home range delineated within the AOI, the Terrestrial Food Web Module predicts a time series of annual average concentrations of contaminants in soil, along with concentrations in the specific plant and prey categories shown in Table 11-1. The Terrestrial Food Web Module uses the same algorithms and contaminant-specific data as the Farm Food Chain Module to calculate concentrations in plants.

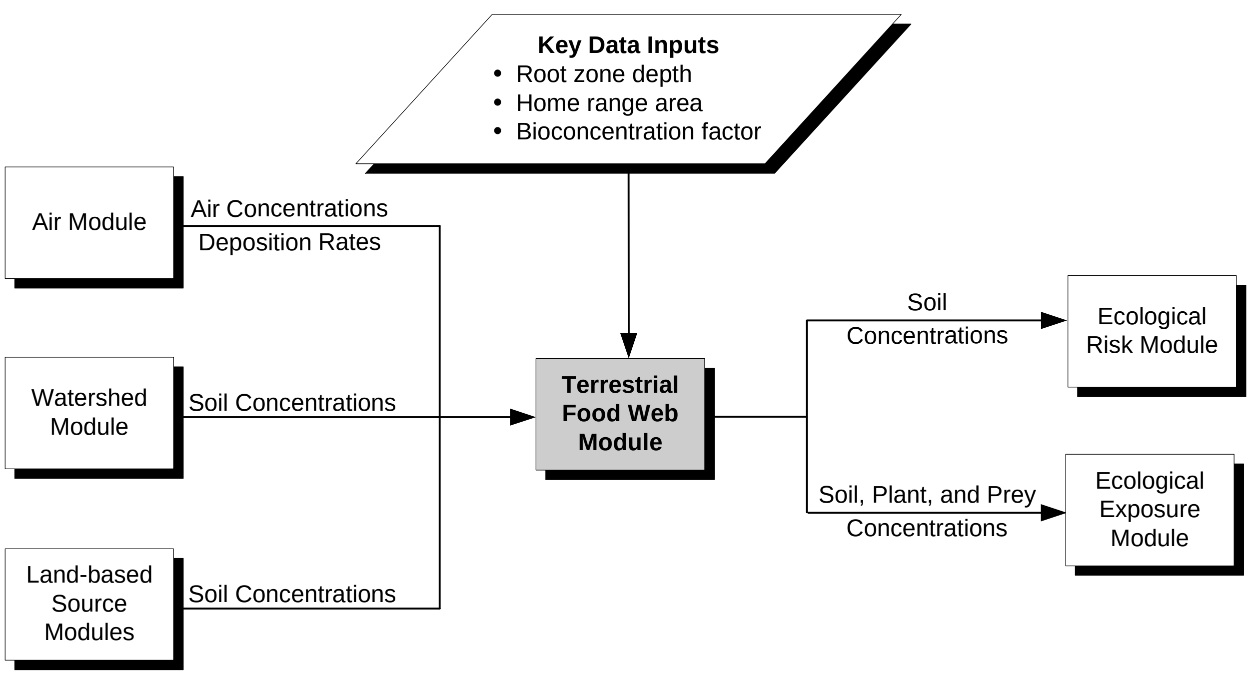

Figure 11-1. Information flow for the Terrestrial Food Web Module in the 3MRA modeling system.

Table 11-1. Terrestrial Plant and Prey Categories in the Terrestrial Food Web

Terrestrial Plant Categories	Terrestrial Prey Categories
Fruits, fruit/seeds (*exposed fruit*)	Worms
Fern(s), fungi, shoots (*exposed vegetation*)	Other soil invertebrates
Forbs, grasses, shrubs (*forage*)	Small mammals
Roots (*root vegetables*)	Small birds
Crops, corn (*silage*)	Small herpetofauna
Seeds/nuts (*grains*)	Herbivorous vertebrates
	Omnivorous vertebrates

The parentheticals shown under the terrestrial plant categories provide the crosswalk between categories of fruits, vegetables, and forage discussed under the Farm Food Chain Module, and the analogous vegetation categories that are eaten by wildlife.

The Terrestrial Food Web Module uses both predictive models and empirical data to calculate contaminant concentrations in terrestrial food items. Specifically, the Terrestrial Food Web Module performs the following four functions:

1. **Calculates contaminant concentrations in soil.** The module calculates spatially averaged soil concentrations for each home range in each habitat. The soil concentrations reported for each site by the Land-based Source Modules and the Watershed Module are catalogued by waste management unit (WMU) and watershed subbasin, respectively. The Terrestrial Food Web Module determines the overall spatial average contaminant concentration based on the proportion of subbasins and/or WMU that overlaps the home range of wildlife species assigned to a given habitat.

2. **Calculates total contaminant concentrations in plants.** The Terrestrial Food Web Module sums plant concentrations across relevant mechanisms, including direct deposition of particle-bound contaminants, vapor-phase uptake, and translocation from soil into the edible parts of plants. The module uses the same approach as described for the Farm Food Chain Module to calculate total contaminant concentrations in plants. The only significant difference in the respective modules is that the Terrestrial Food Web Modules defines plant categories for consumption by wildlife rather than humans.

3. **Calculates contaminant concentrations in soil invertebrates.** Earthworms and other soil invertebrates constitute a significant dietary component for many wildlife species. Contaminant-specific soil-to-biota bioconcentration factors are used to calculate the contaminant concentrations in soil invertebrates based on the spatially averaged, root zone soil concentration for each wildlife species' home range.

4. **Calculates contaminant concentrations in vertebrate prey categories.** Small to medium sized terrestrial vertebrates are eaten by larger predators (e.g., fox, black bear, and red tailed hawk). Contaminant-specific soil-to-vertebrate bioconcentration factors are used to estimate the tissue concentrations of contaminants in vertebrate prey categories based on the spatially averaged, root zone soil concentration for each wildlife species' home range.[1] The module reports the minimum and maximum tissue concentrations in each vertebrate prey category; these outputs allow the Ecological Exposure Module to sample from the spatial variability of possible prey species.

The Terrestrial Food Web Module is applied to the margin habitats in freshwater systems and wetlands and to terrestrial habitats, such as the representative forest or grassland. Simple terrestrial food webs were constructed for each terrestrial habitat to depict the major functional and structural components of healthy terrestrial ecosystems. The components represent major dietary categories for terrestrial wildlife, and represent a broad range of sizes, feeding guilds, and taxa typical of terrestrial food webs. In addition, the terrestrial prey categories reflect available methods and data required to estimate prey concentrations. For example, the category of "small mammals" was developed specifically to take advantage of recent research (Sample et al., 1998) on estimating tissue concentrations of contaminants in small mammals as a function of the soil concentration. This research provides a significant improvement over previous approaches that aggregated a variety of prey items into a single category of "terrestrial invertebrates."

11.2 Conceptual Approach

The Terrestrial Food Web Module was developed to allow for flexibility in calculating contaminant concentrations in soil, plants, and prey eaten by wildlife. Despite the advantages in such a flexible design, it was critical that the Terrestrial Food Web Module perform spatial averaging in a manner that was both ecologically relevant and simple to implement. Therefore, the module was designed to calculate contaminant concentrations in food items and soil in a way that reflects the spatial scale of interest and conserves the role of various species as both predator and prey. Most importantly, the approach takes full advantage of the information available in the site layout file that defines the spatial character of the habitats and species' home ranges assigned to the AOI. Indeed, the habitats and home ranges are a key feature of the 3MRA modeling system ecological risk assessment framework, describing the spatial extent of exposure as well as providing an important metric for reporting ecological risks.

As described in Section 3, the ecological setting for each site layout file includes one or more habitats. Within each habitat, four home range sizes are delineated, and the receptor-specific home ranges are assigned to the smallest one of these four home ranges. For example, the home range of the short-tailed weasel (135,000 m2) fits into home range 3, but not home ranges 2 or 1. Therefore, the short-tailed weasel is always assigned to home range 3. The receptor-specific home ranges were simplified to consist of four size ranges because the current

[1] Root zone, rather than surficial, soil concentrations were used to be more representative of the types of exposures likely to be reflected in soil-to-organism bioaccumulation factors (e.g., direct ingestion of soil invertebrates from deeper soil horizons; ingestion of prey that feed primarily on plant matter).

version of the 3MRA modeling system requires all spatial information to be specified in the site layout file *a priori* (i.e., home ranges cannot be created on the fly).[2] The four home range areas overlap within each habitat so that the predatory-prey relationships are maintained. Thus, physical access by predator species to prey species is built into the site layout file by ensuring that each home range overlaps the other home ranges. From an exposure perspective, a receptor species eats plants, earthworms, and soil invertebrates within its home range, as well as any mobile prey species that cross into its home range. From the perspective of the Terrestrial Food Web Module, the contaminant concentrations in plants, earthworms, and soil invertebrates are spatially consistent within the *receptor species'* home range; the concentrations in other prey categories are spatially consistent within the *prey species'* home range. Consider the following simple example.

In a hypothetical forest habitat shown in Figure 11-2, the short-tailed weasel is assigned to home range 3. Its diet consists of soil invertebrates as well as small vertebrates, including mammals, herpetofauna, and birds. Table 11-2 provides example concentrations of contaminant *y* for soil and prey categories relevant to the weasel's diet calculated by the Terrestrial Food Web Module during a simulation. The calculated concentrations in soil, soil invertebrates, and other prey categories within home range 3 are internally consistent to that home range. That is, the concentration of contaminant *y* in small mammals assigned to home range 3 is 6.7E-07 mg/kg; this concentration includes prey species that are eaten by the short-tailed weasel (e.g., mice, shrews), but does not reflect the concentration for the prey category to which the short-tailed weasel belongs (omnivorous vertebrates). As described in Section 15 (Ecological Exposure Module), the weasel will eat any combination of small mammals, herpetofauna, and small birds in its diet if prey species that belong to these categories are assigned to home ranges that overlap the weasel's home range. However, the weasel may only eat soil invertebrates from within its home range (because these organisms are relatively sessile), and the incidental ingestion of soil is presumed to come exclusively from within the weasels's home range. The concentration of omnivorous vertebrates assigned to home range 3 in this example is not shown in Table 11-2.

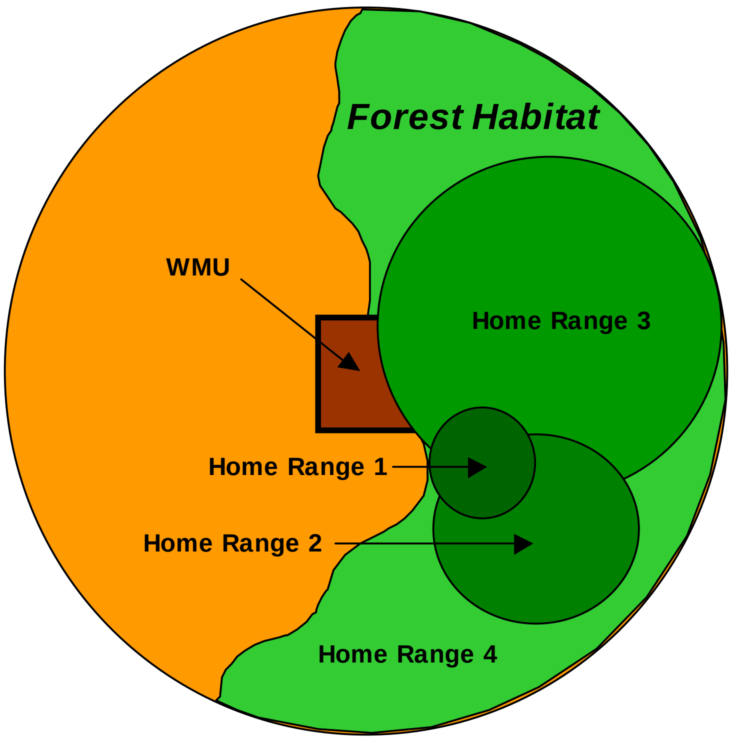

Figure 11-2. Hypothetical forest habitat with four home ranges shown.

[2] The Terrestrial Food Web Module was designed to accept a unique home range for each wildlife species assigned to a habitat. Although this scheme has been simplified by using only four sizes, the module is blind to the fact that home ranges of the same size and geographical location are found in the site layout file.

Table 11-2. Example Exposure Concentrations for Contaminant *y* Calculated by the Terrestrial Food Web Module

Food Item	Home Range 1 (mg/kg)	Home Range 2 (mg/kg)	Home Range 3 (mg/kg)	Home Range 4 (mg/kg)
Soil	2.6E-07	4.9E-07	3.8E-08	3.1E-09
Small mammals	1.3E-06	1.6E-06	6.7E-07	NA
Small herpetofauna	9.4E-07	3.2E-06	NA	NA
Soil invertebrates	2.3E-06	9.1E-06	4.3E-07	1.1E-08
Small birds	2.3E-06	5.9E-06	NA	NA

NA = no receptor species in this prey category were assigned to this home range.

The methods used by the Terrestrial Food Web Module to calculate contaminant concentrations in soil, plants, soil invertebrates (and earthworms), and vertebrate prey species are described in the following sections. Additional detail on the data can be found in Volume II of this report.

11.2.1 Calculate Contaminant Concentrations in Soil

The Terrestrial Food Web Module calculates contaminant concentrations in both surficial soils and root zone soils for each home range. Because the difference between calculated surficial and depth-averaged soil concentrations depends only on the depth of the soil horizon, the same equations are used to calculate the spatially averaged contaminant concentration in each home range. The surficial soil, or top layer (top 1 cm), is relevant to incidental soil ingestion by foraging animals; thus, the surficial soil concentration is used only by the Ecological Exposure Module to calculate the applied dose to terrestrial receptors. The root zone soils (concentrations averaged over 5 cm) represent deeper soils that are relevant to plant uptake through root-to-plant translocation and to direct exposures of soil fauna such as earthworms. Consequently, the root zone soil concentrations are used by the Terrestrial Food Web Module to predict plant and prey concentrations, and are passed to the Ecological Risk Module for use in evaluating risks to the soil community and terrestrial plants (as receptors).

The spatially averaged soil contaminant concentrations may include a contribution from regional watershed subbasins as well as from drainage subbasins associated with the WMU. However, erosion and runoff within the local watershed is only relevant to certain types of WMUs. For aerated tanks and surface impoundments, it is assumed that controls are sufficient to prevent erosion and runoff releases within the drainage subbasin that contains the unit (i.e., there is no erosion or runoff directly from the aerated tank or surface impoundment). For the land application unit, the average soil concentration in each home range may include contributions from both the drainage subbasin and the regional watershed subbasin, depending on the placement of the home range with respect to the WMU. In the hypothetical example in Figure 11-2, the contaminant concentrations in soil for home range 3 and home range 4 could include a contribution from the drainage subbasin associated with the WMU. It was assumed that wildlife could use parts of the land application unit as habitat; however, because the landfill and waste pile are assumed not to serve as suitable habitat for plants or prey species, these units are assumed to not contribute to direct, soil-based exposures to terrestrial receptors. The calculation of spatially averaged contaminant concentrations in soil for each home range is a three-step process (equations apply to both surficial soils and root zone soils). First, the contaminant concentration in soil for the home range due to contributions from watershed subbasins that overlap the home range is calculated as follows:

$$C_{soil}{}^{WS}{}_{HomeRange^{j}} = \sum \left(C_{soil}{}^{WS^{i}} \times Frac^{WS^{i}}_{HomeRange^{j}} \right) \qquad (11\text{-}1)$$

where

$C_{soil}{}^{WS}{}_{HomeRange}{}^{j}$ = annual average contaminant concentration in soil for home range *j* due to contributions from watershed subbasins (mg/kg soil)

$C_{soil}{}^{WSi}$ = annual average contaminant concentration in watershed subbasin *i* (mg/kg soil)

$Frac^{WSi}{}_{HomeRange}{}^{j}$ = fraction of home range *j* impacted by watershed subbasin *i* (unitless).

Second, if the home range overlaps the WMU local drainage subbasins, the contaminant concentration in soil for the home range due to contributions from drainage subbasins associated with the WMU is calculated as follows:

$$C_{soil}{}^{dsb}{}_{HomeRange^{j}} = \sum \left(C_{soil}{}^{dsb^{i}} \times Frac^{dsb^{i}}_{HomeRange^{j}} \right) \qquad (11\text{-}2)$$

where

$C_{soil}{}^{dsb}{}_{HomeRange}{}^{j}$ = annual average contaminant concentration in soil for home range *j* due to contributions from watershed subbasins (mg/kg soil)

$C_{soil}{}^{dsbi}$ = annual average contaminant concentration in watershed subbasin *i* (mg/kg soil)

$Frac^{dsbi}{}_{HomeRange}{}^{j}$ = fraction of home range *j* impacted by watershed subbasin *i* (unitless).

Finally, the total spatially averaged contaminant concentration for soil in the home range is calculated by summing the contributions from the watershed subbasins and the drainage subbasins associated with the WMU as follows:

$$C_{soil}{}^{AVE}{}_{HomeRange^j} = C_{soil}{}^{WS}{}_{HomeRange^j} + C_{soil}{}^{dsb}{}_{HomeRange^j} \tag{11-3}$$

where

$C_{soil}{}^{AVE}{}_{HomeRange^j}$	=	annual spatially averaged contaminant concentration in soil for home range *j* (mg/kg soil)
$C_{soil}{}^{WS}{}_{HomeRange^j}$	=	annual average contaminant concentration in soil for home range *j* due to contributions from watershed subbasins (mg/kg soil)
$C_{soil}{}^{dsb}{}_{HomeRange^j}$	=	annual average contaminant concentration in soil for home range *j* due to contributions from WMU local drainage subbasins (mg/kg soil).

11.2.2 Calculate Total Contaminant Concentrations in Plants

The Terrestrial Food Web Module calculates the total contaminant concentration in the plant categories shown in Table 11-1 by summing the concentrations for all potential exposure pathways for plants. Unlike the Farm Food Chain Module, the Terrestrial Food Web Module does not distinguish between protected and exposed fruits and vegetables, because wildlife are assumed to consume the outer surfaces of fruits and vegetables. Therefore, the exposed fruit and vegetable algorithms from the Farm Food Chain Module are used for all plants in the Terrestrial Food Web Module. For all aboveground plants, the total contaminant concentrations are converted from dry weight (DW) to wet weight (WW) by adjusting for the plant moisture content. These WW concentrations (also referred to as whole weight or fresh weight) are required by the Ecological Exposure Module. This conversion is not needed for root vegetables, because the concentration is calculated in WW directly. The methodology used to calculate contaminant concentrations in plants is described fully in Section 10 (Farm Food Chain Module).

11.2.3 Calculate Contaminant Concentrations Soil Invertebrates

Contaminant concentrations in soil invertebrates are estimated as a function of the root zone soil concentrations in each home range using contaminant-specific soil-to-tissue bioconcentration factors[3] for earthworms and other soil invertebrates. For each home range, the tissue concentrations in earthworms and other terrestrial invertebrates are calculated as follows:

$$C_{invert}{}^{AVE}{}_{HomeRange^j} = C_{soil}{}^{AVE}{}_{HomeRange^j} \times BCF_{invert} \tag{11-4}$$

[3] Bioconcentration factors (BCFs) for terrestrial prey are defined as the ratio of the contaminant concentration in the animal to the contaminant concentration in soil; BCFs are intended to reflect relevant exposure pathways to the study species (e.g., ingestion of contaminated soil and food).

where

$C_{invert}{}^{AVE}{}_{HomeRange}{}^{j}$ = annual average contaminant concentration in soil invertebrate or earthworms for home range *j* (mg/kg tissue)

$C_{soil}{}^{AVE}{}_{HomeRange}{}^{j}$ = annual average contaminant concentration in soil for home range *j* (mg/kg soil)

BCF_{invert} = contaminant-specific bioconcentration factor for soil invertebrates (kg soil / kg tissue).

11.2.4 Calculate Contaminant Concentrations in Vertebrate Prey Categories

The Terrestrial Food Web Module calculates a range of concentrations in vertebrate prey categories (illustrated in Table 11-1) across the four home range sizes. For example, the dietary data for a fox assigned to a terrestrial habitat indicates that part of its diet will consist of small mammals (e.g., rabbits, shrews, mice). The spatial linkages between the fox (predator) and various small mammals (prey) are represented in the site layout by allowing the respective home ranges to overlap. However, the proportion of each species consumed by the fox is unknown; the fox may consume any combination of these animals depending on prey availability, dietary preferences, and numerous other factors that affect prey selection. To address this uncertainty, the Terrestrial Food Web Module estimates the contaminant concentrations for relevant prey categories in each of the home ranges, and reports the minimum and maximum values. These values are used by the Ecological Exposure Module to randomly select (assuming a uniform distribution) an effective concentration for prey categories that represents the full range of concentrations to which a predator may be exposed.

Contaminant concentrations in prey categories are estimated as a function of the root zone soil concentrations in each home range, contaminant-specific soil-to-tissue bioconcentration factors for each prey category, and the fraction of each prey species home range that is contained within the habitat (i.e., the fraction that does not extend beyond the 2 km radius of the AOI). Vertebrate prey categories considered in the module include small mammals, small birds, small herpetofauna, large omniverous vertebrates, and large herbivorous vertebrates. Because these prey types consist of a variety of species (e.g., a Cerulean Warbler, Marsh Wren, and Northern Bobwhite are all considered small birds), the module calculates the concentration of contaminants for species within each prey category and then selects the maximum and minimum values from that range. Because minimum and maximum concentrations are reported for each prey type, the Terrestrial Food Web Module allows the Ecological Exposure Module to represent variability in wildlife diets. Predatory animals may consume prey in different areas of the habitat (changes in foraging patterns) and are highly likely to be opportunistic in their feeding habits (altering diet due to prey availability).

The concentrations in vertebrate prey species (for each home range) are calculated as follows:

$$C_{prey^i HomeRange^j} = C_{soil}{}^{AVE}{}_{HomeRange^j} \times BCF_{prey^i} \times FRAC_{prey^i HomeRange^j} \tag{11-5}$$

where

$C_{prey^i\ HomeRange^j}$ = annual average concentration in vertebrate prey species *i* in home range *j* (mg/kg tissue)

$C_{soil}{}^{AVE}{}_{HomeRange^j}$ = annual average root-zone soil concentration for home range *j* (mg/kg soil)

BCF_{prey^i} = contaminant-specific bioconcentration factor for vertebrate prey species *i* (kg soil / kg tissue)

$Frac_{prey^i\ HomeRange^j}$ = fraction of home range *j* that falls within the habitat for vertebrate prey species *i* (unitless).

The last term in the equation, fraction of home range that falls within the habitat, is used to prorate the contaminant concentrations calculated in vertebrate prey species. The framework underlying the ecological exposure assessment does not assume that 100 percent of the diet originates from the home range, since the home ranges are sometimes larger than the entire habitat area. Consequently, the Terrestrial Food Web Module prorates the vertebrate species-specific tissue concentrations to reflect the fact that a species may not derive all of its food from within the AOI. Although the contaminant concentrations estimated for various vertebrate prey species are spatially consistent within each home range, there is no requirement that concentrations are based exclusively on food and soil from within the species' home range. Nevertheless, the module assumes that some exposure will always occur, and sets the fraction of home range that falls within the habitat to a minimum of 20 percent.

11.3 Module Discussion

11.3.1 Strengths and Advantages

The Terrestrial Food Web Module was developed to predict contaminant concentrations in soil, plants, and terrestrial prey to which receptors may be exposed. The module represents a significant step forward in addressing the spatial and temporal variability of ecological exposures. Specific strengths and advantages of the Terrestrial Food Web Module include the following:

- **Takes full advantage of functionality in the Farm Food Chain Module.** By translating the plant categories described for the Farm Food Chain Module into relevant categories for wildlife, the contaminant concentrations predicted for plants are consistent across both modules. As a result, many of the advantages of the Farm Food Chain Module are also conferred on the Terrestrial Food Web Module. For example, the calculations of contaminant concentrations in plants reflect both the soil and air pathways of exposure, as appropriate, given the properties of a particular chemical constituent. In addition, the Terrestrial Food Web Module also calculates and reports spatially averaged soil concentrations for each habitat and home range delineated at a site; this efficiency reduces the overall run time because average soil concentrations need only be calculated once by the Terrestrial Food Web Module.

- **Maintains spatial attributes of soil, plant, and prey concentrations calculated during the simulation.** A major strength of the Terrestrial Food Web Module is its ability to report concentrations for soil, plants, and terrestrial prey that are spatially consistent with the information in the site layout file. The module calculates these concentrations for each receptor assigned to each habitat, and reports the attributes of that concentration, including space (home range), food category (e.g., small mammals), and time (year of the simulation). The calculated tissue concentrations reflect the spatial boundaries for each ecological receptor assigned to a given habitat. For example, the tissue concentration predicted for a short-tailed weasel (which could be prey for the red-tailed hawk) is prorated by that portion of the weasel's home range that is contained within the habitat. This information is critical to the ecological risk assessment framework in the 3MRA modeling system to ensure that exposures are spatially consistent with the site layout information, and is intended to capture the range of contaminant concentrations in soil, plants, and prey to which any particular receptor may be exposed.

- **Algorithms, methods, and data are consistent with numerous EPA analyses.** The Terrestrial Food Web Module, although highly flexible, was developed to reflect current best practices in use at EPA, such as the Region VI Protocol for Ecological Risk Screening. The module and data inputs were intentionally based on established methods to permit application to a wide variety of chemical contaminants. Alternative approaches, such as fugacity-type models, were considered to be too data intensive (particularly with respect to chemical properties), inconsistent with the performance goals for model run times, and difficult to integrate with other model components. Consequently, the Terrestrial Food Web Module offers significant advantages over other approaches in ease of use for a variety of chemicals, transparency, modeling efficiency, and flexibility in its application and future modifications.

11.3.2 Uncertainty and Limitations

The methodology developed to estimate contaminant concentrations in soil, plants, and prey, as implemented by the Terrestrial Food Web Module, carries certain assumptions and limitations, and acknowledges several important sources of uncertainty. The limitations and uncertainties relevant to calculating forage and produce concentrations discussed under the Farm Food Chain Module are also relevant to the Terrestrial Food Web Module. Thus, the following discussion has been limited to issues exclusively related to the Terrestrial Food Web Module.

- **The Terrestrial Food Web Module is reliant on empirical (or empirically derived) soil-to-tissue bioconcentration factors**. The lack of good quality studies that investigate bioaccumulation in terrestrial systems contributes considerable uncertainty. As explained in detail in Volume II of this report on data collection, values for terrestrial BCFs reflect a broad range of data quality, with some studies reporting only percentiles off of the raw study data (e.g., a 50th percentile from a rank order). Moreover, the bioconcentration factors for a particular prey item vary across species, dietary preferences, seasonal resource requirements, and climatic conditions. For instance, Sample et al. (1998) indicated that vertebrates of varying dietary preferences (i.e., herbivores, omnivores, insectivores) accumulate contaminants to different degrees. A default uptake factor of 1 was used for contaminants for which BCFs could not be identified or calculated. The default value may be unrealistically high for contaminants that are poorly absorbed or are metabolized and excreted by animals. Although biomagnification in the food web has been demonstrated for very few contaminants in terrestrial systems, there is great uncertainty associated with the BCFs derived for prey items in this analysis, particularly in the use of default BCFs.

- **The Terrestrial Food Web Module does not capture the movement of contaminants through the food chain.** Because the module relies on soil-to-tissue BCFs to predict contaminant concentrations in food items, contaminant movement from biological compartment to biological compartment is not simulated. For example, although the module predicts concentrations in plant categories such as seeds and nuts, the contaminant concentrations in herbivorous vertebrates are calculated based on soil-to-vertebrate BCFs rather than biotransfer factors from plant matter into the tissues of herbivores. As a result, the module does not represent contaminant movement in the food chain in the sense that the Aquatic Food Web Module simulates movement among various biological compartments of predator and prey species. The BCF approach, although supported by available data, introduces uncertainty in the calculations of exposure concentration in terrestrial food items. Because few contaminants have been shown to accumulate in terrestrial food webs, it is likely that the BCF approach overestimates exposures to wildlife.

- **The derivation methods for some prey categories rely on extrapolation of soil BCFs from sediment studies.** For invertebrates, BCFs based on sediment exposure were adopted in the absence of data quantifying exposure via soil. This approach introduces uncertainty because these two exposure pathways are not equivalent. The primary literature reports bioconcentration data on various terrestrial insects such as beetles, especially for metals; however, these data are difficult (i.e., costly) to locate through traditional search methods because the data are generally found as secondary assessments conducted within larger site-specific risk analyses. The BCFs adopted from Oak Ridge work (Bechtel Jacobs, 1998) represent the best alternative, given current data limitations. Similarly, the BCFs for herpetofauna (i.e., reptiles and amphibians) were also identified from primary literature searches based on sediment rather than soil exposures. Because only sediment data were identified for herpetofauna, these bioaccumulation factors were adopted until soil-derived uptake factors can be identified; however, there is significant uncertainty in applying sediment exposures to estimate soil uptake values. In addition, the bioaccumulation data on herpetofauna were gleaned almost exclusively from studies on amphibians. Application of amphibian-based BAFs and BCFs to reptiles is associated with great uncertainty, given the physiological differences between these classes.

- **The uptake and accumulation of contaminants within categories of plants (e.g., exposed vegetables) is assumed to be similar**. The algorithms used to estimate biotransfer factors do not distinguish physiological differences across various kinds of plants. For example, the category "forage" includes forbs, grasses, fungi, shrubs, trees, and unclassified plants. Therefore, in estimating biotransfer factors for this category, it is implicitly assumed that the physiological differences in different plant species do not significantly affect contaminant loadings in plant tissues. The use of empirical data on selected plant species (typically crops) also assumes similar mechanisms of uptake and accumulation. Adopting the methodology developed for the Farm Food Chain Module to predict contaminant concentrations in plants that are eaten by wildlife introduces additional uncertainty in the application of those model algorithms.

- **A reasonable averaging depth for root zone soil concentrations is 5 cm.** In view of the multiple purposes of this soil concentration (e.g., to evaluate risks to soil fauna and predict tissue concentrations in prey using soil-based BCFs), and given the performance goals for science modules in the 3MRA modeling system, 5 cm was selected as a depth that is ecologically relevant (with regard to many organisms that live in the soil) and still consistent with the design specifications for the science modules (i.e., run time). Although the root zone depth may be specified by the user, the selection of 5 cm represents a balancing of model performance with ecological relevance. However, this assumption built into the Watershed Module limits the value of information on soil concentrations, and constitutes a source of uncertainty in its application within the Ecological Exposure and Risk Modules. In reality, soil fauna can occupy many different soil horizons, and plant roots (and exposures to soil contaminants) can extend well below 5 cm. For example, some important species of soil organisms live primarily in deeper soil horizons, while others live almost exclusively in the top litter layer of soil. This top layer is highly enriched in organic carbon and humics, and may serve as a significant source of exposure to certain soil species.

11.4 References

Bechtel Jacobs. 1998. *Empirical Models for the Uptake of Inorganic Chemicals from Soil by Plants.* Prepared for the U.S. Department of Energy under contract DE-AC05-98OR22700.

Sample, B.E., J.J. Beauchamp, R.A. Efroymson, and G.W. Suter, II. 1998. *Development and Validation of Bioaccumulation Models for Small Mammals.* Prepared for the U.S. Department of Energy under contract DE-AC05-84OR21400.

12.0 Aquatic Food Web Module

12.1 Purpose and Scope

The Aquatic Food Web Module calculates steady-state contaminant concentrations in aquatic organisms (e.g., fish, benthic invertebrates, aquatic plants) consumed by human and ecological receptors. These concentrations are used as input to the Human Exposure and Ecological Exposure Modules to calculate applied dose to receptors of interest. Figure 12-1 shows the relationship and information flow between the Aquatic Food Web Module and the 3MRA modeling system.

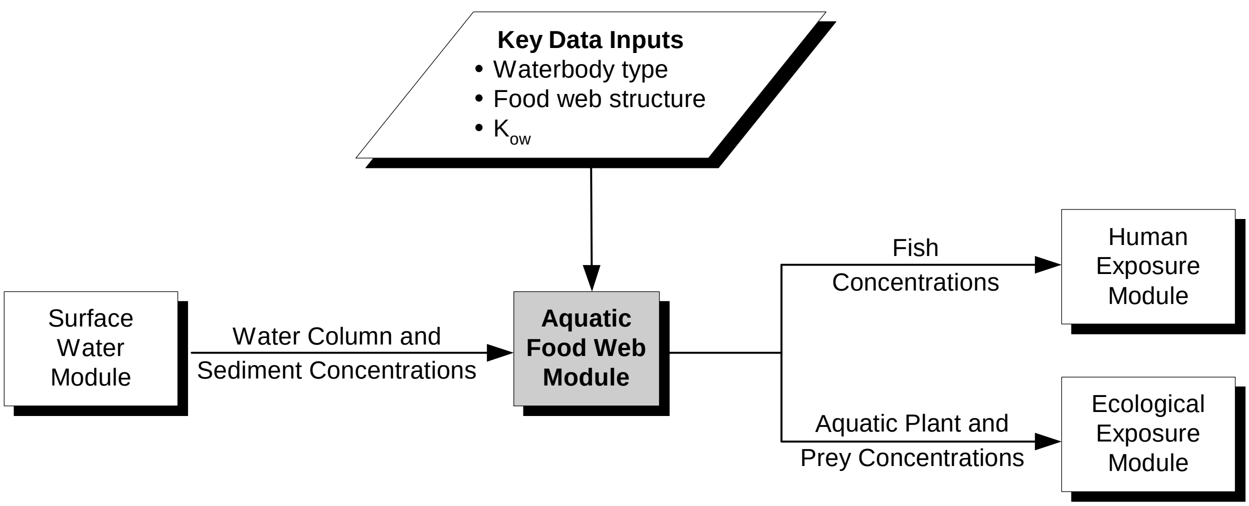

Figure 12-1. Information flow for the Aquatic Food Web Module in the 3MRA modeling system.

For each year in the simulation, the Aquatic Food Web Module predicts annual average contaminant concentrations in aquatic biota in freshwater waterbodies in the area of interest (AOI) considered capable of supporting fish (referred to as “fishable” waterbodies). The model is flexible enough to be applied to different types of waterbodies, including stream reaches, rivers, lakes, ponds, and permanently flooded wetlands. Simple freshwater food webs were constructed for each type of waterbody to depict the major functional and structural components of “healthy” freshwater ecosystems. The components of each food web represent major categories of aquatic biota in freshwater systems: aquatic macrophytes, phytoplankton, periphyton, zooplankton, benthic detritivores, benthic filter feeders, and various feeding guilds of fish. Some of these concentrations are used internally to calculate concentrations in fish, while other concentrations are reported as a time series for use in calculating exposures to wildlife and humans, as well as to calculate ecological hazard (e.g., hazard to sediment dwellers). Thus, the Aquatic Food Web Module determines which data are appropriate for use in a given waterbody and calculates concentrations in the aquatic biota assigned to that waterbody. Specifically, the Aquatic Food Web Module performs the following functions:

1. **Selects food web appropriate for each waterbody.** The Aquatic Food Web Module matches an appropriate food web with each waterbody identified as fishable within the AOI. Eight freshwater food webs were developed to capture the variability in freshwater systems. They represent warmwater streams/rivers, wetlands, ponds, and lakes; and coldwater streams/rivers, wetlands, ponds, and lakes.

2. **Constructs dietary matrix for food web.** The Aquatic Food Web Module uses a constrained, random prey preference sampling approach that selects preference fractions at random between the minimum and maximum, assuming a uniform distribution. This approach allows for the dietary composition to reflect the full range of variability inherent in the diets of freshwater fish.

3. **Calculates contaminant concentrations in food web.** The Aquatic Food Web Module calculates concentrations for the biota assigned to each freshwater food web. The biota categories include

   - Phytoplankton,
   - Periphyton,
   - Zooplankton,
   - Aquatic plants (macrophytes),
   - Benthic filter feeders,
   - Benthic detritivores,
   - Fish in various feeding guilds, and
   - Apex predator fish.

   The model will only calculate concentrations for biota that are assigned to a particular food web and waterbody. The calculations involve mechanistic models or the use of empirical data on bioaccumulation.

4. **Reports contaminant concentrations for fish consumed by wildlife and humans.** Food webs for freshwater aquatic systems typically have a single apex predator species (trophic level 4 [TL4]) and a number of other fish species that occupy different feeding guilds, such as benthic feeders (e.g., catfish). These other species are, for the purposes of exposure assessment, often grouped into the category of trophic level 3 (TL3), indicating that they are both predator and prey in the food web. To predict exposures for wildlife and humans that eat TL3 fish, the model calculates an average contaminant concentration—both wholebody and filet—for fish that fall into the category of TL3. In addition, the Aquatic Food Web Module reports the tissue concentration for the apex predator fish in each waterbody. The wholebody fish concentrations are used by the Ecological Exposure Module, and the filet concentrations are used by the Human Exposure Module.

12.2 Conceptual Approach

12.2.1 Select Food Web Appropriate for Each Waterbody

The 3MRA modeling system was designed to use site-based data to support a national-level assessment strategy. Consequently, an important goal for the Aquatic Food Web Module was to capture the variability in freshwater systems across the contiguous United States, particularly with respect to species composition and dietary preferences. The first step in accomplishing this was to develop a set of representative food webs. The second was to assign each waterbody in the AOI to one of these representative food webs.

Develop Representative Freshwater Food Webs. Upon reviewing literature sources on freshwater systems and food webs, it was apparent that the food web structure and, in many instances, the fish species, are similar across many different freshwater habitats (Schindler et al., 1996). There are common elements to virtually all aquatic communities (e.g., periphyton, benthic detrivores, aquatic plants). Many options to represent variability were considered, ranging from a basic food web consisting of three compartments, to a complex food web that could represent virtually any type of freshwater system. However, the development of a single food web—whether basic or complex—was not consistent with the 3MRA framework goal to use site-based information to support national-scale assessments. Moreover, considerable data were identified to vary the complexity of the food web and represent many different species of fish. To take advantage of these data, freshwater food webs were constructed such that the major functional elements were represented as simply as possible. Several useful tenets from the literature were adopted as guidelines in developing the aquatic food webs. They are summarized in the text box.

Basic tenets in constructing aquatic food webs

- Predator-prey interactions should follow common sense (i.e., larger fish eat smaller fish) and the system should be balanced in the sense that all prey items are connected in the food web.
- Size distinctions within feeding guilds of fish should consider the potential biomass of the most preferred prey item and the interactions with other components of the food web.
- Larger waterbodies tend to support more functional elements, and therefore, are typically more complex than smaller waterbodies (e.g., lakes are more complex than ponds).
- Flowing waters tend to be less complex than still waters and have low plankton density; as a result, zooplankton are not an important food web component.
- Warmwater systems tend to support a more diverse aquatic community than coldwater systems; as a result, they tend to have more functional niches and are more complex.

The resulting freshwater food webs provide a useful framework to model contaminant transport and fate in freshwater waterbodies, offer a reasonable representation of energy flows typical of different habitats, and capture variability in a manner that is appropriate for the application of the 3MRA modeling system. Figure 12-2 shows an example of a freshwater lake food web.

The sources used to construct the freshwater food webs reflect a broad perspective, ranging from biodiversity assessments to game fishing enthusiasts. These data sources were not only used in constructing the food webs, but also in characterizing the fish species and in

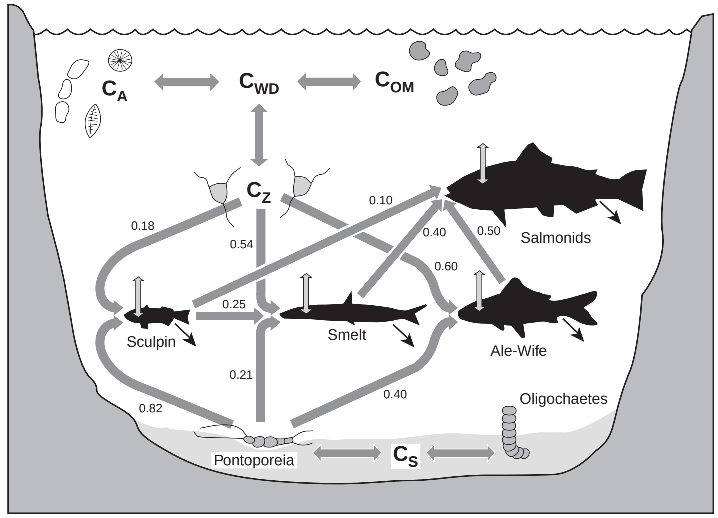

Figure 12-2. Example of simplified food web for freshwater lake (Gobas et al., 1993).

deciding which species of fish are eaten by human receptors. These sources offered a wide range of detail on food webs: from site-specific assessments to more general constructs developed for regional analyses. Many sources included qualitative descriptions of aquatic habitats, as well as indications of fish species that are considered “typical” for these habitats; in particular, the fishing references provided very useful information on the characteristics of fish that inhabit various freshwater systems. Many of these texts also indicated whether the preferred water temperature for a given species of fish was cold water or warm water.

The food web for each type of freshwater system reflects a number of characteristics of the waterbody, such as water temperature, flow (i.e., flowing versus static systems), dominant zones (e.g., pelagic versus littoral zones), and preferences of fish species for certain aquatic systems. The fish species assigned to the eight representative food webs represent a specific functional niche to which the species belong (e.g., feeding guilds; trophic level; size). For example, because the zooplankton density in streams tends to be low, a fish species that primarily feeds on zooplankton is unlikely to be assigned to stream habitats. In contrast, piscivore-dominated lakes are characterized by large-bodied zooplankton with high grazing rates (Schindler et al., 1996). In these lake systems, we would expect to find planktivorous species of fish as an integral part of the food web. Thus, the concept of functional niche is particularly important in the selection of food webs because these niches were used to inform the selection of appropriate fish species and associated data for each habitat (e.g., lipid fraction, body weight, dietary preferences).

The freshwater food webs contain between eight and 12 of the biota types possible in freshwater systems. The following biota are found in freshwater food webs:

- **Periphyton:** algal species typical of freshwater systems that adhere to rocks and detrital material; also includes small crustaceans, which are not modeled in the Aquatic Food Web Module;
- **Phytoplankton:** primary producers in pelagic systems;
- **Aquatic macrophytes:** vascular aquatic plants (e.g., submerged, emergent);
- **Zooplankton:** various invertebrates that graze on phytoplankton;

- **Benthic detrivores:** benthic dwellers that break down detritus in sediment (e.g., amphipods);

- **Benthic filter feeders:** benthic organisms that feed through a filtration mechanism;

- **TL3 benthivore:** TL3 fish whose primary feeding preference is benthic organisms (divided into small, medium, and large);

- **TL3 planktivore:** TL3 fish whose primary feeding preference is zooplankton;

- **TL3 omnivore:** TL3 fish who have no clear feeding preferences (divided into small, medium, and large); and

- **TL4 piscivore:** TL4 piscivorous fish that serve as the apex predator for the community.

Table 12-1 summarizes the food webs constructed for use in modeling contaminant movement in the freshwater food webs. This matrix indicates which food web components are assigned to each aquatic habitat. The presence of a prey item such as zooplankton may not require that an obligate planktivore be assigned to the food web. In wetlands, for example, omnivorous fish tend to feed on zooplankton, as well as on other biota (e.g., periphyton, benthos, detritus); therefore, the biota assignments reflect the goal of accounting for significant predator-prey interactions without imposing artificial constraints on the food web structure. Additional details on the development of habitat-specific food webs (e.g., warmwater wetland) are found in the 1999 background documents (U.S. EPA, 1999).

Assign Each Waterbody in the AOI to a Representative Food Web. The structure of aquatic food web and the fish species used to parameterize the Aquatic Food Web Module were based largely on the type of waterbody. The waterbody characteristics were developed using information from geographical information system (GIS) data sources (e.g., National Wetlands Inventory, or NWI), landscape data on the size of standing waterbodies, and certain conventions used in fish ecology to identify coldwater, stenothermic fish (i.e., coldwater species with narrow tolerance for temperature changes). For example, the threshold adopted for categorizing waters as warm or cold is based on a maximum temperature of 25°C—the water temperature above which coldwater, stenothermic fish cannot survive. The food web structure for warmwater streams is typically more complex than an analogous coldwater stream; as a result, there are frequently more functional niches in a warmwater stream than might be found in a coldwater stream. Stenothermic fish with clear temperature preferences were generally assigned either to warmwater or coldwater systems, but not to both. Other species that are found in both warmwater and coldwater habitats were assigned to both categories.

Table 12-1. Matrix of Biota in Food Webs for Freshwater Systems in 3MRA

	Coldwater habitats				Warmwater habitats			
Biota	**Stream**	**Wetland**	**Pond**	**Lake**	**Stream**	**Wetland**	**Pond**	**Lake**
Periphyton	✔	✔	✔	✔	✔	✔	✔	✔
Phytoplankton		✔	✔	✔		✔	✔	✔
Aquatic macrophytes	✔	✔	✔	✔	✔	✔	✔	✔
Zooplankton		✔	✔	✔		✔	✔	✔
Benthic detrivores	✔	✔	✔	✔	✔	✔	✔	✔
Benthic filter feeders	✔			✔	✔			✔
TL3 benthivore (small)	✔				✔			✔
TL3 benthivore (medium)	✔	✔	✔	✔	✔	✔	✔	
TL3 benthivore (large)								✔
TL3 planktivore (small)			✔					
TL3 planktivore (medium)				✔			✔	
TL3 planktivore (large)								✔
TL3 omnivore (small)		✔	✔	✔		✔	✔	
TL3 omnivore (medium)	✔	✔	✔		✔	✔		✔
TL3 omnivore (large)				✔			✔	✔
TL4 piscivore	✔	✔	✔	✔	✔	✔		✔

12.2.2 Construct Dietary Matrix for Food Web

The aquatic food webs provide the framework for the Aquatic Food Web Module simulations; the food webs identify fish and other biota presumed to be present for each waterbody and indicate the predator-prey interactions. However, most fish are opportunistic feeders, leading to significant variability in the dietary composition, even within feeding guilds. There is also tremendous variability in the dietary preferences of fish associated with life stage, region, prey density, and a host of other conditions. For many contaminants, the primary route of exposure is through gill exchange, and therefore, the dietary preferences are not important contributors to bioaccumulation of contaminants. However, for contaminants shown to biomagnify with trophic level, the dietary composition may have a significant influence on fish contaminant concentrations.

To address these variabilities in dietary composition, a random sampling algorithm was developed to select prey preference fractions for each type of fish assigned to a food web. The model constructs this dietary matrix for each simulation (defined as the combination of a site, waste management unit [WMU], and contaminant) so that the predicted concentrations in fish reflect the substantial variability in the diet. Data obtained from literature sources were evaluated to create a database of prey preference ranges for biota in various food webs. The Aquatic Food Web Module uses the database to (1) construct the food web-specific dietary composition, (2) rank prey items from most preferred to least preferred, and (3) estimate the prey preferences for each biota type (how much of each item is in the total diet). In this context, the dietary composition refers to both the dietary items consumed (e.g., zooplankton, small benthivorous fish) and the fraction of each dietary item consumed by various fish components in the food web.

The approach to construct the dietary matrix and select prey preferences in the food web was based on two objectives: (1) to observe the bounds as defined by the empirical data on prey preferences, and (2) to allow variability within the bounds to be exercised. Estimating prey preferences is accomplished using a constrained, random prey preference sampling algorithm that selects preference fractions at random between the minimum and maximum, assuming a uniform distribution. The algorithm maintains overall dietary preferences and allows for the dietary composition to reflect the full range of variability inherent in the diets of freshwater fish. The algorithm developed to solve this problem treats each dietary fraction as a "resource" to be allocated among the prey items for a particular fish. Before any dietary fractions are assigned for a given fish, the value of the resource remaining to be allocated is 1 (i.e., complete diet). After all dietary fractions have been assigned (zero fractions are allowed), the value of the resource remaining to be allocated is zero. Thus, for a given fish and prey item, the assignment prey preference fraction must consider the minimum and maximum preference values for that prey item, as well as the amount of resource remaining (dietary fraction yet to be assigned). The algorithm used to perform the random sampling is described in the text box on the next page.

12.2.3 Calculate Contaminant Concentrations in Food Web

The Aquatic Food Web Module was developed to be flexible to use empirical data, mechanistic models, or simple regression equations that use physical-chemical properties to calculate contaminant concentrations in food web biota. The choice of method is determined by the type of contaminant modeled, as well as by the availability of suitable empirical data on bioaccumulation. The Aquatic Food Web Module can model the following three groups of contaminants:[1]

- **Hydrophobic, non-ionizable organics.** This group includes dioxin, dioxin-like chemicals and high-molecular-weight, highly halogenated organics that tend to bioaccumulate in aquatic systems. Biota concentrations for these contaminants are estimated using either a mechanistic model or empirical data on bioaccumulation from the surface water column or the sediment. Currently, the Aquatic Food Web Module uses a mechanistic model based on the work of Gobas (e.g., Gobas, 1993) to predict tissue concentrations in aquatic biota. The module uses data on metabolism when available.

[1]The predictive models in the Aquatic Food Web Module can also be applied to ionizable organics. Chemical property values such as the octanol-water partition coefficient (K_{ow}) generated by the chemical properties processor reflect environmental conditions (e.g., pH) that affect ionization in sediment and surface water.

Random Sampling Algorithm Used to Determine Aquatic Prey Preferences

The issue for the aquatic food web is to select prey preferences throughout the food web matrix such that the observed bounds are honored (i.e., the empirical data on prey preferences), yet the allowable variability within the bounds is exercised in a Monte Carlo sense and the diet is complete. Expressed mathematically, the problem is:

$$\begin{gathered} Select \quad P_{ij} \quad i = 1,\ldots,N \quad j = 1,\ldots,M \quad Such\ that \\ Min_{ij} \le P_{ij} \le Max_{ij} \quad i = 1,\ldots,N \quad j = 1,\ldots,M \\ \sum_{j=1}^{M} P_{ij} = 1.0 \quad i = 1,\ldots,N \end{gathered}$$

where

N	=	number of biota types that are fish
M	=	number of prey items
P_{ij}	=	dietary fraction of the prey item for fish i for prey item j
Min_{ij}	=	minimum observed dietary fraction of fish i for prey item j
Max_{ij}	=	maximum observed dietary fraction of fish i for prey item j.

The algorithm that was developed to solve this problem treats P_{ij} as a "resource" to be allocated among the M prey items for a given biota type of fish. Before any dietary fractions are assigned for a given fish i, the value of the resource remaining to be allocated is 1.0 (i.e., complete diet). After all dietary fractions have been assigned (zero fractions are allowed), the value of the resource remaining to be allocated is 0. For a given fish i and prey item j, the assignment (P_{ij}) must consider both the Min_{ij} and the Max_{ij}, as well as the amount of resource remaining (dietary fraction yet to be assigned). The assignment equation for biota type i, assuming a uniform distribution for P_{ij}, is:
with the variables defined as follows:

$$\begin{gathered} LB_{ij} = Maximum[Min_{ij},\ RR_{ij} - \sum_{k=j+1}^{N} Max_{ik}] \\ RR_{ij} = 1.0 - \sum_{k=1}^{j-1} P_{ik} \\ UB_{ij} = Minimum[Max_{ij},\ RR_{ij} - \sum_{k=j+1}^{N} Min_{ik}] \end{gathered}$$

where

LB	=	lower bound of the range
UB	=	upper bound of the range
RND	=	uniform random deviate (0-1).

- **Hydrophilic, non-ionizable organics.** This group includes straight-chain aliphatics, as well as low-molecular-weight organics such as benzene, that do not tend to bioaccumulate in aquatic systems. Biota concentrations for these contaminants can be estimated using a regression model, a mechanistic model, or empirical data on bioaccumulation. Currently, the Aquatic Food Web Module uses a regression model to predict bioaccumulation factors (BAFs), unless the contaminant is considered to be easily metabolized. For metabolizable organic chemicals, the Aquatic Food Web Module uses empirical BAFs to predict tissue concentrations. Tissue concentrations in aquatic plants and benthic filter feeders are calculated based on sediment concentrations and biota sediment accumulation factors (BSAFs).

- **Metals and mercury.** This group includes metals in various valence states as well as mercury as methyl mercury. Biota concentrations are predicted using empirical data on bioaccumulation. For metals, the Aquatic Food Web Module uses median values of BAFs based on a data set that meets specific data quality objectives (see Volume II). For mercury, tissue concentrations of mercury are estimated using empirical BAFs based on dissolved methyl mercury concentrations from the *Mercury Report to Congress* (U.S. EPA, 1997). The *Mercury Report to Congress* provides methyl mercury BAFs for fish in TL3 and TL4. For both metals and mercury, tissue concentrations in aquatic plants and benthic filter feeders are calculated based on sediment concentrations and BSAFs.

Depending on the contaminant of concern, the Aquatic Food Web Module requires inputs on the characteristics of the fish species assigned to each food web (e.g., lipid fraction, body weight, dietary preferences), as well as characteristics of the waterbody (e.g., the fraction organic carbon in bed sediment). The Surface Water Module generates contaminant concentrations in surface water (dissolved and total) and in sediment (dissolved in pore water and total); therefore, the Aquatic Food Web Module is not required to predict contaminant concentrations in the different phases in the environmental media (i.e., sorbed versus freely dissolved phases) but gets these from the Surface Water Module. For all contaminants, the Aquatic Food Web Module calculates tissue concentrations for benthic filter feeders, aquatic plants, and TL3 and TL4 fish. Intermediate concentrations in food items such as zooplankton and benthic detritivores are calculated only for hydrophobic organic chemicals and are used by the mechanistic model as dietary inputs.

The technical approach implemented by the Aquatic Food Web Module to predict contaminant concentrations in biota is summarized below for the three groups of contaminants. More detailed discussions on the algorithms and data sources used by the Aquatic Food Web Module can be found in the 1999 background documents (U.S. EPA, 1999) and in Volume II of this report, respectively.

Hydrophobic, Non-Ionizable Organic Contaminants. For hydrophobic, non-ionizable organic chemicals, the Aquatic Food Web Module is based on the modeling constructs developed by Gobas et al. (1993), Thomann (e.g., Thomann et al., 1992), and a number of other researchers (e.g., Abbott et al., 1995; Campfens and Mackay, 1997; Morrison et al., 1997; and Zaranko et al., 1997). These models were chosen because they were developed specifically for organic chemicals with significant potential to bioaccumulate in the food web, they do not have prohibitive data requirements, they are flexible in their application to different waterbodies and food webs, they have been peer reviewed and validated with field data, and the series of linear equations can be solved using a flexible matrix solution technique. The form of model does not require a system-based solution; the equations, while coupled, can be solved sequentially. However, to accommodate the future use of more complex predator-prey relationships, which may involve true simultaneity, a more generic, system solution was believed desirable and was developed for the Aquatic Food Web Module. In the matrix solution, the contaminant concentrations in fish are predicted as follows:

$$C_{fish}{}^{i} = \frac{[k_1 \times C_{fd} + k_D \sum(Frac_j \times C_j)]}{(k_2 + k_E + k_M + k_G)} \qquad (12\text{-}1)$$

where

$C_{fish}{}^{i}$	=	annual average whole-body concentration in fish *i* (mg/kg wet weight [WW])
k_1	=	rate constant for contaminant uptake from water (L/kg-d)
$C_w{}^{fd}$	=	annual average freely dissolved concentration in surface water (mg/L)
k_D	=	rate constant for contaminant uptake from food (L/d)
$Frac_j$	=	fraction of prey item *j* included in diet (unitless)
C_j	=	annual average concentration in prey item *j* in diet (mg/kg WW)
k_2	=	rate constant for contaminant elimination to water (L/d)
k_E	=	rate constant for elimination by fecal egestion (L/d)
k_M	=	rate constant for metabolic transformation of contaminant (L/d)
k_G	=	rate constant for growth dilution (L/d).

Under steady-state conditions, the contaminant concentrations in periphyton, phytoplankton, zooplankton, and aquatic macrophytes are predicted as follows, assuming that the BCF is satisfactorily approximated by K_{ow}:

$$C_j = C_w^{fd} \left((LipFrac_j\ K_{ow}) + (NonLipFrac_j\ 0.033\ K_{ow}) + WaterFrac_j\right) \qquad (12\text{-}2)$$

where

C_j	=	annual average concentration in prey item *j* (mg/kg WW)
$LipFrac_j$	=	lipid fraction in prey item *j* (unitless)
K_{ow}	=	octanol-water partition coefficient (L/kg lipid)
$C_w{}^{fd}$	=	annual average freely dissolved concentration in surface water (mg/L)
$NonLipFrac_j$	=	nonlipid organic carbon fraction in prey item *j* (unitless)
$WaterFrac_j$	=	water fraction in prey item *j* (unitless).

The tissue concentrations in benthic detrivores and benthic filter feeders are also derived assuming steady-state conditions. As described in Gobas (1993), equilibrium partitioning theory may be used to predict concentrations in benthic organisms as follows:

$$C_j = \frac{C_{sediment} \times \left(\dfrac{\rho_{OC}}{foc_{sediment}} \right)}{\left(\dfrac{\rho_{lip}}{LipFrac_j} \right)} \tag{12-3}$$

where

C_j	=	annual average concentration in prey item *j* in benthos (mg/kg WW)
$C_{sediment}$	=	annual average total concentration in sediment (mg/kg)
ρ_{oc}	=	density of organic carbon in sediment (kg/L)
$foc_{sediment}$	=	fraction of organic carbon in sediment (unitless)
ρ_{lip}	=	density of lipids in benthos (kg/L)
$LipFrac_j$	=	fraction of lipid in prey item *j* in benthos (kg lipid/kg tissue).

Gobas points out that although more detailed models to estimate concentrations in benthos can be derived, this model has been shown to be in better agreement with field data (e.g., see Gobas et al., 1989; Landrum et al., 1992).

Hydrophilic, Non-Ionizable Organic Contaminants. For hydrophilic, non-ionizable organic chemicals (defined operationally as organic chemicals with an octanol-water partition coefficient value below 10,000), the Aquatic Food Web Module calculates concentrations in fish using the following equation:

$$C_{fish}{}^{i} = BAF_i \times C_w^{fd} \tag{12-4}$$

where

$C_{fish}{}^{i}$	=	annual average concentration in fish *i* (mg/kg WW)
BAF_i	=	bioaccumulation factor for fish *i* (L/kg tissue)
$C_w{}^{fd}$	=	annual average freely dissolved concentration in surface water (mg/L).

For these types of organic chemicals, bioaccumulation is primarily a function of gill exchange (rather than accumulation of contaminant through ingestion of contaminated food items). The BAF values are based on empirical data when available. Otherwise, the Aquatic Food Web Module calculates the BAF using the regression algorithm developed by Bertelsen et al. (1998):

$$BAF_i = 10^{(a_{fish} \cdot \log K_{ow} + b_{fish} \cdot \log LipFrac_i + c_{fish})} + WaterFrac_{fish}{}^{i} \tag{12-5}$$

where

BAF_i	=	bioaccumulation factor for fish *i* in trophic level *p* (L/kg tissue)
a_{fish}	=	primary slope term (unitless)
b_{fish}	=	secondary slope term (unitless)
c_{fish}	=	empirical error term (unitless)
$LipFrac_i$	=	lipid fraction in fish *i* (kg lipid/kg tissue)
K_{ow}	=	octanol-water partition coefficient (assume L/kg lipid)
$WaterFrac_{fish}^{i}$	=	fraction of wholebody fish *i* in trophic level *p* that is water (unitless).

The model developed by Bertelsen et al. extends previous work on the bioconcentration of hydrophilic organics in fish presented by Veith et al. (1980); Mackay (1982); Isnard and Lambert (1988); and others. Because gill exchange is considered to be the dominant mechanism by which hydrophilic organics are taken up, a simpler model could be used to predict tissue concentrations in fish. Uptake through the food web is assumed to be negligible; therefore, it is not necessary to calculate the concentration in all of the prey items in the aquatic food web.

Metals and Mercury. The contaminant concentration in fish tissue for metals and mercury is calculated as follows:

$$C_{fish}{}^{i} = BAF_{fish}{}^{i} \times C_w^{fd} \tag{12-6}$$

where

C_{fish}^{i}	=	annual average concentration in fish *i* (mg/kg WW)
BAF_i	=	bioaccumulation factor (dissolved) for fish *i* (L/kg tissue)
C_w^{fd}	=	annual average freely dissolved concentration in surface water (mg/L).

BAFs based on dissolved concentrations are not always available for some contaminants; therefore, the total surface water concentration can also be used with a BAF based on total concentration. However, BAFs based on dissolved concentrations are preferred because they are applicable regardless of the characteristics of a particular waterbody. The tissue concentrations of mercury reflect the freely dissolved concentrations of methyl mercury in the water column, that is, the methyl mercury that is not sorbed to organic matter.

BAFs for metals are estimated exclusively from empirical data. Few mechanistic models are available that can be used in a national-scale analysis to estimate metals transport and accumulation in the food web from surface waters and sediments. Consequently, EPA has devoted considerable effort to identifying studies and developing criteria for selecting appropriate BAFs for metals. The relatively complex environmental behavior of metals in surface water with respect to bioaccumulation and water quality criteria has been a frequent and long-standing topic of discussion in peer-reviewed journals and texts, notably the following:

- Allen, H. E., and D. J. Hansen. 1996. The importance of trace metal speciation to water quality criteria. *Water Environment Research,* 68(1):42-54. January.

- Bergman, Harold L., and Elaine J. Dorward-King (eds.). 1997. *Reassessment of Metals Criteria for Aquatic Life Protection*. Society of Environmental Toxicology and Chemistry Press, Pensacola, FL. Proceedings of the Pellston Workshop on Reassessment of Metals Criteria for Aquatic Life Protection, February 10–14, 1996.

- Chapman, Peter M., Herbert E. Allen, Kathy Godtfredsen, and Michael N. Z'Graggen. 1996. Evaluation of bioaccumulation factors in regulating metals. *Environmental Science & Technology*, 30(10):448A-452A.

- Renner, Rebecca. 1997. Rethinking water quality standards for metals toxicity. *Environmental Science & Technology*, 31(10):466A-468A.

Although uptake and accumulation are not of concern for all metals, the impact of surface water characteristics (particularly dissolved organic carbon) on bioavailability is significant. Several modeling approaches have been developed recently that can be used to predict bioavailability (e.g., the Windermere Humic Aqueous Model, or WHAM), and water effects ratios (WER) provide empirical ratios that can be used to adjust water quality criteria to account for the mitigating effects of natural waters (see Bergman and Dorward-King, 1997, for discussion). Moreover, as shown in Figure 12-3, the effects and accumulation of essential metals change with concentration (i.e., bioaccumulation is nonlinear); thus, a single BAF may be inappropriate.

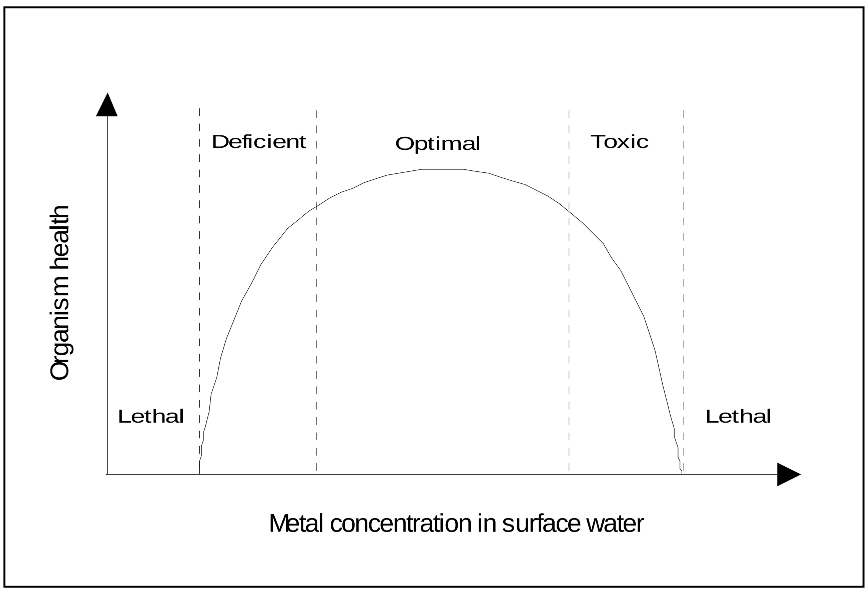

Figure 12-3. Relationship between essential metal concentration and organism health (adapted from Chapman et al., 1996).

The state of the science on metals transport and fate in aquatic systems strongly suggests that the uptake and accumulation of essential metals (e.g., copper, zinc) in fish and other aquatic organisms are fundamentally different than the uptake and accumulation of nonessential metals (e.g., cadmium, lead). Moreover, laboratory studies that report BCFs may have limited relevance to the behavior of metals in the field. The predominant metal species in a "natural" aquatic system may be very different than the metal salt studied in the laboratory, resulting in a very different accumulation profile. The characteristics of the aquatic system and the presence of other cationic metals significantly influence the uptake and accumulation of a single metal species. Although understanding of metals behavior has increased since the Aquatic

Food Web Module was developed, the module relies on empirical data and a simple model to predict metal concentrations in aquatic biota. EPA is currently reviewing recently published studies and data relevant to the development of alternative approaches to predict metal concentrations in aquatic biota (e.g., Bergman et al., 1992).

12.2.4 Report Contaminant Concentrations for Fish

The Human and Ecological Exposure Modules allow receptors to consume fish from any of the water bodies to which they have access. Both human and ecological receptors may consume fish assigned to TL3 and TL4; however, human receptors can only eat TL3 fish that are designated as edible by humans in the fish database. In addition, human receptors are assumed to eat only the filet portion of the fish, whereas ecological receptors are assumed to eat the entire fish. Therefore, the Aquatic Food Web Module performs two processing steps in generating concentrations of contaminants in fish for use by the exposure modules.

First, the Aquatic Food Web Module calculates the filet concentration by adjusting the wholebody concentrations of organic chemicals by the relative lipid content of filet (or muscle) versus the wholebody of the fish. The theory supporting the bioaccumulation of organic chemicals suggests that virtually all of the contaminant accumulates in the lipid tissue. The adjustment factor is calculated by dividing the lipid fraction in filet by the lipid fraction in the wholebody. For example, if the lipid fraction in filet is 3 percent and the lipid fraction in wholebody is 10 percent, the adjustment factor is 0.3 (and the fraction for wholebody would be 0.7). Thus, by adjusting for the differences in lipid content between filet and wholebody, the concentration in filet can be estimated from wholebody tissue concentration.

The calculation for hydrophobic organic chemicals is

$$C_{filet}^{\ i} = C_{fish}^{\ i} \times FiletFrac^{\ i} \tag{12-7}$$

where

C_{filet}^{i}	=	annual average concentration in filet for fish *i* (mg/kg WW)
C_{fish}^{i}	=	annual average concentration in fish *i* (mg/kg wet weight)
$FiletFrac^{i}$	=	the adjustment factor for filet (unitless).

For hydrophilic organic chemicals, the equation for calculating the BAF (Equation 12-5) is parameterized to calculate concentrations in filet directly as described in Bertelsen et al. (1998). For metals and mercury, the wholebody concentrations are presumed to provide a reasonable approximation of filet concentrations given other uncertainties.

Second, the Aquatic Food Web Module calculates a waterbody-specific annual average concentration for all TL3 fish filet presumed to be edible by humans, and all TL3 fish (wholebody) for ecological receptors. Because information on dietary preferences for TL3 fish is not available, this averaging implies that all TL3 fish have an equal probability of being eaten. This implies exposure to the average filet or wholebody concentration across TL3 fish in the waterbody for human and ecological consumption, respectively.

12.3 Module Discussion

12.3.1 Strengths and Advantages

The Aquatic Food Web Module was developed to predict contaminant concentrations in aquatic biota due to long-term chemical releases into surface water. The module offers a number of advantages relative to other approaches that were considered for the 3MRA modeling system. Some of the major strengths of the Aquatic Food Web Module include the following:

- **Applicable to a wide variety of chemicals.** The Aquatic Food Web Module and supporting data can be applied to wide variety of chemical contaminants ranging from hydrophobic organics to metals. The module recognizes the type of contaminant and uses the appropriate data and algorithms to predict tissue concentrations. The module was designed in a modular fashion so that new science could be incorporated to address specific chemical types. For example, the simple approach used to predict mercury concentrations in biota could be upgraded to include a more complex speciation model to simulate mercury behavior in freshwater systems. The ability to handle such a broad range of chemicals in a single module, using either empirical data on bioaccumulation or mechanistic models, is a significant advantage to the 3MRA modeling system.

- **Flexible enough to address any type of aquatic food web.** For bioaccumulative organic chemicals, the solution technique developed for this module is both computationally efficient and flexible. The module will accommodate simple food webs consisting of relatively few components or complex food webs with well-defined predator-prey interactions. As a result, the food webs that were defined for application to a national-scale analysis can be modified, and additional, site-specific food webs may be developed. This flexibility ensures that the Aquatic Food Web Module will be applicable to representative aquatic food webs for national-scale analyses as well as user-defined food webs for site-specific analyses.

- **Based on peer-reviewed, validated models for organic chemicals.** The governing equations for the Aquatic Food Web Module are based on models that have been published in peer-reviewed journals and that have been validated using field and/or laboratory data. In particular, the algorithms used to predict the contaminant concentrations in aquatic biota reflect the state-of-the-science for hydrophobic organic chemicals. The module can use data on metabolic transformation and, because the chemical properties processor can predict properties based on environmental conditions, it may be applied to ionizable organics as well. The level of validation of these equations and their widespread application described in the open literature provides strong support for their use in the 3MRA modeling system.

- **Random sampling algorithms represent dietary variability.** Dietary preferences—both in terms of preferred food items and fraction in the diet—are a potentially important source of variability in the chemical concentrations of hydrophobic organics predicted by the Aquatic Food Web Module. Many species of fish tend to be opportunistic feeders, and the diet may shift significantly depending on the site conditions. To account for variability in the diet, the module includes a random sampling algorithm that selects dietary elements from the database, and constructs a dietary matrix of the preferences such that 100 percent of the diet is accounted for. The ability to represent this source of variability across different aquatic food webs is a substantial improvement over fixing the dietary composition and preferences.

12.3.2 Uncertainty and Limitations

The following uncertainties and limitations are inherent in the Aquatic Food Web Module:

- **The module is implemented assuming steady-state conditions.** The Aquatic Food Web Module cannot be used to evaluate the impacts from storm events, nor can it be used to distinguish the impacts on tissue concentrations from peak events and subsequent averaging from long-term, low-level exposures.

- **The module relies heavily on empirical data for many contaminants.** For contaminants other than dioxin-like compounds and organics, mechanistic models are not used to predict tissue concentrations. Hence, the Aquatic Food Web Module estimates tissue concentrations by multiplying empirical BAFs by water or sediment concentrations. As discussed in Volume II, these BAFs and BSAFs are measured under conditions that may introduce uncertainty for certain environmental settings and species.

- **The module does not allow for separate treatment of essential metals.** Bioconcentration of essential metals is not linear, and modeling approaches are now available to account for nonlinearity. Bioconcentration of essential metals tends to be much greater at low concentrations than at higher concentrations because organisms actively seek to sequester necessary nutrients. Because many metals are regulated in biological systems, the apparent bioconcentration of metals at low concentrations may simply result in metal accumulation at “healthy” levels.

- **The module currently lacks the capability to use sediment concentrations directly in predicting tissue concentrations.** The Aquatic Food Web Module was developed primarily to use dissolved and total contaminant concentrations to predict tissue concentrations. Although sediment concentrations are used in predicting uptake and accumulation into benthic dwellers, the Aquatic Food Web Module lacks the necessary algorithms to use these data directly to predict concentrations in plants or fish. For certain chemicals (e.g., dioxins), it may be useful to build this functionality into the module to provide greater flexibility in data use.

- **The module has not been validated in field studies for all freshwater systems.** Much of the modeling theory on which the Aquatic Food Web Module is based is widely accepted and has been used in numerous analyses. In particular, the methods used to predict concentrations of organics have been validated in coldwater lakes. However, the module has not been validated for other freshwater aquatic habitats, nor has it been validated for application in a national-scale analysis.

12.4 References

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13.0 Human Exposure Module

13.1 Purpose and Scope

The Human Exposure Module calculates the applied dose to human receptors from ingestion and inhalation of contaminated media and food. Detailed information on the Human Exposure Module can be found in the background document (U.S. EPA, 2000). Figure 13-1 shows the relationship and information flow between the Human Exposure Module and the

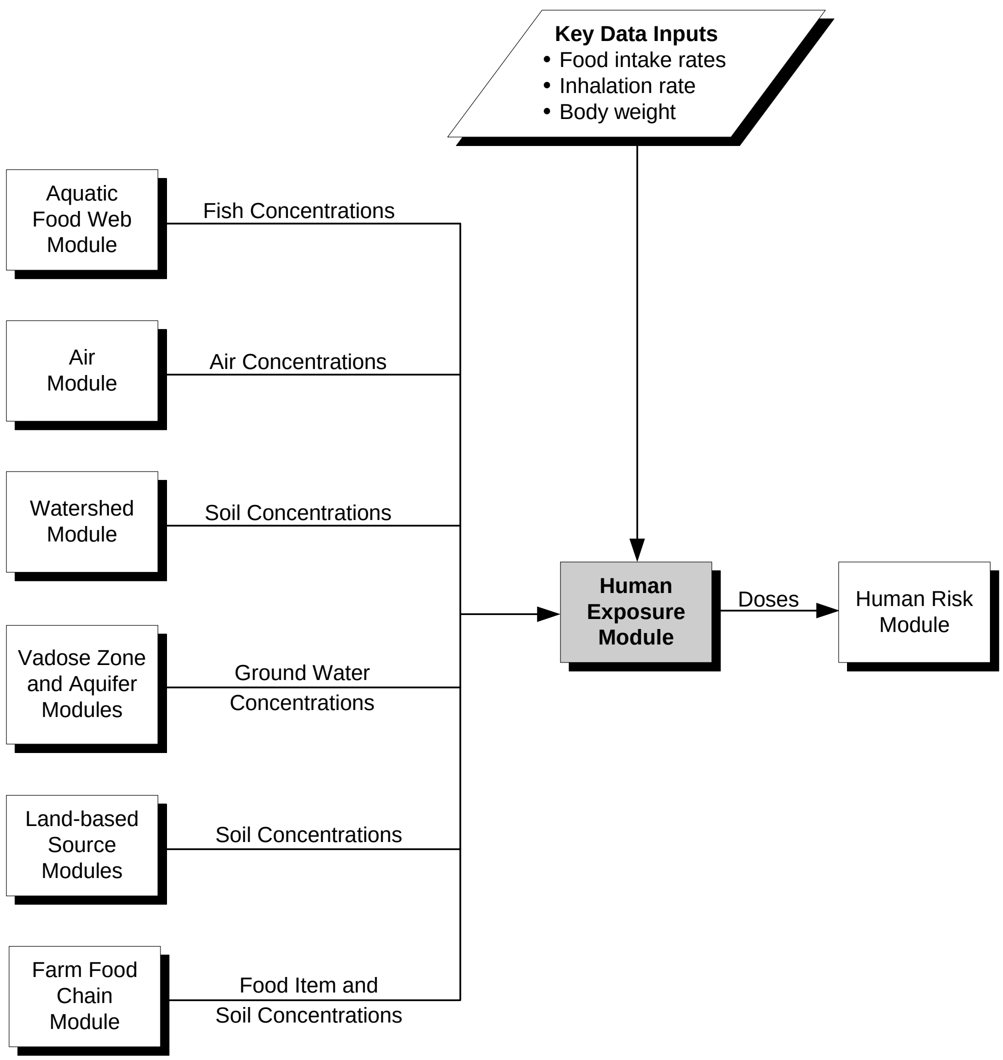

Figure 13-1. Information flow for the Human Exposure Module in the 3MRA modeling system.

3MRA modeling system. The Human Exposure Module uses media and food concentrations calculated in the media and food web modules to calculate applied doses. These doses are used by the Human Risk Module to calculate risk measures.

The purpose of the Human Exposure Module is to calculate the exposure inputs needed by the Human Risk Module to calculate receptor risk and hazard. For carcinogens, risk is calculated from applied dose (in mg/kg-d). For noncarcinogens, hazard via the ingestion pathway is also based on dose, whereas hazard via the inhalation pathway is based on the air concentration to which a receptor is exposed. Thus, the Human Exposure Module has the following functions:

1. **Calculates ambient air concentrations**. The Human Exposure Module calculates areal average ambient air concentrations for farms. Ambient (outdoor) air concentrations for residents are calculated by the Air Module and used by the Human Exposure Module.

2. **Calculates shower air concentration**. The Human Exposure Module calculates shower air concentration from ground water concentration.

3. **Calculates dose from inhalation of carcinogens.** The Human Exposure Module calculates dose from inhalation of ambient (outdoor) air and shower air for carcinogens. Noncarcinogenic risk for inhalation exposures is based directly on air concentration, not dose; thus, the Human Exposure Module does not calculate inhalation dose for noncarcinogens.

4. **Calculates dose from ingestion of contaminants in media or food**. The Human Exposure Module calculates dose for carcinogens and noncarcinogens from ingestion of soil, ground water, produce (fruits and vegetables), beef, milk, and fish.

5. **Calculates dose from ingestion of contaminants in breast milk**. The Human Exposure Module calculates dose to infants from ingestion of contaminated breast milk. This route of exposure is assessed only for dioxin-like chemicals, and only for infants.

Doses and air concentrations are calculated for each contaminant and site, as well as for each

- **Receptor type.** The module calculates exposures for two types of human receptors: residents and farmers. These are distinguished by how they are located—residents at a single exposure point and farmers on an area representing a farm. Residents are further defined by various kinds of behavior that lead to different profiles of exposure, such as home gardening and recreational fishing. Farmers may also be defined by behavior such as recreational fishing. Human receptor types considered in the 3MRA modeling system are a function of the goals of a particular analysis, and are defined by the receptor data used for the analysis.

- **Age cohort.** The human receptors are divided into five age cohorts (infants under 1 year, children aged 1 to 5 years, children aged 6 to 11 years, children aged 12 to 19 years, and adults) that are used to determine the most appropriate exposure factor data for various pathways, such as body weights, inhalation rates, and consumption rates for various food products.

- **Exposure pathway.** The Human Exposure Module considers nine potential exposure pathways. Depending on the exposure inputs entered into the model, all pathways could be considered for all receptor types in a given simulation. Table 13-1 shows the pathways modeled, and indicates which receptor types are modeled for each pathway by default. All pathways are modeled by default for all age cohorts with two exceptions: the breast milk pathway is modeled only for infants (and no other pathways are modeled for infants), and shower inhalation is modeled only for children aged 12 to 19 years and adults. All of these defaults can be changed by the user through the parameterization of the receptor-specific inputs.

Table 13-1. Default Pathways Considered by Receptor Type

	Receptor Type							
Pathway	Resident	Resident Fisher	Home Gardener	Home Gardener Fisher	Beef Farmer	Beef Farmer Fisher	Dairy Farmer	Dairy Farmer Fisher
Air inhalation	✓	✓	✓	✓	✓	✓	✓	✓
Shower inhalation	✓[a]	✓[a]	✓[a]	✓[a]	✓	✓	✓	✓
Ground water ingestion	✓[a]	✓[a]	✓[a]	✓[a]	✓	✓	✓	✓
Soil ingestion	✓	✓	✓	✓	✓	✓	✓	✓
Produce ingestion			✓	✓	✓	✓	✓	✓
Beef ingestion					✓	✓		
Milk ingestion							✓	✓
Fish ingestion		✓		✓		✓		✓
Breast milk ingestion	✓	✓	✓	✓	✓	✓	✓	✓

[a] Ground water and shower pathways are considered for residents and home gardeners only if Census data indicate the presence of private wells in the Census block group. All farms are assumed to have a private well.

- **Location.** The Human Exposure Module calculates exposures for residents for a single point. For the current data set, the point is placed at the centroid of each Census block in the area of interest (AOI). The module calculates spatially averaged exposure for farmers at a single farm in each census block group in the AOI that has Census data for farmers and agricultural land areas.

- **Year.** The module calculates exposures for each year of the simulation. The exposures predicted by the Human Exposure Module are reported as a time series of annual average applied doses or air concentrations. Thus, the equations presented in this section are applied to input data for each year. All temporal averaging for exposure durations exceeding 1 year is done by the Human Risk Module.

13.2 Conceptual Approach

As shown in Table 13-1, receptors are defined by exposures through various pathways; however, not all pathways may be completed at every location for each site, depending on the characteristics at each site (e.g., presence of private drinking water wells or fishable waterbodies). Pathways common to all receptors are inhalation of ambient air and incidental ingestion of soil; residents are assumed to breathe contaminated air and ingest contaminated soil at locations within a site at which contaminant releases are predicted. However, only a subset of receptors have private drinking water wells. This subset is the only group of receptors that could be exposed to contaminated ground water through direct drinking water ingestion and inhalation associated with showering. However, the presence of ground water wells does not necessarily mean that exposures will be estimated by the module. In some situations, the concentration of a contaminant of concern will be zero because of site-specific characteristics. For example, individuals may use drinking water wells and have no exposure because the wells are located outside of the predicted contaminated ground water plume. Similarly, those receptors who use a public water supply (whether from ground water or surface water) drink and shower with treated water that meets all drinking water standards and, thus, are assumed to have no exposure attributable to their water supply.

For the purpose of modeling exposure, the major differences between various receptor types and age cohorts are the specific pathways to which the receptor is potentially exposed and the exposure factors used in the dose calculations. Exposure factors for ingestion rates, inhalation rates, and body weights are randomly selected for each receptor type and each age cohort from distributions can be found in Volume II and are based on data from EPA's *Exposure Factors Handbook* (U.S. EPA, 1997).

With the exception of the shower/bathroom air concentration algorithms, the methodologies and equations used by the Human Exposure Module to convert contaminant concentrations in environmental media or food into human exposure values were obtained from *Methodology for Assessing Health Risks Associated with Multiple Pathways of Exposure to Combustor Emissions* (U.S. EPA, 1998), also known as the MPE methodology, and formerly referred to as the Indirect Exposure Methodology (IEM). The MPE methodology represents the state of the science with respect to providing reliable guidance in the proper conduct of assessments of risks that may result from multimedia, multipathway exposures. The National Center for Environmental Assessment (NCEA) prepared the MPE methodology as an update to EPA's 1990 IEM document. Most of the revisions in the MPE methodology are based on Scientific Advisory Board (SAB) and public comments on IEM. Earlier versions of this document have undergone internal EPA and external peer review. The breast milk exposure pathway algorithms and data were based on MPE and the Dioxin Reassessment (U.S. EPA, 1998, 2000). The shower algorithms were adapted from peer-reviewed sources (McKone, 1987 and Little, 1992).

Exposures from inhalation of ambient air and shower air are expressed differently depending on whether the exposure is being evaluated for a carcinogenic or a noncarcinogenic effect. When inhalation exposures for carcinogenic effects are evaluated, the exposure is expressed as an applied dose in terms of mg/kg-d. For noncarcinogenic effects, the inhalation exposure is expressed as a concentration in mg/m^3. This is because the RfC is in units of mg/m^3 and must be compared to an ambient concentration without explicitly using an inhalation rate or body weight. Ingestion exposures are estimated the same way for carcinogens as for noncarcinogens and are expressed as an applied dose in mg/kg-d.

13.2.1 Calculate Ambient Air Concentrations for Inhalation Exposures

The Human Exposure Module calculates annual average ambient air inhalation concentrations. These are used to calculate dose (also in the Human Exposure Module) for carcinogens and to predict the hazard for noncarcinogens (in the Human Risk Module).

For residents, the air concentration is simply the point estimate for the ambient air concentration at each receptor point, which is calculated by the Air Module. For farms, the Human Exposure Module calculates spatially averaged air concentration over the area of the farm using air concentration data calculated by the Air Module. This concentration is the sum of air vapor and particles (less than 10 μm).

13.2.2 Calculate Shower Air Concentrations for Inhalation Exposures

The Human Exposure Module calculates the annual average air concentration of a contaminant to which the receptor is exposed associated with showering, including both time spent in the shower and time spent in the bathroom immediately after the shower. This concentration is used by the Human Exposure Module for carcinogens to predict dose, and is used directly by the Human Risk Module to predict hazard for noncarcinogens. The primary input is the ground water contaminant concentration calculated by the Aquifer Module.

The concentrations in shower air and bathroom air are related to each other and are determined by solving a system of two coupled differential equations that describe the time-varying changes in average contaminant concentration in the shower and the bathroom, respectively, over the duration of a shower when the shower water is contaminated with contaminant. The change in these air concentrations over time is the result of injecting contaminated spray into an initially clean air environment rather than changes in the inputs over time (e.g., the contaminant concentration in the water, which is constant within a given year). The coupled differential equations are solved using a finite difference approximation. The details of the mathematical derivation of the shower and bathroom air concentrations are described in the U.S. EPA (2000).

The Human Exposure Module calculates the shower and bathroom air concentrations for each of the time steps (e.g., 20 seconds) over the duration of the shower (e.g., 10 minutes) and time spent in the bathroom after the shower (e.g., half an hour). These are then averaged to produce a time-weighted average air concentration for shower-related exposure. This average shower-related concentration is used in subsequent exposure and risk calculations.

13.2.3 Calculate Dose from Inhalation of Carcinogens

Inhalation exposure is calculated for ambient air and shower air for carcinogens. For noncarcinogens, the ambient and averaged shower/bathroom air concentrations described above are used directly by the Human Risk Module.

The applied inhalation dose reflects air concentrations for a single year in the simulation. For each realization (i.e., each time the Human Exposure Module runs), the module will read new data on air concentrations. Although the exposure factors (e.g., body weight, inhalation rate) will remain constant throughout the simulation, the exposure profile will change over time as a function of changing air concentrations.

Inhalation of Contaminated Ambient Air (Carcinogens). Ambient air inhalation exposure is calculated for all receptors that are predicted to be exposed to contaminated ambient air from the Air Module. The following equation is used to calculate an annual average dose from carcinogens inhaled by a human from the ambient air:

$$Dose_{ambient} = \frac{C_{ambient} \times CR_{air}}{BW} \tag{13-1}$$

where

$Dose_{ambient}$	=	annual average applied dose from ambient air inhalation (mg/kg-d)
$C_{ambient}$	=	total annual average ambient air concentration – vapor phase and particulate-bound (PM10) contaminant (mg/m^3)
CR_{air}	=	inhalation rate for age cohort (m^3/d)
BW	=	body weight for age cohort (kg).

Inhalation of Contaminated Air in the Shower/Bathroom (Carcinogens). The annual average dose from shower inhalation exposures is calculated for all receptors that have wells that are predicted to be in the ground water plume contaminated by a WMU. The governing equation is

$$Dose_{shower} = \frac{C_{shower} \times 1000 \times CR_{air} \times T_{shower} \times Ev_{freq}}{BW \times 1440} \tag{13-2}$$

where

$Dose_{shower}$	=	applied dose from inhalation associated with showering (mg/kg-d)
C_{shower}	=	average air concentration in bathroom and shower due to contaminated shower water (mg/L)
1,000	=	units conversion factor (L/m^3)
CR_{air}	=	inhalation rate for age cohort (m^3/d)
T_{shower}	=	time exposed during showering (min/event)
Ev_{freq}	=	event frequency (events/d)
1440	=	averaging time to convert to daily exposure (min/d).

The concentration in the bathroom and shower air is calculated as described in Section 13.2.2.

13.2.4 Calculate Dose from Ingestion of Contaminants in Media or Food

The general equation for calculating annual average dose based on ingestion exposure through a single pathway (e.g., ingestion of contaminated drinking water or ingestion of contaminated produce) is as follows[1]:

$$Dose = \frac{Conc \times CR \times Frac}{BW} \tag{13-3}$$

where

Dose	=	annual average dose of contaminant taken into the body (mg/kg-d)
Conc	=	annual average concentration of a contaminant in the medium or food item (mg/kg or mg/L)
CR	=	consumption rate of the medium or food item (L/d or g/d)
Frac	=	fraction of the total consumption rate that is contaminated (unitless)
BW	=	body weight (kg).

Consumption rates of food items and body weights vary with age cohort. For some pathways, the consumption rate is expressed per unit of body weight; in those cases, the CR/BW terms are combined into one term, which applies to all age cohorts.

The applied dose reflects media and food concentrations for a single year in the simulation. For each realization (i.e., each time the Human Exposure Module runs), the module will read new data on media and food concentrations. Although the exposure factors (e.g., body weight, food ingestion rate) will remain constant throughout the simulation, the exposure profile will change over time as a function of changing media and food concentrations.

The following subsections provide further detail on the exposure calculation for each pathway. Any modifications to Equation 13-3 required are noted.

Ingestion of Contaminated Soil. Dose from ingestion of contaminated soil is calculated for all resident and farm receptors where soils are predicted to be contaminated. Soil concentrations at residential exposure points are calculated by the Watershed Module. Spatially averaged surficial soil contaminant concentrations for farms are calculated by the Farm Food Chain Module.

Ingestion of Contaminated Drinking Water. Exposure from drinking water ingestion is calculated for all farms and for residential receptors in Census block groups that report private

[1] For some pathways, a unit conversion factor may be needed to convert consumption rate to units that will cancel with the concentration units to produce mg/d. For example, if concentration is in mg/kg and consumption rate is in g/d, a factor of 0.001 is needed to convert g to kg in the consumption rate.

wells and are predicted to be contaminated by a ground water plume from a WMU. It is not calculated for residential receptors in Census block groups that report no private wells.

Ingestion of contaminated drinking water is based only on ground water concentrations, which are calculated by the Aquifer Module. Surface water drinking supplies are assumed to be treated in accordance with drinking water regulations.

Ingestion of Contaminated Produce. Ingestion of contaminated produce is evaluated for farmers and home gardeners, who are residents that grow fruits and vegetables in contaminated air and soils. Some fraction of residential receptors are identified from Census data as home gardeners and are assumed to grow produce, and all farm households are assumed to grow produce. Produce concentrations are an input from the Farm Food Chain Module.

Exposures to homegrown fruits and vegetables are calculated for the following five categories of produce:

- Root vegetables,
- Exposed vegetables,
- Protected vegetables,
- Exposed fruit, and
- Protected fruit.

In addition to classifying produce as a fruit or vegetable (so as to account for different consumption rates), these categories also reflect three classifications—exposed, protected, and root—to account for differences in concentrations due to the different mechanisms by which contaminants can accumulate in edible plant tissues. Specifically, the terms "exposed" and "protected" refer to whether or not the edible portion of a fruit or vegetable that grows above ground is exposed to the atmosphere. For instance, apples are exposed fruits, and oranges are protected fruits. Root vegetables grow below the ground where they are not exposed to the atmosphere, so this distinction is not made for root vegetables, which are contaminated by root uptake of contaminants from the soil. Potatoes are an example of root vegetables. Exposure to contaminated produce is the summation of exposures from each of these five categories.

Consumption rates for produce are normalized to body weight; consequently, they are given in g/kg-d instead of g/d. The consumption rate and body weight terms in Equation 13-3 are therefore replaced by a single body-weight-normalized consumption rate term. All produce consumption rates are given in wet weight (WW). Therefore, the produce concentrations must also be per WW.

Ingestion of Contaminated Beef and Milk. Beef and milk ingestion exposure is calculated for farmers in an AOI when beef and dairy cattle are predicted to be contaminated from the release at a WMU. Only beef farmers are assumed to ingest home-raised beef, and only dairy farmers are assumed to ingest home-produced milk. Contaminant concentrations in beef and milk are generated by the Farm Food Chain Module.

Consumption rates for beef and milk are normalized to body weight; consequently, they are given in g/kg-d instead of g/d. Therefore, the consumption rate and body weight terms in

Equation 13-3 are replaced by a single body-weight-normalized consumption rate term. All beef and milk consumption rates are given in WW. Therefore, the beef and milk concentrations must also be per WW.

Ingestion of Contaminated Fish. The recreational fisher has the same exposures as the resident, the home gardener, and the farmer, but is also exposed through fish ingestion. Recreational fishers may have more than one favorite fishing spot. Therefore, the fisher is assumed to fish from up to three fishable reaches within the AOI, and fish ingestion exposure is calculated using an average fish concentration for those fishable reaches. If fewer than three fishable reaches occur in the AOI, then the average is calculated over all the fishable reaches that exist. If more than three exist, three are randomly sampled out of the full set of fishable reaches. The selection of the three fishable reaches is made twice for each site: once for residential receptors and once for farm receptors. The fish tissue concentration includes both trophic level 3 (TL3) and trophic level 4 (TL4) fish, with each predicted to have a different fish tissue concentration in each of the fishable stream reaches in the AOI. The fish tissue concentrations are predicted in the Aquatic Food Web Module.

13.2.5 Calculate Dose from Ingestion of Contaminants in Breast Milk

Infant exposure occurs through the consumption of contaminated breast milk. To calculate infant exposure through breast milk, the maternal exposure through all pathways is summed, and then the resulting breast milk concentration is calculated as follows:

$$C_{milkfat} = \frac{ADD_{mat} \times f_{am} \times f_f}{(ln2)/t_{1/2}^{b} \times f_{fm}} \tag{13-4}$$

where

$C_{milkfat}$	=	annual average concentration in maternal milk fat (mg/kg)
ADD_{mat}	=	average annual total daily dose consumed by the mother (mg/kg-d)
f_{am}	=	fraction of ingested contaminant absorbed by the mother (dimensionless)
f_f	=	proportion of contaminant that is stored in maternal fat (dimensionless)
$t_{1/2}^{b}$	=	biological half-life of contaminant in lactating women (days)
f_{fm}	=	fraction of mother's weight that is fat (dimensionless).

Infant exposures are calculated for eight maternal receptor types: resident, home gardener, beef farmer, dairy farmer, resident/recreational fisher, home gardener/recreational fisher, beef farmer/recreational fisher, and dairy farmer/recreational fisher. The mother is assumed to be an adult (as opposed to a teenager) for the purpose of calculating maternal dose in the infant breast milk pathway.

Infant breast milk exposure is then calculated as

$$Dose_{bmilk} = \frac{[C_{milkfat} \times f_{mbm} + C_{aqueous} \times (1 - f_{mbm})] \times f_{ai} \times CR_{bmilk} \times 0.001}{BW_{infant}} \tag{13-5}$$

where

$Dose_{bmilk}$	=	annual average applied dose from breast milk ingestion (mg/kg-d)
$C_{milkfat}$	=	annual average concentration in maternal milk fat (mg/kg)
f_{mbm}	=	fraction of fat in breast milk (unitless)
$C_{aqueous}$	=	annual average concentration in aqueous phase of maternal milk (mg/kg)
f_{ai}	=	fraction of ingested contaminant absorbed by the infant (dimensionless)
CR_{bmilk}	=	ingestion rate of breast milk (mL/d)
0.001	=	units conversion factor (kg/mL)
BW_{infant}	=	body weight of infant (kg).

Infant exposure to contaminants via breast milk is calculated only for dioxin-like chemicals. The concentration in milk fat is calculated as shown in Equation 13-4. The concentration in the aqueous phase of breast milk is set to zero because dioxin-like compounds do not accumulate in the aqueous phase. If this pathway were extended to additional constituents, the aqueous-phase concentration might be important. However, data are currently inadequate to extend this pathway to other constituents.

13.3 Module Discussion

13.3.1 Strengths and Advantages

The major strengths and advantages of the Human Exposure Module include the following:

- **Based on peer-reviewed EPA-ORD methodology.** With the exception of the shower/bathroom air concentration models, the methods and equations used by the Human Exposure Module to calculate exposure or applied dose for each environmental media or food are based on Methods for Assessing Health Risks Associated with Multiple Pathways of Exposure to Combustor Emissions (U.S. EPA, 1998) also known as the MPE methodology and formerly referred to as the Indirect Exposure Methodology (IEM). The MPE methodology represents the state of the science with respect to providing reliable guidance in the proper conduct of assessments of risks that may result from multimedia, multipathway exposures. EPA's National Center for Environmental Assessment (NCEA) prepared the MPE methodology as an update to EPA's 1990 IEM document. Most of the revisions in the MPE methodology are based on Scientific Advisory Board (SAB) and public comments on IEM. Earlier versions of this document have undergone internal EPA and external peer review. The breast milk exposure pathway algorithms and data were based on MPE (U.S. EPA, 1998) and the Dioxin Reassessment (U.S. EPA, 2000).

- **Accounts for exposures attributable to a specific microenvironment.** Inhalation exposure to volatile chemicals from contaminated ground water during showering is a specific pathway of concern. Therefore, a specific model that includes volatilization of contaminants during showering and the build-up of a contaminant's concentration in the bathroom during and after a showering activity was incorporated into the Human Exposure Module. The shower and bathroom algorithms were adapted from peer-reviewed sources (McKone, 1987 and Little, 1992).

- **Accounts for spatial variability in exposure across an AOI.** The Human Exposure Module carries forward all spatial information used in the 3MRA modeling system. This provides for the analysis of exposure variability across the AOI. By using census block centroids as the point of analysis, the module accounts for the variability in population across the site; in addition, the number of specific locations across the AOI can range from a few dozen to several hundred.

- **Accounts for different behaviors that may lead to increases in exposure.** Different behavior such as fishing, farming, and gardening are used to identify different groups within the population that may engage in certain activities that can lead to higher exposures. For example, a certain percentage of the population gardens. Most gardeners eat most of the produce that they grow. If these crops are contaminated then the gardening activity increases their exposures over the exposures of their neighbors who might not garden.

- **Accounts for variability in exposure due to differences in age.** Within a specific receptor type, the calculated applied dose varies due to the differences in the exposure factors used for the different age cohorts.

- **Accounts for inter-individual variability within a receptor-type/age cohort.** Statistical distributions were developed for intake rates and body weights for each age cohort within each receptor type. Use of these distributions within the Monte Carlo framework provides a way of assessing the interindividual variability due to exposure factor variability.

13.3.2 Uncertainty and Limitations

Uncertainties and limitations associated with the Human Exposure Module include the following:

- **Human receptors are assumed to be stationary.** It is assumed in characterizing exposure that human receptors both reside and work at the receptor location identified for them during site characterization (i.e., the farm area for farmers or residential exposure area for nonfarmers). The point of exposure is, in general, the Census block centroid for a resident or home gardener and the centroid of a farm for farmers. This assumption may result in either an overestimate or underestimate of exposure, because individuals may reside at the identified location within the study area but commute to work areas outside of the study area, or commute to more or less contaminated areas within the study area.

- **All residential receptors are assumed to be located at the Census block centroid**. The spatial point estimates of average daily doses for residential receptors assume all receptors in a Census block are located at the centroid of the Census block. Thus, to the extent that some receptors in fact reside or spend appreciable time at more highly contaminated areas within the block, their average daily doses will be underestimated. The converse is also true; that is, the average daily dose at the centroid will overestimate dose in other areas where concentrations are lower. If contaminant concentrations decrease approximately linearly with distance away from the waste management unit (WMU), there is probably little, if any, bias introduced by the centroid assumption. If concentrations decrease nonlinearly (e.g., first-order air deposition), the centroid assumption may overestimate the true average daily dose across the Census block to an unknown extent. The effect of any such bias will also be influenced by the size of the Census block—relatively larger blocks have greater potential for a bias.

- **Only one farm per Census block group reporting farms is modeled**. The number of farms modeled in the study AOI (which affects the farm-receptor population possibly at risk) is the number of Census block groups that make up the AOI that also contain farms. For example, if the AOI includes two Census block groups, and both Census block groups contain farming land use, then two farms will be modeled within the AOI. This could introduce a risk-conservative bias if the AOI does not in fact contain a farm.

- **Estimated exposures due to fish ingestion are subject to random sampling error for both farm and residential receptors.** Residential and farming fishers are assumed to be mobile and catch fish from up to three randomly selected fishable reaches throughout the AOI. These selected reaches may or may not reflect actual preferred fishing locations in the AOI. There is no reason to expect any systematic bias in estimated fish ingestion exposure.

- **Incremental exposures are estimated.** No provision is made for considering background exposures for the purpose of generating aggregate or total risk, hazard quotient, or margin of exposure (MOE) estimates for modeled receptors.

- **No reduction of contaminant concentration in food items is assumed to occur through food preparation.** Washing of fruits and vegetables, and cooking of produce, beef, and milk may reduce contaminant concentrations in foods. This is not accounted for in the model.

13.4 References

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14.0 Human Risk Module

14.1 Purpose and Scope

The Human Risk Module calculates risk measures for a given constituent, waste concentration, and site. Detailed information on the Human Risk Module can be found in the background document (U.S. EPA, 2000). Figure 14-1 shows the relationship and information flow between the Human Risk Module and the 3MRA modeling system. The Human Risk Module uses the annual average daily doses calculated in the Human Exposure Module to calculate receptor risk statistics. These risk statistics are used by the Exit Level Processors to determine national-level risk distributions.

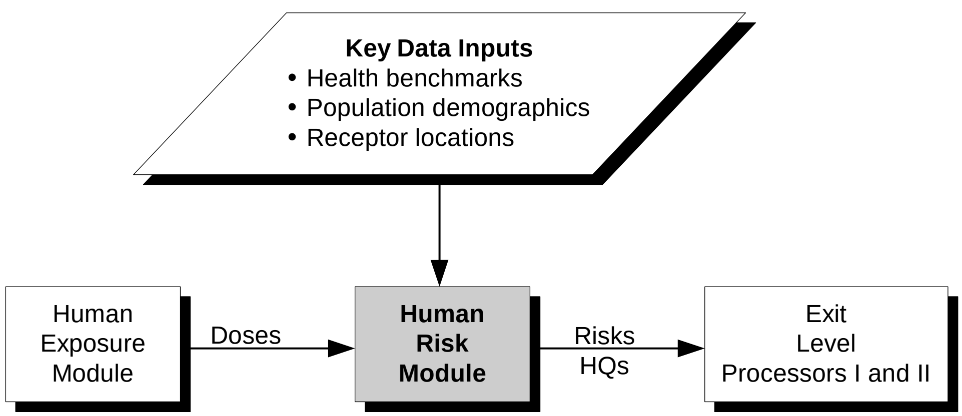

Figure 14-1. Information flow for the Human Risk Module in the 3MRA modeling system.

For each constituent, waste concentration, and site, the Human Risk Module generates risk estimates for each receptor location in the area of interest (AOI) and then calculates the number of receptors that fall within a specified risk or hazard range to describe the distribution of risks for the population at each site. The module also determines the timing of maximum risks. The Human Risk Module has the following functions:

1. **Calculates risk measures.** The Human Risk Module calculates cancer risk, noncancer hazard quotient (HQ), and noncancer margin of exposure (MOE) (for breastfeeding infants only). Depending on the constituent, the Human Risk Module calculates risk, HQ, or both. MOE is only calculated for breastfeeding infants for dioxin-like chemicals. These calculated risk measures are specific to a receptor type, an age cohort, an exposure pathway, a receptor location, and a specific exposure period (identified by starting year). The Human Risk Module

also aggregates risks and HQs from individual exposure pathways (e.g., ground water ingestion) to determine risk for groups of pathways (e.g., all ingestion pathways).

2. **Processes results for decision making.** The Human Risk Module puts exposed and unexposed population in the AOI into risk bins to estimate the number of receptors that experience risk within a specified range. Each risk bin is a range of risks or HQs. For any given exposure pathway and exposure period, the Human Risk Module uses Census data on population for each receptor location to determine the number of people of each receptor type and age cohort that experience risk from the specified pathway in the specified exposure period at risk levels within that bin. These populations are summed across receptor locations that have risks within the same bin. For each exposure pathway or pathway group, the Human Risk Module estimates total risk by multiplying the population at a location by the risk for that location, and uses this to determine the exposure period for which the total risk or HQ across all receptor types and age cohorts is the greatest. This estimate of total risk is not intended as a final risk measure, but is used only to identify the timing of maximum risk. The exposure period is identified by the year in which the risk averaged over a specified exposure period starts.

The scope of the Human Risk Module includes nine exposure pathways, four exposure pathway groupings, four receptor types, and five age cohorts, as follows:

Exposure pathways:

- Air inhalation,
- Shower inhalation,
- Ground water ingestion,
- Soil ingestion,
- Fruit and vegetable ingestion,
- Beef ingestion,
- Dairy ingestion,
- Fish ingestion, and
- Breast milk ingestion (infants only).

Exposure pathway groupings:

- All inhalation pathways,
- All ingestion pathways,
- All ground water pathways (ground water ingestion and shower inhalation), and
- All ingestion and inhalation pathways combined (if appropriate for the constituent).

Receptor types:

- Residents,
- Gardeners,
- Farmers, and
- Fishers.

Age cohorts:

- Infants under 1 year (breastfeeding pathway only)
- Children aged 1 to 5 years,
- Children aged 6 to 11 years,
- Children aged 12 to 19 years, and
- Adults.

The module uses the above groupings in calculations, but risk results are output only for the different exposure pathways and pathway groupings; these results are summarized across receptor types and age cohorts.

14.2 Conceptual Approach

The Human Risk Module calculates risk or HQ by pathway, receptor type, and age cohort; aggregates pathway risks across pathway groupings; bins population into risk or HQ bins; estimates total population risk; and determines the timing of maximum risks. The Human Risk Module can calculate risks and HQs for the whole AOI or within radial distance rings within the AOI if they are defined. These functions are described in the following subsections.

14.2.1 Calculate Risk Measures

The first major function of the Human Risk Module is to calculate cancer risk and/or HQ for a given receptor type, age cohort, exposure pathway, receptor location, and exposure period.

Cancer Risk Calculations. The governing equation used to calculate increased incremental cancer risk over a lifetime attributable to a lifetime exposure to a contaminant at a given dose (exposure) is

$$Risk \;=\; CSF \times Dose \qquad (14\text{-}1)$$

where

Risk	=	lifetime risk (probability units)
CSF	=	contaminant-specific cancer slope factor $(mg/kg\text{-}d)^{-1}$
Dose	=	annual average daily contaminant dose (mg/kg-d), expressed as an average daily dose over a lifetime.

This equation reflects a daily, lifetime exposure. However, the exposure duration may vary, as may the number of days per year a receptor is actually exposed (exposure frequency).

Assuming that cancer risk is linearly related to exposure duration and exposure frequency, Equation 14-1 can be modified to account for this as follows:

$$Risk = \frac{ExpDur}{LifeTime} \times \frac{ExpFreq}{365} \times CSF \times ADD \tag{14-2}$$

where

ExpDur	=	duration of exposure (yr)
Lifetime	=	average lifetime (yr)
ExpFreq	=	exposure frequency (d/yr)
CSF	=	cancer slope factor $(mg/kg\text{-}d)^{-1}$
ADD	=	average daily dose over specified exposure period (mg/kg-d).

Exposure duration and exposure frequency may be set to any reasonable value so long as exposure duration is less than or equal to the lifetime value chosen and exposure frequency is less than or equal to the maximum of 365 d/yr. The exposure duration is currently set to 9 years, which corresponds to the recommended value for median residence time presented in the *Exposure Factors Handbook* (U.S. EPA, 1997a,b,c), and to set exposure frequency to 350 d/yr.

The ADD is calculated as follows:

$$ADD = \frac{\sum_{t=1}^{ExpDur} Dose_t}{N} \tag{14-3}$$

where

ADD	=	average daily dose over entire exposure period (mg/kg-d).
t	=	year
ExpDur	=	exposure duration (yr)
$Dose_t$	=	annual average daily dose in year *t* (mg/kg-d)
N	=	number of years in the exposure duration (unitless).

In addition to the straight averaging of dose over the exposure period, a child receptor will age out of his or her initial cohort as the exposure duration progresses. For example, a child beginning the exposure duration at age 3 will, over a 9-year exposure period, increase in age to 11 years old. The annual doses are based on exposure factors (such as body weight or intake rate) that are age-cohort-specific. Therefore, the Human Risk Module ages each child receptor through the appropriate age cohorts as exposure progresses. Child receptors for a specific starting cohort are assumed to begin exposure at the midpoint of the age cohort (e.g., 3 years for a child aged 1 to 5 years). The age of a child receptor is monitored at each year over the exposure duration, and the dose is set in accordance with the relevant cohort as the child ages through them.

The Human Risk Module calculates a time series of risks averaged over the specified exposure duration, with each average risk representing a different starting year within the period of calculation. Calculations continue until the exposure medium concentration drops to less than 1 percent of the peak concentration, up to a maximum of 10,000 years. If the exposure duration were 9 years and the concentration dropped to 1 percent in year 100, the Human Risk Module would calculate a time series of 92 annual average risks, for years 1 through 9, 2 through 10, and so forth, up to years 92 through 100. These averages are identified by starting year (so in this example, year 1, year 2, and so forth, through year 92), but the risk for “year 1” is the risk for the 9-year period starting in year 1.

The Human Risk Module calculates cancer risks for any given contaminant only if an appropriate health benchmark is available. For ingestion pathways, the calculations are performed if an oral cancer slope factor (CSF) is available; for inhalation pathways, the calculations are performed if an inhalation CSF is available. If neither an inhalation nor an ingestion CSF is available, then no cancer risks are calculated for the contaminant.

Noncancer HQ Calculations. The governing equation used to calculate noncancer HQs depends on the route of exposure. For ingestion exposures, the equation is

$$HQ = \frac{ADD}{RfD} \tag{14-4}$$

where

HQ	=	hazard quotient for exposure period (unitless)
ADD	=	average daily dose over exposure period (mg/kg-d)
RfD	=	reference dose (mg/kg-d).

The ADD is calculated as shown in Equation 14-3 for carcinogens.

For inhalation exposures, the equation is

$$HQ = \frac{C_{avg}}{RfC} \tag{14-5}$$

where

HQ	=	hazard quotient over exposure period (unitless)
C_{avg}	=	average air concentration over exposure period (mg/m^3)
RfC	=	reference air concentration (mg/m^3).

The average air concentration may be an ambient air concentration or a shower/bathroom air concentration, depending on the pathway, and is calculated as

$$C_{avg} = \frac{\Sigma_{t}^{t + ExpDur - 1} \times Conc_t}{ExpDur} \tag{14-6}$$

where

C_{avg}	=	average air concentration over exposure period (mg/m^3)
t	=	first year of exposure period
ExpDur	=	exposure duration (yr)
$Conc_t$	=	annual average air concentration in year *t* (mg/m^3).

HQs are calculated for a specified exposure duration. They do not vary with exposure frequency, but are based on the assumption that the receptor has regular, chronic exposure. Exposure duration may be set to any reasonable value less than or equal to a typical lifetime. The exposure duration is currently set to 9 years, which corresponds to the recommended value for median residence time presented in the *Exposure Factors Handbook* (U.S. EPA, 1997a,b,c).

As for risk, the Human Risk Module calculates a time series of HQs averaged over the specified exposure duration, with each HQ representing a different starting year within the period of calculation. Calculations continue until the exposure medium concentration drops to less than 1 percent of the original WMU concentration, up to a maximum of 10,000 years.

The Human Risk Module calculates noncancer HQs for any given contaminant only if an appropriate health benchmark is available. For ingestion pathways, the calculations are performed if an oral reference dose (RfD) is available; for inhalation pathways, the calculations are performed if an inhalation reference concentration (RfC) is available. If neither an RfD nor an RfC is available, then no noncancer HQs are calculated for the contaminant.

Noncancer MOE Calculations. For infant exposure to breast milk, effects are typically quantified as an MOE rather than a risk or HQ. The MOE for infants is calculated as

$$MOE = \frac{Dose}{BM} \tag{14-7}$$

where

MOE	=	margin of exposure (unitless)
Dose	=	annual average applied dose from breast milk ingestion (mg/kg-d)
BM	=	contaminant-specific benchmark for breast milk exposure based on background exposure levels (mg/kg-d)

This calculation is currently performed only for 2,3,7,8-TCDD for infant exposures through the breast milk pathway. MOE estimates are single-year averages based on 1-year average doses. The Human Risk Module calculates an average MOE for each year in the calculation period.

14.2.2 Process Results for Decision Making

The Human Risk Module generates risk estimates for a variety of receptor types, age cohorts, exposure pathways, and receptor locations for each year modeled. The module processes and aggregates these results so that decision makers can answer a variety of questions about potential human risks. Specifically, the Human Risk Module (1) aggregates risk measures across pathways, (2) places population into risk bins, and (3) determines the timing of the maximum risk or hazard.

Aggregate Risk across Pathways. The 3MRA modeling system calculates risks and HQs for individual pathways. However, many receptors are exposed via multiple pathways, so the 3MRA modeling system also aggregates risk across pathways, where appropriate. The Human Risk Module performs the following aggregations:

- **Aggregation across ingestion pathways.** The Human Risk Module sums risks and HQs across the ingestion pathways for ground water, soil, fruits and vegetables, beef, milk, and fish.

- **Aggregation across inhalation pathways.** The Human Risk Module sums risks and HQs across both inhalation pathways: ambient air and shower.

- **Aggregation across all pathways.** When health effects can be combined across routes of exposure, the Human Risk Module aggregates risks for all pathways (both ingestion and inhalation).

- **Aggregation across ground water pathways.** When health effects can be combined across routes of exposure, the Human Risk Module aggregates risks for both ground water pathways: ground water ingestion and shower inhalation.

Whether or not it is appropriate to aggregate risks or HQs across routes of exposure depends on the contaminant and the health effects it causes via different routes of exposure. Cancer and noncancer effects (risks and HQs) are never combined.

Place Population into Risk Bins. Calculated risks or HQs by pathway are a measure of possible individual risk. The second major function of the Human Risk Module is to calculate population risk measures. For each pathway or aggregation of pathways, the Human Risk Module calculates the number of people in each receptor group and age cohort that experience various risk or HQ levels across the AOI or in a distance ring. These population counts are the basis of a series of cumulative frequency histograms, each specific to a receptor type, age cohort, exposure pathway (or aggregation of pathways), and exposure period. Thus, the full collection of cumulative histograms is a time series of conditional histograms for risk and/or HQ. Although the time series of histograms appears to be annual, the histograms are actually based on a multiyear moving average for risks and HQs, as discussed in the section on calculating risk. The length of the averaging time depends on the specified exposure duration.

The 3MRA modeling system currently uses 1990 Census data, 1992 agricultural census data, and 1996 fishing survey data to estimate receptor/age-cohort-specific population estimates at each residential or farm location for each site, as described in Volume II of this report. These populations are associated with census blocks. For the purpose of calculating and assigning carcinogenic risk and noncarcinogenic hazard, all of the population in a Census block is assumed to be located at the centroid of the block.

The 3MRA modeling system has seven bins for risk and four bins for HQ and MOE. The bins for risk are

- **Bin 1:** 0 to 5×10^{-9}
- **Bin 2:** $>5 \times 10^{-9}$ to 7.5×10^{-8}
- **Bin 3:** $>7.5 \times 10^{-8}$ to 7.5×10^{-7}
- **Bin 4:** $>7.5 \times 10^{-7}$ to 2.5×10^{-6}
- **Bin 5:** $>2.5 \times 10^{-6}$ to 7.5×10^{-6}
- **Bin 6:** $>7.5 \times 10^{-6}$ to 5×10^{-5}
- **Bin 7:** $>5 \times 10^{-5}$.

The HQ/MOE bins are

- **Bin 1:** 0 to 0.05
- **Bin 2:** > 0.05 to 0.5
- **Bin 3:** > 0.5 to 5.0
- **Bin 4:** > 5.0.

All bin ranges are inclusive of the upper bound.

For any exposure period, and given a receptor type, age cohort, and pathway (or pathway aggregation), a cumulative risk histogram is generated that contains the total population corresponding to the receptor/cohort combination, across all receptor locations in the AOI that experiences risks falling within each risk bin range. An HQ cumulative histogram is constructed in a similar fashion.

As an example, assume 150 adult residents reside collectively at all residential receptor locations within an AOI. Assume that for a given contaminant and exposure pathway for a specified exposure period starting in year 1, they fall into the bins as shown in Table 14-1. The conditional, cumulative population for these two example exposure periods would be as shown in Table 14-2.

Table 14-1. Example HQ Counts for Hypothetical Sites

	Population in Each HQ Range			
	Bin 1	**Bin 2**	**Bin 3**	**Bin 4**
Exposure starting in...	**<0.05**	**0.05 to 0.5**	**0.5 to 5**	**>5**
Year 1	30	45	75	0
Year 2	15	45	60	30
Year 3	6	24	60	60

Table 14-2. Example Cumulative Frequency at Hypothetical Site

Exposure starting in...	Percent of Population in this HQ Range or Lower			
	Bin 1	Bin 2	Bin 3	Bin 4
Year 1	20	50	100	
Year 2	10	40	80	100
Year 3	4	20	60	100

Determine Timing of Maximum Risk. The third major function of the Human Risk Module is to determine the timing of maximum risk and/or HQ across all receptor/cohort individuals for a given exposure pathway or aggregation of pathways. The time series of risk histograms is analyzed to determine that year in which the maximum total risk and/or HQ over time occurs.

Specifically, the Human Risk Module first estimates the total risk/HQ for all individuals at a site by multiplying the population at each receptor location by the calculated risk or HQ at that location. This is then summed across all locations, receptor types, and age cohorts. The Human Risk Module calculates this total risk for each exposure period in the time series. The module then determines the exposure period for which this total risk is the highest.

14.3 Module Discussion

14.3.1 Strengths and Advantages

The major strengths and advantages of this module include the following:

- **Provides coverage for key receptor populations and exposure pathways**. The Human Risk Module supports risk characterization for four types of resident receptors and four types of farmer receptors. Together, this set of receptor populations includes the majority of those typically considered in evaluating multipathway exposure and risk. Further, in modeling each of these receptor populations, the 3MRA modeling system provides coverage for key exposure pathways related to receptor behavior (e.g., ambient air inhalation, crop ingestion). To enhance the representativeness for risk estimates, the modeling of specific exposure pathways is linked to known receptor activity characterized using demographic (or other relevant) data. The 3MRA modeling system also includes five age cohorts in modeling risk for each receptor population to reflect age-specific differences in exposure and risk. Specifically, cohort aging is considered in modeling risk such that the ingestion or inhalation rates used for a given receptor are adjusted as that receptor “ages” into the next age cohort to reflect the exposure parameters relevant for that older age group.

- **GIS-based representative modeling of population-level risk.** The Human Risk Module uses a GIS-based framework to support spatially representative modeling of risk for modeled receptors. The use of U.S. Census, Census of Agriculture and GIRAS land use data to place residents, drinking water wells, farmers, farms, and recreational fishers for modeling risk supports a more representative analysis of population-level exposure which explicitly considers the spatial distribution of these receptors across AOIs.

- **Generation of time series of risk profiles for modeled receptor populations provides coverage for contaminants with different temporal exposure profiles.** The Human Risk Module generates a time series of risk estimates for each combination of receptor/pathway/location/simulation year/chemical-endpoint that allows the temporal profile of risk over longer simulation periods (up to 10,000 years) to be evaluated. Providing detailed tracking of temporal risk profiles over longer modeling periods can be important in assessing multipathway risk for pathways involving contaminants with widely varying fate and transport profiles (e.g., an inhalation toxicant that produces risk shortly after WMU release, versus a contaminant with low mobility that can take many years to reach an off-site well).

- **Use of cumulative risk histograms provides a ready means for identifying the year of maximum risk given widely varying risk profiles for different receptor/pathway/contaminant combinations.** The Human Risk Module produces cumulative risk histograms to characterize the distribution of risk across specific receptor populations for a given exposure pathway/simulation year/contaminant combination. These cumulative risk histograms can be used as the basis for identifying the simulation year which has the maximum cumulative risk for a given regulatory percentile of the population (e.g., the maximum cumulative risk for the 95th percentile of beef farmers resulting from beef ingestion). The 3MRA modeling system includes an automated procedure for querying the entire time series of cumulative risk distributions to identify this maximum risk year and then outputs the risk distributions for all exposure pathways modeled for that receptor population for that year. Not only does this procedure provide a ready means to identify the significant maximum risk year and extract the full risk distributions generated for that year, this approach also represents an effective means of data reduction, which is a key issue given the large number of simulation years and receptor/pathway/location/contaminant combinations that can be modeled. Cumulative risk distributions can be generated for user-specified distance rings within the AOI. This provides a distance-differentiated set of risk metrics that can be used to support decision making.

14.3.2 Uncertainty and Limitations

The following limitations or uncertainties are inherent in the Human Risk Module:

- **Risk/HQ/MOE estimates are aggregated for certain receptor types.** The four receptor types considered by the Human Risk Module (resident, residential gardener, farmer, fisher) are fewer than the number of receptor types simulated by the Human Exposure Module in order to maintain output storage requirements at reasonable levels. The Human Risk Module internally aggregates dairy farmers and beef farmers into a single farmer (dairy and beef) and aggregates all of the Human Exposure Module's receptor type-specific fishers (e.g., resident fisher) into a single fisher receptor. Some resolution is lost by this aggregation; for example, the risks specific to farmers who drink contaminated milk, but do not consume contaminated beef, would not be available.

- **Synergistic or antagonistic effects among multiple contaminants or individual contaminant species on risk/HQ/MOE are not considered.** The Human Risk Module considers only one contaminant at a time, and the risk associated with that contaminant is implicitly considered to be independent of the risks posed by other contaminants.

- **Cancer slope factors do not vary with cohort age.** Age-specific differences in exposure responses are not considered.

- **The fraction of residents in a Census block that ingest ground water is assumed to equal the fraction of residents in the Census block group that have ground water wells.** The Census data report the number of households within a Census block group that are served by ground water wells. However, this information is available only at the Census block group level, and it is not possible to determine from Census data alone whether individual Census blocks within a block group with wells have wells or not. The Human Risk Module calculates at the Census block level when considering residential exposure areas. Consequently, the actual fraction of residents on wells for any individual residential exposure area is uncertain. The assumption is made that the fraction of residents in an exposure area (a Census block) consuming ground water is equal to the fraction of the population in the Census block group that are served by ground water wells. That is not to say that all well water is contaminated. Only those wells lying within the ground water plume from the WMU source are potentially contaminated. To the extent that the fraction of residents on wells differs among Census blocks within the plume and outside the plume, population risks may be over- or underestimated.

14.4 References

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U.S. EPA (Environmental Protection Agency). 1997c. *Exposure Factors Handbook. Volume III-Activity Factors*. EPA/600/P-95/002Fa. Office of Research and Development, Washington, DC. Website at http://www.epa.gov/ncea/exposfac.htm. August.

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15.0 Ecological Exposure Module

15.1 Purpose and Scope

The Ecological Exposure Module calculates the applied contaminant dose (in mg/kg-d) to ecological receptors that may be exposed via ingestion of contaminated plants, prey, and media (i.e., soil, sediment, and surface water). The Ecological Exposure Module uses various input contaminant concentrations from the Surface Impoundment, Surface Water, Terrestrial Food Web, and Aquatic Food Web Modules. Detailed information on the Ecological Exposure Module can be found in the background document (U.S. EPA, 2000). Figure 15-1 shows the relationship and information flow between the Ecological Exposure Module and the 3MRA modeling system.

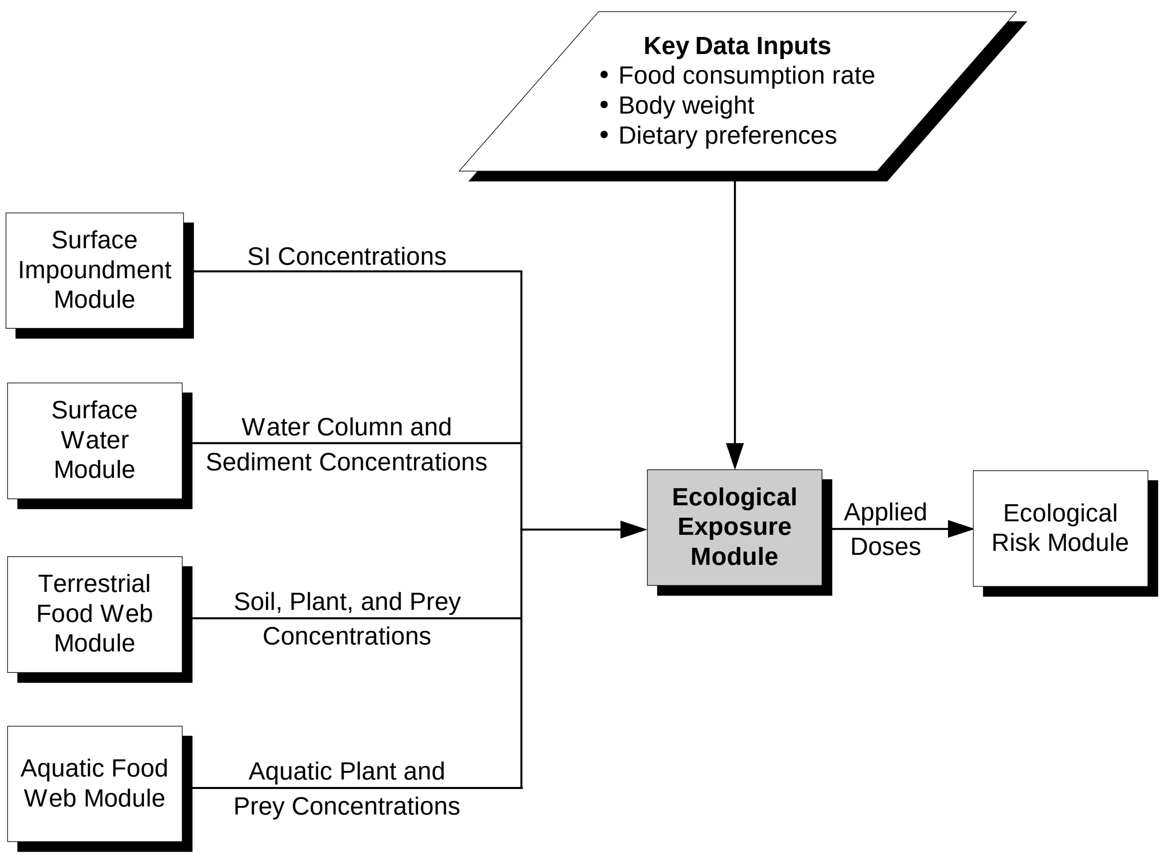

Figure 15-1. Information flow for the Ecological Exposure Module in the 3MRA modeling system.

The methodology and equations used to calculate the applied dose to mammals and birds assigned to habitats within the AOI are consistent with the principles and guidelines described in the *Guidelines for Ecological Risk Assessment* (U.S. EPA, 1998). The basic forms of these equations have been used by OSW and other EPA programs to predict applied doses in a variety of ecological risk analyses, and they are similar to the exposure equations recommended by other non-EPA risk assessors (see, for example, the *Methods and Tools for Estimation of the Exposure of Terrestrial Wildlife to Contaminants,* Sample et al., 1997). The Ecological Exposure Module performs the following three functions:

1. **Constructs a dietary matrix for each receptor for each habitat in the AOI.** The Ecological Exposure Module creates a diet for each ecological receptor based on dietary preferences.

2. **Calculates applied doses for animals in terrestrial habitats.** Using the dietary matrix and the media,[1] plant, and prey concentrations calculated by the Terrestrial Food Web Module, the Ecological Exposure Module calculates applied doses for each avian and mammalian receptor species assigned to terrestrial habitats in the AOI.

3. **Calculates applied doses for animals in margin habitats (wetland or waterbody).** Using the dietary matrix and the media,[1] plant, and prey concentrations calculated by the Aquatic Food Web and Terrestrial Food Web Modules, the Ecological Exposure Module calculates applied doses for each avian and mammalian receptor species assigned to margin habitats in the AOI.

15.2 Conceptual Approach

The conceptual approach to predicting ecological exposures in the 3MRA modeling system addresses several major sources of variability in ecological exposures, such as environmental characteristics of different ecosystems, spatial resolution of contaminant concentrations, dietary composition, and receptor-specific exposure factors. Specifically, the approach addresses variability through (1) the development of representative habitats; (2) selection of receptors based on ecological region; (3) the recognition of opportunistic feeding and foraging behavior using probabilistic methods; (4) the creation of a dietary scheme specific to region, habitat, and receptor; and (5) the application of appropriate graphical tools to capture spatial variability in exposure.

The Ecological Exposure Module simulates exposures for birds and mammals assigned to 11 types of habitats representing waterbody margin habitats for streams/rivers, lakes, and ponds; terrestrial habitats; and wetland margin habitats. The 3MRA modeling system exposure factor database contains data on 57 species, including mammals (23), birds (22), and selected herpetofauna (12). This suite of receptors covers a wide range of feeding strategies from obligate herbivores to opportunistic feeders. Because toxicological data were considered

[1] Constituent concentrations in surface impoundments may also be used to calculate exposure if the receptor's home range overlaps the impoundment.

insufficient to develop ingestion benchmarks for herpetofauna for the contaminants included in this version of the 3MRA modeling system, the Ecological Exposure Module does not predict applied doses to these receptors. However, the module was developed to calculate exposures to herpetofauna as toxicological data become available.

For each year in the simulation, the Ecological Exposure Module predicts the applied dose to mammals and birds assigned to waterbody margin, wetland, and terrestrial habitats delineated within the area of interest (AOI). The model is flexible enough to be applied to any habitat, allows a unique home range to be defined for each receptor in the AOI, and constructs a receptor-specific diet based on information in the ecological exposure factor database. Simple food webs, constructed to represent each type of habitat, provide the framework for creating a matrix of dietary preferences for each simulation. That is, the diet of avian and mammalian receptors is constructed based on the food web specific to each habitat, and the dietary preferences can be varied with each simulation. Data on dietary preferences are used to determine which of the 17 possible categories of food are eaten by each receptor and how much of each food category is eaten.

The ecological exposure framework is largely based on the desire to provide an appropriate level of resolution for national-scale analyses. The Ecological Exposure Module performs the necessary calculations to predict contaminant exposures for birds and mammals; the underlying framework defines the spatial scale for the exposure modeling, as well as the habitats and receptors that are assigned to the AOI.

15.2.1 Criteria for the Ecological Exposure Module

The Ecological Exposure Module was designed to satisfy three important criteria that respond to reviews of the earlier efforts to conduct a national-level ecological risk assessment. Those criteria are as follows:

- The module must capture the wide variability in ecological systems in a manner that is appropriate given the availability of data to characterize and evaluate ecological exposure and risk.

- The module must define spatial boundaries for ecological exposures at a scale that takes full advantage of the spatial resolution offered by the 3MRA modeling system and is meaningful with respect to predicting ecological risks.

- The module must allow for the site-based assignment of ecological receptors that reflect the major trophic elements and feeding strategies relevant to exposure, as well as regional characteristics that influence the composition of ecological communities.

To satisfy these criteria, the Ecological Exposure Module contains the following three critical elements:

- A representative habitat scheme,
- Habitat-specific food webs, and

- Appropriate receptor species for each habitat and food web.

Within the context of these criteria, the Ecological Exposure Module selects the media, plant, and prey contaminant concentrations to which each receptor is exposed and calculates the applied dose to chemical stressors across space and time. The Ecological Exposure Module was developed to allow for flexibility in calculating exposures in habitats, home ranges, and receptors that can be defined at virtually any scale desired by the user. For instance, a home range can be defined as a spatially unique area for each receptor, or many receptors can share the same home range. Regardless of how the spatial extent of exposure is defined, the module will calculate a “spatially consistent” applied dose, allowing each receptor to ingest plants, prey, or environmental media within its home range. Similarly, the representative habitats can be delineated for virtually an unlimited number of sizes and shapes provided that the geographical coordinates can be identified. Although a receptor species can be assigned to multiple habitats within an AOI, the Ecological Exposure Module calculates a time series of applied doses to each receptor within each habitat once and only once. That is, a given receptor may not be assigned to multiple home ranges within a habitat.

Representative Habitat Scheme. The representative habitats shown in Table 15-1 were designed to be general enough to encompass a broad range of ecological systems in the conterminous 48 states and can be used to define the spatial boundaries of exposure for any site in the 3MRA modeling system. The representative habitats capture important characteristics of a variety of environmental settings that determine what plants and animals are likely to be present and what exposure pathways are likely to be of interest. The representative habitat scheme supports a level of detail commensurate with the data and models available to predict exposures to chemical stressors. For example, there are many important ecological distinctions between coniferous and deciduous forests; however, the fate and transport models in the 3MRA modeling

Table 15-1. Representative Habitats for 3MRA

Terrestrial Habitats
Grassland
Shrub/scrub
Forest
Crops/pasture
Residential
Waterbody Margin Habitats
River/stream
Lake
Pond
Wetland Margin Habitats
Permanently Flooded Grassland
Permanently Flooded Shrub/scrub
Permanently Flooded Forest

system are not capable of producing forest-specific simulations, and data on potential forest receptors are insufficient to generate a meaningful distinction in predicted exposures and risks to receptors found in these two types of forest habitats. Consequently, the forest habitat represents both deciduous and coniferous forests in the 3MRA modeling system.

The type of habitat determines which species are present at a site because the habitat provides essential resources such as food; shelter; nesting sites or materials; and appropriate sites for behaviors such as courtship, mating, roosting, or hibernation. The criteria used to develop the representative habitats were based on a survey of existing ecological classifications and reflect the importance of the physical setting in terms of essential resources (Bailey, 1996; Bourgeron and Engelking, 1994; Cowardin et al., 1979; Davis and Simon, 1995; Demarchi, 1996; Drake and Faber-Langendoen, 1997; Federal Geographic Data Committee, 1997; Küchler, 1964; Omernik, 1987; Sawyer and Keeler-Wolf, 1995; Shafale and Weakley, 1990; USDA Forest Service, 1994; U.S. FWS, 1998; Viereck and Elbert, 1991; Weakley et al., 1998; and Whitney, 1985). The following subsections provide a very brief summary of the criteria and resulting habitats that were defined to represent the wide variability in ecological systems; additional detail is provided in U.S. EPA (1999b).

Criteria for Terrestrial and Wetland Habitats. The primary criteria for defining terrestrial and wetland habitats are soil moisture and vegetation structure. Soil moisture, or degree of saturation, affects soil chemistry, general vegetation structure, and habitat suitability. Soil moisture is differentiated based on the following three categories:

- Terrestrial—well-aerated, nonsaturated soils;
- Intermittently flooded—periodically saturated or inundated but aerated for some periods during the growing season; and
- Permanently flooded—saturated or inundated throughout most years.

These characteristics are quite general and do not include many of the abiotic parameters often associated with ecological classification systems, such as latitude, climate, topography, elevation, or soil type. Nevertheless, vegetation type is directly affected by these abiotic parameters and is, therefore, often used as a general indicator of many abiotic characteristics. Vegetation structure refers to the stature, spacing, and relative stem size of the dominant vegetation. It describes the primary producers and indicates the appropriateness of the habitat for use by major trophic levels. Generally accepted categories of vegetation structure include grasses, herbs, shrub/scrub, forest, and cropland. Each category has dominant vegetation with distinct height and density that, in turn, supports a distinct suite of fauna.

In addition to the general classifications cited in U.S. EPA (1999b), primary sources for defining wetland habitats include Christensen et al. (1988), Damman and French (1987), Glaser (1987), Gosselink and Turner (1978), Kadlec and Knight (1996), Larsen (1982), Mitsch and Gosselink (1993), Niering (1985), Norquist (1984), Sharitz and Gibbons (1982), Verry (1997), Windell et al. (1986), and Winter (1989).

Criteria for Freshwater Margin Habitats. Using waterbody margin habitats in the exposure assessment allows for the inclusion of freshwater aquatic plants and prey that are integral parts of the food web for terrestrial receptors in habitats that include fishable waterbodies. A brief review of the literature supports the use of two criteria, energy/flow and size, as a simple but effective classification approach. Although aquatic classifications consist of more complex divisions, they fundamentally include these two criteria. The energy/flow criterion distinguishes between still water and flowing water, and the size criterion addresses the intrinsic differences between large and small systems, such as net primary production, diversity of habitat, and length and complexity of food webs. No commonly used size cutoffs were found in the literature. For lakes versus ponds, the waterbody margin habitats use EPA's Environmental Monitoring and Assessment Program (EMAP) 10-hectare cutoff for small versus large lakes; thus, surface waterbodies with a surface area greater than 10 hectares were classified as lakes, and those below 10 hectares were classified as ponds. For streams versus rivers, the generally accepted stream order concept was considered, and stream order 5[2] was initially proposed as a cutoff between streams (small flowing waterbodies) and rivers (large flowing waterbodies). Based on simple mass balance calculations of probable contaminant loadings to surface waterbodies, it was readily apparent that the predicted contaminant concentrations in streams larger than stream order 5 would effectively be diluted below detectable levels. Therefore, streams and rivers of order 5 and below were represented as a single habitat, and those of order 6 and above are not included in the ecological exposure assessment. References consulted for the development of the waterbody margin habitats include the habitat classifications cited in U.S. EPA (1999), as well as in Davis and Simon (1995), USDA (1998), and Caduto (1990).

Habitat-Specific Food Webs. The representative habitats are used not only to define the spatial boundaries for site-based exposures, but also to provide the ecological basis for development of food webs that reflect important exposure pathways to chemical stressors. The habitat-specific food webs were designed to be simple enough to allow for parameterization in the model, but flexible enough to capture a full range of feeding strategies and exposure pathways. The food webs were developed based on generally accepted concepts about food webs and natural community dynamics (Anderson, 1997; Begon and Mortimer, 1981; Caduto, 1990; Davis and Simon, 1995; Kadlec and Knight, 1996; Sample et al., 1997; Schoener, 1989; Schoenly and Cohen, 1991; Suter, 1993; Tanner, 1978; U.S. EPA, 1993, 1994). In describing the predator-prey interactions in the representative habitats, a number of sources were consulted for general habitat information on receptor species, including a wide variety of field guides, nature guides, wildlife encyclopedias, and species-specific monographs that describe the habitats known to be frequented or used by the species. All dietary items reported as commonly eaten were considered in constructing the food webs and, of course, in developing the ecological exposure factors described in Volume II on data collection.

Terrestrial Habitat Food Webs. As suggested in Figure 15-2, the receptors assigned to the terrestrial food web include primary producers (vascular plants); soil biota; and birds; mammals, and herpetofauna across three trophic levels. Trophic level 1 (TL1) consists of species that consume only plants (i.e., the herbivores) and that are potential prey for higher

[2] The Strahler (1957) stream ordering system is used throughout this document.

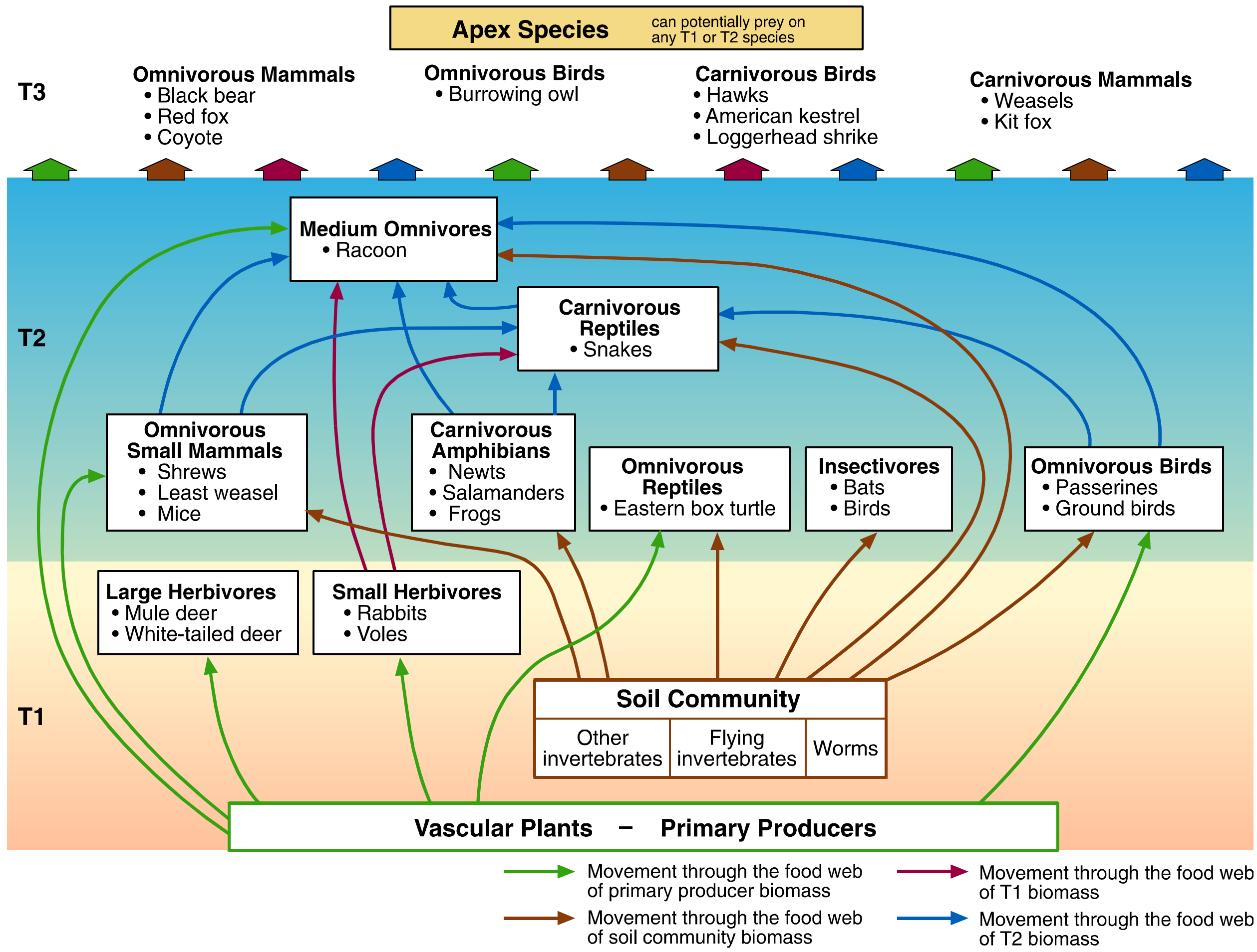

Figure 15-2. Simple terrestrial food web showing example receptors.

trophic level species. TL1 species include small and large mammals and invertebrates. The soil community is a unique subset within TL1 and includes invertebrate soil organisms that live in direct contact with soil, thus reflecting a unique exposure pathway. Within this conceptual framework, the soil community is both a source of food for certain receptors and a receptor group evaluated by the Ecological Risk Module. The dynamics within soil communities are, in fact, very complex and include herbivores, omnivores, and carnivores at several trophic levels within the soil community. However, modeling this complexity within the 3MRA modeling system is well beyond the level of resolution that can be used in a national assessment strategy.

Trophic level 2 (TL2) includes species that consume plants and/or animals (omnivores and carnivores) and are themselves eaten by larger predators. The TL2 species include a wide array of small- to medium-sized mammals, birds, and herpetofauna. For example, the TL2 carnivores include species of reptiles and amphibians that eat soil invertebrates, as well as specialized feeding guilds such as insectivores. The species included in TL2 represent several faunal classes, functional groups, and size ranges. Opportunistic feeders in TL2, such as raccoons, increase the complexity of the web by feeding on virtually any TL1 or TL2 prey. Trophic level 3 (T3) consists of apex species, or those that do not have any predators (other than humans) in the habitat. Apex species include several faunal classes of receptors, such as large mammals (black bears) and raptors (Cooper's hawks), and tend to be among the widest ranging receptors in the habitat.

Waterbody/Wetland Margin Habitat Food Webs. As suggested by Figure 15-3, the margin habitats for surface waterbodies (e.g., streams, lakes) include elements of both aquatic systems and terrestrial systems. As with the terrestrial habitats, the receptors assigned to the aquatic and wetland food webs include primary producers (vascular aquatic plants and algae); sediment and surface water biota; and birds, mammals, and herpetofauna across three trophic levels. These receptors may be exposed through the ingestion of aquatic biota (e.g., fish, benthic invertebrates, and aquatic plants) and surface water, as well as through the incidental ingestion of sediment. In addition, receptors assigned to margin habitats may be exposed through the ingestion of soil, terrestrial plants, and prey items described in the terrestrial habitat food web section (e.g., raccoons). For receptors in these habitats that feed strictly on aquatic food items (e.g., muskrats and mink), the food web is assumed to be relatively simple, consisting of the four-compartment aquatic food web directly linked to the receptor. For example, the osprey eats fish almost exclusively and, therefore, is an apex predator at the top of this simplified aquatic food web.

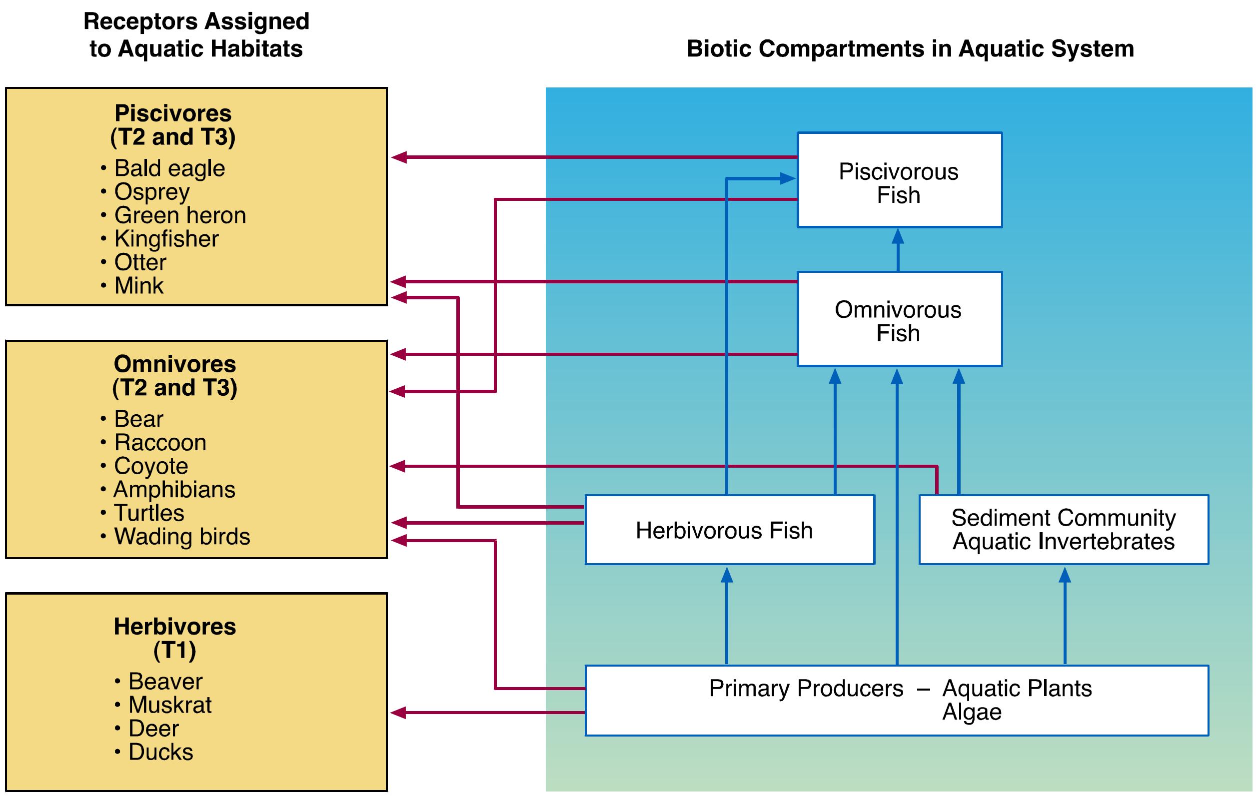

Figure 15-3. Simple margin food web showing both aquatic and terrestrial components.

For omnivores and more opportunistic species in the waterbody margin habitats, the food web is essentially the same as that for strictly terrestrial species, with aquatic prey available as additional potential food items. Many of these species (e.g., raccoon or black bear) are equally successful whether or not aquatic prey are available; however, these species are opportunistic and will take advantage of any prey that are readily available.

The receptor assignments for wetland habitats raised some unique questions with respect to flooding regimes and the implications on habitat classification and receptors. Habitat information for wetland species rarely indicates the degree or frequency of flooding when describing wetland habitats. Therefore, it was difficult to differentiate between species using intermittently flooded as opposed to permanently flooded wetland habitats. In many cases, the food or prey items attributed to a species were used as an indicator. For example, if a source reported that a species fishes in wetland habitats, that species was assigned to the appropriate permanently flooded wetland habitat, which, as defined in this framework, supports fish and other aquatic life. Because of the predominant use in the source literature of terms such as swamps and marshes, which imply a relatively long flood duration, fewer species were assigned to the intermittently flooded wetland habitats. Indeed, intermittently flooded wetlands are generally less discernible from surrounding uplands than are permanently flooded wetlands and, thus, are reported less frequently in general wildlife habitat literature. These differences are reflected in the smaller receptor groups associated with the intermittently flooded wetlands. Intermittently flooded wetlands are not delineated as independent habitats in the 3MRA modeling system. Consequently, receptors assigned to intermittently flooded wetlands are simply added to the list of receptors assigned to the terrestrial habitat if the intermittently flooded wetland is located within the spatial boundaries of the terrestrial habitat.

Ecological Receptors. Within the framework of habitats and food webs, receptor species were selected using a weight-of-evidence approach intended to support analyses of various habitats across the conterminous 48 states. Most importantly, receptor species were chosen to reflect a full range of exposures and, as a group, represent all of the faunal classes, trophic levels, and feeding strategies that are typical of terrestrial and aquatic margin habitats. The simple food webs created for the terrestrial and aquatic margin habitats provided the context for receptor selection and are used to define the relationships between predators and prey. The receptors selected to populate the food webs may be both predator and prey in a given habitat. For example, small omnivores, such as mice or shrews, may consume a variety of plants and animal prey items in lower trophic levels. Given their position in the food web, mice or shrews might also be eaten by apex predators, such as the coyote or red-tailed hawk. Key determinants for receptor selection are summarized as follows:

- **Geographic distribution.** National applicability was achieved primarily by selecting species that are widely distributed throughout the conterminous 48 states and then adding species to cover as many ecological regions as possible. Section level data from Bailey’s ecoregions (Bailey, 1996) were digitized and included in the 3MRA geographic information system (GIS). Only those species documented to occur in the section where a site was located were included in the exposure assessment.

- **Availability of wildlife exposure factors.** The majority of the receptors selected for the representative national data set included species for which wildlife exposure factors were readily available. The main sources for ecological exposure factor data were the *Wildlife Exposure Factors Handbook* (U.S. EPA, 1993), *Methods and Tools for Estimation of the Exposure of Terrestrial Wildlife to Contaminants* (Sample et al., 1997), and the U.S. Army Corps of Engineers’ *Species Profile Series* (various dates and authors, e.g., Lane and Mitchell, 1997).

- **Faunal class.** Major faunal classes—mammals, birds, reptiles, and amphibians—generally occur throughout all trophic levels. Faunal class distinctions are useful in selecting receptors because species within similar groups are known to respond similarly to environmental disturbance. Although the 3MRA modeling system contains redundancy within each faunal class, exposure factors were not available for species in all classes at all trophic levels. As a result, certain species are underrepresented.

- **Functional niche.** Receptors experience different exposures based on dietary composition. To ensure that all potential exposure pathways were reflected in each habitat, receptors were selected to cover the full spectrum of dietary preferences and feeding strategies (e.g., herbivores, carnivores, omnivores, and insectivores).

To summarize, receptor species were chosen to reflect a full range of exposures and, as a group, represent all of the faunal classes, trophic levels, and feeding strategies that are typical of terrestrial and aquatic margin habitats, respectively. Table 15-2 provides a complete list of receptor species cross-referenced by representative habitat.

Table 15-2. Representative Habitats for Receptor Species

	Representative Habitats										
	Terrestrial					**Waterbody Margin**			**Wetland Margin**		
Species	**Grasslands**	**Shrub/Scrub**	**Forest**	**Cropland**	**Residential**	**River/Stream**	**Lake**	**Pond**	**PF Grasslands**	**PF Shrub/Scrub**	**PF Forest**
Alligator snapping turtle						✔	✔	✔			✔
American kestrel	✔		✔	✔	✔						
American robin	✔	✔	✔	✔	✔						
American woodcock	✔	✔	✔	✔		✔			✔	✔	✔
Bald Eagle						✔	✔	✔	✔	✔	✔
Beaver						✔	✔	✔	✔	✔	✔
Belted kingfisher						✔	✔	✔	✔	✔	✔
Black bear		✔	✔			✔	✔	✔			✔
Black-tailed jackrabbit	✔	✔		✔							
Bullfrog						✔	✔	✔	✔		✔
Burrowing owl	✔			✔							
Canada goose							✔	✔			
Cerulean warbler			✔								✔

(continued)

Table 15-2. Representative Habitats for Receptor Species (continued)

Species	Representative Habitats										
	Terrestrial					Waterbody Margin			Wetland Margin		
	Grasslands	Shrub/Scrub	Forest	Cropland	Residential	River/Stream	Lake	Pond	PF Grasslands	PF Shrub/Scrub	PF Forest
Cooper’s hawk			✔	✔	✔						
Coyote	✔	✔	✔	✔						✔	✔
Deer mouse	✔	✔	✔	✔							
Eastern newt			✔				✔		✔	✔	✔
Eastern cottontail rabbit	✔	✔	✔	✔	✔						
Eastern box turtle	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Flatwoods salamander					✔			✔	✔	✔	✔
Gopher frog	✔	✔	✔					✔			
Great blue heron						✔	✔	✔	✔	✔	✔
Great Basin pocket mouse	✔	✔									
Green heron						✔	✔	✔	✔	✔	✔
Green frog						✔	✔	✔	✔	✔	✔
Herring gull						✔	✔	✔	✔	✔	
Kit fox	✔	✔									
Least weasel	✔		✔	✔							
Lesser scaup							✔	✔	✔		✔
Little brown bat				✔	✔		✔	✔			
Loggerhead shrike	✔	✔	✔	✔							
Long-tailed weasel	✔		✔	✔							
Mallard						✔	✔	✔	✔	✔	✔
Marsh wren									✔	✔	
Meadow vole	✔			✔	✔						
Mink						✔	✔	✔	✔		
Mule deer	✔	✔	✔						✔	✔	✔
Muskrat						✔	✔	✔			
Northern water snake						✔	✔	✔	✔	✔	✔
Northern bobwhite	✔	✔	✔	✔							
Osprey						✔	✔	✔	✔	✔	

(continued)

Table 15-2. Representative Habitats for Receptor Species (continued)

Species	Representative Habitats										
	Terrestrial					Waterbody Margin			Wetland Margin		
	Grasslands	Shrub/Scrub	Forest	Cropland	Residential	River/Stream	Lake	Pond	PF Grasslands	PF Shrub/Scrub	PF Forest
Painted turtle						✔	✔	✔			
Pine vole	✔		✔								
Prairie vole	✔			✔							
Raccoon	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Racer	✔	✔									
Red fox	✔	✔	✔	✔					✔	✔	✔
Red-tailed hawk	✔	✔		✔	✔						
River otter						✔			✔	✔	
Short-tailed shrew	✔		✔	✔	✔						
Short-tailed weasel	✔		✔	✔							
Snapping turtle							✔	✔			
Southern hognose snake			✔	✔							
Spotted sandpiper						✔	✔	✔			✔
Tree swallow	✔	✔		✔	✔						
Western meadowlark	✔	✔		✔							
White-tailed deer	✔	✔	✔			✔	✔	✔			

Within the framework of representative habitats, food webs, and receptors, the Ecological Exposure Module constructs dietary matrices and calculates exposures. These functions are described in the following sections.

15.2.2 Construct a Dietary Matrix

To predict a time series of applied doses for wildlife, the Ecological Exposure Module must first construct a dietary matrix for each receptor based on the dietary preferences contained in the ecological exposure database. The database contains each receptor's minimum and maximum preference values for each of the dietary items shown in Table 15-3. These categories are sampled to generate a percentage of the diet within the range of the minimum and maximum preference values in the database for a particular receptor and dietary item.

Table 15-3. Categories of Dietary Items for Ecological Exposure Assessment

Category	Dietary Item
Terrestrial Prey	Earthworms
	Other soil invertebrates
	Small mammals
	Small birds
	Small herpetofauna
	Medium omnivores
	Herbivores
Aquatic Prey	Benthic filter feeders
	TL3 fish
	TL4 fish
Vegetation	Aquatic plants
	Fruits, fruit/seeds (single item)
	Fern(s), fungi, dicot & monocot shoots
	Forbs, grasses, shrubs
	Roots
	Crops, corn
	Seeds/nuts
Environmental Media	Soil
	Sediment
	Surface water

The Ecological Exposure Module rank orders categories of dietary items from most preferred to least preferred (based the maximum preference values), and the categories are sampled starting with the most preferred category first and continuing through the ranked categories in order until the diet is complete. The sampling approach was designed to address the wide variability in animal diets indicated by available data and, at the same time, observe trends in dietary preferences indicated by those data. There is no requirement that a receptor consume every food item in the list; however, the diet is constrained by two rules: (1) any food item with a non-zero minimum preference value must be part of the diet,[3] and (2) the summation of dietary preferences must equal 100 percent of the entire diet.

[3] In general, the three or four dietary items with the highest maximum values have non-zero minimum values. As a result, the sampling routine usually completes 100 percent of the diet quickly.

Table 15-4 provides an example of the construction of a dietary matrix for a raccoon assigned to a stream margin habitat. The raccoon's dietary composition is constructed based on the prey preference data on various plants, prey, and media shown, and includes both terrestrial and aquatic food items. Based on the preference data, the Ecological Exposure Module might construct the dietary matrix shown in the Example Dietary Matrix column of Table 15-4.

Table 15-4. Example of Dietary Preferences for Raccoon

Food Item	Minimum Preference Value (% of diet)	Maximum Preference Value (% of diet)	Example Dietary Matrix (% of diet)	Cumulative Diet (% of diet)
Soil invertebrates	0	90	45	45
Fruits	25	86	26	71
Forbs	10	66	12	83
Small mammals	0	35	5	88
Benthic organisms	0	25	7	95
TL3 fish	0	23	5	100
TL4 fish	0	23	0	
Small birds	0	19	0	
Earthworms	0	10	0	
Grain	0	10	0	
Roots	0	10	0	
Silage	0	10	0	

Information on receptor species' dietary composition comes from a wide range of data sources and is of two general types. Some data consist of reported quantities of certain items eaten by particular individuals in a localized or site-specific study. These data consist of measured stomach contents, nest or burrow contents, or counts of items observed to be eaten during a particular time span. The principal sources for these data are the *Exposure Factors Handbook* (U.S. EPA, 1993) and Sample et al. (1997). The second type of information consists of qualitative reports of items documented to be eaten. These reports reflect a compilation of observations and measurements for the species in general, and are reported as descriptions of the species' potential diet. The principal sources of this type of data are the Army Corps of Engineers' *Species Profile Series*, the American Society of Mammalogists' *Mammalian Species Series*, and various field guides and handbooks (e.g., Willner et al., 1980).

When only qualitative data were available, the assignment of estimated dietary fractions was based primarily on a set of decision rules implemented by a senior ecologist to maintain consistency in interpreting qualitative descriptions. References to a diet item that implied a single most significant component, such as "primary food source," "bulk of the diet," or "consumes mostly," were assigned a minimum of 50 percent dietary composition. Items that are of secondary importance but that would always make up at least some portion of the diet were assigned a minimum of 10 percent. Descriptions implying occasional sources of food were given a minimum of zero and a maximum of 10 percent. A few species have relatively limited diets and eat only one dietary item. For example, the cerulean warbler eats insects almost exclusively. In these cases, the exclusive diet item (insects) was assigned a minimum of 95 percent and a maximum of 100 percent. The quantified dietary profiles were entered into the exposure factor database as maximum and minimum preference values for each prey category.

15.2.3 Calculate Applied Dose for Receptors in Terrestrial Habitats

The applied dose for mammals and birds assigned to terrestrial habitats is calculated as a function of the contaminant concentrations in soil, drinking water (e.g., streams or surface impoundments), and contaminated plants and prey. The calculation is a summation of time-dependent exposures to contaminated media, plants, and prey. The following example presents the methods and equations for estimating the applied dose to the short-tailed weasel in a forest habitat. The discussion is organized around five elements required to support the exposure calculation:

1. Spatial boundaries that define exposure;
2. Development of the dietary matrix;
3. Contribution to exposure from drinking water;
4. Contribution to exposure from soil, plants, and terrestrial prey; and
5. Calculation of total applied dose.

Spatial Boundaries for Exposure. All of the contaminant concentrations to which a receptor is exposed are spatially consistent; that is, each receptor may only be exposed to contaminant concentrations that overlap with its home range. However, it is not assumed that 100 percent of the diet originates from the home range; the Ecological Exposure Module prorates exposure based on the ratio between the area of the home range and the total habitat area. For example, if the home range for the weasel (~135,000 m^2) is 20 percent larger than the forest habitat, the calculated exposure is prorated by a factor of 0.8 that effectively reduces the applied dose to 80 percent of the dose that would have occurred had the weasel taken all of its food from within the AOI.

Development of Dietary Matrix. The Ecological Exposure Module uses the dietary preference database to construct the short-tailed weasel's diet for each iteration. The database includes all the relevant prey categories and the range of reported dietary preferences. As suggested in Table 15-5, the dietary matrix can vary considerably from iteration to iteration. The values shown in the last column comprise 100 percent of the weasel's diet for one iteration. Note that the data set follows the two constraints for the prey preference random sampling algorithm: (1) the diet includes all items with non-zero minimum preference values, and (2) the dietary preferences sum to 100 percent.

Table 15-5. Example Dietary Preferences for Short-Tailed Weasel

Food Item	Minimum Preference Value (%)	Maximum Preference Value (%)	Value for Realization (%)
Small mammals	50	80	63
Small herpetofauna	0	45	0
Soil invertebrates	0	25	13
Small birds	0	25	24
Total diet			100

Exposure Contribution from Ingestion of Surface Water. The Ecological Exposure Module estimates exposure due to surface water ingestion by calculating the total average concentration (dissolved plus particle-bound) for all surface waters found within the home range. This average includes any reach and/or surface impoundment that intersects the home range. Thus, it is implicitly assumed that the receptor does not prefer one waterbody over another. For example, if the weasel's home range overlaps two stream reaches with calculated surface water concentrations of 2.2E-06 mg/L and 9.3E-07 mg/L, respectively, the average drinking water concentration for contaminant *y* is 1.6E-06 mg/L.

Exposure Contribution from Ingestion of Soil, Plants, and Prey. The Ecological Exposure Module reads concentrations in plants, prey, and soil from the Terrestrial Food Web Module. For prey, these concentrations are spatially averaged for the prey's home range. For soil and soil invertebrate concentrations, the spatial averaging is defined by the home range of the short-tailed weasel. That is, the weasel is assumed to consume soil and soil invertebrates from within its home range, and the concentrations of these food items are assumed to be homogenous within the weasel's home range. For concentrations in small vertebrates, the weasel may take any prey that overlap its home range. Consequently, any small mammalian, avian, or herpetofaunal species that are assigned to the habitat are potential prey for the short-tailed weasel. This may include species within the weasel's home range, as well as species that simply overlap the weasel home range; thus, there is exposure variability associated with the prey selection (i.e., the home range of the prey species) due to potential differences in tissue concentrations in small vertebrates that may be eaten. To address this variability, the Ecological Exposure Module reads the minimum and maximum concentrations in mobile prey categories, such as small mammals, and randomly selects a value from that range. Thus, the exposure concentrations reflect the full range of prey combinations that could be eaten by the short-tailed weasel. Table 15-6 provides an example exposure profile for contaminant *y* for the weasel (calculated by the Terrestrial Food Web Module and passed to the Ecological Exposure Module). The diet includes incidental ingestion of soil, as well as the ingestion of contaminated soil invertebrates, small mammals, birds, and herpetofauna. A single concentration for soil, plants, and "local" food items, such as earthworms, is passed to the Ecological Exposure Module; however, for terrestrial prey categories that are relatively mobile, the Ecological Exposure Module receives a range of minimum and maximum prey concentrations and chooses one from within that range. A uniform distribution for prey concentrations is assumed to represent free access by predators to any prey that have access to its home range.

Table 15-6. Example Exposure Concentrations to Contaminant *y* for Short-Tailed Weasel

Food Item	Minimum Concentration (mg/kg)	Maximum Concentration (mg/kg)	Value for Iteration (mg/kg)
Soil (incidental)	3.8E-06	3.8E-06	3.8E-06
Small mammals	1.3E-06	6.7E-05	1.3E-05
Small herpetofauna	5.4E-07	5.4E-07	5.4E-07
Soil invertebrates	8.4E-06	8.4E-06	8.4E-06
Small birds	2.3E-08	7.8E-07	7.7E-07

Calculation of Total Applied Dose. After the Ecological Exposure Module constructs the dietary matrix and selects (from the outputs of other modules) exposure concentrations for relevant media, plants, and prey items to which the receptor has access (i.e., that overlap with the receptor's home range), the appropriate ecological exposure factors are used to calculate the total applied dose for each year in the simulation as shown in Equation 15-1 (all concentrations are converted to a wet weight basis).

$$Dose^{i}_{rec} = \frac{\sum\left(IR^{i}_{food}\,C^{j}_{food}\,FracPrey^{i\ \ j}_{food}\right) + \left(IR^{i}_{food}\,C_{soil}\,FracSoil^{i}\right) + \left(IR^{i}_{water}\,C_{water}\right)}{BodyWt^{1}}\,Frac^{i}_{HomeRange} \quad (15\text{-}1)$$

where

$Dose^{i}_{rec}$	=	total applied dose for receptor *i* (mg/kg-day)
IR^{i}_{food}	=	total ingestion rate of food for receptor *i* (kg/day)
C_{food}^{j}	=	concentration food item *j* to which receptor *i* may be exposed (mg/kg tissue)
$FracPrey^{i}_{food}{}^{j}$	=	dietary fraction of food item *j* for receptor *i* (unitless)
C^{i}_{soil}	=	average concentration in surficial soil in home range for receptor *i* (mg/kg)
$FracSoil^{i}$	=	dietary fraction of soil ingested by receptor *i* (unitless)
IR^{i}_{water}	=	ingestion rate of drinking water for receptor *i* (L/day)
C_{water}	=	average concentration in surface water within home range for receptor *i* (mg/L)
$BodyWt^{i}$	=	body weight of receptor *i* (kg)
$Frac^{i}_{HomeRange}$	=	fraction of home range for receptor *i* within habitat (unitless).

The applied dose reflects media and food concentrations for a single year in the simulation. For each realization (i.e., each time the Ecological Exposure Module runs), the module will read new data on media, prey, and food concentrations and construct a new dietary matrix from the dietary preference database. Although the exposure factors (e.g., body weight, food ingestion rate) and physical description (e.g., fraction of home range in the habitat) will remain constant throughout the simulation, the exposure profile will change over time as a function of changing media and food concentrations, as well as the random selection of dietary items.

15.2.4 Calculate Applied Dose for Receptors in Margin Habitats

The applied dose to mammals and birds assigned to margin habitats is calculated using essentially the same methodology described for terrestrial habitats. Equation 15-1 is modified to add the incidental ingestion of sediment, as appropriate, to the exposure pathways that contribute to the total applied dose. The list of dietary preferences for margin receptors is typically longer than that for terrestrial receptors because many species of mammals and birds that live margin habitats consume both aquatic biota and terrestrial biota. However, there are two subtle differences worth noting with respect to the exposure calculations for mammals and birds that are exposed to contaminated media and aquatic biota from waterbodies in margin habitats.

1. **Exposure to Aquatic Biota within a Margin Habitat.** As described in Sections 8 and 12, contaminant concentrations in sediment, surface water, and aquatic biota are calculated for each stream reach and surface waterbody within the AOI. The Surface Water Module reports these contaminant concentrations on a reach-specific and waterbody-specific basis. That is, even though stream reaches may be connected, the Surface Water Module may predict different contaminant concentrations. As a result, the exposure concentrations in sediment, surface water, and aquatic biota must be averaged prior to being used as inputs to Equation 15-1. The calculated concentrations in media and food are averaged for those reaches and waterbodies that overlap the home range for any mammalian or avian receptor assigned to a margin habitat. For example, the home range for an osprey assigned to a stream habitat might include three fishable stream reaches. The concentration in TL4 fish eaten by the osprey would be the average TL4 fish concentration across all three reaches. The underlying assumption is that a receptor may take food from any fishable reaches or waterbodies within its home range; therefore, the exposure concentration is an effective average for the receptor's home range.

2. **Exposure to Aquatic Biota in Adjacent Margin Habitat.** Margin receptors are presumed to rely on those fishable reaches that occur within their home range as the source of both aquatic food items and drinking water. However, a receptor's access to fishable reaches is not strictly constrained within its own habitat. In many instances, stream corridors are located adjacent to wetlands and, because receptors are likely to use reaches in each of these habitats as a source of food, adjacent margin habitats are delineated so that "reach crossover" can occur freely (i.e., habitats and home ranges overlap reaches in each habitat). For example, a stream margin located adjacent to a permanently flooded forest wetland would likely be delineated such that the habitat includes a part of both waterbodies. Consequently, the home range of a kingfisher assigned to the stream margin could

overlap both the stream and wetland, allowing this receptor to eat fish from either waterbody. The fish concentration to which the kingfisher is exposed would be the average tissue concentration, by trophic level, across the wetland and stream reaches that intersect its home range. The 3MRA modeling system models wetlands as distinct reaches, and, in this example, a forested wetland and a stream corridor would each be delineated and modeled separately by the system. The ecological connection that allows the kingfisher to take fish from both the stream and wetland is not reflected in the modeling simulation performed by the Surface Water Module.

The Ecological Exposure Module performs the averaging functions described above based on the data describing the habitats and home ranges for margin receptors and uses those average concentrations as inputs in Equation 15-1.

15.3 Module Discussion

15.3.1 Strengths and Advantages

The Ecological Exposure Module was developed to predict spatially explicit exposures to chemical contaminants for mammalian and avian receptors. Designed to address long-term, low-level exposures to chemical contaminants, the module offers several advantages for national-scale applications, and is flexible enough to perform site-specific analyses. The major strengths of the module include the following:

- **Exposure profiles calculated as total applied dose.** The vast majority of ecotoxicological data relevant to long-term exposures are reported as applied doses. Therefore, the Ecological Exposure Module was developed to report a time series of annual average applied doses so that the module would be broadly applicable to a large number of chemical contaminants. Although other metrics for exposure were considered (e.g., dietary concentrations, body burden), the module was designed to calculate applied dose so that EPA could take full advantage of a substantial body of data on reproductive and developmental effects associated with chronic exposures. This approach is consistent with EPA guidelines and the practice of ecological exposure assessment in numerous peer-reviewed ecological risk assessments conducted by EPA.

- **Maintains predator-prey relationships for soil, plant, and prey concentrations in space and time.** A major strength of the Ecological Exposure Module is its ability to calculate the total applied dose for each receptor assigned to each habitat within the AOI. For receptors in margin habitats, this is particularly important because the diet may include food items from both the terrestrial and aquatic food webs, as well as incidental ingestion of contaminated sediment and surface water. The module maintains the spatial relationship for all contaminated media and food items so that the aggregate exposure is calculated consistently for each year of the simulation. In addition to these advantages, the module avoids unnecessary conservatism by prorating exposure based on size of the receptor's home range, explicitly recognizing that feeding is not constrained by the habitat boundaries.

- **Dietary variability is addressed in the selection of food items.** The Ecological Exposure Module addresses this important source of variability by (1) sampling from the minimum and maximum concentrations predicted by the Terrestrial Food Web Module for appropriate food categories, and (2) using a random sampling algorithm to select dietary preference fractions for each food category. The former method explicitly recognizes that the food categories (e.g., small mammals) are intentionally general and that the specific prey items may vary substantially depending on feeding opportunities. The latter method, which is similar to the sampling algorithms used in the Aquatic Food Web Module, recognizes that the dietary proportions will be highly variable with season and other environmental conditions. Representing this variability (rather than assigning fixed, point estimates for diet) represents a significant improvement to the state-of-the-science for national-scale analyses.

- **Designed to evolve with changing assessment goals, new science, and improved data.** Given the state-of-the-science on predicting ecological exposure concentrations, one of the primary design goals for the Ecological Exposure Module was flexibility. The module currently predicts contaminant concentrations for 17 food categories; however, the module could be easily modified to include additional categories or to collapse existing categories into fewer categories to further reduce run time or to better reflect the data for a particular chemical constituent. Similarly, the module currently handles each receptor home range as a unique spatial element, even though the system aggregates these home ranges into four bin sizes. The Ecological Exposure Module can handle either level of resolution based on how the site layout file is constructed. Lastly, the module was designed with future modifications in mind that address the evolving science and data for ecological risk assessment (e.g., calculating receptor-specific lifetime average doses rather than annual average doses).

15.3.2 Uncertainty and Limitations

The methodology developed to characterize potential ecological exposures, as implemented by the Ecological Exposure Module, carries certain assumptions and limitations, and acknowledges several important sources of uncertainty.

- **The physical description of the spatial dimensions of ecological exposure are fixed for each site.** The Ecological Exposure Module calculates the applied doses to receptors for a single random placement of four home range sizes within habitats delineated at each site. Although the representative habitats were delineated using available GIS coverages to represent potential habitats at a given site, the habitats that may actually be present at any site may be very different from those delineated in the site layout. Despite the fact that the home ranges were chosen at random and retain predator-prey interactions, these are fixed in the site layout file.

- **The 3MRA system is designed to model a single chemical stressor released within the AOI for each simulation.** Given the design goals for the 3MRA modeling system (e.g., to support national-level assessment strategies of WMUs, waste streams, and contaminants), exposure to multiple contaminants or contaminant releases outside of the AOI are not considered. In addition, background concentrations of contaminants are not considered in developing exposure estimates, nor are other potential nonchemical stressors, such as habitat fragmentation. This introduces an unquantifiable uncertainty in the prediction of total exposure and risk.

- **The exposure calculations are highly dependent on reliable estimates of plant and prey concentrations, as well as random selection of plant/prey dietary preferences represented in the dietary matrix.** Ultimately, the veracity of the exposure estimates is a function of the data used in predicting tissue concentrations in dietary plants and prey. As discussed in Section 11 on the Terrestrial Food Web Module, there is significant uncertainty associated with the use of empirical data and default factors used to predict the uptake and accumulation of contaminants from media into biota.

- **The exposure profiles generated by the Ecological Exposure Module are based on the annual average concentrations in food items and media.** Consequently, concentration spikes due to episodic events (e.g., rain storms) or elevated WMU source releases following waste additions are not evaluated. In addition, the annual average approach does not capture elevated exposures during critical life stages. More specifically, the module predicts exposures only for adult animals; intrayear contaminant exposures to juveniles, often with very different dietary preferences, are not predicted.

- **Exposure doses are adjusted to account for the proportion of a species' required home range provided at a site.** However, no adjustment is made to account for differences in species diversity in small versus large habitat patches or in disturbed versus undisturbed areas. The 3MRA modeling system assumes that receptor species occur in their assigned representative habitats regardless of a site's position in the landscape. In fact, it is probably unlikely that all of the receptor species, particularly those less adapted to human impacts and development, would be present. Moreover, when the habitat patches at a site are small, it is questionable whether the entire receptor group would use the habitat.

- **The dietary preferences for each receptor remain constant throughout each year in the simulation.** The Ecological Exposure Module constructs the dietary preferences for each receptor based on dietary data covering one or more seasons. Some of the seasonal variability in the diet is captured indirectly by the hierarchical algorithm used to determine the dietary preferences. However, the algorithm is implemented on data across multiple seasons and, therefore, does not necessarily reflect seasonal differences. Similarly, the estimation of ingestion rates was based on average, gender-neutral body weights and does not account for differences in size, season, habitat, or activity level. However, prey preferences are represented by distributions that are intended to reflect some of the natural variation in wildlife feeding behaviors.

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16.0 Ecological Risk Module

16.1 Purpose and Scope

The Ecological Risk Module calculates the hazard quotient (HQ) for species of mammals, birds, and herpetofauna and, through inference, for communities of organisms that live in close contact with soil (including plants and soil invertebrates), sediment (benthic invertebrates), and surface water (e.g., fish, aquatic invertebrates, algae, and aquatic plants). The HQ is the ratio of the annual average exposure (in units of concentration or dose) to a benchmark for ecological effects (in units of concentration or dose) and quantifies the potential for contaminants to elicit an adverse ecological response once released into the environment. To calculate the HQs, the Ecological Risk Module uses input concentrations from the Surface Water and Terrestrial Food Web Modules, as well as applied doses calculated by the Ecological Exposure Module, and compares those values to ecotoxicological benchmarks (EBs, in units of dose) and chemical stressor concentration limits (CSCLs, in units of concentration), as appropriate. The module calculates an HQ for every ecological receptor assigned to habitats within the area of interest (AOI) for a given site. Detailed information on the Ecological Risk Module can be found in the background document (U.S. EPA, 2000). Figure 16-1 shows the relationship and information flow between the Ecological Risk Module and the 3MRA modeling system.

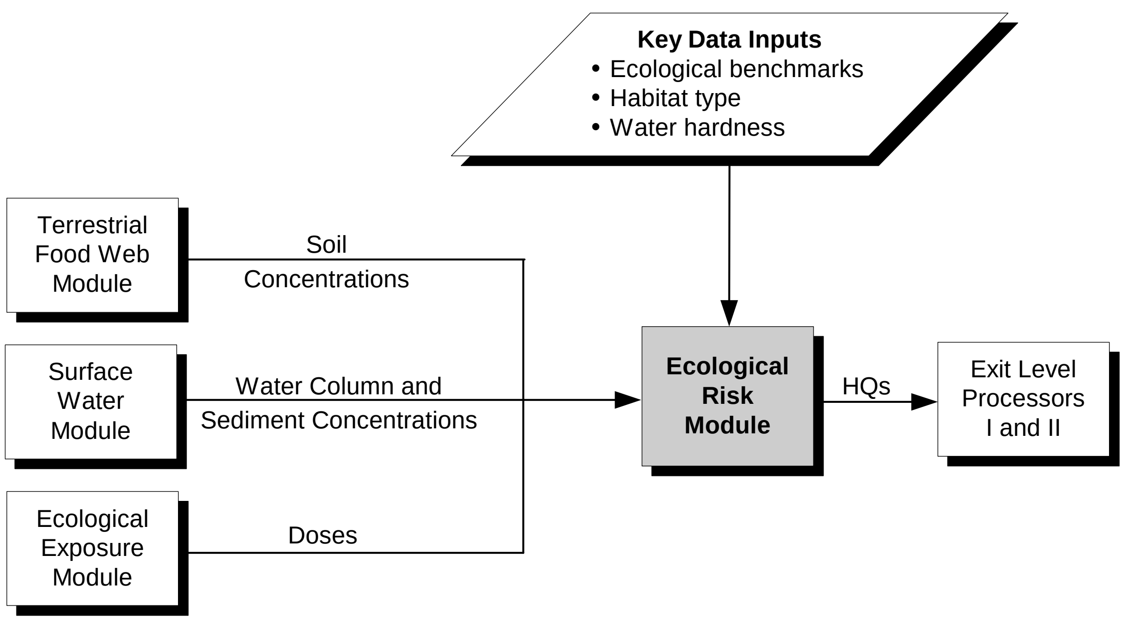

Figure 16-1. Information flow for the Ecological Risk Module in the 3MRA modeling system.

The conceptual approach to characterizing the potential for adverse ecological effects depends on the assessment endpoints used. Assessment endpoints are explicit expressions of the actual environmental values that are to be protected (U.S. EPA, 1998). Candidates for assessment endpoints often include threatened or endangered species, functional attributes that support food sources or flood control, or aesthetic values such as the existence of charismatic species (e.g., eagles) that have special value to society (U.S. EPA, 1998). The assessment endpoints must be defined with respect to the valued ecological entity (e.g., a particular species), and an attribute of that entity that is to be protected (e.g., reproductive fitness). For the 3MRA modeling system, the assessment endpoints were selected to represent multiple levels of biological organization (e.g., populations, communities), key functional elements of natural communities (e.g., primary producers), and biota throughout the trophic continuum. Although the 3MRA modeling system does not restrict the selection of assessment endpoints for future applications, the Ecological Risk Module and supporting databases were developed to evaluate three primary assessment endpoints:

- Survival of species that comprise key structural and functional elements of soil, freshwater, and benthic (sediment) communities;
- Reproductive fitness and survival of mammalian, avian, and herpetofaunal wildlife populations; and
- Growth and survival of primary producers (e.g., plants) in terrestrial and freshwater systems.

These endpoints are ecologically relevant to the habitat types used to represent ecological variability in the 3MRA modeling system (see Section 15.2) and are sensitive to a broad range of chemical stressors.

Table 16-1 describes the specific assessment endpoints addressed by the 3MRA modeling system in terms of: (1) the significance of the ecological entity (i.e., the reason EPA wants to protect it), (2) the ecological receptor(s) that represent that entity, (3) the characteristic(s) of the entity that is important to protect, and (4) the measure of effect used to quantify the potential for an adverse response. Measures of effect such as fecundity and mortality were chosen to support the development of the EBs and CSCLs based on their relevance to the assessment endpoints.

The Ecological Risk Module performs two major functions:

1. **Calculate Hazard Quotients (HQs).** The Ecological Risk Module calculates HQs for each receptor at each site according to spatial, temporal, and environmental characteristics of the site.

2. **Process the HQ Results for Decision Making.** The Ecological Risk Module tracks various attributes such as taxa and habitat type to process the HQ results for decision making. This processing includes placing results in bins and determining the timing of maximum risks.

Table 16-1. Assessment Endpoints and Measures of Effects for the 3MRA Modeling System

Assessment Endpoint	Ecological Significance	Example Receptors	Characteristic(s)	Measure(s) of Effect
survival and reproductive fitness of mammalian wildlife populations	■ multiple trophic levels and apex predators ■ multiple exposure pathways represented ■ species with large foraging ranges ■ charismatic species (e.g., eagles) ■ sensitive species (e.g., mammals and dioxins)	deer mouse, meadow vole, red fox	reproductive success in adult animals; survival and development of offspring	chronic/subchronic MATL for physiological effects relevant to reproductive success (e.g., gonadotoxicity; number of viable offspring; egg production) and development (e.g., malformations; growth rate)
survival and reproductive fitness of avian wildlife populations		red-tailed hawk, northern bobwhite		
survival of amphibian populations	■ unique habitat niches (e.g., partially aquatic and terrestrial) ■ includes species that are particularly sensitive to chemical stressors, particularly metals	frog, newt	lethality and deformity	primarily acute LC_{50}s for lethality resulting from early life stage exposures; EC_{20}s for developmental effects used when available
reproductive success and survival of reptile populations		snake, turtle	reproductive success in adult animals; survival and development of offspring	none identified for chemical stressors of concern
survival of species that comprise key structural and functional elements of the soil community	■ high levels of exposure through direct contact ■ base of the food web in terrestrial systems ■ vital habitat for decomposers, soil aerators ■ essential for nutrient cycling	nematodes, arthropods, springtails, annelids, mites	mortality, growth, survival, reproductive success	95% of soil species below low effects concentration at 50^{th} percentile confidence interval; when data were insufficient, LOEC values for earthworms, microbes used
growth and survival of terrestrial plants	■ primary producers of energy ■ food base for herbivores ■ essential habitat for many wildlife species ■ large fraction of the earth's biomass	soy beans, alfalfa, rye grass	plant growth and yield	10^{th} percentile from data on LOEC from studies on plant growth, seed germination; when data were insufficient, lowest LOEC value was selected
survival of species that comprise key structural and functional elements of the freshwater aquatic community	■ high levels of exposure through direct contact ■ includes several species that are very sensitive to chemical stressors (e.g., daphnids) ■ important food source for wildlife that live in waterbody margin	fish, aquatic invertebrates	mortality, growth, survival, reproductive success	National Ambient Water Quality Criteria (NAWQC) for aquatic life (95% of aquatic species); alternate water quality criteria developed using abbreviated data sets

(continued)

Table 16-1. (continued)

Assessment Endpoint	Ecological Significance	Example Receptors	Characteristic(s)	Measure(s) of Effect
survival of species that comprise key structural and functional elements of the sediment community	■ sediment serves as sink for persistent contaminants, resulting in high levels of exposure ■ habitat for early life stages (e.g., midge larvae) ■ nutrient processing and decomposition	protozoa, flat worms, ostracods	mortality, growth, survival, reproductive success, and community measures such as abundance, diversity	*metals* - threshold effects levels from LOEC data associated with community endpoints (e.g., species abundance) *organics* - water quality criteria adjusted for sorption to organic carbon in sediment
growth and survival of aquatic plants and algae	■ primary producers of energy ■ base of food source in the aquatic system ■ substrate for other organisms in the water column ■ essential habitat for developing organisms	algae and vascular aquatic plants (e.g., duckweed)	growth, cell numbers, mortality, biomass, root length	algae - EC_{20} and EC_{50} for growth, decreased cell numbers, reduction in carbon fixation vascular plants - lowest LOEC for endpoints relevant to growth, biomass, root number

MATC maximum allowable toxicant level
LOEC lowest observed effect concentration
LC_{50} lethal concentration to 50 percent of the organisms
EC_{50} effective concentration to elicit a response in 50 percent of the organisms
EC_{20} effective concentration to elicit a response in 20 percent of the organisms

Because the assessment endpoints evaluated include both populations of wildlife species and assemblages of species that represent communities, the calculated HQs for different receptors represent different measures of effect at several levels of biological organization. As a result, it is critical to understand the rationale behind the development of the EBs and CSCLs included in the current 3MRA sample data base. Therefore, although these benchmarks are not calculated by the Ecological Risk Module, the development of these benchmarks is discussed in this section.

16.2 Conceptual Approach

The Ecological Risk Module quantifies the potential for adverse ecological effects by calculating HQs for individual receptors such as raccoons, aquatic plants, or the soil community. The HQs provide a "bright line" metric for risks to the individual organisms that represent wildlife populations and, based on statistical inference, the risks to narrowly defined communities (e.g., the sediment community). The Ecological Risk Module does not estimate population-level risks, defined as the likelihood that some percentage of the individuals in a population will sustain an adverse effect, nor does it characterize the risks to communities. Although population-level models have long been used by ecologists to evaluate the response of species populations to certain types of stressors, the data requirements and implications on model run time prohibited such an approach in this version of the 3MRA modeling system. Thus, the Ecological Risk Module uses specific measures of effect (e.g., a reduction in viable offspring) to generate HQs that may be used to *infer* whether the potential for an adverse ecological response is above levels of concern. That is, a target HQ of 1 serves as the indicator for adverse ecological responses; a value greater than 1 indicates that the ecological response is above a level for concern, and a value less than 1 indicates that the potential for adverse ecological effects is at a *de minimis* level for the receptor for which the HQ was calculated. The implications of this approach underscore the importance of developing EBs and CSCLs that reflect the assessment endpoints for the modeling application, tracking the attributes of each HQ that is calculated, and processing the HQ results in a transparent manner to support decision making.

16.2.1 Development of EBs and CSCLs

Receptor-specific benchmarks[1] (EBs and CSCLs) were developed for use in the Ecological Risk Module. As indicated previously, these benchmarks were based on measures of effects considered appropriate to support risk inferences for ecological receptors at various levels of biological organization, including individual organisms, wildlife species populations, and communities. In identifying appropriate studies from which to develop EBs and CSCLs, study selection criteria were developed to ensure consistency in the interpretation of study results and to satisfy data quality objectives. The study selection criteria address the desire for consistency across EPA programs as well as within the representative national data set, the appropriateness of the study given the assessment endpoints used in the 3MRA modeling system; and the quality

[1] The term " benchmarks" is used in this discussion to refer to both EBs for mammalian and avian species and CSCLs for selected receptors (e.g., amphibian species) and communities.

of the study with respect to endpoint selection, dose-response information, and appropriate use of extrapolation techniques (e.g., tools for statistical inference).

The EBs represent a *de minimis* level of effect for mammals and birds exposed through the ingestion of contaminated media, plants, and prey. In order of importance, the study selection criteria considered the following:

1. Relevance of study endpoints to population-level effects;

2. Whether the study contained adequate data to demonstrate a dose-response relationship;

3. Appropriateness of study design with respect to the exposure route (e.g., gavage versus dietary exposure) and duration (e.g., acute versus chronic);

4. Quality of the study as determined by the use of appropriate dosing regimes, statistical tools, etc.; and

5. Consistency with other EPA programs, such as Superfund and the Office of Water.

The CSCLs (except those for amphibians and plants) are intended to represent a *de minimis* level of effect for communities that live in close contact with the soil, sediment, or surface water. The study selection criteria for these receptors considered the following:

1. Acceptance of the benchmark by other EPA programs (e.g., National Ambient Water Quality Criteria);

2. Consistency with EPA guidelines on study selection for aquatic toxicity data;

3. Relevance of study to species identified as key functional elements of the community;

4. Relevance of study endpoints to address community-level effects (e.g., growth, survival, abundance);

5. Whether the study contained adequate data to demonstrate a dose-response relationship; and

6. Quality of the study data with respect to the design (e.g., field versus laboratory), and appropriate use of statistical tools to characterize effects (e.g., confidence levels).

For amphibians, the extensive database on acute and subchronic aqueous exposures to developing organisms was used to derive CSCLs for surface water contact at sensitive life stages over short durations (e.g., less than 7 days). For terrestrial plants, studies on growth, seed germination, and other relevant endpoints were used to derive CSCLs for exposure durations that roughly correspond to a typical growing season (e.g., 3 months). The study data on vascular aquatic plants was very limited, and short-term studies on algal growth were frequently selected to evaluate primary producers in aquatic systems.

The following subsections provide a brief description of benchmark development for the following receptor groups:

- Mammals and birds,
- Amphibians,
- Reptiles,
- Soil community,
- Terrestrial plants,
- Freshwater aquatic community,
- Sediment community, and
- Aquatic plants and algae.

The discussions highlight the minimum requirements for data used to support benchmark development and the calculation methods used. To provide the context of each description, the assessment endpoint and measure of the effect for each receptor group is presented at the beginning of each subsection. A comprehensive description of the data collection efforts and methodology developed to derive ecological benchmarks may be found in Volume II of this report.

Mammals and Birds.

- Assessment Endpoint: *survival and reproductive fitness of mammalian and avian wildlife populations.* The characteristics to be protected are: (1) the reproductive success of adult animals, and (2) the growth and development of offspring.

- Measure of Effect: *a de minimis threshold for developmental and reproductive toxicity in mammalian and avian wildlife species.* The threshold was calculated as the geometric mean of the No Observed Adverse Effect Level (NOAEL) and the Lowest Observed Adverse Effect Level (LOAEL), defined as the maximum acceptable toxicant level (MATL) in the 3MRA database. Implicit in this calculation is the assumption that the toxicological sensitivity is lognormal.

For mammals and birds, ecotoxicological data were evaluated to identify the most appropriate studies to support EB development. Studies meeting the data quality objectives summarized above (e.g., sufficient dose-response data), were reviewed and the study with the most sensitive endpoint was selected to derive receptor-specific benchmarks. Once the MATL was calculated from the NOAEL and LOAEL, the MATLs for mammals and birds were scaled for each receptor species using the relative body weights of the test and receptor species. The scaling equation is based on the default methodology EPA proposes for carcinogenicity assessments and reportable quantity documents for adjusting animal data to an equivalent human dose (U.S. EPA, 1992), and is widely used in ecological risk assessments (see, for example, Sample et al., 1998). Research indicates that the cross-species scaling equation for mammals is inappropriate for direct application to birds (Mineau et al., 1996). Therefore, an avian scaling factor developed by Mineau was used to derive EBs for birds.

Amphibians.

- <u>*Assessment Endpoint*</u>*: survival of amphibian populations.* The characteristics to be protected were the survival, growth, and development through early life stages.

- <u>*Measure of Effect*</u>: *an acute LC_{50} for lethality or developmental effects resulting from early life stage exposures*.

For amphibians, the available data on toxicity were limited almost exclusively to acute studies on lethality and, in some cases, growth and developmental effects. Amphibians appear to be highly sensitive to a number of toxicants (e.g., trace metals) during the developmental stages of their life cycle. After a review of several compendia presenting amphibian toxicity data (e.g., U.S. EPA, 1996; Power et al., 1989) as well as primary literature sources, it was determined that the lack of standard methods on endpoints, species, and test durations would preclude the development of a CSCL for chronic exposures. Available data typically involved aqueous exposure during early life stages and, as a result, the only exposure pathway that could be evaluated for amphibians was direct contact with contaminated surface waters. The CSCL for early life stage effects was calculated as the geometric mean of acute LC_{50} data (lethal concentration to 50 percent of the organisms). A few general guidelines were followed in selecting acute studies for developing the CSCL: (1) test duration was usually less than 15 days, (2) only studies on lethality were included to calculate the geometric mean, and (3) exposure occurred during early life stages (i.e., embryo, larvae, and tadpole).

Reptiles.

- <u>*Assessment Endpoint*</u>: *survival and reproductive fitness of reptile populations.* The characteristics to be protected are: (1) the reproductive success of adult animals, and (2) the growth and development of offspring.

- <u>*Measure of Effect*</u>: *a de minimis threshold for developmental and reproductive toxicity to reptile species.* The threshold is calculated using the same approach as that taken for mammals and birds.

Toxicity data relevant to reptile exposures could not be identified for the chemical stressors in the representative national data set. Additional research is needed to: (1) identify more recent data for applicability to reptile exposures and (2) determine whether methods can be developed to extrapolate benchmarks across species.

Soil Community.

- <u>*Assessment Endpoint*</u>*: survival of species that comprise key structural and functional elements of the soil community*. The characteristics to be protected include mortality, growth, survival, and reproductive success of selected taxa.

- *Measure of Effect: concentration in soil that, for 95 percent of the species, will be below the low effects concentration at the 50th percentile confidence interval.* The CSCLs for the soil community were typically derived at a 95 percent protection level using primarily low effects data for the endpoints of interest.

Ecotoxicological data on lowest observed effects concentrations (LOECs) were reviewed for soil biota for a number of functional categories in soil systems (e.g., decomposers, predators) to derive the soil community CSCLs. Thus, criteria were required both for individual study reviews as well as to evaluate the entire data set for completeness. Criteria developed for study selection included four categories of exposure: (1) topical application; (2) surface-soil application, in which the soil organisms are placed onto a treated surface; (3) mixed-soil application, in which the soil organisms are placed into a soil that was mixed with a contaminant; and (4) food application (i.e., contaminant mixed with organic food source). The endpoints for soil species were selected based on relevance to species populations and, in order of preference, included reproduction, growth, mortality, population increase/decrease, sexual development, mobility, and regeneration.

The first, and preferred, method was based on a community-level approach analogous to that used to develop the National Ambient Water Quality Criteria (NAWQC) for the protection of aquatic biota. Like the NAWQC, the soil community CSCLs are intended to protect the structure and function of the community and its critical role in the terrestrial food web (e.g., nutrient processing). A detailed discussion of the calculations used to develop a community-based soil CSCL are provided in Volume II of this report.[2] The second method used to derive soil CSCLs required the identification of LOECs for earthworms and microbial endpoints despite the obvious limitations in relying on a single species to infer adverse effects to the community. Nevertheless, earthworms have been recognized for the critical role they play in promoting soil fertility, releasing nutrients, and providing aeration and aggregation of soil, as well as being an important food source for higher trophic level organisms. In addition, their constant contact with soil media and permeable epidermis makes them more susceptible to contaminant exposures. Likewise, microbial communities play a key functional role in soil fertility, decomposition processes, and nutrient cycling, providing nutrients in available forms to plants. Microbial CSCLs were used only when they indicated a significantly higher sensitivity to a particular contaminant than the corresponding earthworm toxicity data. The earthworm/microbial CSCLs were derived using one of two approaches: (1) if more than 10 studies were identified reporting a LOEC, the 10th percentile value was selected as the CSCL, or (2) if fewer than 10 values were identified, the lowest LOEC was selected as the CSCL.

Terrestrial Plants.

- Assessment Endpoint: *growth and survival of terrestrial plants*. The characteristics to be protected include the growth and yield of terrestrial plants.

[2] The discussion in Volume II uses no observed effects concentrations (NOECs) rather than LOECs. The methodology presented is entirely accurate; however, EPA has elected to use the less conservative LOECs to derive the soil CSCLs based on a comparison between NOEC-derived CSCLs and typical background concentrations.

- <u>*Measure of Effect*</u>: *soil concentrations related to growth, yield, seedling emergence, and germination endpoints*. The lowest observed effects data on phytotoxicity were rank ordered, and the plant CSCL was estimated as the 10^{th} percentile value.

The development of CSCLs for terrestrial plants primarily included endpoints relevant to growth and yield (e.g., seed germination, seedling emergence). Data collection and review activities were focused on these endpoints because they are ecologically significant responses and because the database of phytotoxicity studies provides sufficient coverage of these types of effects (Efroymson et al., 1997). However, very few contaminants have toxicity data for a sufficient number of terrestrial species to represent even a simple plant community, including short-lived and long-lived plants, flowering and nonflowering plants, high seed producers, and plants with extensive root systems (Eijsackers, 1994). Consequently, the data quality requirements presented in Efroymson et al. (1997) regarding study preferences (e.g., field studies were preferred over greenhouse studies) were adopted for development of soil CSCLs for plants. The terrestrial plant CSCLs were derived using the same approach as described previously for the earthworm/microbial CSCLs (i.e., based on the number of studies).

Freshwater Aquatic Community.

- <u>*Assessment Endpoint*</u>: *survival of species that comprise key structural and functional elements of the freshwater aquatic community*. The characteristics to be protected include mortality, growth, survival, and reproductive success of selected taxa.

- <u>*Measure of Effect*</u>: *surface water concentration that, for 95 percent of the species, will be below the low effects concentration at 50^{th} percentile confidence interval.* Typically, the NAWQC was chosen as the CSCL for this receptor group.

The CSCLs for the freshwater community reflect endpoints ranging from mortality to growth and reproductive effects. As with the soil community, criteria for individual study selection were established, as well as criteria applicable to the data set as a whole. The minimum data requirements to derive a CSCL for the aquatic community were based on the *Guidelines for Deriving Numerical National Water Quality Criteria for the Protection of Aquatic Organisms and Their Uses* (Stephan et al., 1985), and, for data sets that did not meet those requirements, the Tier II guidelines proposed in the *Water Quality Guidance for the Great Lakes System and Correction; Proposed Rule* (58 FR 20802). These methods require the compilation of appropriate acute and chronic toxicity data on adverse effects to aquatic biota for specific taxa of the freshwater community. Specifically, either the Final Chronic Value (FCV) developed using the above guidance or the criterion continuous concentration (CCC) developed for the Great Lakes Water Quality Initiative (GLWQI) was selected as the CSCL for the freshwater aquatic community. If neither a CCC nor an FCV was available, a Secondary Chronic Value (SCV) was calculated using Tier II methods developed through the GLWQI (Stephan et al., 1985; Suter and Tsao, 1996).

Sediment Community.

- <u>*Assessment Endpoint*</u>*: survival of species that comprise key structural and functional elements of the sediment community*. The characteristics to be protected include mortality, growth, survival, and reproductive success of selected taxa.

- <u>*Measure of Effect*</u>*: two types of measures were used, one for organics (based on the NAWQC) and one for metals (similar to the approach for earthworms and microbes).* The sediment quality criteria based on partitioning theory and the similarity in toxicological sensitivity between water column and sediment biota were used to derive the CSCLs for organic chemicals. Empirical data from field studies were used to derive the CSCLs for metals.

Two methods were applied in developing the CSCLs for the benthic community (e.g., worms, amphipods). For metals and ionic organic chemicals, CSCLs were based on either: (1) threshold effects levels (TELs) developed by the Florida Department of Environmental Protection (FDEP), which are the upper limit of the range of sediment concentrations for endpoints on survival, species diversity, and abundance; or (2) the 10^{th} percentile effects concentrations (ER-L) developed by National Oceanic and Atmospheric Administration’s (NOAA) National Status and Trends (NS&T) Program. The TELs were preferred over the ER-L values because: (1) the same database was used for both the NOAA criteria and the FDEP criteria development; (2) in most cases, the FDEP criteria were based on more appropriate measurement endpoints; and (3) the marine TELs developed by the FDEP were found to be analogous to TELs observed in freshwater organisms (Smith et al., 1996).

For nonionic organic chemicals, the CSCL derivation method was based on an equilibrium partitioning relationship between sediment and surface water. This method calculates the sediment CSCL based on the surface water FCV or SCV, assuming that the equilibrium partitioning between sediment and the water column is a function of the fraction organic carbon. In calculating the baseline sediment CSCL for nonionic chemicals, the fraction organic carbon was assumed to be 1, and the organic carbon partitioning coefficients were adopted as reported in Jones et al. (1997). However, because the HQs are intended to represent the ecological hazard for the environmental characteristics of each site, the Ecological Risk Module adjusts the sediment CSCL for the site-specific fraction organic carbon prior to calculating the sediment community HQ value.

Aquatic Plants and Algae.

- <u>*Assessment Endpoint*</u>: *growth and survival of aquatic plants and algae in freshwater systems*. The characteristics to be protected are different for aquatic plants and algae, and include root length and biomass, and growth and cell numbers, respectively.

- <u>*Measure of Effect*</u>: *surface water concentrations related to gross measures of health (e.g., biomass) for the algal community and a variety of endpoints for aquatic plants (e.g., number of fronds, root number, plant number, root length).*

For algae, the effective concentration to 20 percent of the population (EC_{20}) was selected as the threshold for an adverse response. For plants, the lowest LOEC for endpoints of interest was chosen as the CSCL because of the paucity of data and the importance of vascular plants to maintain a healthy aquatic ecosystem.

The CSCLs developed for primary producers in aquatic systems include data on both vascular aquatic plants and algae. Algae were included in this receptor group because they have a relatively long history of toxicity testing and have often been shown to be more sensitive than vascular aquatic plants to chemical stressors (Klaine and Lewis, 1995). For algae, the measure of effect was based on the EC_{20} and EC_{50} values related to growth inhibition, decreased cell numbers, and reduction in carbon fixation as common responses measured in algal toxicity tests. For aquatic plants such as duckweed (e.g., *Lemna minor*), the endpoints included LOECs for development of fronds, biomass, root number and length, and plant number. The toxicological data relevant to this receptor group were identified in the open literature and from data compiled in *Toxicological Benchmarks for Screening Potential Contaminants of Concern for Effects on Aquatic Biota: 1996 Revision* (Suter and Tsao, 1996).

16.2.2 Calculate Hazard Quotients

The Ecological Risk Module calculates an HQ for receptors assigned within the AOI for a given site. The module may calculate more than one HQ for a receptor if the receptor is assigned to more than one habitat; however, the module calculates only one HQ per habitat for any receptor.[3] In calculating an HQ, the Ecological Risk Module checks first to determine whether an EB or CSCL (as appropriate) is available for a particular receptor. An HQ is not reported for a receptor if no benchmark is available; however, the lack of an ecological benchmark does not indicate the lack of ecological risk.

The Ecological Risk Module calculates HQs for each year, habitat, receptor, and site. Annual average media concentrations and applied doses are used to represent chronic, long-term exposures to chemical stressors released from waste management units. Although the HQ calculation is essentially the same for any receptor, the spatial scale and environmental characteristics relevant to the HQ calculations are handled differently for different habitat/receptor/constituent combinations. Therefore, it is useful to organize the discussion of HQ calculations around three basic groups of receptors:

1. **Freshwater Receptors (freshwater aquatic community, sediment community, amphibians, and aquatic plants and algae).** The module calculates HQs for these receptors based on appropriate CSCLs, determines the average concentration across the habitat or ephemeral water body, and adjusts certain CSCLs for environmental conditions prior to calculating the HQ.

2. **Soil receptors (soil community and terrestrial plants).** The module calculates HQs for these receptors based on the depth-averaged soil concentration and the

[3] There is one exception to this rule: more than one HQ can be calculated and reported for the soil community and terrestrial plants. This exception is explained in Section 16.2.2.2.

soil CSCLs, and reports an HQ for each receptor with a unique spatial definition (based on the receptor home ranges) in each habitat.

3. **Mammals, birds, and reptiles.** The module calculates HQs for mammals and birds using the applied doses calculated by the Ecological Exposure Module and the receptor-specific EBs. The spatial scale for each mammalian and avian HQ calculated by the module is based on the home range of the species. As noted earlier, insufficient data were available for the current 3MRA data set to calculate reptile HQs.

Freshwater Receptors. By definition, freshwater habitats (i.e., waterbody and wetland margins) include at least one fishable surface waterbody. Prior to calculating HQs for these receptors, the Ecological Risk Module identifies which habitats are waterbody margin or wetland, determines the number of reaches that are connected in each habitat,[4] and calculates the average surface water and sediment concentration for the habitat. For surface water, the module calculates average concentrations for both the total (i.e., dissolved plus bound fraction) and freely dissolved phases for each constituent.

For metals, the module adjusts the surface water CSCL to account for the effect of water hardness on toxicity. This adjustment is made only for cationic metals for which toxicological sensitivity has been shown to correlate with water hardness *and* for which a quantitative expression of that relationship has been developed by EPA. For nonionic organics, the module adjusts the sediment CSCL for the site-specific fraction organic carbon to ensure that the partitioning reflects the site conditions.

Using the average surface water and sediment concentrations and the adjusted CSCLs (if appropriate), the Ecological Risk Module calculates HQs for the aquatic community, aquatic plants or algae, amphibians, and the benthic community as follows:

$$HQ_{receptor_{habitat\ i}} = \frac{C_{medium_{habitat\ i}}}{CSCL_{receptor}} \tag{16-1}$$

where

$HQ_{receptor\_habitat^i}$	=	hazard quotient for receptor in habitat i (unitless)
$C_{medium\_habitat^i}$	=	concentration in surface water or sediment in habitat i (mg/L or mg/kg)
$CSCL_{receptor}$	=	chemical stressor concentration limit for receptor (mg/L or mg/kg).

The Ecological Risk Module uses two conventions that pertain to the use of surface water CSCLs to calculate HQs. First, the Ecological Risk Module uses the freely dissolved chemical

[4] The 3MRA modeling system treats all waterbodies, including streams, lakes, ponds, and permanently flooded wetlands, as reaches.

concentration if the corresponding CSCL also reflects freely dissolved chemical. The total water concentration is used to calculate the HQ if no freely dissolved CSCL is available (typically, surface water CSCLs are available only for total surface water concentrations). Second, the module calculates HQs for amphibians in intermittently flooded wetlands, reach order 2 streams, or any other transitory bodies of water that can serve as a breeding ground for amphibians. Indeed, an estimated 30 percent and 50 percent of all species of caudates (salamanders) and anurans (frogs), respectively, use temporary ponds for breeding, and many of these species reproduce in them exclusively (Freda, 1991).

Soil Receptors. For the soil community and terrestrial plants, the Ecological Risk Module calculates an HQ for each unique home range in each habitat delineated at the site. Four possible unique home ranges are currently implemented in the 3MRA modeling system (although, as indicated in Section 15, the 3MRA modeling system can process a unique home range for *every* receptor). For terrestrial habitats at many sites, the two largest home range sizes are larger than the habitat delineated within the AOI. As discussed in Section 15, these home ranges were constrained by the size of the habitat and, therefore, have identical spatial characteristics. In these instances, the HQs for these home ranges would be identical and reflect the same spatial averaging within the habitat and reporting them all would be redundant. Therefore, for most receptors, only one HQ is calculated per habitat, with the exception of plants and soil community receptors. This exception accounts for the fact that wildlife are tightly coupled to their environment, and adverse effects to the base of the food web may be highly significant ecologically. Consequently, the HQs for the soil community and terrestrial plants are calculated for unique spatial averages (defined by the home range) within each habitat. The HQs for soil receptors are calculated as the ratio of the root zone soil concentration (depth-averaged to 5 cm) to the soil CSCL for soil biota and plants, respectively, as follows:

$$HQ_{receptor_{habitat\ i}} = \frac{C_{soil_{HomeRange\ j}}}{CSCL_{receptor}} \tag{16-2}$$

where

$HQ_{receptor\_HomeRange^i}$	=	hazard quotient for receptor in home range i (unitless)
$C_{soil\_HomeRange^i}$	=	root zone soil concentration in home range i (mg/kg)
$CSCL_{receptor}$	=	chemical stressor concentration limit for receptor (mg/kg).

For any terrestrial habitat delineated within the AOI, between 1 and 4 HQs will be calculated for the soil community and terrestrial plants, respectively, provided that the chemical database contains CSCLs for both receptor types.

Mammals, Birds, and Reptiles. The Ecological Risk Module calculates HQs for mammals and birds as the ratio of the applied dose from the Ecological Exposure Model to the receptor-specific EB. Thus, Equation 16-3 has the same general form as the previous two HQ equations; however, the applied dose is used to represent the environmental exposure through the ingestion pathway, and the EB is used as the ecological benchmark for endpoints relevant to population growth and survival.

$$HQ_{receptor_{habitat\ i}} = \frac{AppliedDose_{receptor}}{EB_{receptor}} \tag{16-3}$$

where

$HQ_{receptor\_habitat\,i}$	=	hazard quotient for receptor in habitat i
$AppliedDose_{receptor}$	=	applied dose to receptor (mg/kg-d)
$EB_{receptor}$	=	ecological benchmark for receptor (mg/kg-d).

The spatial characteristics of the HQ are defined by the home range for each mammalian and avian receptor.[5] Thus, the HQ values calculated for the same receptor assigned to two different terrestrial habitats at a site are likely to reflect the differences in soil, plant, and prey concentrations associated with the respective home ranges.

16.2.3 Process HQ Results for Decision Making

The HQs calculated by the Ecological Risk Module provide a risk metric with essentially a binary outcome: an HQ above 1 indicates that ecological risks are above levels of concern, and an HQ below 1 indicates that ecological risks are below levels of concern. Naturally, this approach implies that an HQ much greater than 1 (say, two orders of magnitude) represents a more serious ecological threat than an HQ of 2; however, the 3MRA modeling system and data do not quantify the *probability* that an adverse effect will occur. For example, an HQ of 20 calculated for a drop in egg production for the osprey does not imply that the probability of this effect is 10 times more likely than an HQ of 2, only that the magnitude of the effect is likely to be more serious. Similarly, because the actual dose-response relationship is not used in the HQ calculations, the HQ of 20 does not imply that the effect would be 10 times more severe than for osprey with an HQ of 2. The HQ approach provides limited information on the probability and significance of ecological risks for decision-making purposes; the implications of calculated receptor-specific HQs as to the quality of ecological systems as a whole simply cannot be inferred. The sometimes unpredictable nature of community dynamics, as well as the presence of other stressors (e.g., habitat alteration), are such that the potential ecosystem effects associated with, say, a reduction in reproductive fitness of a single receptor can be difficult to predict, particularly when the modeling simulation may run hundreds or even thousands of years.

To address this limitation for national-scale applications of the 3MRA modeling system, the Ecological Risk Module processes the calculated HQs so that decision makers can answer a variety of questions about potential ecological risks. Specifically, the Ecological Risk Module (1) assigns attributes appropriate to each HQ, including the hazard bin, distance to waste management unit (WMU), receptor taxa, and habitat type, and (2) identifies the critical year in the simulation at which ecological risks (i.e., HQs) are collectively at the highest level.

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[5] The Ecological Risk Module is also designed to calculate HQs for reptiles; however, insufficient ecotoxicity data were identified to support an EB for reptile species.

Assigning Attributes to HQs. One of the most important attributes of the HQs calculated by the Ecological Risk Module is the hazard range (referred to as a hazard "bin") to which the HQ is assigned. The HQs for all receptors assigned to habitats within the AOI are placed into one of five hazard bins identified by EPA as useful to decision-makers. The HQ bins are defined as

- **Bin 1:** ≤ 0.1
- **Bin 2:** > 0.1 to ≤ 1
- **Bin 3:** >1 to ≤ 10
- **Bin 4:** >10 to ≤ 100
- **Bin 5:** >100.

As an example, suppose that a particular site has a total of 25 mammalian receptor species assigned across the three habitats: 10 mammals in a forest habitat, 7 mammals in a grassland habitat, and 8 mammals in a stream margin habitat. For a simulation that runs three years, assume that the HQs calculated for the mammalian receptors are as shown in Table 16-2. These HQ calculations reflect the applied doses to mammalian wildlife due to the ingestion of contaminated media, plants, and prey. Notice that the hazard profile changes with each year in the simulation; in this hypothetical example, the level of exposure and resulting hazard is increasing over time.

Table 16-2. Example HQ Counts for Mammals at Hypothetical Site

	Number of Mammal Species in Each HQ Range				
	Bin 1	**Bin 2**	**Bin 3**	**Bin 4**	**Bin 5**
Exposure starting in...	**≤ 0.1**	**>0.1 to ≤1**	**>1 to ≤ 10**	**>10 to ≤ 100**	**>100**
Year 1	19	5	1	0	0
Year 2	15	7	2	1	0
Year 3	12	9	1	2	1

The Ecological Risk Module uses the HQ counts to create a cumulative frequency histogram: the results expressed as the cumulative percentile of the total number of HQs. Therefore, the conditional, cumulative frequency percentiles for mammalian risks (defined by the HQ) for each year in the simulation would be as shown in Table 16-3.

Table 16-3. Example Cumulative Frequency HQ Histogram for Mammals at Hypothetical Site

	Percent of Mammal Species in this HQ Range or Lower				
Exposure starting in...	**Bin 1**	**Bin 2**	**Bin 3**	**Bin 4**	**Bin 5**
Year 1	76	96	100		
Year 2	60	88	96	100	
Year 3	48	84	88	96	100

Aggregating the HQs into hazard ranges serves two important functions in the 3MRA modeling system. First, converting the results to cumulative hazard allows the module to greatly reduce the amount of data that it produces. Because the Ecological Risk Module calculates HQs for every receptor in every year in the simulation, the system produces a tremendous amount of data. Second, the use of hazard ranges recognizes that the state of ecological risk assessment science (for national applications) lacks the data to validate the level of accuracy in risk estimates. Accumulating the ecological hazard quotients implicitly acknowledges the limitations in predicting ecological risks using the HQ approach. EPA developed these bin ranges to be meaningful in terms of decision making, increasing from below levels of concern to high levels of concern.

In addition to the hazard bin, each HQ is associated with a series of attributes that allow ecological risks to be characterized in different ways. EPA identified five attributes considered relevant to interpreting the potential significance for adverse ecological effects within the context of regulatory decision making. These attributes support analyses at different scales of organization with respect to space as well as trophic position, and include:

- Habitat type (e.g., forest, grassland, pond, stream, permanently flooded forest),
- Habitat group (i.e., terrestrial, freshwater, and wetland),
- Receptor group (e.g., mammals, amphibians, soil community),
- Trophic level (i.e., producers, TL1, TL2, TL3, top predators), and
- Distance to WMU (e.g., within 1 km of WMU).

The maximum HQ across the site is also reported, along with its ecological risk attributes. This metric was added for use in pass/fail analyses that may be needed to prioritize sites for further refinement or assessment.

In summary, for each year in the simulation, the Ecological Risk Module does all of the necessary accounting to construct cumulative frequency histograms, expressed as a percentile of the total HQs across all receptors or by attribute. The cumulative frequency data contain the HQ values for each attribute of interest (e.g., receptor group, habitat type) so that the results may be accumulated to respond to different questions (e.g., what are the risks to mammals at a site). The cumulative frequency data on HQs across all sites serves as the input data set for the Exit Level Processor (ELP) of the 3MRA modeling system. As described in the overview in Section 1.0, the ELP is a system level tool used to develop cumulative frequency distributions of ecological risks for national analyses.

Identifying the Critical Year in the Simulation. Having produced the time series of cumulative frequencies for HQs, the Ecological Risk Module identifies the year during which the total hazard is a maximum for each of the ecological attributes listed above (i.e., habitat type, habitat group, receptor group, and trophic level; all HQs are specific to a distance from the WMU). This is the critical year in the simulation (referred to as TCrit); in processing ecological risk results for national applications, the ELP uses only the HQ results for the TCrit year for each site included in the simulation. The Ecological Risk Module identifies the TCrit year for each of the first four ecological attributes listed above as a function of distance to the WMU. For example, the module produces the cumulative frequency HQ histogram for all receptor groups (e.g., mammals, birds, aquatic community) for the entire site, and for distance intervals within

1 km of the WMU, and between 1 and 2 km of the WMU.[6] In addition, the module identifies the TCrit for the maximum HQ across all receptors assigned to the site.

The Ecological Risk Module identifies the TCrit for each attribute in three basic steps. First, the module calculates the total HQ (by attribute) for each year in the simulation by summing the HQs across all receptors that share that attribute. Second, the module rank orders the time series of total HQs. Third, the module selects the Ecological Regulatory Percentile (ERP) and identifies the TCrit year that matches the total HQ at that percentile of the rank ordering. The Ecological Regulatory Percentile is a user-specified percentile that indicates the level of protection required to select TCrit. For example, if the user specified 95 percent as the Ecological Regulatory Percentile, the Ecological Risk Module would identify TCrit as the output year when the total HQ value is greater than 95 percent of all the other total HQ values reported for the simulation. In the current implementation of the 3MRA modeling system, the default value for Ecological Regulatory Percentile is 100 percent; therefore, the module identifies the maximum total HQ and reports that year as TCrit for the appropriate attribute.

16.3 Module Discussion

16.3.1 Strengths and Advantages

The Ecological Risk Module calculates HQs for all receptors assigned to habitats within the AOI. The module offers a number advantages, including the underlying ecological benchmarks as well as site-specific adjustments to the benchmarks that reflect the environmental characteristics of the site. Examples of these advantages include the following:

- **The ecological benchmarks reflect the management goals and assessment endpoints for national-scale analyses.** The ecological benchmarks in the 3MRA modeling system were developed specifically to evaluate the ecological risks associated with long-term, low-level contaminant releases into the environment. The relevance of these benchmarks to national-scale applications of the 3MRA modeling system is a major advantage of this module. Although a variety of different benchmarks could be used by the module to calculate ecological hazard, the benchmark database ensures that the resulting HQs will be highly relevant to establishing exit levels that are protective of the environment. The rationale, confidence levels, and data quality objectives are transparent for ecological benchmarks that are incorporated into the 3MRA modeling system benchmark database.

- **Benchmarks are adjusted for site-specific conditions as appropriate.** For certain benchmarks and chemicals, environmental characteristics exert a strong influence on toxicity. For example, the water quality CSCL for certain metals is sensitive to water hardness. For each simulation, the surface water CSCL is

[6] Although the Ecological Risk Module reports hazard quotients for all three distance intervals, EPA has determined that distance to WMU is of limited value in characterizing ecological risks. The vast majority of habitats delineated for sites of interest overlap the 1 km distance from the WMU. As a result, it would be double counting to report these risks for both the 1 km and 1-2 km distance intervals.

adjusted for the site-specific water hardness to better represent the potential toxicity given the specific environmental setting. Similarly, the sediment CSCL for organic chemicals is adjusted for the site-specific organic carbon fraction in the sediment. The ability to modify benchmarks during a simulation allows the module to be applied to virtually any site, and adjusts the toxicity according to the site conditions.

- **HQs retain attributes critical to characterizing the nature of ecological risk estimates.** Each HQ calculated by the module retains all of its attributes such as the receptor group (e.g., mammals) and habitat type (e.g., forest). In addition, the Ecological Risk Module reports the maximum HQ calculated at each site (along with its attributes) to allow for a rapid screening function or coarse ranking of sites. This information on HQ attributes is critical to allow decision makers to evaluate the results using different kinds of ecological indicators. For example, it may be desirable to evaluate concentration levels specifically for wetlands, because of their ecological importance. Similarly, the user may want to understand the nature of the risk estimates with regard to the receptors most likely at risk, or the habitats most likely at risk. Consequently, the module reports this information to retain flexibility in the characterization of predicted ecological risks.

- **Binning HQ greatly reduces the volume of data.** Although the module calculates HQs for each receptor at a site, these HQs are binned into discrete ecological hazard bins. Because these bins are developed across each of the possible attributes (e.g., trophic level, receptor group), the binning provides a very efficient method of storing and manipulating these data without losing any of the relevant information. This approach offers a very practical solution to the potential storage issues associated with storing time series of HQs for dozens of receptors across hundreds of sites. Moreover, the use of hazard bins is a very appropriate technique to characterize ecological risk, given the current state-of-the-science for national-scale analyses.

16.3.2 Uncertainty and Limitations

The methodology implemented by the Ecological Risk Module to characterize the potential for adverse ecological effects carries certain assumptions and limitations, and acknowledges several important sources of uncertainty. A number of the limitations and uncertainties discussed in Section 15.3 on the Ecological Exposure Module also apply to the Ecological Risk Module. Although parts of this section are redundant with the discussion in Section 15.3, the primary focus of the following discussion is on the implications for risk estimation and characterization for all ecological receptors, not just those receptors for which ingestion exposures were calculated.

- **The HQ estimates are based exclusively on an annual time step; that is, only annual average applied doses and media concentrations are used in calculating the HQ.** The ecological HQ estimates are based on annual averages. This time step represents much longer-than-lifetime exposures for some receptors and substantially less-than-lifetime for other receptors. As indicated in Section 15.3, concentration spikes due to episodic events (e.g., rain storms) or elevated WMU source releases following waste additions are not evaluated. Even though the measures of effect often reflect critical life stage endpoints, the use of annual average doses and concentrations to predict risk may not adequately reflect effects associated with elevated, short-term exposures. This "one-time-step-fits-all" approach, although necessary to address run time issues, does limit the applicability of the HQ estimates for certain receptors.

- **The HQs are not calculated at the population or community level; ecological risks must be inferred to higher levels of biological organization**. Ecosystems are enormously complex, and our understanding of even simple community dynamics is limited. Data on chemical stressors are seldom available above the level of an individual organism; that is, the study endpoints focus on individual organisms rather than processes crucial to assemblages of organisms. Although the measures of effect are relevant to higher levels of organization, the comparison of point estimates of simulated contaminant doses or concentrations for individual receptors with ecological benchmarks does not support the quantification of population or community risks. Our ability to address community-level effects associated with contaminant releases into the environment is even more limited; the CSCLs developed to evaluate risks to communities are derived by statistical inference based on toxicity data for individual organisms. This limitation in the data and technology (i.e., the inability to run population-level models in the 3MRA modeling system) introduces significant uncertainty in the risk estimates.

- **The effects of multiple stressors (chemical and nonchemical) are not considered in developing estimates of potential ecological risk**. Given the design goals for the 3MRA modeling system (i.e., to support national-level assessment strategies of WMUs, waste streams, and contaminants), exposure to multiple contaminants, contaminant releases outside of the AOI, and effects associated with multiple stressors are not considered. In addition, background concentrations of constituents were not considered in developing exposure estimates, nor were other potential nonchemical stressors such as habitat fragmentation. Data availability on the antagonistic and synergistic effects associated with multiple stressors are extremely limited at this time (with the possible exception of narcotic contaminants in aqueous systems), and prevented the development of a multistressor analytical approach for the universe of constituents in the current data set. Data limitations notwithstanding, the inability to consider multiple stressors is a limitation in our ability to interpret the risk results generated by this module. This is a significant but unquantifiable uncertainty inherent in the HQ calculations.

- **The HQ estimates reflect different endpoints at varying levels of effect**. The ecological benchmarks address a variety of receptors (e.g., soil fauna, mammals, plants) and, because the quality and quantity of relevant data vary widely across receptors, the HQs generated by the Ecological Risk Module represent different levels of knowledge regarding the exposure and toxicity of chemical stressors. The variability in supporting data suggests that the level of confidence in the exemption criteria is dependent on the quantity and quality of available data. The HQ methodology—the ratio of an exposure to a benchmark—is applied uniformly across all ecological receptors. However, the data supporting the HQ calculation vary in that they include endpoints from lethality to reproductive fitness and address population- and community-level effects by inference. To some extent, the HQ estimates for different receptor groups represent different risk metrics. The interpretation of these HQ estimates is, therefore, limited by our understanding of the potential ecological significance of the measures of effect as well as overall confidence in the data used to support the calculations.

- **The data used to support the development of sediment CSCLs for metals have come under significant scrutiny because of concerns regarding the applicability of the tests to predict sediment effects.** The sediment CSCLs for metals are ultimately based on data from the National Oceanic and Atmospheric Administration's (NOAA) National Status and Trends (NS&T) data. The NOAA data generally rely on observed effects to sediment biota exposed to sediment samples. The sediment samples are contaminated with mixtures of metals and, as a result, the observed effects cannot be definitively attributed to a specific metal. Thus, the effects levels reported for specific metals may in fact reflect effects of a mixture of metals. HQs based on these sediment CSCLs may overestimate the potential for adverse effects to the sediment community.

- **The HQs calculated for the aquatic and benthic communities are resolved at the habitat, rather than reach level**. There is some uncertainty associated with calculating risks to aquatic life across an entire habitat (as defined within the study area). Species of fish such as brown trout tend to use certain segments of stream habitats; therefore, HQs at the reach level may be more appropriate. Conversely, establishing artificial boundaries between stream reaches is contrary to the goals of the assessment strategy, namely, to evaluate ecological risks using the habitat as the fundamental unit. In short, the spatial scale adopted for evaluating freshwater communities of water column and sediment biota introduces uncertainty in HQ for these receptors. Risks may be underestimated for stream reaches with elevated contaminant concentrations; conversely, risks may be overestimated for a stream habitat with only a single stream reach above levels of concern.

16.4 References

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